

A Cheatsheet for Algebraic Number Theory

Contents

1	Rings and Fields	5
1.1	DVR and Dedekind Domains	5
1.1.1	Absolute Values and Discrete Valuation Ring	5
1.1.2	Dedekind Domains	10
2	Extensions	19
2.1	Algebraic Extensions	19
2.1.1	Etale Algebra	19
2.1.2	Dedekind Extensions	28
2.1.3	Galois Extensions	44
2.2	Extensions of Complete DVR	50
2.2.1	Completions	50
2.2.2	Extensions of Complete DVRs-Unramified	64
2.2.3	Extensions of Complete DVRs-Totally Ramified	66
2.2.4	Ramification of the Extensions of the Complete DVRs	75
3	Global Fields	83
3.1	Global Fields and Primes	83
3.1.1	Global Fields	83
3.1.2	Number Fields	88
4	Local Class Field Theory	101
4.1	Artin Map and Local Class Field Theory	101
4.1.1	The Prime Number Theorem	101
4.1.2	Dirichlet Theorem	105
4.1.3	Kronecker-Weber Theorem	116
4.2	Cohomologies and Local Class Field Theory	119
4.2.1	Cohomologies	119
4.2.2	Tate Cohomologies	129
4.3	Local Class Field Theory	131
4.3.1	Direct Construction of LCFT	131

4.3.2	The Existence Theorem and LCFT	136
4.3.3	Brauer Group	139
4.4	Local Class Field Theory and Global Class Field Theory	139
4.4.1	Hilbert Symbols and Quadratic Extensions	139
4.4.2	The First Inequality	142
4.4.3	The Second Inequality	142
A	Representation Theory	143
A.1	Abstract Representation Theory	143
A.1.1	Semisimple Rings	143
A.1.2	The Representation Ring	153
A.1.3	Induction and Frobenius Reciprocity	154
A.2	Finite Group Representation	159
A.2.1	Introduction	159
A.2.2	Characters in $\text{char}(k) = 0$	161
A.2.3	Characters in $\text{char}(k) = p$	171
A.3	Compact Group Representation	182

Chapter 1

Rings and Fields

1.1 DVR and Dedekind Domains

1.1.1 Absolute Values and Discrete Valuation Ring

Definition 1.1.1. An absolute value on a field k is a map $|\cdot| : k \rightarrow \mathbb{R}_+$ s.t. for all $x, y \in k$,

1. $|x| = 0$ iff $x = 0$
2. $|xy| = |x||y|$
3. $|x + y| \leq |x| + |y|$

If $|x + y| \leq \max(|x|, |y|)$ also holds, then the absolute value is nonarchimedean; otherwise it is archimedean.

e.g. The trivial absolute value on k is the map $|\cdot| : k \rightarrow \mathbb{R}_+$ s.t. $|x| = \begin{cases} 1 & x \neq 0 \\ 0 & x = 0 \end{cases}$.

Lemma 1.1.1. An absolute value $|\cdot|$ on a field k is nonarchimedean iff $|1 + \dots + 1| \leq 1, \forall n \geq 1$.

pf. If $|\cdot|$ is nonarchimedean, $|1 + \dots + 1| \leq \max(|1 + \dots + 1|, |1|) \leq 1$ by the inductive hypothesis. Hence for all $n \geq 1$, $|1 + \dots + 1| \leq 1$.

If $|\cdot|$ is not nonarchimedean, $\exists a, b$ s.t. $|a + b| > \max(|a|, |b|)$. Then $|1 + a/b| > 1$ so the inequality does not hold for all $n \geq 1$.

Corollary 1.1.1. In a field of positive characteristic every absolute value is nonarchimedean, and the only absolute value on a finite field is the trivial one.

Definition 1.1.2. Two absolute values $|\cdot|, |\cdot|'$ on the same field k are equivalent if there exists $\alpha \in \mathbb{R}_{++}$ for which $|x|' = |x|^\alpha, \forall x \in k$.

Definition 1.1.3. For a prime p , the p -adic valuation $v_p : \mathbb{C} \rightarrow \mathbb{Z}$ is defined by $v_p(\pm \Pi_q q^{e_q}) \equiv e_q$ and $v_p(0) \equiv \infty$. The p -adic absolute value on $\mathbb{Q}$ is defined by $|x|_p \equiv p^{-v_p(x)}$ where $|0|_p = p^{-\infty} = 0$.

Theorem 1.1.1 (Ostrowski's Theorem). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_p$ for some $p \leq \infty$.

pf. Let $|\cdot|$ be any nontrivial absolute value on $\mathbb{Q}$. Since it is nontrivial, $\exists p \in \mathbb{Z}$ prime s.t. $|p| \neq 1$. Then by the definition, given $|p| = c, |p^n| = |p|^n = c^n$. Then $|\cdot| = (|\cdot|_p)^\alpha$ s.t. $\alpha = -\log_p c \in \mathbb{R}_{++}$.

Theorem 1.1.2 (Product Formula). For every $x \in \mathbb{Q}^\times$, we have $\prod_{p \leq \infty} |x|_p = 1$.

pf. $\prod |x|$ over all kinds of absolute value on $\mathbb{Q}$ is equivalent to $\prod_{p \leq \infty} |x|_p^{\sum_{r \in \mathbb{R}_{++}} r}$ by the Ostrowski's theorem. Then $\prod_{p \leq \infty} |x|_p = (\text{products of all absolute values on } \mathbb{Q} \text{ for } x)^{-\sum_{r \in \mathbb{R}_{++}} r} \rightarrow 1$

Definition 1.1.4. A valuation on a field k is a group homomorphism $k^\times \rightarrow \mathbb{R}$ s.t. for all $x, y \in k$, we have $v(x+y) \geq \min(v(x), v(y))$. We may extend v to a map $k \rightarrow \mathbb{R} \cup \{\infty\}$ by defining $v(0) \equiv \infty$. For any $0 < c < 1$, defining $|x|_v \equiv c^{v(x)}$ yields a nonarchimedean absolute value. The image of v in $\mathbb{R}$ is the value group of v . We say that v is a discrete valuation if its value group is equal to $\mathbb{Z}$. Given a field k with valuation v , the set $A \equiv \{x \in k : v(x) \geq 0\}$ is the valuation ring of k . A discrete valuation ring (DVR) is an integral domain that is the valuation ring of its fraction field w.r.t. a discrete valuation. Such a ring A cannot be a field since $v(\text{Frac } A) = \mathbb{Z} \neq \mathbb{Z}_+ = v(A)$.

$x \in k^\times, v(1/x) = v(1) - v(x) = -v(x)$ so at least one of x and $1/x$ has nonnegative valuation and lies in A . Thus $x \in A$ is invertible iff $v(x) = 0 \leftrightarrow A^\times = \{x \in k : v(x) = 0\}$.

Definition 1.1.5. A valuation ring is an integral domain A with fraction field k with the property that $\forall x \in k$, either $x \in A$ or $x^{-1} \in A$.

For a valuation ring A , $\exists \pi \in A$ s.t. $v(\pi) = 1$ called a uniformiser (since $v(A) = \mathbb{Z}_+$). If we fix a uniformiser π , $\forall x \in k^\times, x = u\pi^n$ for unique u, n , i.e. A is a UFD. Every nonzero ideal of A is equal to $(\pi^n) = \{a \in A : v(a) \geq n\}$ for some integer $n \geq 0$, i.e. A is a PID. Thus ideals of A are totally ordered and the ideal $\mathfrak{m} = (\pi) = \{a \in A : v(a) > 0\}$ is the unique maximal ideal of A .

Definition 1.1.6. A local ring is a commutative ring with a unique maximal ideal. The residue field of a local ring A with maximal ideal $\mathfrak{m}$ is the field $A/\mathfrak{m}$.

e.g.

- For the p -adic valuation $v_p: \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$, $\mathbb{Z}_{(p)} \equiv \{\frac{a}{b} : a, b \in \mathbb{Z}, p \nmid b\}$ is the valuation ring with maximal ideal $\mathfrak{m} = (p)$. The residue field $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{Z}/p\mathbb{Z} \cong \mathbb{F}_p$.
- For any field k , the valuation $v: k((t)) \rightarrow \mathbb{Z} \cup \{\infty\}$ on the field of Laurent series over k defined by $v(\sum_{n \geq n_0} a_n t^n) = n_0$ where $a_{n_0} \neq 0$ has valuation ring $k[[t]]$.

Theorem 1.1.3. *For an integral domain A , TFAE*

- A is a DVR
- A is a noetherian valuation ring that is not a field.
- A is a local PID that is not a field
- A is an integrally closed noetherian ring of dimension 1.
- A is a regular noetherian local ring of dimension 1.
- A is a noetherian local ring whose maximal ideal is nonzero and principal.
- A is a maximal noetherian ring of dimension 1.

Definition 1.1.7. *Given a ring extension $A \subseteq B$, an element $b \in B$ is integral over A if it is a root of a monic polynomial in $A[x]$. The ring B is integral over A if all of its elements are integral over A .*

Proposition 1.1.1. *Let $\alpha, \beta \in B$ be integral over $A \subseteq B$. Then so are $\alpha + \beta, \alpha\beta$.*

pf. Let $f \in A[x]$ $f(x) = a_0 + a_1x + \dots + x^m$.
 $g \in A[y]$ $g(y) = b_0 + b_1y + \dots + y^n$. s.t. $f(\alpha) = g(\beta) = 0$. Consider the case
 $A = \mathbb{Z}[a_0, \dots, a_{m-1}, b_0, \dots, b_{n-1}]$, $B = \frac{A[x, y]}{(f, g)}$ with α, β equal to the images of x and y in B respectively. Let k be the algebraic closure of the fraction field A , and let $\alpha_1, \dots, \alpha_m$ be the roots of f in k and $\beta_1, \dots, \beta_n$ be the roots of g in k . The polynomial $h(z) = \prod_{i,j} (z - (\alpha_i + \beta_j))$ has coefficients that may be expressed as polynomials in the symmetric functions of the α_i, β_j equivalently the coefficients a_i, b_j of f and g respectively. Thus $h \in A[z]$ and $h(x + y) = 0$ so $x + y$ are integral over A . Similarly for $h(z) = \prod_{i,j} (z - \alpha_i \beta_j)$, xy can be shown to be integral over A . Then any $A' \subseteq B'$ with homomorphisms $A \rightarrow A', B \rightarrow B'$ defined by $a_i \mapsto a_i, b_i \mapsto b_i, x \mapsto \alpha, y \mapsto \beta$, if $x + y, xy \in B$ are integral over A , then so are $\alpha + \beta, \alpha\beta \in B'$. Thus the proposition follows.

Definition 1.1.8. *Given a ring extension B/A , the ring $\tilde{A} = \{b \in B : b \text{ is integral over } A\}$ is the integral closure of A in B . When $\tilde{A} = A$, A is said to be integrally closed in B . For a domain A , its integral closure is its integral closure in its fraction field and A is integrally closed if it is integrally closed in a fraction field.*

Proposition 1.1.2. *If $C/B/A$ is a tower of ring extensions in which B is integral over A and C is integral over B , then C is integral over A .*

Corollary 1.1.2. *If B/A is a ring extension, then the integral closure of A in B is integrally closed in B .*

Proposition 1.1.3. *The ring $\mathbb{Z}$ is integrally closed.*

pf. Let $r/s \in \mathbb{Q}$ be a root of a polynomial $f \in \mathbb{Z}[x]$ s.t. $(r, s) = 1$. Then $(r/s)^n + a_{n-1}(r/s)^{n-1} + \dots + a_0 = 0 \rightarrow r^n + a_{n-1}sr^{n-1} + \dots + a_0s^n = 0 \leftrightarrow r^n = -s(a_{n-1}r^{n-1} + \dots + a_0s^{n-1})$, which implies $s|r$. But they are coprime, so $s = \pm 1$ and therefore $r/s \in \mathbb{Z}$.

Corollary 1.1.3. *Every UFD is integrally closed. In particular every PID is integrally closed.*

e.g. Consider the ring $\mathbb{Z}[\sqrt{5}]$. Then $x^2 - x - 1 \in \mathbb{Z}[\sqrt{5}][x]$ has a root $\phi = \frac{1+\sqrt{5}}{2} \in \text{Frac}\mathbb{Z}[\sqrt{5}]$ but not in $\mathbb{Z}[\sqrt{5}]$ so it is not integrally closed and therefore it is not a UFD.

Proposition 1.1.4. *Every valuation ring is integrally closed.*

pf. Let A be a valuation ring with fraction field k and let $\alpha \in k$ be integral over A . Then $\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0, a_1 \dots a_{n-1} \in A$. If $\alpha \notin A, \alpha^{-1} \in A$ since A is a valuation ring. Multiplying the equation by $\alpha^{-(n-1)} \in A, \alpha = -a_{n-1} - a_{n-1}\alpha^{-1} - \dots - a_0\alpha^{1-n} \in A$, a contradiction. Thus A is integrally closed.

Definition 1.1.9. *A number field K is a finite extension of $\mathbb{Q}$. The ring of integers $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K .*

Proposition 1.1.5. *Let A be an integrally closed domain with fraction field K . Let α be an element of a finite extension L/K , and let $f \in K[x]$ be its minimal polynomial over K . Then α is integral over A iff $f \in A[x]$.*

pf. If $f \in A[x]$, α is integral over A . Suppose α is integral over A and let $g \in A[x]$ be a monic polynomial s.t. $g(\alpha) = 0$. In $\bar{K}[x]$, we may factor $f(x) = \prod (x - \alpha_i)$. For each α_i , we have a field embedding $K(\alpha) \rightarrow \bar{K}$ that sends α to α_i and fixes K . As elements of $\bar{K}$, we have $g(\alpha_i) = 0$, so $\forall \alpha_i \in \bar{K}$ is integral over A and lies in the integral closure $\tilde{A}$ of A in $\bar{K}$. Each coefficient of $f \in K[x]$ can be expressed as a sum of products of the α_i and is therefore an element of the ring $\tilde{A}$ that also lies in K . But $A = \tilde{A} \cap K$, since A is integrally closed in its fraction field K .

- Localisation of Rings

Let A be a commutative ring and let S be a multiplicative subset of A . The **localisation** of A w.r.t. S is a ring $S^{-1}A$ equipped with a ring homomorphism $\iota : A \rightarrow S^{-1}A$ that maps S into $(S^{-1}A)^\times$ and satisfied the universal

property :
$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ & \searrow \iota & \nearrow \exists! \\ & S^{-1}A & \end{array}$$
 if $\phi : A \rightarrow B$ is a ring homomorphism

with $\phi(S) \subseteq B^\times$, then there is a unique ring homomorphism $S^{-1}A \rightarrow B$ that makes the diagram commutes.

→ Construct $S^{-1}A$ as the quotient of $A \times S$ modulo the equivalence relation $\sim$ s.t.

$(a, s) \sim (b, t) \leftrightarrow \exists u \in S$ s.t. $(at - sb)u = 0$. Then use a/s to denote the equivalence class of (a, s) and define $\iota(a) \equiv a/1$. If $\phi : A \rightarrow B$ is a ring homomorphism with $\phi(S) \subseteq B^\times$. $\phi = \pi \circ \iota$ where π is defined by $\pi(a/s) \equiv \phi(a)\phi(s)^{-1}$. Then such homomorphism $\phi(a)\phi(s)^{-1} = \pi(\iota(a))\pi(\iota(s))^{-1} = \pi(\iota(a)\iota(s)^{-1}) = \pi((a/1)(1/s)) = \pi(a/s)$ so π is unique.

If A is an integral domain, the map $\iota : A \rightarrow S^{-1}A$ is injective, allowing to identify A with its image $\iota(A) \subseteq S^{-1}A$. So we can view $S^{-1}A$ as an intermediate ring that lies between A and its fraction field $\text{Frac}A$.

If $\phi : A \rightarrow B$ is a ring homomorphism and $\mathfrak{b}$ is a B -ideal, then $\phi^{-1}(\mathfrak{b})$ is an A -ideal called the contraction of $\mathfrak{b}$ to A , denoted $\mathfrak{b}^c$. If A is a subring of B and ϕ is an inclusion map, then $\mathfrak{b}^c = \mathfrak{b} \cap A$.

If $\mathfrak{a}$ is an A -ideal, the B -ideal generated by $\phi(\mathfrak{a})$ is the extension of $\mathfrak{a}$ to B , denoted $\mathfrak{a}^e$.

→ if A is an integral domain, and $B = S^{-1}A$ and if $a_1, \dots, a_n \in A$ generate $\mathfrak{a}$ as an A -ideal, then they also generate $\mathfrak{a}^e = \mathfrak{a}B$ as a B -ideal since $\mathfrak{a}^e = \mathfrak{a}B \equiv \{ab : a \in \mathfrak{a}, b \in B\}$.

→ **If A is a noetherian ring/PID, then so is its localisation.**

Theorem 1.1.4. *Let S be a multiplicative subset of an integral domain A . There is a one-to-one correspondence between the prime ideals of $S^{-1}A$ and the prime ideals of A that do not intersect S given by the inverse map $\mathfrak{q} \mapsto \mathfrak{q} \cap A, \mathfrak{p} \mapsto \mathfrak{p}S^{-1}A$.*

e.g.

- For a field k , let $A = k[x]$ and $\mathfrak{p} = (x - 2)$. Then $A_{\mathfrak{p}} = \{f \in k(x) : f \text{ is defined at } 2\}$. Since A is a PID, so is $A_{\mathfrak{p}}$ with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}} = \{f \in k(x) : f(2) = 0\}$, hence it is a DVR. The residue field is $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong k$.

- Let $p \in \mathbb{Z}$, $\mathbb{Z}_{(p)} = \{a/b : a, b \in \mathbb{Z}, p \nmid b\}$. Since $\mathbb{Z}$ is a PID so is $\mathbb{Z}_{(p)}$ and therefore it is a DVR. The residue field is $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{F}_p$.

- Localisation of Modules

Let A be a commutative ring, let S be a multiplicative subset of A , and let M be an A -module. Then the **localisation** $S^{-1}M$ of M w.r.t. S is an $S^{-1}A$ -module equipped with an A -module homomorphism $\iota : M \rightarrow S^{-1}M$ with the universal property: If N is an $S^{-1}A$ -module and $\phi : M \rightarrow N$ is an A -module homomorphism, then ϕ factors uniquely through $S^{-1}M$.

→ Construct $S^{-1}M$ as the quotient of the $S^{-1}A$ -module $M \times S$ modulo the same equivalence relation above. Or $S^{-1}M$ can be constructed as the base change of M from A to $S^{-1}A$, i.e. $S^{-1}M \equiv M \otimes S^{-1}A$.

If A lies in a field K and M is an A -module that is contained in a K -vector space, $\forall s \in A, s \neq 0$ is injective and we can view M as an A -submodule of any of its localisations $S^{-1}M$.

Proposition 1.1.6. *Let A be a subring of a field K , and let M be an A -module contained in a K -vector space V . Then $M = \cap_{\mathbf{m}} M_{\mathbf{m}} = \cap_{\mathbf{p}} M_{\mathbf{p}}$ where $\mathbf{m}$ ranges over the maximal ideals of A , $\mathbf{p}$ ranges over the prime ideals of A and the intersection takes place in V .*

pf. $M \subseteq \cap_{\mathbf{m}} M_{\mathbf{m}}$ is immediate.

Suppose $x \in \cap_{\mathbf{m}} M_{\mathbf{m}}$ and consider the A -ideal $\mathbf{a} = \{a \in A : ax \in M\}$. For each maximal ideal $\mathbf{m}$, $x = m/s$ for some $m \in M, s \in A \setminus \mathbf{m}$. Then $sx \in M, s \in \mathbf{a}$ and $s \notin \mathbf{m}$ so $\mathbf{a} \not\subseteq \mathbf{m}$. Thus $\mathbf{a}$ must be a unit ideal, i.e. $1 \in \mathbf{a}$ and $x \cdot 1 = x \in M \rightarrow \cap_{\mathbf{m}} M_{\mathbf{m}} \subseteq M$.

Since maximals are primes and for all prime ideal is contained in maximal ideals, $\cap_{\mathbf{m}} M_{\mathbf{m}} = \cap_{\mathbf{p}} M_{\mathbf{p}}$.

→ When $K = \text{Frac} A, V = K$, M is an A -submodule of K . Every ideal I of A is an A -submodule of K , and can be localised as above. Thus the localisation of I at a prime ideal $\mathbf{p}$ of A is the same thing as the extension of I to the localisation of A at $\mathbf{p} \rightarrow I_{\mathbf{p}} = \{i/s : i \in I, s \in A \setminus \mathbf{p}\} = \{ia/s : i \in I, s \in A \setminus \mathbf{p}\} = IA_{\mathbf{p}}$,

Corollary 1.1.4. *Let A be an integral domain. Every ideal I of A is equal to the intersection of its localisations at the maximal ideals of A , and also to the intersection of its localisations at the prime ideals of A .*

1.1.2 Dedekind Domains

Proposition 1.1.7. *Let A be a noetherian domain. TFAE*

1. For every nonzero prime ideal $\mathfrak{p} \subset A$, the local ring $A_{\mathfrak{p}}$ is a DVR.
2. The ring A is integrally closed and $\dim A \leq 1$.

pf. If A is a field, both 1 and 2 hold, so assume A is not a field, and $K = \text{Frac} A$. $\dim A$ is the supremum of the length of all chains of prime ideals. Thus by the first theorem, every chain of prime ideals $(0) \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_l$ extends to a corresponding chain in $A_{\mathfrak{p}}$ of the same length. Conversely, every chain in $A_{\mathfrak{p}}$ contracts to a chain in A of the same length. Thus $\dim A = \sup\{\dim A_{\mathfrak{p}} : \mathfrak{p} \in \text{Spec} A\} = 1$, since every $A_{\mathfrak{p}}$ is a DVR or a field, in which case $\dim A_{\mathfrak{p}} = 1$ and $\dim A_{\mathfrak{p}} = 0$ respectively. Any $x \in K$ that is integral over A is integral over every $A_{\mathfrak{p}}$, since they all contain A , and $A_{\mathfrak{p}}$ is integrally closed since they are DVRs or fields. So $x \in \bigcap_{\mathfrak{p}} A_{\mathfrak{p}} = A$ and therefore A is integrally closed.

Suppose $x \in K$ is integral over $A_{\mathfrak{p}}$. Then $x^n + \frac{a_{n-1}}{s_{n-1}}x^{n-1} + \cdots + \frac{a_0}{s_0} = 0$ for some $a_0, \dots, a_{n-1} \in A, s_0, \dots, s_{n-1} \in A \setminus \mathfrak{p}$. Define $s = s_0 \cdots s_{n-1} \in S$. Multiplying both sides by s^n shows that sx is integral over A , hence $sx \in A$, since A is integrally closed. But $sx/s = x \in A_{\mathfrak{p}}$ so $A_{\mathfrak{p}}$ is integrally closed, given A is integrally closed. Also noetherian property, dimension, and reduced property are inherited from A . Thus $A_{\mathfrak{p}}$ is an integrally closed noetherian local domain of dimension at most 1. Thus it is a DVR.

Definition 1.1.10. A noetherian domain satisfying either of the equivalent properties of the previous proposition is called a Dedekind domain.

Corollary 1.1.5. Every PID is a Dedekind domain. In particular $\mathbb{Z}$ is a Dedekind domain, as is $k[x]$ for any field k .

Definition 1.1.11. A fractional ideal of a noetherian domain A is finitely generated A -submodule of its fraction field.

Lemma 1.1.2. Let A be a noetherian domain with fraction field K , and let $I \subseteq K$ be an A -module. Then I is finitely generated iff $aI \subseteq A$ for some nonzero $a \in A$.

pf. If $r_1/s_1, \dots, r_n/s_n$ generate I as an A -module, then $\exists a = s_1 \cdots s_n \in A$ s.t. $aI \subseteq A$.

If $aI \subseteq A$, then aI is an ideal of A , hence finitely generated, given A is noetherian.

If $a_1, \dots, a_n$ generate aI as an ideal, $a_1/a, \dots, a_n/a$ generate I .

→fractional ideals are A -submodule of K for which there exists a nonzero $r \in A$ s.t. $rI \subseteq A$ even when A is not noetherian.

Corollary 1.1.6. Every fractional ideal of A can be written in the form $\frac{1}{a}I$ for some nonzero $a \in A$ and the ideal I .

Definition 1.1.12. A fractional ideal of A is principal if it is generated by one element, i.e. it has the form xA for some $x \in A$.

Ideal Operations

- $I + J \equiv (i + j : i \in I, j \in J)$
- $IJ \equiv (ij : i \in I, j \in J)$
- $(I : J) \equiv \{x \in K : xJ \subseteq I\}$

Lemma 1.1.3. *Let I and J be fractional ideals of a noetherian domain A with $J \neq 0$. Then $(I : J)$ is a fractional ideal of A .*

pf. Suppose $I, J \subseteq A$ are ideals. Then for any nonzero $j \in J$, $j(I : J) \subseteq I \subseteq A$ so $(I : J)$ is finitely generated by the previous lemma. If I, J are not ideals, pick $a, b \in A$ s.t. $aI \subseteq A, bJ \subseteq A$ by the previous lemma. Then $(I : J) = (abI : abJ), abI, abJ \subseteq A$. Then by the previous argument it is finitely generated. Thus $(I : J)$ is an A -submodule of the fraction field of A s.t. it is finitely generated. So the lemma follows.

Definition 1.1.13. *A fractional ideal I is invertible if $IJ = A$ for some fractional ideal J .*

Lemma 1.1.4. *A fractional ideal I of A is invertible iff $I(A : I) = A$.*

pf. Suppose I is invertible with $IJ = A$. Then $jI \subseteq A$ for all $j \in J$ so $J \subseteq (A : I)$. Also $A = IJ \subseteq I(A : I) \subseteq A$ thus $I(A : I) = A$.
If $I(A : I) = A$, by the previous lemma $(A : I)$ is a fractional ideal, it is invertible.

Definition 1.1.14. *The ideal group $\mathcal{I}_A$ of a noetherian domain A is the group of invertible fractional ideals.*

→ every nonzero fractional ideal (x) is invertible, since $(x)^{-1} = (x^{-1})$. The set of nonzero principal ideals $\mathcal{P}_A$ is a subgroup of $\mathcal{I}_A$.

Definition 1.1.15. *Let A be a noetherian domain. The quotient $cl(A) \equiv \mathcal{I}_A / \mathcal{P}_A$ is the ideal class group of A ; it is also called the Picard group of A and denoted $Pic(A)$.*

e.g. If A is a DVR with uniformizer π , then its nonzero fractional ideals are the principal fractional ideals $(\pi^n), n \in \mathbb{Z}$. Also $(\pi^m)(\pi^n) = (\pi^{m+n})$. Thus the ideal group of A is isomorphic to $\mathbb{Z}$ and $\mathcal{P}_A = \mathcal{I}_A \rightarrow cl(A)$ is trivial.

Lemma 1.1.5. *Let I, J be fractional ideals of a noetherian domain A , and let $\mathfrak{p}$ be a prime ideal of A . Then $I_{\mathfrak{p}}, J_{\mathfrak{p}}$ are fractional ideals of $A_{\mathfrak{p}}$ as are $(I + J)_{\mathfrak{p}} = I_{\mathfrak{p}} + J_{\mathfrak{p}}, (IJ)_{\mathfrak{p}} = I_{\mathfrak{p}}J_{\mathfrak{p}}, (I : J)_{\mathfrak{p}} = (I_{\mathfrak{p}} : J_{\mathfrak{p}})$. The same applies if we localise w.r.t. any multiplicative subset S of A .*

pf. $I_{\mathbf{p}} = IA_{\mathbf{p}}$ is a finitely generated $A_{\mathbf{p}}$ -module, since I is a finitely generated A -module. Thus a fractional ideal of $A_{\mathbf{p}}$ and similarly for $J_{\mathbf{p}}$

$$(I + J)_{\mathbf{p}} = (I + J)A_{\mathbf{p}} = IA_{\mathbf{p}} + JA_{\mathbf{p}} = I_{\mathbf{p}} + J_{\mathbf{p}}$$

$$(IJ)_{\mathbf{p}} = (IJ)A_{\mathbf{p}} = I_{\mathbf{p}}J_{\mathbf{p}}$$

$$(I : J)_{\mathbf{p}} = \{x \in K : xJ \subseteq I\}_{\mathbf{p}} = \{x \in K : xJ_{\mathbf{p}} \subseteq I_{\mathbf{p}}\} = (I_{\mathbf{p}} : J_{\mathbf{p}}).$$

Theorem 1.1.5. *Let I be a fractional ideal of a noetherian domain A . Then I is invertible iff its localisation at every maximal ideal of A is invertible, equivalently, iff its localisation at every prime ideal of A is invertible.*

pf. Suppose I is invertible. Then $I(A : I) = A$ and for any maximal ideal $\mathbf{m}$, $I_{\mathbf{m}}(A_{\mathbf{m}} : I_{\mathbf{m}}) = A_{\mathbf{m}}$ by the previous lemma. Thus $I_{\mathbf{m}}$ is invertible.

Suppose $I_{\mathbf{m}}$ is invertible for every maximal ideal $\mathbf{m}$. Then $I_{\mathbf{m}}(A_{\mathbf{m}} : I_{\mathbf{m}}) = A_{\mathbf{m}}$ for every maximal $\mathbf{m}$. Then we get $I(A : I) = \cap_{\mathbf{m}} (I(A : I))_{\mathbf{m}} = \cap_{\mathbf{m}} I_{\mathbf{m}}(A_{\mathbf{m}} : I_{\mathbf{m}}) = \cap_{\mathbf{m}} A_{\mathbf{m}} = A$ by the previous lemma. So I is invertible.

Corollary 1.1.7. *In a Dedekind domain, every nonzero fractional ideal is invertible.*

pf. If A is a Dedekind domain, then all of its localisation at maximal ideals are DVRs, hence PIDs and in a PID every nonzero fractional ideal is invertible. Thus by the previous theorem every nonzero fractional ideal of A is invertible.

Lemma 1.1.6. *A nonzero fractional ideal I in a noetherian local domain A is invertible iff it is principal.*

pf. If I is principal, it is invertible.

Let I be an invertible fractional ideal, and let $\mathbf{m}$ be the maximal ideal of A . Then we have $II^{-1} = A$ so $\sum_i a_i b_i = 1, a_i \in I, b_i \in I^{-1}, a_i b_i \in II^{-1} = A$. Then one of the products $a_i b_i$ must be a unit, otherwise the sum would not be a unit. Wlog, $a_1 b_1$ is a unit. $\forall x \in I, a_1 b_1 x \in (a_1)$ since $b_1 x \in A$. Thus $x = (a_1 b_1)^{-1} a_1 b_1 x \in (a_1)$ since $(a_1 b_1)^{-1} \in A$. Thus $I \subseteq (a_1) \subseteq I$, which implies $I = (a_1)$ is principal.

Corollary 1.1.8. *A nonzero fractional ideal in a noetherian domain A is invertible iff it is locally principal, that is its localisation at every maximal ideal of A is principal.*

An integral domain s.t. every nonzero ideal is invertible is a Dedekind domain.

pf. Let $(a_1) \subsetneq (a_1, a_2) \subsetneq \dots$ be an ascending chain of ideals. Then there exists $(a_0)^{-1} \supsetneq (a_1, a_2)^{-1} \supsetneq \dots$ which implies it terminates, i.e. it is Noetherian. For any prime $\mathbf{p}$, $A_{\mathbf{p}}$ is a local noetherian domain, and therefore all the invertible ideal is principal, which implies $A_{\mathbf{p}}$ is a DVR. Hence A is a Dedekind domain.

Lemma 1.1.7. *Let A be a Dedekind domain and let $a \in A$ be nonzero. The set of prime ideals that contain a is finite.*

pf. Define $S \equiv \{I \in \mathcal{I}_A : (a) \subseteq I \subseteq A\}$, $T \equiv \{I \in \mathcal{I}_A : A \subseteq I \subseteq (a)^{-1}\}$, subsets of the ideal group $\mathcal{I}_A$. Then since $A \in S \cap T$, they are both nonempty and partially ordered by inclusion. The elements of S are all ideals and we have bijections $\phi_1 : S \rightarrow T$, $\phi_2 : T \rightarrow S$ with $I \mapsto I^{-1}$, $I \mapsto aI$ with ϕ_1 order-reversing and ϕ_2 order-preserving. The composition $\phi \equiv \phi_2 \circ \phi_1$ is thus an order-reversing permutation of S . Since A is noetherian, the set S satisfies the a.c.c. By applying the order-reversing permutation ϕ , S also satisfies d.c.c.. If a lies in infinitely many distinct prime ideals $\mathbf{p}_1, \mathbf{p}_2, \dots$, then $\mathbf{p}_1 \supseteq \mathbf{p}_1 \cap \mathbf{p}_2 \supseteq \dots$ is a descending chain of ideals of S that must stabilise. Thus for n sufficiently large, $\mathbf{p}_1 \dots \mathbf{p}_{n-1} \subseteq \mathbf{p}_1 \cap \dots \cap \mathbf{p}_{n-1} = \mathbf{p}_1 \cap \dots \mathbf{p}_n \subseteq \mathbf{p}_n$. The prime ideal $\mathbf{p}_n$ contains the product $\mathbf{p}_1 \dots \mathbf{p}_{n-1}$ so it must contain one of the factors of $\mathbf{p}_1 \dots \mathbf{p}_{n-1}$, which contradicts $\dim A \leq 1$.

Corollary 1.1.9. *Let I be a nonzero ideal of a Dedekind domain A . The number of prime ideals of A that contain I is finite.*

pf. Apply the previous lemma to all $a \in I$.

e.g. The Dedekind domain $A = \mathbb{C}[t]$ contains uncountably many nonzero prime $\mathbf{p}_r = (t - r)$, $r \in \mathbb{C}$. But any nonzero $f \in \mathbb{C}[t]$ lies in only finitely many of them, i.e. f has finitely many roots.

Let $\mathbf{p}$ be a nonzero prime ideal in a Dedekind domain A with fraction field K , let π be a uniformizer for the discrete valuation ring $A_{\mathbf{p}}$, and let I be a nonzero fractional ideal of A . The localisation $I_{\mathbf{p}}$ is a nonzero fractional ideal of $A_{\mathbf{p}}$, hence of the form (π^n) , $n \in \mathbb{Z}$ that does not depend on the choice of uniformizer. Extend the valuation $v_{\mathbf{p}} : K \rightarrow \mathbb{Z} \cup \{\infty\}$ to fractional ideals by defining $v_{\mathbf{p}}(I) \equiv n$, $v_{\mathbf{p}}((0)) \equiv \infty$. The map $v_{\mathbf{p}} : \mathcal{I}_A \rightarrow \mathbb{Z}$ is a group homomorphism (If $I_{\mathbf{p}} = (\pi^m)$, $J_{\mathbf{p}} = (\pi^n)$, then $(IJ)_{\mathbf{p}} = (\pi^m)(\pi^n) = (\pi^{m+n})$).

Lemma 1.1.8. *Let $\mathbf{p}$ be a nonzero prime ideal in a Dedekind domain A . If I is an ideal of A , then $v_{\mathbf{p}}(I) = 0$ iff $\mathbf{p}$ does not contain I . In particular, if $\mathbf{q}$ is any nonzero prime ideal different from $\mathbf{p}$, then $v_{\mathbf{q}}(\mathbf{p}) = v_{\mathbf{p}}(\mathbf{q}) = 0$.*

pf. If $I \subseteq \mathbf{p}$, $v_{\mathbf{p}}(I) \geq v_{\mathbf{p}}(\mathbf{p}) = 1$ is nonzero. If $I \not\subseteq \mathbf{p}$, $\exists a \in I \setminus \mathbf{p}$. Then $0 = v_{\mathbf{p}}(a) \geq v_{\mathbf{p}}(I) \geq v_{\mathbf{p}}(A) = 0$ since $(a) \subseteq I \subseteq A$. The prime ideals $\mathbf{p}, \mathbf{q}$ are nonzero, hence maximal, given $\dim A \leq 1$. Thus neither contains the other, so by the previous argument, the lemma follows.

Corollary 1.1.10. *Let A be a Dedekind domain with fraction field K . For each nonzero fractional ideal I , we have $v_{\mathbf{p}}(I) = 0$ for all but finitely many prime ideals $\mathbf{p}$. In particular, if $x \in K^\times$, $v_{\mathbf{p}}(x) = 0$ for all but finitely many $\mathbf{p}$.*

pf. For $I \subseteq A$, by the previous lemma and the previous corollary, the corollary follows. For $I \not\subseteq A$, $I = \frac{1}{a}J$, $a \in A$, J an A -ideal. Then $v_{\mathbf{p}}(I) = v_{\mathbf{p}}(J) - v_{\mathbf{p}}(a) = 0 - 0 = 0$ for all but finitely many prime ideals $\mathbf{p}$. This holds in particular for $I = (x)$, $\forall x \in K^\times$.

Theorem 1.1.6. *Let A be a Dedekind domain. The ideal group $\mathcal{I}_A$ of A is the free abelian group generated by its nonzero prime ideal $\mathbf{p}$. The isomorphism $\mathcal{I}_A \cong \oplus_{\mathbf{p}} \mathbb{Z}$ is given by the inverse maps*

$$\begin{array}{ccc} I & \mapsto & (\dots, v_{\mathbf{p}}(I), \dots) \\ \Pi_{\mathbf{p}} \mathbf{p}^{e_{\mathbf{p}}} & & (\dots, e_{\mathbf{p}}, \dots) \end{array}$$

pf. By the previous corollary, the first map is well-defined. For each nonzero prime ideal $\mathbf{p}$, the maps $I \mapsto v_{\mathbf{p}}(I)$ and $e_{\mathbf{p}} \mapsto \mathbf{p}^{e_{\mathbf{p}}}$ are group homomorphisms, and it follows that the maps in the theorem are both group homomorphisms.

If $v_{\mathbf{p}}(I) = v_{\mathbf{p}}(J)$, $I_{\mathbf{p}} = J_{\mathbf{p}}$. So if this holds for every prime $\mathbf{p}$, $I = \cap_{\mathbf{p}} I_{\mathbf{p}} = \cap_{\mathbf{p}} J_{\mathbf{p}} = J$. So the first map is injective.

By the previous lemma, for any vector $(\dots, e_{\mathbf{p}}, \dots)$ in the image, we have $v_{\mathbf{q}}(\Pi_{\mathbf{p}} \mathbf{p}^{e_{\mathbf{p}}}) = \sum_{\mathbf{p}} e_{\mathbf{p}} v_{\mathbf{q}}(\mathbf{p}) = e_{\mathbf{q}}$, which implies $\Pi_{\mathbf{p}} \mathbf{p}^{e_{\mathbf{p}}}$ is the pre-image of $(\dots, e_{\mathbf{p}}, \dots)$. Thus the first map is surjective and its inverse is the second map.

→ When A is a DVR, the isomorphism in the previous theorem becomes a discrete valuation map $v_{\mathbf{p}}: \mathcal{I}_A \rightarrow \mathbb{Z}$ where $\mathbf{p}$ is the unique maximal ideal of A .

Corollary 1.1.11. *In a Dedekind domain, every nonzero fractional ideal I has a unique factorisation $I = \Pi_{\mathbf{p}} \mathbf{p}^{v_{\mathbf{p}}(I)}$ into nonzero prime ideals $\mathbf{p}$.*

Every integral domain with UFD is a Dedekind domain

pf. Let A be any integral domain with UFD. Since every ideal is a finite product of prime ideals, a.c.c. holds and hence noetherian. If $\dim A > 1$, $\exists \mathbf{p}, \mathbf{q}$ s.t. $(0) \subsetneq \mathbf{p} \subsetneq \mathbf{q}$. Since $\mathbf{q}$ contains $\mathbf{p}$ and both are primes, $\mathbf{q} = \mathbf{p}$ a contradiction. Hence it is a Dedekind domain.

A Noetherian domain A with "to contain is to divide" is a Dedekind domain.

pf. If $\dim A > 1$, $\exists \mathbf{p}, \mathbf{q}$ s.t. $(0) \subsetneq \mathbf{p} \subsetneq \mathbf{q}$. Then $\mathbf{q} | \mathbf{p}$. Then $\exists a \in \mathbf{q} \setminus \mathbf{p}$ nonzero s.t. $\exists b \in A$ with $ab \in \mathbf{p}$ a contradiction since $\mathbf{p}$ is a prime. Hence $\dim A \leq 1$. Let x be integral over A . If $x \notin A$, $x = \frac{a}{s}$, $a, s \in A$, $(a, s) = 1$. Then there is a minimal polynomial for $x \rightarrow (a/s)^n + \alpha_{n-1}(a/s)^{n-1} + \dots + \alpha_0 = 0, \forall \alpha_i \in A$. By multiplying both sides by s^n and rearranging it, the above equation becomes $a^n = -s(\alpha_{n-1}a^{n-1} + \alpha_{n-2}a^{n-2}s + \dots + \alpha_0s^{n-1})$, which implies $s|a$. But $(s, a) = 1$ so $s = 1$ and $x \in A$ and it is integrally closed. Thus it is a Dedekind domain.

$x \in I \leftrightarrow (x) \subseteq I \leftrightarrow v_{\mathbf{p}}(x) \geq e_{\mathbf{p}}, \forall \mathbf{p}$ and therefore $I = \{x \in K : v_{\mathbf{p}}(x) \geq e_{\mathbf{p}}, \forall \mathbf{p}\}$.

Corollary 1.1.12. *A Dedekind domain is a UFD iff it is a PID, equivalently iff its class group is trivial.*

pf. Every PID is a UFD.

Let $\mathbf{p}$ be a nonzero prime ideal in a Dedekind domain A that is a UFD. Let $a \in \mathbf{p}$, nonzero, and let $a = p_1 \dots p_n$ be the unique factorisation of a into irreducible elements. Then $\mathbf{p}$ contains and therefore divides $(a) = (p_1) \dots (p_n)$ so $\mathbf{p}$ divides some (p_i) , which is necessarily a prime ideal. But A has dimension 1, so $\mathbf{p} = (p_i)$. Given the fact that every prime ideals are principal and that A is a UFD, A is a PID.

Definition 1.1.16. *Two A -ideals I, J are said to be relatively prime or coprime if $I + J = A$.*

Lemma 1.1.9. *Let A be a Dedekind domain and let I, I' be nonzero ideals. There exists an ideal J coprime to I' s.t. IJ is principal.*

pf. Let $\mathbf{p}_1, \dots, \mathbf{p}_n$ be the nonzero prime ideals dividing I' , which is finite by the previous corollary. For $1 \leq i \leq n$, define $\mathbf{a}_i \equiv \mathbf{p}_1 \dots \mathbf{p}_{i-1} \mathbf{p}_{i+1} \dots \mathbf{p}_n$ and choose $a_i \in I$ so that $a_i \in \mathbf{a}_i I, a_i \notin \mathbf{p}_i I$. Then $\mathbf{a}_i I \cap \mathbf{p}_i I \subsetneq \mathbf{a}_i I$ because $v_{\mathbf{p}_i}(\mathbf{a}_i I \cap \mathbf{p}_i I) = v_{\mathbf{p}_i}(\mathbf{p}_i I) > v_{\mathbf{p}_i} = v_{\mathbf{p}_i}(\mathbf{a}_i I)$ so such a_i exists. Each a_i is necessarily nonzero, and satisfies $v_{\mathbf{p}_i}(a_i) = v_{\mathbf{p}_i}(I)$ since $v_{\mathbf{p}_i}(a_i) \geq v_{\mathbf{p}_i}(\mathbf{a}_i I) = v_{\mathbf{p}_i}(I), v_{\mathbf{p}_i}(a_i) < v_{\mathbf{p}_i}(\mathbf{p}_i I) = v_{\mathbf{p}_i}(I) + 1$ and for $i \neq j, v_{\mathbf{p}_j}(a_i) \geq v_{\mathbf{p}_j}(\mathbf{p}_j I) > v_{\mathbf{p}_j}$. Define $a \equiv \sum_i a_i$, so that $v_{\mathbf{p}_i}(a) = v_{\mathbf{p}_i}(a_i) = v_{\mathbf{p}_i}(I), \forall i \in [1, n]$ by the nonarchimedean triangular equality. Thus $v_p(a) = v_p(I)$ for all prime $\mathbf{p} | I'$. $(a) \subseteq I$ and therefore it is divisible by I given A a Dedekind domain. So $(a) = IJ$ for some ideal J . For each prime $\mathbf{p} | I', v_p(J) = v_p(a) - v_p(I) = 0$ so J is coprime to I' and $IJ = (a)$.

NONARCHIMEDEAN TRIANGLE EQUALITY : if $|\cdot|$ is a nonarchimedean absolute value on a field k and $|x| \neq |y|$, then $|x + y| = \max(|x|, |y|)$.

pf. Assume the contrary. Then $|x + y| < \max(|x|, |y|) = |x|$ by assume wlog $\max(|x|, |y|) = |x|$. Then $1 < \frac{|x|}{|x + y|}$ since it is a field. Then $1 > \frac{|x + y|}{|x|} = 1 + \frac{|y|}{|x|} \leftrightarrow \frac{|y|}{|x|} < 0$ a contradiction. Hence the equality follows.

An integral domain, with the property that $\forall I$ ideals of $A, \exists J$ another ideal s.t. IJ is principal, is a Dedekind domain.

pf. $IJ = (a)$. Then since every principal ideal is invertible, $\exists (a)^{-1}$. Multiplying both sides gives $IJ(a)^{-1} = A$. Thus every nonzero ideal I in A is invertible. Hence it is a Dedekind domain.

Corollary 1.1.13 (Finite approximation). *Let I be a nonzero fractional ideal in a Dedekind domain A and let $\mathbf{p}_1, \dots, \mathbf{p}_n$ be a finite set of nonzero prime ideals of A . Then I contains an element x for which $v_{\mathbf{p}_i}(x) = v_{\mathbf{p}_i}(I)$ for $1 \leq i \leq n$.*

pf. Let $I = \frac{1}{s}J, s \in A, J$ an A ideal. Then we can pick $a \in J$ so that $v_{\mathbf{p}_i}(a) = v_{\mathbf{p}_i}(J)$ for all $i \in [1, n]$. If we let $x = a/s$, $v_{\mathbf{p}_i}(x) = v_{\mathbf{p}_i}(a) - v_{\mathbf{p}_i}(s) = v_{\mathbf{p}_i}(J) - v_{\mathbf{p}_i}(s) = v_{\mathbf{p}_i}(I)$.

Corollary 1.1.14. *Let I be a nonzero ideal in a Dedekind domain A . The quotient ring A/I is a PID.*

pf. Let $\phi : A \rightarrow A/I$ be the quotient map, let $\bar{J}$ be an (A/I) -ideal and let $J \equiv \phi^{-1}(\bar{J})$ be its inverse image in A . Then $I \subseteq J, \bar{J} \cong J/I$ as (A/I) -modules. Then by the finite approximation, $\exists a \in J$ s.t. $v_p(a) = v_p(J)$ for all nonzero prime ideals $\mathbf{p}|I$. For every nonzero prime ideal $\mathbf{p}$,

$v_p(J) \leq v_p(I), v_p((a) + I) = \begin{cases} \min(v_p(a), v_p(I)) = v_p(a) = v_p(J) & \mathbf{p}|I \\ \min(v_p(a), v_p(I)) = v_p(I) = 0 = v_p(J) & \mathbf{p} \nmid I \end{cases}$. So $(a) + I = J$. Thus $\bar{J} \cong J/I = ((a) + I)/I = \phi((a)) = (\phi(a))$.

If every quotient of A by a nonzero ideal is a PID for an integral domain A , then it is a Dedekind domain.

pf. For any ideals $I \subsetneq J, A/J \subseteq A/I$. Let $(x), (y)$ be maximal principal ideals for $A/I, A/J$ respectively. Then $(x) \subseteq (y)$. This implies $y|x$ so to contain is to divide holds.

Definition 1.1.17. *A ring that has only finitely many maximal ideals is called semilocal.*

Corollary 1.1.15. *Every semilocal Dedekind domain is a PID.*

pf. If we let I' be the product of all prime ideals in A , then by the previous lemma, any ideal I will have $J = A$ s.t. $IJ = I$ principal.

Theorem 1.1.7. *Let I be a nonzero ideal in a Dedekind domain A and let $a \in I$ be nonzero. Then $I = (a, b)$ for some $b \in I$.*

pf. $(a) \subseteq I \rightarrow I$ divides (a) so $II' = (a)$ for some nonzero ideal I' . By the previous lemma, there is an ideal J coprime to I' s.t. IJ is principal, so $IJ = (b)$ for some $b \in I$. then $((a), (b)) = (II'IJ) = I$ so $(a, b) = I$.

If for every nonzero ideal I in $A, 0 \neq a \in I, I = (a, b)$ for some $b \in I$, then A is a Dedekind domain.

pf. It is noetherian, since any nonzero ideal I in A are generated by two elements. If $\dim A > 1, \exists p, q$ prime s.t. $(0) \subsetneq p \subsetneq q$. Since p, q both are nonzero ideals in

$A, p = (a_p, b_p), q = (a_q, b_q)$. Since A is an integral domain and both are primes, a_q or b_q divides both a_p, b_p . But then p is not a prime ideal, a contradiction. Hence $\dim A \leq 1$ and therefore it is an integral domain.

Chapter 2

Extensions

2.1 Algebraic Extensions

2.1.1 Etale Algebra

Definition 2.1.1. A polynomial f in $K[x]$ is separable if $(f, f') = (1)$. Otherwise f is inseparable.

→ If f is separable, then it splits into distinct linear factors over $\bar{K}$, where it has $\deg f$ distinct roots.

Definition 2.1.2. Let L/K be an algebraic field extension. An element $\alpha \in K$ is separable over K if it is the root of a separable polynomial in $K[x]$. The extension L/K is separable if $\forall \alpha \in K$ is separable over K ; otherwise it is inseparable.

Lemma 2.1.1. An irreducible polynomial $f \in K[x]$ is inseparable iff $f' = 0$.

pf. Let $f \in L[x]$ be irreducible. Then f is neither zero nor a unit, so $\deg f > 0$. If $f' = 0$, $(f, f') = f \notin K^\times$ and f is inseparable.

If f is inseparable, then $g \equiv (f, f')$ is a nontrivial divisor of f and f' . Given f irreducible, $\deg g = \deg f$, but $\deg f' < \deg g$ so g cannot divide f' unless $f' = 0$.

Corollary 2.1.1. Let $f \in K[x]$ be irreducible and let $p \geq 0$ be the characteristic of K . Then $f(x) = g(x^{p^n})$ for some irreducible separable polynomial $g \in K[x]$ and integer $n \geq 0$ that are uniquely determined by f .

pf. If f is separable, $g = f$ and the theorem holds. Otherwise, $f(x) \equiv \sum f_r x^r$ is inseparable and $f'(x) = \sum r f_r x^{r-1} = 0$ by the previous lemma. This can occur only if $p > 0, f_r = 0, \forall r \geq 0$ not divisible by p . So $f(x) = g(x^p)$ for some $g \in K[x]$. If g is separable, nothing to prove. Otherwise, we can repeat this procedure until g is separable.

Corollary 2.1.2. *If $\text{char} K = 0$, then every algebraic extension of K is separable.*

Lemma 2.1.2. *Let $L = K(\alpha)$ be an algebraic field extension contained in an algebraic closure $\bar{K}$ of K and let $f \in K[x]$ be the minimal polynomial of α over K . Then $\#Hom_K(L, \bar{K}) = \#\{\beta \in \bar{K} : f(\beta) = 0\} \leq [L : K]$, with equality iff α is separable over K .*

pf. For each element of $Hom_K(L, \bar{K})$ is uniquely determined by the image of α , which must be a root β of $f(x)$ in $\bar{K}$. The number of these roots is equal to $[L : K] = \deg f$ precisely when f and therefore α is separable over K .

Definition 2.1.3. *Let L/K be a finite extension of fields. The separable degree of L/K is*

$[L : K]_s \equiv \#Hom_K(L, \bar{K})$. *The inseparable degree of f is $[L : K]_i \equiv [L : K]/[L : K]_s$.*

Theorem 2.1.1. *Let L/K be an algebraic field extension and let $\phi_K : K \rightarrow \Omega$ be a homomorphism to an algebraically closed field Ω . Then ϕ_K extends to a homomorphism $\phi_L : L \rightarrow \Omega$.*

pf. Define a partial ordering on the set $\mathcal{F}$ of pairs (F, ϕ_F) for which F/K is a subextension of L/K and $\phi_F : F \rightarrow \Omega$ extends ϕ_K by defining $(F_1, \phi_{F_1}) \leq (F_2, \phi_{F_2})$ whenever $F_1 \subseteq F_2$ and ϕ_{F_2} extends ϕ_{F_1} . Given any totally ordered subset $\mathcal{C}$ of $\mathcal{F}$, let E be the field $\cup\{F : (F, \phi_F) \in \mathcal{C}\}$ and define $\phi_E : E \rightarrow \Omega$ for $x \in F \subseteq E$ by $x \mapsto \phi_F(x)$. Then (E, ϕ_E) is a maximal element of $\mathcal{C}$ and therefore by Zorn's lemma, $\mathcal{F}$ contains a maximal element (M, ϕ_M) .

CLAIM: $M = L$

pf. Assume contrary. Then $\exists \alpha \in L \setminus M$, so consider the field $F = M(\alpha) \subseteq L$ properly containing M , and extend ϕ_M to $\phi_F : F \rightarrow \Omega$ by letting $\phi_F(\alpha)$ be any root of $\alpha_M(f)$ in Ω where $f \in M[x]$ is a minimal polynomial of α over M and $\alpha_M(f)$ is the image of f in $\Omega[x]$ obtained by applying ϕ_M to each coefficient. Then (M, ϕ_M) is strictly dominated by (F, ϕ_F) , a contradiction to the maximality of M . Hence $M = L$.

Lemma 2.1.3. *Let $L/F/K$ be a tower of finite extensions of fields and $\bar{K}$ be an algebraic closure of K that contains L . Then $\#Hom_K(L, \bar{K}) = \#Hom_K(F, \bar{K})\#Hom_F(L, \bar{K})$.*

pf. When $F = K$ or $L = F$, lemma trivially holds. So assume $K \subsetneq F \subsetneq L$ and decompose the extensions $L/F, F/K$ into finite towers of non-trivial simple extensions

$$K = K_0 \subsetneq K_1 \subsetneq \cdots \subsetneq K_m = F = K_m \subsetneq K_{m+1} \subsetneq \cdots \subsetneq K_n = L$$

where $K_i = K_{i-1}(\alpha_i)$, $\forall i \in [1, n]$. Then $\#Hom_K(L, \bar{K}) = \#Hom_{K_0}(K_n, \bar{K}) = \prod_i \#Hom_{K_{i-1}}(K_i, \bar{K}) = \#Hom_{K_0}(K_m, \bar{K})\#Hom_{K_m}(L, \bar{K}) = \#Hom_K(F, \bar{K})\#Hom_F(L, \bar{K})$ if the second equality holds.

CLAIM: $\#Hom_{K_0}(K_n, \bar{K}) = \prod_i \#Hom_{K_{i-1}}(K_i, \bar{K})$.

pf. Consider the map $\Phi : \text{Hom}_K(K_n, \bar{K}) \rightarrow \bar{K}^n$ This map is injective since $\phi : K_n = K_0(\alpha_1, \dots, \alpha_n) \rightarrow \bar{K}$ is uniquely determined by the images $\phi(\alpha_1), \dots, \phi(\alpha_n)$. So given $h_i \equiv \#\text{Hom}_{K_{i-1}}(K_i, \bar{K})$, if the image of Φ has the cardinality of $\prod_i h_i$, then the CLAIM holds. For any polynomial $f \in \bar{K}[x]$, let $r(f) \equiv \{\beta \in \bar{K} : f(\beta) = 0\}$ be the set of its $\bar{K}$ -roots. $\forall \sigma \in \text{Aut}_K(\bar{K})$ induces a bijection $r(f) \rightarrow r(\sigma(f))$ with $\#r(f) = \#r(\sigma(f))$. Let f_i be the minimal polynomial of α_i over K_{i-1} . By the previous lemma $h_i = \#r(f_i)$ and $\forall \phi \in \text{Hom}_K(K_n, \bar{K})$, $h_i = \#r(\phi(f_i))$ since by the previous theorem, we can extend ϕ to an element of $\text{Hom}_K(\bar{K}, \bar{K}) = \text{Aut}_K(\bar{K})$. Let $\phi_0 : K \rightarrow \bar{K}$ be the inclusion map. $\forall \phi \in \text{Hom}_K(K_n, \bar{K})$, we may define a compatible sequence of pairs (ϕ_i, β_i) with $\phi_i \in \text{Hom}_K(K_i, \bar{K})$, satisfying $\phi_i|_{K_{i-1}} = \phi_{i-1}$, $\beta_i \in r(\phi_{i-1}(f_i))$ by putting $\phi_i = \phi|_{K_i}$ and $\beta_i = \phi(\alpha_i)$. Then $(\beta_1, \dots, \beta_n)$ uniquely determine ϕ and the sequence of pairs (α_i, β_i) , so the set of the compatible sequences of pairs is in bijection with the image of Φ . So initially choose $\beta_1 \in r(f_1)$ and let $\phi_1(\alpha_1) = \beta_1$, which uniquely determine $\phi_1 \in \text{Hom}_K(K_1, \bar{K})$. There are exactly h_1 choices. Extend this sequence by choosing $\beta_2 \in r(\phi_1(f_2))$ and let $\phi_2(\alpha_2) = \beta_2, \phi_2|_{K_1} = \phi_1$. They uniquely determine $\phi_2 \in \text{Hom}_K(K_2, \bar{K})$. There are exactly h_2 such choices. By repeating this procedure, we find there are exactly $\prod_i h_i$ sequences of pairs (α_i, β_i) corresponding to some $\phi \in \text{Hom}_K(K_n, \bar{K})$. So the claim follows.

Corollary 2.1.3. *Let $L/F/K$ be a tower of finite extensions of fields. Then $[L : K]_s = [L : F]_s [F : K]_s$ and $[L : K]_i = [L : F]_i [F : K]_i$.*

pf. The first equality follows by the previous lemma. The second follows by the first equality and the identities $[L : K] = [L : F][F : K]$, $[L : K] = [L : K]_i [L : K]_s$.

Theorem 2.1.2. *Let L/K be a finite extensions of fields. TFAE*

1. L/K is separable
2. $[L : K]_s = [L : K]$
3. $K = K(\alpha)$ for some $\alpha \in K$ separable over K
4. $L \cong K[x]/(f)$ for some monic irreducible separable polynomial $f \in K[x]$.

pf.

- $iii \leftrightarrow iv$

Let f be the minimal polynomial of α . Then $iii \rightarrow iv$.

Let α be the image of x in $K[x]/(f)$. Then $iv \rightarrow iii$.

- $ii \leftrightarrow iii$ by the previous lemma

- $i \rightarrow iii$ by the primitive element theorem
- $iii \rightarrow i$

Let $L = K(\alpha)$ with α separable over K . $\forall \beta \in K$, $L = K(\beta)(\alpha)$. Then α is separable over $K(\beta)$ since its minimal polynomial over $K(\beta)$ divides the minimal polynomial over K , which is separable. Then by the previous lemma, $[L : K]_s = L : K$, $[L : K(\beta)]_s = [L : K(\beta)]$. Then $[K(\beta) : K]_s = [K(\beta) : K]$ so β is separable over K . Since β is arbitrary, i follows.

Corollary 2.1.4. *Let L/K be a finite extension of fields. The $[L : K]_s \leq [L : K]$ with equality iff L/K is separable*

pf. Decompose L/K into a finite tower of simple extensions and proceed by induction on the number of extensions. Then it follows.

Corollary 2.1.5. *Let $L/F/K$ is a tower of finite extensions of fields. Then L/K is separable iff both $L/F, F/K$ separable.*

pf. If $L/K, K/F$ are separable, by the previous corollary $[L : F]_s = [L : F]$, $[F : K]_s = [F : K]$ and therefore $[L : K]_s = [L : K]$ and the corollary follows. The converse is trivial.

Corollary 2.1.6. *Let $L/F/K$ be a tower of algebraic field extensions. Then L/K is separable iff both $L/F, F/K$ are separable.*

pf. Assume $L/F, F/K$ are separable. $\forall \beta \in L$, if $\beta \in F$, nothing to prove. Otherwise, Let M/K be the subextension of F/K generated by the coefficients of the minimal polynomial $f \in F[x]$ of β over F . Then it is a finite separable extension of K and $M(\beta)$ is also a finite separable extension of M since the minimal polynomial of β over $M(\beta)$ is f , which is separable. Then by the previous corollary, $M(\beta)$ is separable over K .

The converse is trivial.

Corollary 2.1.7. *Let L/K be an algebraic field extension and let $F = \{\alpha \in L : \alpha \text{ is separable over } K\}$. Then F is a separable field extension of K .*

pf. If α, β are separable over K , then $K(\alpha), K(\alpha, \beta)$ are separable extensions of K by the previous corollary. Thus for all elements of $K(\alpha, \beta)$ including $\alpha\beta, \alpha + \beta$ are separable over K and lies in F . So it is a field and by construction it is separable.

Definition 2.1.4. *Let L/K be an algebraic field extension. The field F in the above corollary is the separable closure of K in L . When L is an algebraic closure of K , it is simply called a separable closure of K .*

Definition 2.1.5. *A field K is perfect if every algebraic extension of K is separable.*

Theorem 2.1.3. *A field K of characteristic $p > 0$ is perfect iff $K = K^p$, i.e. every element of K is a p th power, equivalently the map $x \mapsto x^p$ is an automorphism.*

pf. If $K \neq K^p$, $\forall \alpha \in K \setminus K^p$, $x^p - \alpha$ is irreducible and the extension $K[x]/(x^p - \alpha)$ is inseparable and therefore K is not perfect.

If $K = K^p$, let $f \in L[x]$ be irreducible. Then $f(x) = g(x^{p^n})$ for some separable $g \in K[x]$. If $n > 0$, $f(x) = g(x^{p^n}) = \tilde{g}(x^{p^{n-1}})^p$, where $\tilde{g}$ is the polynomial obtained from g by replacing each coefficient with its p th root, a contradiction to the irreducibility of f . So $n = 0$, $f = g$ is separable. Since every irreducible polynomial in $K[x]$ is separable, every algebraic extension of K is separable, so K is perfect.

Corollary 2.1.8. *Every finite field is a perfect field.*

pf. If a field K has cardinality p^n , $\#K^\times = p^n - 1$, thus $\alpha = \alpha^{p^n} = (\alpha^{p^{n-1}})^p$ for all $\alpha \in K$. So every element of K is a p th power.

Definition 2.1.6. *A field K is separably closed if K has no nontrivial finite separable extensions. Equivalently, K is equal to its separable closure in any algebraic closure of K .*

Definition 2.1.7. *An algebraic extension L/K is purely inseparable, if $[L : K]_s = 1$.*

e.g. If $K = \mathbb{F}_p(t)$, $L = K[x]/(x^p - t) = \mathbb{F}_p(t^{1/p})$, then L/K is a purely inseparable extension of degree p .

Proposition 2.1.1. *Let K be a field of characteristic $p > 0$. If L/K is purely inseparable of degree p , then $L = K(a^{1/p}) \cong K[x]/(x^p - a)$ for some $a \in K \setminus K^p$.*

pf. $\forall \alpha \in L \setminus K$, inseparable over K , its minimal polynomial over K is of the form $f(x) = g(x^{p^n})$ with f monic. Then $1 < \deg f \leq [L : K] = p$, so g must be a monic polynomial of degree 1, i.e. $g(x) = x - a$. Then $f(x) = x^p - a$, $a \notin K^p$, given f irreducible. Then $[L : K(\alpha)] = 1 \rightarrow L = K(\alpha) \cong K[x]/(x^p - a)$.

Theorem 2.1.4. *Let L/K be an algebraic extension and let F be the separable closure of K in L . Then L/F is purely inseparable.*

pf. If L/K is separable, $F = L$ and the theorem follows. Otherwise, $\text{char} K = p \neq 0$. Fix an algebraic closure $\bar{K}$ of K that contains L . Let $\alpha \in L \setminus F$ have minimal polynomial f over F . Then $f(x) = g(x^{p^n})$, $g \in F[x]$ irreducible and separable. Then we must have $\deg g = 1$ since otherwise the roots of g would be separable over F and therefore over K , but not lie in the separable closure F of K in L . Thus $f(x) = x^{p^n} - a$ for some $a \in F$. Since $p > 0$, $f(x) = x^{p^n} - a^{p^n} = (x - \alpha)^{p^n}$. There is thus only one F -homomorphism from $F(\alpha)$ to $\bar{K}$. Since α is arbitrary, $\#Hom_F(L, \bar{K}) = 1$, so L/F is purely inseparable.

Corollary 2.1.9. *Every algebraic extension L/K can be uniquely determined into a tower of algebraic extensions $L/F/K$ with F/K separable and L/F purely inseparable.*

Corollary 2.1.10. *The inseparable degree of any finite extension of fields is a power of the characteristic.*

Definition 2.1.8. *Let K be a field. An étale K -algebra is a K -algebra L that is isomorphic to a finite product of separable field extensions of K . The dimension of an étale K -algebra is its dimension as a K -vector space. When the dimension is finite, L is said to be a finite étale K -algebra. A homomorphism of étale K -algebras is a homomorphism of K -algebras.*

→ for any separable polynomial $f \in K[x]$, $L = K[x]/(f)$. Then L becomes a commutative K -algebra whose elements are all separable over K . By the CRT, if f is separable, then the K -algebra L is isomorphic to a direct product of finite separable extensions of $K \rightarrow f = f_1 \dots f_n$ is a factorisation of f into irreducibles in $K[x]$, $\frac{K[x]}{(f)} = \frac{K[x]}{(f_1 \dots f_n)} \cong \frac{K[x]}{(f_1)} \times \dots \times \frac{K[x]}{(f_n)}$ and by the separability of f , f_i s are pairwise coprime. Thus L is a K -algebra that is isomorphic to a finite product of separable field extensions $K[x]/(f_i)$ of K .

Definition 2.1.9. *A ring R is simple if it is nonzero and has no nonzero proper ideals. A ring R is semisimple if it is isomorphic to a nonempty finite product of simple rings $\prod R_i$.*

→ If R is commutative, it is simple iff it is a field and it is semisimple iff it is a finite product of fields. Thus étale algebras are semisimple algebras.

Proposition 2.1.2. *Let $A = \prod K_i$ be a K -algebra written as a product of field extensions K_i/K . Every surjective homomorphism $\phi : A \rightarrow B$ of K -algebras corresponds to the projection of A onto a subproduct of its factors.*

pf. $\ker \phi$ is a subproduct of $\prod K_i$ and therefore $A \cong \ker \phi \times \text{im } \phi$ and $B = \text{im } \phi$ is isomorphic to the complementary subproduct.

Corollary 2.1.11. *The decomposition of an étale algebra into field extensions is unique upto permutation and isomorphism of factors.*

pf. Let A be an étale K -algebra and suppose A is isomorphic to two products of field extensions of $K \rightarrow \prod_{i=1}^m K_i \cong A \cong \prod_{j=1}^n L_j$. Composing with isomorphisms yields surjective K -algebra homomorphisms $\begin{matrix} \pi_i : \prod L_j \rightarrow K_i \\ \pi_j : \prod K_i \rightarrow L_j \end{matrix}$. By the previous proposition, each K_i must be isomorphic to one of the L_j and vice versa. Thus the uniqueness follows.

Definition 2.1.10. Let $\phi : A \rightarrow B$ be a homomorphism of rings, so B is an A -module. Let M be any A -module. The tensor product of A -modules $M \otimes_A B$ is a B -module called the base change of M from A to B . If M is an A -algebra, then its base change to B is a B -algebra.

$\rightarrow \forall \phi : A \rightarrow B$ determines a functor from the category of A -modules to the category of B -modules. It has an adjoint functor called the restriction of scalars that converts B -module M into an A -module by the rule $am = \phi(a)m$.

If a ring homomorphism $\phi : A \rightarrow B$ is an inclusion, we have a ring extension B/A .

Proposition 2.1.3. Suppose L is a finite etale K -algebra and K'/K is any field extension. Then $L \otimes_K K'$ is a finite etale K' -algebra of the same dimension as L .

pf. If L is not a field, it is a product of fields. Then by the surjection, we can focus on one factor of L , which is a field. So wlog assume L is a field. Then $L \cong K[x]/(f)$ for some separable $f \in K[x]$ and if $f = f_1 \dots f_m$ is the factorisation of f in $K'[x]$, $L \otimes_K K' \cong K'[x]/(f) \cong \prod_i K'[x]/(f_i)$, in which each factor $K'[x]/(f_i)$ is a finite separable extension of K' . Thus $L \otimes_K K'$ is a finite etale K' -algebra and $\dim_K L = \deg f = \dim_{K'} K'[x]/(f)$ so the dimension is preserved.

e.g. Any finite real vector space V is a finite etale $\mathbb{R}$ -algebra, and its base change from $\mathbb{R}$ to $\mathbb{C}$ complex vector space $V \otimes_{\mathbb{R}} \mathbb{C}$ is a finite etale $\mathbb{C}$ -algebra of the same dimension.

Corollary 2.1.12. Let $L \cong K[x]/(f)$ be a finite separable extension of a field K defined by an irreducible separable polynomial $f \in K[x]$. Let K'/K be any field extension and let $f = f_1 \dots f_m$ be the factorisation of f into distinct irreducible polynomials $f_i \in K'[x]$. Then we have an isomorphism of finite etale K' -algebras $L \otimes_K K' \cong \prod_i K'[x]/(f_i)$, where each $K'[x]/(f_i)$ is a finite separable field extension of K' .

Definition 2.1.11. An element $\alpha \in R$ is nilpotent if $\alpha^n = 0$. A ring R is reduced if it contains no nonzero nilpotents.

Theorem 2.1.5. Let L be a commutative K -algebra of finite dimension and assume that the dimension of L is less than the cardinality of K . Then TFAE:

1. L is a finite etale K -algebra
2. Every element of L is separable over K
3. $L \otimes_K K'$ is reduced for every extension K'/K

4. $L \otimes_K K'$ is semisimple for every extension K'/K
5. $L = K[x]/(f)$ for some separable $f \in K[x]$

pf.

- $i \rightarrow ii$

Let $L = \prod K_i$ with K_i/K separable and consider $\alpha = (\alpha_1, \dots, \alpha_n) \in L = \prod K_i$ is separable over K with separable minimal polynomial $f_i \in K[x]$ and α is a root of $f \equiv \text{LCM}(f_1, \dots, f_n)$, which is separable. Thus α is separable.

- $ii \rightarrow iii$

If $\alpha \in L$ is nonzero and separable over K , it cannot be nilpotent, since otherwise the minimal polynomial of the nilpotent x^n , $n > 1$ is not separable. Separability is preserved under the base change so the iii follows.

- $iii \leftrightarrow iv$

Lemma 2.1.4. *Let K be a field. A commutative K -algebra of finite dimension is semisimple iff it is reduced.*

pf. If A is semisimple, it is reduced, since otherwise we could project a nonzero nilpotent of A to a nonzero nilpotent in a field, a contradiction. Every ideal of a commutative K -algebra A is also a K -vector space, and therefore when $\dim_K A$ is finite, A satisfies both a.c.c. and d.c.c. and is therefore noetherian and artinian. Thus it has finitely many maximal ideals $M_1, \dots, M_n$ and the intersection of maximal ideals is equal to the nilradical of A . Taking the product of the projection map $A \rightarrow A/M_i$ yields a surjective ring homomorphism $\phi : A \rightarrow \prod A/M_i$ from A to a product of fields. If A is reduced, $\ker \phi = \cap M_i = \{0\}$, i.e. it is an isomorphism. Thus A is semisimple.

- $iv \rightarrow i$

Assume L is semisimple by taking $K' = K$. Wlog assume L is a field. By base-changing to the separable closure of K in L , we can reduce to the case that L/K is a purely inseparable field extension. If $L = K$, nothing to prove. Otherwise $\exists \alpha \in L$ inseparable and the minimal polynomial for α has the form $f(x) = x^{p^n} - a$ for some $a \in K, n \geq 1$. Consider $\gamma \equiv \alpha \otimes 1 - 1 \otimes \alpha \in L \otimes_K L$. Then $\gamma \neq 0$ since $\gamma \notin K$, but $\gamma^{p^n} = \alpha^{p^n} \otimes 1 - 1 \otimes \alpha^{p^n} = a \otimes 1 - 1 \otimes a = 0$ so γ is a nilpotent and $L \otimes_K L$ is not reduced, contradiction to iii .

- $v \rightarrow i$ by the previous corollary

- $i \rightarrow v$

Suppose $L = \prod L_i$ with each L_i/K finite separable extensions of K . Pick a monic irreducible separable polynomial $f_1(x)$ so that $L_1 \cong K[x]/(f_1)$ and do the same for all $i \in [2, n]$, ensuring each polynomial f_j , we pick is not equal to $f_i \forall i < j$. This can be done by replacing $f_j(x)$ with $f_j(x + a)$, $a \in K^\times$, if necessary. The existence is guaranteed by the cardinality condition. The polynomials $f_1, \dots, f_n$ are the coprime and separable and therefore their product f is separable and $L = K[x]/(f)$.

Proposition 2.1.4. *Suppose L is a finite etale K -algebra and Ω is a separably closed field extension of K . There is an isomorphism of finite etale Ω -algebras*

$$\begin{aligned} L \otimes_K \Omega &\rightarrow \prod_{\sigma \in \text{Hom}_K(L, \Omega)} \Omega \\ \beta \otimes 1 &\mapsto (\sigma(\beta))_\sigma \text{ for each } \beta \in L. \end{aligned}$$

pf. Reduce to the case where $L = K[x]/(f)$ is a separable field extension and then we may factor $f(x) = \prod (x - \alpha_i)$ over Ω with all α_i are distinct. Then there is a bijection between $\text{Hom}_K(K[x]/(f), \Omega)$ and the set $\{\alpha_i\}$ since $\forall \sigma \in \text{Hom}_K(K[x]/(f), \Omega)$ is determined by $\sigma(x) \in \{\alpha_i\}$ and for each α_i , $x \mapsto \alpha_i$ determines a K -algebra homomorphism $\sigma_i \in \text{Hom}_K(K[x]/(f), \Omega)$. Then we have Ω -algebra isomorphisms

$$\frac{K[x]}{(f)} \otimes_K \Omega \rightarrow \frac{\Omega[x]}{(f)} \rightarrow \prod_i \frac{\Omega[x]}{(x - \alpha_i)} \rightarrow \prod_i \Omega$$

which maps $x \otimes 1 \mapsto x \mapsto (\alpha_1, \dots, \alpha_n) \mapsto (\sigma_1(x), \dots, \sigma_n(x))$. Since the element $x \otimes 1$ generates $L \otimes_K \Omega$, $\beta \otimes 1 \mapsto (\sigma(\beta))_\sigma$ for every $\beta \in L$.

Definition 2.1.12. *Let B/A be a commutative ring extension in which B is a free A -module of finite rank. The norm $N_{B/A}(b)$ and trace $T_{B/A}(b)$ of b are the determinant and trace of the A -linear multiplication-by- b map $B \rightarrow B$ defined by $x \mapsto bx$.*

If A is a field and B is a finite A -algebra, B is a finite A -vector space, hence a free A -module of finite rank. Then we can compute the norm and trace by picking a basis for B as an A -module and computing the matrix of the multiplication-by- b map w.r.t. this basis.

If $B_1/A, B_2/A$ are two ring extensions that are free A -modules of finite rank, then

$$N_{B_1 \times B_2/A}(x) = N_{B_1/A}(x_1) N_{B_2/A}(x_2), \quad T_{B_1 \times B_2/A} = T_{B_1/A}(x_1) + T_{B_2/A}(x_2), \quad \forall x = (x_1, x_2) \in B_1 \times B_2.$$

Lemma 2.1.5. *Let B/A be ring extension with B free of rank n over A , and let $\phi : A \rightarrow A'$ be a ring homomorphism. The base change $B' = B \otimes_A A'$ of B to A' is a free A' -module of rank n and for every $b \in B$, $\phi(N_{B/A}(b)) = N_{B'/A'}(b \otimes 1)$, $\phi(T_{B/A}(b)) = T_{B'/A'}(b \otimes 1)$.*

pf. Let $b \in B, (b_1, \dots, b_n)$ be a basis for B as an A -module, and $M = (m_{ij}) \in A^{n \times n}$ be the matrix of $B \rightarrow_{\times B} B$ w.r.t. this basis. Then $(b_1 \otimes 1, \dots, b_n \otimes 1)$ is a basis for B' as an A' -module and $M' = (\phi(m_{ij})) \in A'^{n \times n}$ is the matrix of $B' \rightarrow_{\times B \otimes 1} B'$ and therefore

$$\phi(N_{B/A}(b)) = \phi(\det M) = \det M' = N_{B'/A'}(b \otimes 1)$$

$$\phi(T_{B/A}(b)) = \phi(\text{tr} M) = \text{tr} M' = N_{B'/A'}(b \otimes 1) \quad .$$

Theorem 2.1.6. *Let K be a field with separable closure Ω and let L be a finite etale K -algebra. $\forall \alpha \in L, N_{L/K}(\alpha) = \prod_{\sigma \in \text{Hom}_K(L, \Omega)} \sigma(\alpha), T_{L/K}(\alpha) = \sum_{\sigma \in \text{Hom}_K(L, \Omega)} \sigma(\alpha)$.*

pf. Let n be the rank of L as a K -module. Then $N_{L/K}(\alpha) = N_{(L \otimes_K \Omega)/\Omega}(\alpha \otimes 1) = N_{\Omega^n/\Omega}(\sigma_1(\alpha), \dots, \sigma_n(\alpha))$
 $= \prod \sigma_i(\alpha)$ by the previous lemma with isomorphism $L \otimes_K \Omega \rightarrow \prod_{\sigma} \Omega = \Omega^n$ sends $\alpha \otimes 1$ to $(\sigma_1(\alpha), \dots, \sigma_n(\alpha))$. Using the standard basis for Ω^n , the matrix of multiplication-by- $(\sigma_1(\alpha), \dots, \sigma_n(\alpha))$ is the diagonal matrix with $\sigma_i(\alpha)$ in the i th diagonal entry. Similarly, $T_{L/K}(\alpha) = T_{(L \otimes_K \Omega)/\Omega}(\alpha \otimes 1) = T_{\Omega^n/\Omega}(\sigma_1(\alpha), \dots, \sigma_n(\alpha)) = \sum \sigma_i(\alpha)$.

Proposition 2.1.5. *Let L/K be a finite extension, let $\bar{K}$ be an algebraic closure of K containing L . Let $\alpha \in L^\times$ have minimal polynomial $f \in K[x]$ with factorisation $f(x) = \prod_{i=1}^d (x - \alpha_i)$ in $\bar{K}[x]$, and let $e = [L : K(\alpha)]$. Then*

$$N_{L/K}(\alpha) = \prod \alpha_i^e, T_{L/K} = e \sum \alpha_i$$

In particular, if $f(x) = \sum a_i x^i, N_{L/K}(\alpha) = (-1)^{de} \alpha_0^e, T_{L/K}(\alpha) = -e a_{d-1}$.

pf. By the previous theorem $N_{K(\alpha)/K}(\alpha) = \prod \sigma(\alpha) = \prod \alpha_i$
 $T_{K(\alpha)/K} = \sum \sigma(\alpha) = \sum \alpha_i$. Since $e = [L : K(\alpha)]$, we can view the matrix as a block diagonal matrix s.t. one block for $K(\alpha)$ is repeated by e times (given not necessarily separable). Then $N_{L/K}(\alpha) = \prod \alpha_i^e, T_{L/K} = e \sum \alpha_i$ hold.
 For a polynomial $f(x) = x^d + a_{d-1}x^{d-1} + \dots + (-1)^d a_0, \prod \alpha_i = -a_{d-1}, \sum \alpha_i = (-1)^d a_0$. Then by the previous statement, $N_{L/K}(\alpha) = (-1)^{de} \alpha_0^e, T_{L/K} = e a_{d-1}$ follows.

Corollary 2.1.13. *Let A be an integrally closed domain with fraction field K and let L/K be a finite extension. If $\alpha \in L$ is integral over A , then $N_{L/K}(\alpha) \in A, T_{L/K}(\alpha) \in A$.*

Theorem 2.1.7 (Transitivity of Norm and Trace). *Let $A \subseteq B \subseteq C$ be rings with C free of finite rank over B , and B free of finite rank over A . Then C is free of finite rank over A and $N_{C/A} = N_{B/A} \circ N_{C/B}, T_{C/A} = T_{B/A} \circ T_{C/B}$.*

2.1.2 Dedekind Extensions

Definition 2.1.13. *Let A be a commutative ring and M an A -module. The dual module $M^\vee$ is the A -module $\text{Hom}_A(M, A)$ with scalar multiplication $(af)(m) =$*

$af(m), a \in A, f \in \text{Hom}_A(M, A), m \in M$. If $\phi : M \rightarrow N$ is an A -module homomorphism, the dual homomorphism.

$$\begin{aligned} \phi^\vee : N^\vee &\rightarrow M^\vee \\ g &\mapsto g \circ \phi \end{aligned}$$

$\forall g \in N^\vee, m \in M$.

Lemma 2.1.6. *Let A be a commutative ring. For all A -module M, N , the A -modules $(M \oplus N)^\vee$ and $M^\vee \oplus N^\vee$ are canonically isomorphic.*

pf. $\phi \mapsto (m \mapsto \phi(m, 0), n \mapsto \phi(0, n))$
 $(\phi, \psi) \mapsto ((m, n) \mapsto \phi(m) + \psi(n))$ are inverses of each other, hence the lemma follows.

Definition 2.1.14. *A A -module M is reflexive if $M^{\vee\vee} \cong M$.*

If A is a field and M is a finitely generated A -module, then M is a vector space of finite dimension, it is reflexive. (e.g. invertible fractional ideals)

Proposition 2.1.6. *Let A be an integral domain with fraction field K and let M be a nonzero A -submodule of K . Then $M^\vee \cong (A : M) \equiv \{x \in K : xM \subseteq A\}$; in particular if M is an invertible fractional ideal, then $M^\vee \cong M^{-1}, M^{\vee\vee} \cong M$.*

pf. For any $x \in (A : M)$, $m \mapsto xm$ is an A -linear map from M to A , hence $x \in M^\vee$ and this defines an A -module homomorphism $\phi : (A : M) \rightarrow M^\vee$, since the map $x \mapsto (m \mapsto xm)$ is itself A -linear. Since $M \subseteq K$ is a nonzero A -module, it contains $0 \neq a \in A$. If $f \in M^\vee, m = b/c \in M, f(m) = f(\frac{b}{c}) = \frac{ac}{ac} f(\frac{b}{c}) = \frac{b}{ac} f(\frac{ac}{c}) = \frac{b}{ac} f(a) = \frac{f(a)}{a} m$ by the A -linearity of f . Thus f corresponds to multiplication by $x = \frac{f(a)}{a}$, which lies in $(A : M)$ since $xm = f(m) \in A, \forall m \in M$. This map $f \mapsto f(a)/a$ defines an A -module homomorphism $M^\vee \rightarrow (A : M)$ inverses to ϕ , and therefore it is an isomorphism. When M is an invertible fractional ideal, $M^\vee \cong (A : M) = M^{-1}$, so the last statement holds.

Theorem 2.1.8. *Let A be a commutative ring and let M be a free A -module of rank n . Then $M^\vee$ is also a free A -module of rank n , and each basis $(e_1, \dots, e_n)$ of M uniquely determines a dual basis $(e_1^\vee, \dots, e_n^\vee)$ of $M^\vee$ with the property $e_i^\vee = \delta_{ij} \equiv \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$.*

pf. If $n = 0, M = M^\vee = 0$ and therefore the theorem holds.
 $n \geq 1$, fix an A -basis $\mathbf{e} \equiv (e_1, \dots, e_n)$ for M . Then for all $\mathbf{a} \equiv (a_1, \dots, a_n) \in A^n$, define $f_a \in M^\vee$ by setting $f_a(e_i) = a_i$ and extending A -linearly. The map $\mathbf{a} \mapsto f_a$ gives an A -module homomorphism $A^n \rightarrow M^\vee$ with inverse $f \mapsto (f(e_1), \dots, f(e_n))$ and is therefore an isomorphism. Thus $M^\vee \cong A^n$ is a free A -module of rank n .
Let $e_i^\vee \equiv f_{\hat{i}}$, where $\hat{i} \equiv (0, \dots, 0, 1, 0, \dots, 0) \in A^n$ has a 1 in the i th position. Then $\mathbf{e}^\vee \equiv (e_1^\vee, \dots, e_n^\vee)$ is a basis for $M^\vee$, since $(\hat{1}, \dots, \hat{n})$ is a basis for A^n and $e_i^\vee(e_j) = \delta_{ij}$. The basis $\mathbf{e}^\vee$ is uniquely determined by $\mathbf{e}$ since it must be the image of $(\hat{1}, \dots, \hat{n})$ under the isomorphism $\mathbf{a} \mapsto f_a$ determined by $\mathbf{e}$.

Definition 2.1.15. Let A be a commutative ring and M an A -module. A bilinear pairing on M is an A -linear $\langle \cdot, \cdot \rangle: M \times M \rightarrow A$. Explicitly $\forall u, v, w \in M, \lambda \in$

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$

$$A, \quad \langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle .$$

$$\langle \lambda u, v \rangle = \langle u, \lambda v \rangle = \lambda \langle u, v \rangle$$

→ If $\langle v, w \rangle = \langle w, v \rangle$, then it is symmetric, if $\langle v, w \rangle = -\langle w, v \rangle$, it is skew-symmetric, and if $\langle v, v \rangle = 0$, it is alternating.

The pairing $\langle \cdot, \cdot \rangle$ induces an A -module homomorphism $\phi: M \rightarrow M^\vee$
 $m \mapsto (n \mapsto \langle m, n \rangle)$.
 If $\ker \phi = \{0\}$, it is nondegenerate, and if ϕ is an isomorphism, it is perfect.

Proposition 2.1.7. Let A be a commutative ring and let M be a free A -module of rank n with a perfect pairing $\langle \cdot, \cdot \rangle$. For each A -basis $(e_1, \dots, e_n)$ of M , there is a unique basis $(e'_1, \dots, e'_n)$ for M s.t. $\langle e'_i, e_j \rangle = \delta_{ij}$.

pf. If M is a free A -module with basis $(e_1, \dots, e_n)$ and $\langle \cdot, \cdot \rangle$ is a perfect pairing, by the inverse of the isomorphism $\phi: M \rightarrow M^\vee$ applied to the dual basis $(e_1^\vee, \dots, e_n^\vee)$, we can obtain a basis $(e'_1, \dots, e'_n)$ for M s.t. $\langle e'_i, e_j \rangle = \delta_{ij}$; $e'_i = \phi^{-1}(e_i^\vee)$. Then $e_i^\vee = \phi(e'_i) = m \mapsto \langle e'_i, m \rangle$ and $\langle e'_i, e_j \rangle = \phi(e'_i)(e_j) = e_i^\vee(e_j) = \delta_{ij}$.

If there is another basis $(f'_1, \dots, f'_n)$ for M with the same property, then for each i , $\langle e'_i - f'_i, e_j \rangle = \delta_{ij} - \delta_{ij} = 0$, $\forall e_j$ and therefore $\langle e'_i - f'_i, m \rangle = 0$, $\forall m \in M$. Then $e'_i - f'_i \in \ker \phi = \{0\}$ and therefore uniqueness follows.

Definition 2.1.16. Let A be an integral domain with fractional field K and let V be a K -vector space of finite dimension. A A -lattice in V is a finitely generated A -submodule M of V that spans V as a K -vector space.

Definition 2.1.17. Let A be a noetherian domain with fraction field K , and let V be a K -vector space of finite dimension with a perfect pairing $\langle \cdot, \cdot \rangle$. If M is an A -lattice in V , its dual lattice is the A -module $M^* \equiv \{x \in V : \langle x, m \rangle \in A, \forall m \in M\}$.

Theorem 2.1.9. Let A be a noetherian domain with fraction field K , let V be a K -vector space with a perfect pairing $\langle \cdot, \cdot \rangle$ and let M be an A -lattice in V . The dual lattice M^* is an A -lattice in V isomorphic to $M^\vee$.

pf. $\mathbf{e} \equiv (e_1, \dots, e_n)$ be a K -basis for V that lies in M and let $\mathbf{e}' \equiv (e'_1, \dots, e'_n)$ be the unique K -basis for V given by the previous proposition that satisfies $\langle e'_i, e_j \rangle = \delta_{ij}$. Write a finite set S of generators for M in terms of the basis $\mathbf{e}$ with coefficients in K and let d be the product of all denominators that appear.

CLAIM: $d\mathbf{e}'$ lies in M^*

pf. For each $e'_i, m \in S$ if we put $m = \sum_j m_j e_j$, then $\langle d e'_i, m \rangle = d \langle e'_i, \sum_j m_j e_j \rangle = d \sum_j m_j \langle e'_i, e_j \rangle = d \sum_j m_j \delta_{ij} = d m_i \in A$ and therefore $d e'_i \in M^*$.

Thus M^* contains a basis $d\mathbf{e}'$ for V .

CLAIM: M^* is finitely generated

pf. Define $N \equiv \{\sum a_i e_i, \forall a_i \in A\} \cong A^n$ to be the free A -submodule of M spanned by $\mathbf{e}$. The A -module N contains a basis for V and is finitely generated and therefore it is an A -lattice in V . The K -basis $\mathbf{e}'$ for V lies in N^* , since $\langle e'_i, e_j \rangle = \delta_{ij} \in A$. Given $x \in N^*$ write $x = \sum x_i e'_i$, then $\langle x, e_i \rangle = x_i \langle e'_i, e_j \rangle = x_i \in A$ since $x \in N^*$. Thus x lies in the A -span of $\mathbf{e}'$. Thus N^* is a free A -module of rank n and in particular a finitely generated module over a noetherian ring, i.e. it is a noetherian module. Since $N \subseteq M$, $M^* \subseteq N^*$. Since every submodule of noetherian module is finitely generated, so is M^* .

Hence M^* is an A -lattice.

CLAIM: $M^* \cong M^\vee$.

pf. There are A -module homomorphism $\phi: M^* \rightarrow M^\vee$ and the

$$\psi: M^\vee \rightarrow M^* \text{ which are inverses to each other.}$$

Indeed, $\forall x = \sum x_i e'_i \in M^*, \psi(\phi(x)) = \sum \phi(x)(e_i) e'_i = \sum \langle x, e_i \rangle e'_i = \sum_{i,j} x_j \langle e'_j, e_i \rangle e'_i = \sum_i x_i e'_i = x$
 $\forall f \in M^\vee, \phi(\psi(f))(m) = \phi(\sum f(e_i) e'_i)(m) = \sum \phi(f(e_i) e'_i)(m) = \sum \langle f(e_i) e'_i, \sum_j m_j e_j \rangle = f(m).$

Corollary 2.1.14. *Let A be a noetherian domain with fraction field K . If M_1, M_2 are A -lattices in K -vector spaces V_1, V_2 with perfect pairing $\langle \cdot, \cdot \rangle_1, \langle \cdot, \cdot \rangle_2$ respectively, then $\langle \cdot, \cdot \rangle_1 + \langle \cdot, \cdot \rangle_2$ defines a perfect pairing on $V_1 \oplus V_2$ and $(M \oplus N)^* \cong M^* \oplus N^*$.*

Corollary 2.1.15. *Let A be a noetherian domain with fraction field K , let V be a K -vector space with a perfect pairing $\langle \cdot, \cdot \rangle$, and let M be a free A -lattice in V with A -basis $(e_1, \dots, e_n)$. The dual lattice M^* is a free A -lattice in V that has a unique A -basis $(e_1^*, \dots, e_n^*)$ that satisfies $\langle e_i^*, e_j \rangle = \delta_{ij}$.*

pf., By the previous theorem with $N = M, e_i^* = e'_i$.

Lemma 2.1.7. *Let A be a noetherian domain with fraction field K , let V be a K -vector space of finite dimension with a perfect pairing $\langle \cdot, \cdot \rangle$, let M be an A -lattice in V , and let S be a multiplicative subset of A . Then $S^{-1}M$ and $S^{-1}M^*$ are $(S^{-1}A)$ -lattices in V satisfying $(S^{-1}M)^* = S^{-1}M^*$.*

pf. $S^{-1}M, S^{-1}M^*$ contain a basis for V since M, M^* do and both are finitely generated as $S^{-1}A$ -modules since M, M^* are finitely generated A -modules. Thus both are $S^{-1}A$ -lattices.

Let $m_1, \dots, m_n$ be A -module generated for M . Then they are also generators of $S^{-1}M$ as an $S^{-1}A$ -module. If $x \in (S^{-1}M)^*$, then for each $m_i, \langle x, m_i \rangle = a_i/s_i$ for

some $a_i \in A, s_i \in S$ and if we put $s = s_1 \dots s_n, \langle sx, m_i \rangle \in A, \forall m_i$ and therefore $\forall m \in M$. Thus $sx \in M^*, x \in S^{-1}M^*$.

If $x = y/s \in S^{-1}M^*, y \in M^*, s \in S$, then $\langle y, m_i \rangle \in A, \forall m_i$ and $\langle x, m_i \rangle = \langle y, m_i \rangle / s \in S^{-1}A$ for all m_i , hence for all $m \in S^{-1}M$ and therefore $x \in (S^{-1}M)^*$.

Proposition 2.1.8. *Let A be a Dedekind domain with fraction field K , let V be a K -vector space of finite dimension with a symmetric perfect pairing $\langle \cdot, \cdot \rangle$ and let M be an A -lattice in V . Then $M^{**} = M$.*

pf. Let $\mathfrak{p}$ be a maximal ideal of A . Then $(M^{**})_{\mathfrak{p}} = M_{\mathfrak{p}}^{**}$ by the previous lemma. So wlog replace A with $A_{\mathfrak{p}}$, which is a DVR. Then A is a PID and M, M^* are both torsion-free modules over PID, hence free A -modules. Choose an A -basis $(e_1, \dots, e_n)$ for M , and let $(e_1^*, \dots, e_n^*)$ be the unique dual A -basis for M^* that satisfies $\langle e_i^*, e_j \rangle = \delta_{ij}$ by the previous corollary. If we let $(e_1^{**}, \dots, e_n^{**})$ be the unique A -basis for M^{**} that satisfies $\langle e_i^{**}, e_j^* \rangle = \delta_{ij}$. By the uniqueness $e_i^{**} = e_i, \forall i$ and therefore $M^{**} = M$.

Proposition 2.1.9. *Let A be a Dedekind domain with fraction field K , let L/K be a finite extension, and let B be the integral closure of A in L . Then every element of L can be written as $b/a, a \in A, b \in B$. In particular, B spans L as a K -vector space and L is the fraction field of B .*

pf. Let $\alpha \in L$. By multiplying the minimal polynomial of α in $K[x]$ by the product of the denominators of its coefficients, we obtain a polynomial in $A[x] \rightarrow g(x) = a_n x^n + \dots + a_1 x + a_0$ with $a_n \neq 0$, that has α as a root. By Replacing x with x/a_n , and multiplying both sides by a_n^{n-1} , we get $a_n^{n-1} g(x/a_n) = x^n + a_{n-1} x^{n-1} + \dots + a_n^{n-2} a_1 x + a_n^{n-1} a_0$, a monic polynomial with coefficients in A that has $a_n \alpha \in L$ as a root. Thus $a_n \alpha \in B$, given B integral closure of A in L and $\alpha = b/a_n, b \in B, a_n \in A$ as claimed. Thus B generates L as K -vector space and $B \subseteq L \subseteq \text{Frac} B$ implies $L = \text{Frac} B$, since no smaller field can contain B .

Proposition 2.1.10. *Let A be a Dedekind domain with fraction field K , let L/K be a finite extension of fields, and let B be the integral closure of A in L . Then $N_{L/K}(b) \in A, T_{L/K}(b) \in A, \forall b \in B$.*

pf. The minimal polynomial $f = \sum_{i=0}^d a_i x^i \in K[x]$ of b has coefficients in A and therefore $N_{L/K}(b) = (-1)^{de} a_0^e \in A, T_{L/K}(b) = -e a_{d-1} \in A$.

Definition 2.1.18. *Let B/A be a ring extension with B a free A -module of finite rank. The trace pairing on B is the map $B \times B \rightarrow A$ defined by $\langle x, y \rangle_{B/A} \equiv T_{B/A}(xy)$.*

Theorem 2.1.10. *Let L be a commutative K -algebra of finite dimension. The trace pairing*

$\langle \cdot, \cdot \rangle_{L/K}$ is a symmetric bilinear pairing. It is a perfect pairing iff L is a finite etale K -algebra.

pf. Bilinearity follows from the K -linearity of the trace map $T_{L/K}$ and symmetry is immediate.

If L is not reduced, the proposition holds, since it is not etale and the trace pairing is degenerate (for x nilpotent, the map $y \mapsto T_{L/K}(xy)$ must be a zero map, since every xy is also nilpotent and the trace of any nilpotent element z is zero).

If L is reduced, then it is semisimple and thus a product of fields. Wlog we can reduce to the case where L is a field since the trace pairing on a product of field extensions is nondegenerate iff the trace pairing on each factor is nondegenerate and a product of field extensions is etale iff each factor is etale. $T_{L/K}$ is a zero map iff the field extension L/K is inseparable. If $T_{L/K}$ is a zero map, then the trace pairing is clearly degenerate. Otherwise, pick $z \in L$ for which $T_{L/K}(z) \neq 0$. Then $\forall x \in L^\times, \langle x, z/x \rangle_{L/K} = T_{L/K}(z) \neq 0$ so $x \mapsto \langle x, y \rangle_{L/K}$ is not the zero map and the trace pairing is nondegenerate.

($T_{L/K}$ is a zero map iff the field extension L/K is inseparable.)

pf. If L/K is inseparable, for $\alpha \in L \setminus K$, its minimal polynomial in K is $x^{p^n} - a$ and therefore $T_{L/K} = \sum \alpha_i = -a_{d-1} = 0$.

Conversely, if L/K is separable, $\exists \alpha \in L \setminus K$ with its minimal polynomial f s.t. $f(\alpha) \neq 0$. Then $\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0 \Leftrightarrow a_{n-1} = -(\alpha + a_{n-2}\alpha^{-1} + \dots a_0\alpha^{n-1}) \in K$ is not 0).

AKLB means A is a Dedekind domain, $\text{Frac} A = K$, L/K is a finite separable extension of K , and B is the integral closure of A in L .

Proposition 2.1.11. *Assume AKLB. Then B is an A -lattice in L , and in particular B is finitely generated A -module.*

pf. By the proposition 2.2.9, B spans L as a K -vector space, so it contains a basis $(e_1, \dots, e_n)$ for L as a K -vector space. Let $M \subseteq B$ be the A -span of $(e_1, \dots, e_n)$. /then M is an A -lattice in L contained in B and it has a dual lattice M^* that contains the A -module $B^* \equiv \{x \in L : \langle x, b \rangle_{L/K} \in A \ \forall b \in B\}$. Then by the proposition 2.2.10, $B \subseteq B^*$ and therefore $M \subseteq B \subseteq B^* \subseteq M^*$. Then M^* is an A -lattice in L and hence finitely generated, hence noetherian. Thus its A -submodule B is finitely generated and thus an A -lattice in L by the theorem 2.2.9.

→ If L/K is inseparable, B needs not be finitely generated as an A -module.

Lemma 2.1.8. *Let B/A be an extension of domains with B integral over A and let $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ be primes of B . Then $\mathfrak{q}_0 \cap A \subsetneq \mathfrak{q}_1 \cap A$ and $\dim A \geq \dim B$.*

pf. Wlog replace B with $B/\mathfrak{q}_0$ and replace $A, \mathfrak{q}_0, \mathfrak{q}_1$ with their image in $B/\mathfrak{q}_0$. Then $\mathfrak{q}_0 = (0)$ and $\mathfrak{q}_1$ is a nonzero prime ideal. Let $\alpha \in \mathfrak{q}_1$ be nonzero. Its minimal polynomial $x^n + a_{n-1}x^{n-1} + \dots + a_0$ over K has coefficients in A , since $\alpha \in \mathfrak{q}_1 \subseteq B$ is integral over A . $a_0 \neq 0$ since if $a_0 = 0$, that polynomial is not irreducible. Thus

we have $a_0 = -\alpha a_1 - \cdots - \alpha^n \in \mathfrak{q}_1$ thus $0 \neq \mathfrak{q}_1 \cap A$. So $\mathfrak{q}_1 \cap A$ is not the zero ideal and therefore properly contains $\mathfrak{q}_0 \cap A = \{0\}$. We can apply this result repeatedly to any chain of distinct prime ideals in B to get a corresponding chain of distinct ideals in A . Then $\dim A \geq \dim B$.

Theorem 2.1.11. *Let A be a Dedekind domain with fraction field K , let L/K be a finite separable extension and let B be the integral closure of A in L . Then B is a Dedekind domain.*

pf. B is an integrally closed domain by definition. Also B is finitely generated over the noetherian ring A by the proposition 2.2.11. and therefore it is noetherian. Also B has dimension at most 1 since $\dim B \leq \dim A \leq 1$ by the previous lemma. Hence B is a Dedekind domain.

Corollary 2.1.16. *The ring of integers of a number field is a Dedekind domain.*

Assume AKLB. Then B is a Dedekind domain by the previous theorem. Every nonzero ideal in a Dedekind domain can be uniquely factored into prime ideals. If A has dimension 0, so does B so assume $\dim A = 1$, in which case $\dim B = 1$ as well. Thus In AKLB setup, we implicitly assume $A \neq K$.

Given a prime $\mathfrak{p}$ of A , consider the ideal $\mathfrak{p}B$, which may not be a prime, But since B is a Dedekind domain, it can be uniquely factored into nonzero prime ideals. Thus $\mathfrak{p}B = \prod_{\mathfrak{q}} \mathfrak{q}^{e_{\mathfrak{q}}}$ where $\mathfrak{q}$ ranges over primes of B and $e_{\mathfrak{q}} \geq 0$ are zero for all but finitely many primes $\mathfrak{q}$. The prime ideal $\mathfrak{q}$ for which $e_{\mathfrak{q}} > 0$ are said to lie over or above the prime ideal $\mathfrak{p}$.

Lemma 2.1.9. *Let A be a ring of dimension one contained in a Dedekind domain B . Let $\mathfrak{p}$ be a prime of A and let $\mathfrak{q}$ be a prime of B . Then $\mathfrak{q} | \mathfrak{p}$ iff $\mathfrak{q} \cap A = \mathfrak{p}$.*

pf. If $\mathfrak{q}$ divides $\mathfrak{p}B$, then it contains $\mathfrak{p}B$, since B is a Dedekind domain. Thus $\mathfrak{q} \cap A$ contains $\mathfrak{p}B \cap A$ which contains $\mathfrak{p}$. The ideal $\mathfrak{p}$ is maximal and $\mathfrak{q} \cap A \neq A$, $\mathfrak{q} \cap A = \mathfrak{p}$

If $\mathfrak{q} \cap A = \mathfrak{p}$, $\mathfrak{q} = \mathfrak{q}B$ contains $(\mathfrak{q} \cap A)B = \mathfrak{p}B$. Thus given B a Dedekind domain, $\mathfrak{q}$ divides $\mathfrak{p}B$.

$\rightarrow \exists \begin{matrix} \text{Spec } B & \rightarrow & \text{Spec } A \\ \mathfrak{q} & \mapsto & \mathfrak{q} \cap A \end{matrix}$, a surjective map.

The primes of A are all maximal, given $\dim A = 1$, and therefore each has an associated residue field $A/\mathfrak{p}$ and similarly for primes $\mathfrak{q}$ of B . If $\mathfrak{q}$ lies above $\mathfrak{p}$, we may regard the residue field $B/\mathfrak{q}$ as a field extension of $A/\mathfrak{p}$. A ring homomorphism of fields $A/\mathfrak{p} = A/(\mathfrak{q} \cap A) \rightarrow B/\mathfrak{q}$ is injective since the map $A \rightarrow B \rightarrow B/\mathfrak{q}$ has the kernel $\mathfrak{q} = A \cap \mathfrak{q}$.

Definition 2.1.19. *Assume AKLB and let $\mathfrak{p}$ be a prime of A . The exponent $e_{\mathfrak{q}}$ in the factorisation $\mathfrak{p}B = \prod_{\mathfrak{q} | \mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}}}$ is the ramification index of $\mathfrak{q}$ and the degree*

$f_{\mathbf{q}} = [B/\mathbf{q} : A/\mathbf{p}]$ of the corresponding residue field extension is the residue degree of $\mathbf{q}$.

Lemma 2.1.10. *Let A be a Dedekind domain with fraction field K , let $M/L/K$ be a tower of finite separable extension and let B and C be the integral closure of A in L and M respectively. Then C is the integral closure of B in M and if τ is a prime of M lying above a prime $\mathbf{q}$ of L lying above a prime $\mathbf{p}$ of K , then $e_{\tau/\mathbf{p}} = e_{\tau/\mathbf{q}}e_{\mathbf{q}/\mathbf{p}}, f_{\tau/\mathbf{p}} = f_{\tau/\mathbf{q}}f_{\mathbf{q}/\mathbf{p}}$.*

pf. If $\mathbf{q}$ divides $\mathbf{p}B$, then it contains $\mathbf{p}B$ by the "to divide is to contain", and therefore $\mathbf{p} \subseteq \mathbf{p}B \cap A \subseteq \mathbf{q} \cap A$. The ideal $\mathbf{p}$ is maximal and $\mathbf{q} \cap A \neq A$, $\mathbf{a} \cap A = \mathbf{p}$ if $\mathbf{p} \cap A = \mathbf{p}$, $\mathbf{q} = \mathbf{q}B$ certainly contains $(\mathbf{q} \cap A)B = \mathbf{p}B$ and B is a Dedekind domain, so $\mathbf{q}$ divides $\mathbf{p}B$.

$\rightarrow \exists \begin{array}{ccc} \text{Spec } B & \rightarrow & \text{Spec } A \\ \mathbf{q} & \mapsto & \mathbf{q} \cap A \end{array}$ which is surjective (for any nonzero $\mathbf{p} \in \text{Spec } A, \mathbf{p}B \neq 0$ and it is not a unit ideal, and therefore divisible by at least one $\mathbf{q} \in \text{Spec } B$).

Definition 2.1.20. *Assuem AKLB, and let $\mathbf{p}$ be a prime of A . The exponent $e_{\mathbf{q}}$ in the factorisation $\mathbf{p}B = \prod_{\mathbf{q}|\mathbf{p}} \mathbf{q}^{e_{\mathbf{q}}}$ is the ramification index of $\mathbf{q}$ and the degree $f_{\mathbf{q}} = [B/\mathbf{q} : A/\mathbf{p}]$ of the corresponding residue field extension is the residue degree of $\mathbf{q}$.*

Lemma 2.1.11. *Let A be a Dedekind domain with fraction field K , let $M/L/K$ be a tower of finite separable extension, and let B, C be the integral closures of A in L, M respectively. Then C is the integral closure of B in M and if τ is a prime of M lying above a prime $\mathbf{q}$ of L lying above a prime $\mathbf{p}$ of K , then $e_{\tau/\mathbf{p}} = e_{\tau/\mathbf{q}}e_{\mathbf{q}/\mathbf{p}}, f_{\tau/\mathbf{p}} = f_{\tau/\mathbf{q}}f_{\mathbf{q}/\mathbf{p}}$.*

pf. The integral closure of B in M lies in C and it contains C since $A \subseteq B$. Thus we have a tower of Dedekind extensions $C/B/A$. If $\tau|\mathbf{q}|\mathbf{p}$, then the factorisation of $\mathbf{p}C$ in C refines the factorisation of $\mathbf{p}B$ in B , so $e_{\tau/\mathbf{p}} = e_{\tau/\mathbf{q}}e_{\mathbf{q}/\mathbf{p}}$. The residue field embedding $A/\mathbf{p} \rightarrow C/\tau$ factors as $A/\mathbf{p} \rightarrow B/\mathbf{q} \rightarrow C/\tau, f_{\tau/\mathbf{p}} = f_{\tau/\mathbf{q}}f_{\mathbf{q}/\mathbf{p}}$.

Lemma 2.1.12. *Assuem AKLB, and let $\mathbf{p}$ be a prime of A . The dimension of $B/\mathbf{q}B$ as an $A/\mathbf{p}$ -vector space is equal to the dimension of L as a K -vector space.*

pf. Let $A_{\mathbf{p}} \equiv S^{-1}A, B_{\mathbf{q}} \equiv S^{-1}B$ be localisations of A and B respectively, where $S = A \setminus \mathbf{p}$. Then $A_{\mathbf{p}}/\mathbf{p}A = S^{-1}A/\mathbf{p}S^{-1}A \cong A/\mathbf{p}$ and $B_{\mathbf{p}}/\mathbf{p}B_{\mathbf{p}} \cong S^{-1}B/(\mathbf{p}S^{-1}B) \cong B/\mathbf{p}B$. Assume wlog A is a DVR, and therefore it is a PID. B is a finitely generated as an A -module and as an integral domain containing A it must be torsion-free. By the structure theorem for finitely generated modules over a PID, B is free of finite rank over A . Also B spans L as a K -vector space, so any A -basis for B is a K -basis for L . Thus $B \cong A^n$. Then $\mathbf{p}B \cong \mathbf{p}A^n = (\mathbf{p}A)^n \rightarrow B/\mathbf{p}B \cong A^n/(\mathbf{p}A)^n \cong (A/\mathbf{p})^n$ is a free $A/\mathbf{p}$ -module of dimension n .

Theorem 2.1.12. Assume AKLB. For each prime $\mathfrak{p}$ of A , $\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}} = [L : K]$.

pf. $B/\mathfrak{p}B \cong \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}}}$. Then by the previous proposition
 $[L : K] = [B/\mathfrak{p}B : A/\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} [B/\mathfrak{q}^{e_{\mathfrak{q}}} : A/\mathfrak{p}]$ (by the CRT)
 $= \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} [B/\mathfrak{q} : A/\mathfrak{p}]$ (given $B/\mathfrak{q}^{e_{\mathfrak{p}}}$ a $B/\mathfrak{q}$ vector space of dimension $e_{\mathfrak{q}}$)
 $= \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}}$.

For $\pi \in \mathfrak{q}$ a uniformiser of $B_{\mathfrak{q}}$, the images of $(\pi^0, \dots, \pi^{e_{\mathfrak{q}}-1})$ in $B/\mathfrak{q}^{e_{\mathfrak{q}}}$ are a $B/\mathfrak{q}$ -basis for $B/\mathfrak{q}^{e_{\mathfrak{q}}}$.

Corollary 2.1.17. Assume AKLB, and let $\mathfrak{p}$ be a prime of A . Let $g_{\mathfrak{p}} \equiv \#\{\mathfrak{q}|\mathfrak{p}\}$ denote the cardinality of the fibre above $\mathfrak{p}$. Then $g_{\mathfrak{p}}$ is an integer in the interval $[1, n]$, where $n = [L : K]$ as are $e_{\mathfrak{q}}, f_{\mathfrak{q}}$ for each $\mathfrak{q}|\mathfrak{p}$.

Definition 2.1.21. Assume AKLB, let $\mathfrak{p}$ be a prime of A .

- L/K is totally ramified at $\mathfrak{q}$ if $e_{\mathfrak{q}} = [L : K]$
- L/K is unramified at $\mathfrak{q}$ if $e_{\mathfrak{q}} = 1$ and $B/\mathfrak{q}$ is a separable extension of $A/\mathfrak{p}$.
- L/K is unramified above $\mathfrak{p}$ if it is unramified at all $\mathfrak{q}|\mathfrak{p}$, equivalently if $B/\mathfrak{p}B$ is a finite etale algebra over $A/\mathfrak{p}$

. When L/K is unramified above $\mathfrak{p}$, we say that

- $\mathfrak{p}$ remains inert in L if $\mathfrak{q} = \mathfrak{p}B$ is prime
- $\mathfrak{p}$ splits completely in L if $g_{\mathfrak{p}} = [L : K]$

e.g. Let $A \equiv \mathbb{R}[x], K = \text{Frac} A = \mathbb{R}(x), L = \mathbb{R}\sqrt{x^3 + 3x}$. The integral closure of A in L is the Dedekind domain $B = \mathbb{R}[x, y]/(y^2 - x^3 - 3x)$. Then $[L : K] = 2$.

-The prime $(x-1)$ factors in B into $(x-1) = (x-1, y-2)(x-1, y+2)$ since $y^2 - 4 = x^3 + 3x - 5 \in (x-1)$. Thus $e_{(x-1, y-2)} = 1, f_{(x-1, y-2)} = [B/(x-1, y-2) : A/(x-1)] = [\mathbb{R} : \mathbb{R}] = 1$. so L/K is unramified above $(x-1)$ and $(x-1)$ splits completely.

-The prime $(x+1)$ remains prime in B since $y^2 = -1$ has no solution in $\mathbb{R}$. Thus $e_{(x+1)} = 1, f_{(x+1)} = [B/(x+1) : A/(x+1)] \cong [\mathbb{C} : \mathbb{R}] = 2$. so L/K at $(x+1)$ is unramified above $\mathfrak{p}$ and $\mathfrak{p}$ remains inert.

-The prime (x) factors in B as $(x) = (x, y)^2, e_{(x, y)} = 2, f_{(x, y)} = 1$. so L/K is totally ramified at (x) .

For a ring extension B/A in which B is a free A -module of finite rank. the norm map is defined as $N_{B/A} : B \rightarrow A$
 $b \mapsto \det(B \rightarrow B)$. If A is a Dedekind domain, even when M is not free, $A_{\mathfrak{p}}$ is a DVR and therefore a PID, $M_{\mathfrak{p}}$ is a torsion-free

$A_{\mathbf{p}}$ -module, given it lies in a K -vector space and $A_{\mathbf{p}} \subseteq K$, and any finitely generated torion-free module over a PID is free. Thus its localisation $M_{\mathbf{p}}$ at any prime $\mathbf{p}$ of A is a free $A_{\mathbf{p}}$ -module, i.e. M is a locally free module.

Definition 2.1.22. *Let A be a Dedekind domain with fraction field K , let V be an n -dimensional K -vector space, let M, N be A -lattices in V and let $\mathbf{p}$ be a prime of A . Then $A_{\mathbf{p}}$ is a PID and we must have $M_{\mathbf{p}} \cong A_{\mathbf{p}}^n \cong N_{\mathbf{p}}$ by the above statement. Choose an $A_{\mathbf{p}}$ -module isomorphism $\phi_{\mathbf{p}} : M_{\mathbf{p}} \rightarrow N_{\mathbf{p}}$ and let $\hat{\phi}_{\mathbf{p}}$ denote the unique K -linear map $V \rightarrow V$ extending $\phi_{\mathbf{p}}$. The linear map $\hat{\phi}_{\mathbf{p}}$ is an isomorphism and therefore has nonzero determinant. The module index $[M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}} \equiv (\det \hat{\phi}_{\mathbf{p}})$.*

→ this does not depend on the choice of $\phi_{\mathbf{p}}$, since any other choice can be written as $\phi_1 \phi_{\mathbf{p}} \phi_2$, where $\phi_1 : M_{\mathbf{p}} \rightarrow M_{\mathbf{p}}, \phi_2 : N_{\mathbf{p}} \rightarrow N_{\mathbf{p}}$ that necessarily have the unit determinant.

The module index $[M : N]_A$ is the A -module $[M : N]_A \equiv \cap_{\mathbf{p}} [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$, where $\mathbf{p}$ ranges over primes of A and the intersection takes places in K . → $[M : N]_A$ is an A -submodule of K .

Proposition 2.1.12. *Let A be a Dedekind domain with fraction field K and let M, N be A -lattices in a K -vector space of finite dimension. The module index $[M : N]_A$ is a nonzero fractional ideal of A whose localisation at each prime $\mathbf{p}$ of A is equal to the local module index $[M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$.*

pf. The finitely generated A -module M is locally free, then there exists nonzero $a_1, \dots, a_r \in A$ generating unit ideal s.t. each $M[1/a_i]$ is a free $A[1/a_i]$ -module. Similarly we have $b_1, \dots, b_s \in A$ generating the unit ideal s.t. each $N[1/b_j]$ is a free $A[1/b_j]$ -module. For any pair a_i, b_j , if we localise at the multiplicative set $S_{ij} \equiv \{a_i^m b_j^n : m, n \in \mathbb{Z}_+\}$, then $S_{ij}^{-1}M, S_{ij}^{-1}N$ are free $S_{ij}^{-1}A$ -module and $([S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A})_{\mathbf{p}} = [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$, for all primes $\mathbf{p}$ of A that do not contain either a_i or b_j , since we can fix a global $S_{ij}^{-1}A$ -module isomorphism $\phi : S_{ij}^{-1}M \rightarrow S_{ij}^{-1}N$ that induces $A_{\mathbf{p}}$ -module isomorphisms $\phi_{\mathbf{p}} : M_{\mathbf{p}} \rightarrow N_{\mathbf{p}}$ with $(\det \hat{\phi})_{\mathbf{p}} = (\det \hat{\phi}_{\mathbf{p}})$. If a_i or b_j is contained in $\mathbf{p}$, $[S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A}$ is a unit ideal. Additionally, since $\{a_i\}, \{b_j\}$ both generate the unit ideal, for every prime $\mathbf{p}$, $\exists a_i, b_j$ s.t. they do not lie in $\mathbf{p}$. Thus

$$[M : N]_A = \cap_{\mathbf{p}} [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}} = \cap_{ij} [S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A}.$$

Let $I_{ij} \equiv [S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A}$. Each I_{ij} is a nonzero principal fractional $S_{ij}^{-1}A$ -ideal, and therefore we can choose a single $\alpha \in K^\times$ s.t. αI_{ij} is an $S_{ij}^{-1}A$ -ideal. The intersection of the αI_{ij} lies in $\cap_{ij} S_{ij}^{-1}A = A$ and therefore it is an A -submodule of A , hence an ideal and finitely generated because A is noetherian. Thus $[M : N]_A$

is a fractional ideal of A . It is nonzero since it contains the product of the generators of the I_{ij} .

The localisation of the intersection of a finite set of A -modules is equal to the intersection of their localisation, thus $([M : N]_A)_{\mathbf{p}} = (\cap_{ij} [S_{ij}^{-1}M : S_{ij}^{-1}N]_{S_{ij}^{-1}A})_{\mathbf{p}} =$

$$\cap_{ij} ([S_{ij}^{-1}M : S_{ij}^{-1}M]_{S_{ij}^{-1}A})_{\mathbf{p}}$$

$$= [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}.$$

$$\rightarrow [M : N]_A \in \mathcal{I}_A$$

$\rightarrow [M : N]_A [N : M]_A = [M : M]_A = A$ thus $[M : N]_A$ and $[N : M]_A$ are inverses in the ideal group $\mathcal{I}_A$.

$\rightarrow$ When $N \subseteq M$, $[M : N]_A \subseteq A$ is actually an ideal since in the case we can express a basis for $N_{\mathbf{p}}$ as $A_{\mathbf{p}}$ -linear combinations of a basis for $M_{\mathbf{p}}$ and the matrix for $\hat{\phi}_{\mathbf{p}}$ will have entries in $A_{\mathbf{p}}$.

If M, N are nonzero fractional ideals of A , then $M[M : N]_A = M(N : M) = N$ and if $N \subseteq M$, $I \equiv [M : N]_A \subseteq A$ is an ideal and $MI = N = NA \leftrightarrow M/N \cong A/I$ as the quotients of A -modules. Then $I = \{a \in A : aM \subseteq N\}$ is an annihilator of M/N , which is a cyclic A -module. Conversely, if $M/N \cong A/I$ for nonzero fractional ideals $N \subseteq M$, $I = [M : N]_A$.

Theorem 2.1.13. *Let A be a Dedekind domain with fraction field K , and let $N \subseteq M$ be A -lattices in a K -vector space V of dimension r for which the quotient module M/N is a direct sum of cyclic A -modules : $M/N \cong A/I_1 \oplus \cdots \oplus A/I_n$ where $I_1, \dots, I_n$ are nonzero ideals of A . Then $[M : N]_A = I_1 \dots I_n$.*

pf. Let $\mathbf{p}$ be a prime of A , let π be a uniformiser for $A_{\mathbf{p}}$ and let $e_j = v_{\mathbf{p}}(I_j)$ for $j \in [1, n]$. Pick a basis for $M_{\mathbf{p}}$ and an isomorphism $\phi_{\mathbf{p}} : M_{\mathbf{p}} \rightarrow N_{\mathbf{p}}$ so that $M_{\mathbf{p}}/N_{\mathbf{p}} = \text{coker } \phi_{\mathbf{p}}$. The matrix of $\phi_{\mathbf{p}}$ is an $r \times r$ matrix over the PID $A_{\mathbf{p}}$ with nonzero determinant. Thus it has UDV form with $U, V \in GL_r(A_{\mathbf{p}}), D = \text{diag}(\pi^{d_1}, \dots, \pi^{d_r})$ for some uniquely determined nonnegative integers $d_1 \leq \dots \leq d_r$. Then $A_{\mathbf{p}}/(\pi^{e_1}) \oplus \cdots \oplus A_{\mathbf{p}}/(\pi^{e_n}) \cong M_{\mathbf{p}}/N_{\mathbf{p}} = \text{coker } \phi \cong A_{\mathbf{p}}/(\pi^{d_1}) \oplus \cdots \oplus A_{\mathbf{p}}/(\pi^{d_r})$. By the structure theorem for modules over a PID, the nontrivial summands on each side are precisely the invariant factors of $M_{\mathbf{p}}/N_{\mathbf{p}}$, possibly in different orders. Thus $\sum_j e_j = \sum_i d_i$ and by the definition of the module index $[M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}} = (\det \phi_{\mathbf{p}}) = (\det D) = (\pi^{\sum d_i}) = (\pi^{\sum e_i}) = (\pi_{\mathbf{p}}^{e_1}) \dots (\pi_{\mathbf{p}}^{e_n}) = (I_1 \dots I_n)_{\mathbf{p}}$. Thus $[M : N]_A = I_1 \dots I_n$, since the localisations are $([M : N]_A)_{\mathbf{p}} = [M_{\mathbf{p}} : N_{\mathbf{p}}]_{A_{\mathbf{p}}}$, and $(I_1 \dots I_n)_{\mathbf{p}}$ coincide for every prime $\mathbf{p}$.

Definition 2.1.23. Assume AKLB. The ideal norm is the map
$$\begin{array}{ccc} N_{B/A} : \mathcal{I}_B & \rightarrow & \mathcal{I}_A \\ I & \mapsto & [B : I]_A \end{array}.$$
 We extend $N_{B/A}$ to the zero ideal by defining $N_{B/A}((0)) = (0)$.

Proposition 2.1.13. Assume AKLB and let $\alpha \in L$. Then $N_{B/A}((\alpha)) = (N_{L/K}(\alpha))$.

pf. The case $\alpha = 0$ is immediate. So assume $\alpha \in L^\times$. Then $N_{B/A}((\alpha)) = [B : \alpha B]_A = \cap_{\mathbf{p}} [B_{\mathbf{p}} : \alpha B_{\mathbf{p}}]_{A_{\mathbf{p}}} = (\det(L \rightarrow L)) = (N_{L/K}(\alpha))$, since each $B_{\mathbf{p}} \rightarrow \alpha B_{\mathbf{p}}$ is an isomorphism of free $A_{\mathbf{p}}$ -modules that are $A_{\mathbf{p}}$ -lattices in L .

Proposition 2.1.14. *Assume AKLB. The map $N_{B/A} \mathcal{I}_B \rightarrow \mathcal{I}_A$ is a group homomorphism.*

pf. Let $\mathbf{p}$ be a maximal ideal of A . Then $A_{\mathbf{p}}$ is a DVR and $B_{\mathbf{p}}$ is a semilocal Dedekind domain, hence a PID. Thus every element of $\mathcal{I}_{B_{\mathbf{p}}}$ is a principal ideal (α) for some $\alpha \in L^\times$, and by the previous proposition, $N_{B_{\mathbf{p}}/A_{\mathbf{p}}} : \mathcal{I}_B \rightarrow \mathcal{I}_{A_{\mathbf{p}}}$ is a group homomorphism, since $N_{L/K}$ is. For any $I, J \in \mathcal{I}_B$, $N_{B/A}(IJ) = \cap_{\mathbf{p}} N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(I_{\mathbf{p}} J_{\mathbf{p}}) = \cap_{\mathbf{p}} N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(I_{\mathbf{p}}) N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(J_{\mathbf{p}}) = N_{B/A}(I) N_{B/A}(J)$.

Corollary 2.1.18. *Assume AKLB. For all $I, J \in \mathcal{I}_B$, $[I : J]_A = N_{B/A}(I^{-1}J) = N_{B/A}((J : I))$.*

pf. Because B is a Dedekind domain, $(J : I) = I^{-1}J$ and therefore the second equality follows. Also $[I : J]_A = [I : B]_A [B : J]_A = [B : I]_A^{-1} [B : J]_A = N_{B/A}(I^{-1}) N_{B/A}(J) = N_{B/A}(I^{-1}J)$ by the previous proposition.

Corollary 2.1.19. *Assume AKLB and let I be a fractional ideal of B . The ideal norm of I is the fractional ideal of A generated by the image of I under the field norm $N_{L/K}$, that is, $N_{B/A}(I) = (N_{L/K}(\alpha) : \alpha \in I)$.*

pf. Let $J \equiv (N_{L/K}(\alpha) : \alpha \in I)$. Then for any nonzero prime $\mathbf{p}$ of A , the localisation of the ideal $N_{B/A}(I) = [B : I]_A$ at $\mathbf{p}$ is $[B_{\mathbf{p}} : I_{\mathbf{p}}]_{A_{\mathbf{p}}} = N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(I_{\mathbf{p}})$. The fractional ideal $N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(I_{\mathbf{p}})$ of $A_{\mathbf{p}}$ is principal, so $N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(I_{\mathbf{p}}) = J_{\mathbf{p}}$ by the proposition, and $N_{B/A}(I) = \cap_{\mathbf{p}} N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(I_{\mathbf{p}}) N_{B_{\mathbf{p}}/A_{\mathbf{p}}}(J_{\mathbf{p}}) = \cap_{\mathbf{p}} J_{\mathbf{p}} = J$.

Theorem 2.1.14. *Assume AKLB and let $\mathbf{q}$ be a prime lying above $\mathbf{p}$. Then $N_{B/A}(\mathbf{q}) = \mathbf{p}^{f_{\mathbf{q}}}$ where $f_{\mathbf{q}} = [B/\mathbf{q} : A/\mathbf{p}]$ is the residue field degree of $\mathbf{q}$.*

pf. The $(A/\mathbf{p})$ -vector space $B/\mathbf{q}$ has dimension $f_{\mathbf{q}}$; as a quotient of A -modules, we have $B/\mathbf{q} \cong A/\mathbf{p} \oplus \cdots \oplus A/\mathbf{p}$, an $f_{\mathbf{q}}$ -fold direct sum of cyclic A -modules $A/\mathbf{p}$, and therefore we can apply the previous theorem. Thus $N_{B/A}(\mathbf{q}) = [B : \mathbf{q}]_A = \mathbf{p} \cdots \mathbf{p} = \mathbf{p}^{f_{\mathbf{q}}}$.

Corollary 2.1.20. *Assume AKLB. For $I \in \mathcal{I}_A$, $N_{B/A}(IB) = I^n$ where $n = [L : K]$.*

pf. Since $N_{B/A}$ and $I \mapsto IB$ are group homomorphisms, we can reduce to the case where $I = \mathbf{p}$ is a nonzero prime. Then $N_{B/A}(\mathbf{p}B) = N_{B/A}(\prod_{\mathbf{q}|\mathbf{p}} \mathbf{q}^{e_{\mathbf{q}}}) = \prod_{\mathbf{q}|\mathbf{p}} N_{B/A}(\mathbf{q})^{e_{\mathbf{q}}} = \prod_{\mathbf{q}|\mathbf{p}} \mathbf{p}^{e_{\mathbf{q}} f_{\mathbf{q}}} = \mathbf{p}^{\sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{q}} f_{\mathbf{q}}} = \mathbf{p}^n$. e.g.

- The Ideal Norm in Number Fields

Let $A = \mathbb{Z}, K = \mathbb{Q}, B = \mathcal{O}_L$ the ring of integers of the number field L . Then for $\mathfrak{q}$ a nonzero prime ideal of $\mathcal{O}_L$, $N(\mathfrak{q}) = (\mathfrak{p}^f)$, where $p \in \mathbb{Z}$ is the unique prime in $\mathfrak{q} \cap \mathbb{Z}$ and f is the degree of the finite field $B/\mathfrak{q}$ as an extension of $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$.

Proposition 2.1.15. *Let L be a number field with ring of integers $\mathcal{O}_L$. For any nonzero $\mathcal{O}_L$ -ideal α , $N(\alpha) = ([\mathcal{O}_L : \alpha])$. If $\beta \subseteq \alpha$ are nonzero fractional ideals of $\mathcal{O}_L$, then $[\alpha : \beta]_{\mathbb{Z}} = ([\alpha : \beta])$.*

pf. The ring $\mathcal{O}_L$ is a free $\mathbb{Z}$ module of rank $n \equiv [L : \mathbb{Q}]$. It is free because it is a torsion-free module over $\mathbb{Z}$, which is a PID with rank n because it contains a $\mathbb{Q}$ -basis for L . The same is true of any nonzero fractional ideal of $\mathcal{O}_L$; thus $\alpha\mathcal{O}_L$ spans L , because the multiplication by α -map $L \rightarrow {}_{\alpha}L$ is an invertible $\mathbb{Q}$ -linear transformation. Fix $\mathbb{Z}$ -basis for $\mathcal{O}_L$ and the nonzero $\mathcal{O}_L$ -ideal α . Let $\Phi \in \mathbb{Z}^{n \times n}$ be the matrix whose columns express each basis element of α in terms of the basis for $\mathcal{O}_L$. Multiplication by Φ defines a $\mathbb{Z}$ -module isomorphism from $\mathcal{O}_L$ to α , since it maps the basis for $\mathcal{O}_L$ to the basis for α . Thus $[\mathcal{O}_L : \alpha]_{\mathbb{Z}} = (\det \Phi)$; for every prime $p \in \mathbb{Z}$, Φ is defined as a $\mathbb{Z}_{(p)}$ -module isomorphism $\phi_{(p)} : (\mathcal{O}_L)_{(p)} \rightarrow \alpha_{(p)}$ with $\det \hat{\phi}_{(p)} = \det \Phi$, given the fact that any $\mathbb{Z}$ -basis for a free $\mathbb{Z}$ -module M is also a $\mathbb{Z}_{(p)}$ -basis for the free $\mathbb{Z}_{(p)}$ -module $M_{(p)}$. Then if we identify $\mathcal{O}_L$ with $\mathbb{Z}^n$, $|\det \Phi|$ is the volume of a fundamental parallelepiped for α , viewed as a sublattice of $\mathbb{Z}^n$, the absolute value of the determinant of Φ is equal to the index of α in $\mathcal{O}_L$. Thus $([\mathcal{O}_L : \alpha]) = (\det \Phi) = [\mathcal{O}_L : \alpha]_{\mathbb{Z}} = N(\alpha)$, so the proposition follows.

For any $\alpha \in L^\times$, $[\alpha : \beta] = [a\alpha : a\beta]$, $[\alpha : \beta]_{\mathbb{Z}}$. Then we have a tower of free $\mathbb{Z}$ -modules $\beta \subseteq \alpha \subseteq \mathcal{O}_L \rightarrow [\mathcal{O}_L : \alpha][\alpha : \beta] = [\mathcal{O}_L : \beta]$ so by replacing both sides with the $\mathbb{Z}$ -ideals they generate, $N(\alpha)([\alpha : \beta]) = N(\alpha)N(\alpha^{-1}\beta) = N(\beta)$. so the rest of the proposition follows.

$\rightarrow \mathbb{Z}$ has only ± 1 as units, so we can identify each fractional ideal with a positive rational number and view the absolute norm $N : \mathcal{I}_{\mathcal{O}_L} \rightarrow \mathcal{I}_{\mathbb{Z}}$ as a homomorphism $N : \mathcal{I}_{\mathcal{O}_L} \rightarrow \mathbb{Q}_{++}^\times$ from the ideal group of $\mathcal{O}_L$ to the multiplicative group of positive rational numbers.

In the usual AKLB setup, by the primitive element theorem, when L/K is a finite separable extension, $L = K(\alpha)$, $\exists \alpha \in L$ and also $\alpha \in B$ but this does not mean $B = A[\alpha]$.

Definition 2.1.24. *Extensions L/K for which $B = A[\alpha]$ for some $\alpha \in B$ are said to be monogenic.*

→ then B is a free A -module and A -basis for B is an integral power basis $\{1, \alpha, \dots, \alpha^{n-1}\}$.

Theorem 2.1.15 (Dedekind-Kummer). *Assuem AKLB with $L = K(\alpha)$, $\alpha \in B$. Let $f \in A[x]$ be the minimal polynomial for α , let $\mathfrak{p}$ be a prime of A and let $\bar{f} = \bar{g}_1^{e_1} \dots \bar{g}_r^{e_r}$ be its factorisation into monic irreducibles in $(A/\mathfrak{p})[x]$. Let $\mathfrak{q}_i \equiv (\mathfrak{p}, g_i(\alpha))$, where $g_i \in A[x]$ is any lift of $\bar{g}_i$ in $(A/\mathfrak{p})[x]$ under the reduction map $A[x] \rightarrow (A/\mathfrak{p})[x]$. If $B = A[\alpha]$, then $\mathfrak{p}B = \mathfrak{q}_1^{e_1} \dots \mathfrak{q}_r^{e_r}$, is the prime factorisation of $\mathfrak{p}B$ in B and the residue field degree of $\mathfrak{q}_i$ is $\deg \bar{g}_i$.*

pf. We have $B = A[\alpha] \cong A[x]/(f)$ and therefore $\frac{B}{\mathfrak{q}_i} = \frac{A[\alpha]}{(\mathfrak{p}, g_i(\alpha))} \cong \frac{A[x]}{(f, \mathfrak{p}, g_i)} \cong \frac{(A/\mathfrak{p})[x]}{(\bar{f}, \bar{g}_i)} \cong \frac{(A/\mathfrak{p})[x]}{\bar{g}_i}$, which is a field, given $\bar{g}_i$ irreducible. Thus $\mathfrak{q}_i$ is a prime above $\mathfrak{p}$ with residue field degree $f_{\mathfrak{q}_i} = \deg \bar{g}_i$ as claimed.

The ideal $\Pi_i \mathfrak{q}_i^{e_i} = \Pi_i (\mathfrak{p}, g_i(\alpha))^{e_i} = \Pi (\mathfrak{p}B + (g_i(\alpha)))^{e_i}$ is divisible by $\mathfrak{p}B$ since $\Pi_i (g_i(\alpha))^{e_i} \equiv (f(\alpha)) \equiv (0) \pmod{\mathfrak{p}B}$. The $\bar{g}_i(x)$ are distinct elements of $(A/\mathfrak{p})[x]/(f) \cong A[x]/(\mathfrak{p}, f) \cong A[\alpha]/\mathfrak{p}A[\alpha]$ and therefore $g_i(\alpha)$ are distinct module $\mathfrak{p}B$. Thus the prime ideals $\mathfrak{q}_i$ are distinct and $e_i \geq e_{\mathfrak{q}_i}$ and $\{\mathfrak{q}|\mathfrak{p}\} \subseteq \{\mathfrak{q}_i\}$ in order for $\Pi \mathfrak{q}_i^{e_i}$ to be divisible by $\mathfrak{p}B$. Thus $\{\mathfrak{q}_i\} = \{\mathfrak{q}|\mathfrak{p}\}$ since $\mathfrak{q}_i$ is a prime above $\mathfrak{p}$. Now $N_{B/A}(\Pi \mathfrak{q}_i^{e_i}) = \Pi N_{B/A}(\mathfrak{q}_i)^{e_i} = \Pi (\mathfrak{p}^{f_{\mathfrak{q}_i}})^{e_i} = \mathfrak{p}^{e_i \deg \bar{g}_i} = \mathfrak{p}^{\deg f} = \mathfrak{p}^{[L:K]}$ so $e_i = e_{\mathfrak{q}_i}$.

Definition 2.1.25. *Let S/R be an extension of commutative rings. The conductor of R in S is the largest S -ideal that is also an R -ideal, equivalently it is the largest ideal of S contained in R . It can be written as $\mathfrak{c} \equiv \{\alpha \in S : \alpha S \subseteq R\} = \{\alpha \in R : \alpha S \subseteq R\}$. If R is an integral domain, the conductor of R is the conductor of R in its integral closure.*

Lemma 2.1.13. *Let R be a noetherian domain. The conductor of R in its integral closure S is nonzero iff S is finitely generated as an R -module.*

If R has nonzero conductor, its integral closure S is a fractional ideal of R that is also a ring, i.e. $S = \frac{1}{r}I$, $r \in R$, I is an R -ideal, and the conductor $\mathfrak{c}$ is the set of denominator $r \in R$ for which $S = \frac{1}{r}I$.

Definition 2.1.26. *An order $\mathcal{O}$ is a noetherian domain of dimension one whose conductor is nonzero, equivalently, whose integral closure is finitely generated as an $\mathcal{O}$ -module.*

Every Dedekind domain that is not a field is an order. The integral closure of an order is a Dedekind domain, but not every ring whose integral closure is a Dedekind domain is an order. But in AKLB, B is finitely generated over A and therefore over every intermediate ring between A and B , including all those whose integral closure is B .

Definition 2.1.27. Let A be a noetherian domain with fraction field K , and let L be a K -algebra of finite dimension. An A -order in L is an A -lattice that is also a ring.

Proposition 2.1.16. Assume AKLB and let $\mathcal{O}$ be a subring of L . Then $\mathcal{O}$ is an A -order in L iff it is an order with integral closure of B .

pf. Under AKLB assumption, $\dim A = 1 = \dim B$ since $A = B \cap K, \mathcal{O} \subseteq L$ is an A -module containing 1, so it contains A . Suppose $\mathcal{O}$ is an A -order in L . Then $\mathcal{O}$ is an A -lattice, hence finitely generated as an A -module, and therefore integral over A . Thus $\mathcal{O}$ lies in the integral closure B of A in L . The fraction field of $\mathcal{O}$ is a K -vector space spanning $4L$, hence equal to $4L$, so $\mathcal{O}$ and B have the same fraction field and B is the integral closure of $\mathcal{O}$. Thus $\mathcal{O}$ is a domain of dimension 1, and it is noetherian because it is a finitely generated module over the noetherian ring A . The integral closure B of $\mathcal{O}$ is finitely generated over A , hence over $\mathcal{O}$; hence it is an order.

Suppose $\mathcal{O}$ is an order with the integral closure B . It is an A -submodule of the noetherian A -module B , hence finitely generated over A . It contains a K -basis for L because L is its fraction field. Thus $\mathcal{O}$ is an A -lattice in L that is also a ring and therefore is an A -order in L .

Definition 2.1.28. An A -order in L is a maximal order if it is not properly contained in any other A -order in L .

→ if A is a Dedekind domain, every A -order in L lies in a maximal order, and under the AKLB assumption, B is the unique maximal order.

Proposition 2.1.17. In any order $\mathcal{O}$, only finitely many primes contain the conductor.

pf. The map
$$\begin{array}{ccc} \text{Spec } B & \rightarrow & \text{Spec } \mathcal{O} \\ \mathfrak{q} & \mapsto & \mathfrak{q} \cap \mathcal{O} \end{array}$$
 is surjective, since the fact that B is integrally closed ensures every prime $\mathfrak{p}$ of $\mathcal{O}$ has at least one prime $\mathfrak{q}$ lying above it. If a prime $\mathfrak{q}$ of B contains the conductor $\mathfrak{c}$, so does $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}$ and conversely. In B , we can factor $\mathfrak{c}$ into a product of powers of finitely many primes $\mathfrak{q}$, and therefore only finitely many prime $\mathfrak{p}$ of $\mathcal{O}$ contain $\mathfrak{c}$.

Lemma 2.1.14. Let $\mathcal{O}$ be an order with integral closure B and conductor $\mathfrak{c}$ and let $\mathfrak{p}$ be a prime of $\mathcal{O}$ not containing $\mathfrak{c}$. then $\mathfrak{p}B$ is prime of B .

pf. Let $\mathfrak{q}$ be a prime of B , lying above $\mathfrak{p}$, so that $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}$, and pick an element $s \in \mathfrak{c}$ not in $\mathfrak{p}$. Then if $a/b \in \mathcal{O}_{\mathfrak{p}}$ with $a \in \mathcal{O}, b \in \mathcal{O} \setminus \mathfrak{p}$, then $b \in B \setminus \mathfrak{q}$ and therefore $a/b \in B_{\mathfrak{q}} \rightarrow \mathcal{O}_{\mathfrak{p}} \subseteq B_{\mathfrak{q}}$
If $a/b \in B_{\mathfrak{q}}$ with $a \in B, b \in B \setminus \mathfrak{q}$, then $sa \in \mathcal{O}, sb \in \mathcal{O} \setminus \mathfrak{p}$ so $(sa)/(sb) = a/b \in \mathcal{O}_{\mathfrak{p}} \rightarrow$

$B_{\mathbf{q}} \subseteq \mathcal{O}_{\mathbf{p}}$ and therefore $\mathcal{O}_{\mathbf{p}} = B_{\mathbf{q}}$. Note that $\mathbf{q}' \mid \mathbf{p}$ implies $B_{\mathbf{q}'} = \mathcal{O}_{\mathbf{p}} = B_{\mathbf{q}}$ which also implies $\mathbf{q}' = \mathbf{q}$. So there is only one prime $\mathbf{q}$ lying above $\mathbf{p}$. Thus $\mathbf{p}B = \mathbf{q}^e$, $e \geq 1$. Since we must have $\mathbf{p}\mathcal{O}_{\mathbf{p}} = \mathbf{q}B_{\mathbf{q}}$, $\mathbf{q}^e B_{\mathbf{q}} = \mathbf{q}B_{\mathbf{q}} \rightarrow e = 1$.

Corollary 2.1.21. *Let $\mathcal{O}$ be an order with integral closure B and conductor $\mathbf{c}$. The restriction of the map $\text{Spec}B \rightarrow \text{Spec}\mathcal{O}$ defined by $\mathbf{q} \mapsto \mathbf{q} \cap \mathcal{O}$ to the prime ideals not containing $\mathbf{c}$ is a bijection with inverse $\mathbf{p} \mapsto \mathbf{p}B$.*

Theorem 2.1.16. *Let $\mathcal{O}$ be an order with integral closure B and conductor $\mathbf{c}$, and let $\mathbf{p}$ be a prime of $\mathcal{O}$. TFAE*

1. $\mathbf{p}$ does not contain $\mathbf{c}$
2. $\mathcal{O} = \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}$
3. $\mathbf{p}$ is invertible
4. $\mathcal{O}_{\mathbf{p}}$ is a DVR
5. $\mathbf{p}\mathcal{O}_{\mathbf{p}}$ is principal.

If any of these equivalent properties hold, then $\mathbf{p}B$ is a prime of B .

pf.

- $i \rightarrow ii$
 $\forall x \in \mathcal{O}, x\mathbf{p} \subseteq \mathbf{p}$ and therefore $x \in \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}$. If $x \in \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}$, $x = a/b$ with $a, b \in \mathcal{O}$, $(a, b) = \mathbf{s.t.}$ there exists a minimal monic polynomial satisfying $(a/b)^n + \alpha_{n-1}(a/b)^{n-1} + \dots + \alpha_0 = 0 \leftrightarrow a^n = -b(\alpha_{n-1}a^{n-1} + \dots + b^{n-1}\alpha_0) \leftrightarrow b \mid a$ this is only possible when $b = 1$ given $\mathbf{p}$ does not contain $\mathbf{c}$ and therefore $x \in \mathcal{O}$ and therefore $\mathcal{O} = \{x \in B : x\mathbf{p} \subseteq \mathbf{p}\}$
- $ii \rightarrow i$
 If $\mathbf{c} \subseteq \mathbf{p}$, $\forall x \in B, xB = \mathbf{c}$ which implies $B = \mathcal{O}$ a contradiction. hence $\mathbf{c} \not\subseteq \mathbf{p}$.
- $i \rightarrow iii$
 $\exists x \in \mathbf{c} \setminus \mathbf{p}$ by i . Also there exists $\mathcal{O}$ -ideal I s.t. $\frac{1}{x}I = B$ by definition. Then $\mathcal{O} = \{x' \in B : x'\mathbf{p} \subseteq \mathbf{p}\}$ and therefore $\mathcal{O} = \{x' \in \frac{1}{x}I : x'\mathbf{p} \subseteq \mathbf{p}\}$. Then $I\mathbf{p} \subseteq \mathcal{O}$ and $\forall y \in \mathcal{O} = \{x' \in \frac{1}{x}\mathcal{O} : x'\mathbf{p} \subseteq \mathbf{p}\}$, $y = \frac{r}{x}$ with $r \in \mathcal{O}$ w.t. $\frac{r}{x}\mathbf{p} \subseteq \mathbf{p}$. Thus $x = \frac{rp'}{x} = a\bar{p}$, $a \in \mathcal{O}$ and $r \in I$, $\frac{p'}{x} \in \mathbf{p}$ and therefore $\mathbf{p}$ is invertible.
- $iii \rightarrow iv$
 Since $\mathcal{O}$ is an order, $\mathcal{O}_{\mathbf{p}}$ is a local ring and noetherian with dimension 1. Its unique maximal ideal is $\mathbf{p}$. Assume it is not principal. Then $\mathbf{p} = (a, b)$, $a, b \in \mathcal{O}$. Since $\mathbf{p}$ is invertible, $\exists J$ s.t. $J\mathbf{p} = \mathcal{O}$. Then $\exists c \in J$ s.t. $c(ra + r'b) = u$ where

u is a unit in $\mathcal{O}$. Thus $u^{-1}cra + u^{-1}cr'b = 1$ which means a, b are coprime and therefore $0 \subsetneq (a) \subsetneq (a, b) = \mathfrak{p}$ a contradiction. Hence it is principal and therefore it is a DVR.

- $iv \leftrightarrow v$

If the unique maximal of $\mathcal{O}_{\mathfrak{p}} = \mathfrak{p}\mathcal{O}_{\mathfrak{p}}$ is principal, $\mathcal{O}_{\mathfrak{p}}$ is a PID. Thus it is a local PID and therefore is a DVR.

$\mathcal{O}_{\mathfrak{p}}$ is a DVR and therefore it is a PID so its maximal is principal.

- $iv \rightarrow i$

Assume the contrary, i.e. $\mathfrak{c} \subseteq \mathfrak{p}$. If $\mathfrak{c}$ is prime, then $\mathfrak{p} = \mathfrak{c}$ and $\mathcal{O}_{\mathfrak{p}} = B$. If not, $\exists a \in \mathfrak{p} \setminus \mathfrak{c}$. For such an a , $(a) \neq \mathfrak{p}$. Also $\forall b \in \mathfrak{c}$, $(b) = \mathfrak{p}$, i.e. $\mathfrak{p}\mathcal{O}_{\mathfrak{p}}$ is not principal hence $\mathcal{O}_{\mathfrak{p}}$ is not a DVR, a contradiction.

Orders in Dedekind domains also have a geometric interpretation. If $\mathcal{O}$ is an order, the curve $X = \text{Spec } \mathcal{O}$ have a singularity at each closed point P corresponding to a maximal of $\mathcal{O}$ that contains the conductor. Taking the integral closure B of $\mathcal{O}$ yields a smooth curve $Y = \text{Spec } B$ with the same function field as X and a morphism $Y \rightarrow X$ that looks like a bijection above non-singular point. The curve Y is called the normalisation of X .

Definition 2.1.29. Let A be a noetherian domain and let J be an ideal of A . A fractional ideal I of A is prime to J if $IA_{\mathfrak{p}} = A_{\mathfrak{p}}$ for all prime ideals $\mathfrak{p}$ that contain J . The set of invertible fractional ideals prime to J is denoted $\mathcal{I}_A^J$, which is a subgroup of the ideal group $\mathcal{I}_A$.

Theorem 2.1.17. Let $\mathcal{O}$ be an order with integral closure B . Let $\mathfrak{I}$ be any ideal of B contained in the conductor of $\mathcal{O}$. The map $\mathfrak{q} \mapsto \mathfrak{q} \cap \mathcal{O}$ induces a group isomorphism from $\mathcal{I}_B^{\mathfrak{c}}$ to $\mathcal{I}_{\mathcal{O}}^{\mathfrak{c}}$ and both groups are isomorphic to the free abelian group generated by their prime ideals. In particular, every fractional ideal of $\mathcal{O}$ prime to the conductor has a unique factorisation into prime ideals $\prod \mathfrak{p}_i^{e_i}$ which matches the factorisation $IB = \prod \mathfrak{q}_i^{e_i}$ with $\mathfrak{p}_i = \mathfrak{q}_i \cap \mathcal{O}$.

pf. The B -ideal $\mathfrak{c}$ lies in the conductor of $\mathcal{O}$ and is therefore also an $\mathbf{O}$ -ideal, so the subgroups $\mathcal{I}_B^{\mathfrak{c}}$ and $\mathcal{I}_{\mathcal{O}}^{\mathfrak{c}}$ are well defined and the map $\mathfrak{q} \mapsto \mathfrak{q} \cap \mathcal{O}$ gives a bijection between the sets of prime ideals contained in these subgroups by the previous corollary. So the theorem follows.

2.1.3 Galois Extensions

In AKLB setup, add another assumption that L/K is normal, hence Galois, and let $G \equiv \text{Gal}(L/K)$ to denote the Galois group. Then we denote it by AKLBG.

Definition 2.1.30. Let G be a group. A left G -module is an abelian group M equipped with a left G -action that commutes with its group operation; in additive notation, we have $\sigma(a + b) = \sigma(a) + \sigma(b)$, $\forall \sigma \in G, a, b \in M$. One similarly defines a right G -module as an abelian group with a right G -action that commutes with the group operation.

Theorem 2.1.18. Assume AKLBG. For each fractional ideal I of B and $\sigma \in G$, define $\sigma(I) \equiv \{\sigma(x) : x \in I\}$. The set $\sigma(I)$ is a fractional ideal of B , and this defines a group action on $\mathcal{I}_B$ that makes it a left G -module. Moreover, the restriction of this action to $\text{Spec}B$ makes it a G -set.

pf. **CLAIM:** $\sigma(B) = B, \forall \sigma \in G$

pf. $\forall b \in B$ is integral over A hence $f(b) = 0$ for some monic polynomial $f \in A[x]$ and $0 = \sigma(0) = \sigma(f(b)) = f(\sigma(b)) \rightarrow \sigma(b) \in B$, given B is an integral closure of A . Thus $\sigma(B) \subseteq B$. Similarly, $\sigma^{-1}(B) \subseteq B \leftrightarrow B \subseteq \sigma(B)$ and therefore the claim follows.

Each $\sigma \in G = \text{Gal}(L/K)$ is a field automorphism of L and thus commutes with addition and multiplication. Thus if $I \subseteq B$ is a finitely generated B -module, $\sigma(I)$ is a finitely generated $\sigma(B)$ -module and therefore B -module, i.e. I is a fractional ideal. Clearly $\sigma((0)) = (0)$, $\forall \sigma \in G$ so G permutes $\mathcal{I}_B$, the group of nonzero fractional ideals. Also $(\sigma\tau)(I) = \{(\sigma\tau)(x) : x \in I\} = \{\sigma(\tau(x)) : x \in I\} = \{\sigma(y) : y \in \tau(I)\} = \sigma(\tau(I))$ so G is a left action of $\mathcal{I}_B$.

Let $I, J \in \mathcal{I}_B, \sigma \in G$. Each $x \in IJ$ has the form $x = \sum_i a_i b_i, a_i \in I, b_i \in J$ and $\sigma(x) = \sum_i \sigma(a_i) \sigma(b_i) \in \sigma(I) \sigma(J)$ thus $\sigma(IJ) \subseteq \sigma(I) \sigma(J)$. Similarly, $\sigma^{-1}(\sigma(I) \sigma(J)) \subseteq IJ \leftrightarrow \sigma(I) \sigma(J) \subseteq \sigma(IJ)$ and therefore $\sigma(IJ) = \sigma(I) \sigma(J)$ implying that $\mathcal{I}_B$ is a left G -module.

Let $\mathbf{p}$ be a prime of B and let $\sigma(\mathbf{p}) = \mathbf{q}_1^{e_1} \dots \mathbf{q}_n^{e_n}$ be the unique factorisation of $\sigma(\mathbf{p})$ in B . Applying σ^{-1} to both sides yields $\mathbf{p} = \sigma^{-1}(\mathbf{q}_1)^{e_1} \dots \sigma^{-1}(\mathbf{q}_n)^{e_n}$ and therefore $n = 1$ and $e_1 = 1$, given $\mathbf{p}$ prime and the G -action of $\mathcal{I}_B$ restricts to a G -action on $\text{Spec}B$ and $\text{MaxSpec}B$.

Corollary 2.1.22. Assume AKLBG. For each prime $\mathbf{p}$ of A , the group G acts transitively on the set $\{\mathbf{q}|\mathbf{p}\}$; in other words, the orbits of the G -action on $\text{Spec}B$ are the fibres of the contraction map $\text{Spec}B \rightarrow \text{Spec}A$.

pf. Consider any $\sigma \in G$. For $\mathbf{q}|\mathbf{p}$, we have $\mathbf{p}B \subseteq \mathbf{q}$ and $\sigma(\mathbf{p}B) \subseteq \sigma(\mathbf{q})$ so $\sigma(\mathbf{q})|\mathbf{p}$. Thus $\{\mathbf{q}|\mathbf{p}\}$ is closed under the action of G . Let $\{\mathbf{q}|\mathbf{p}\} = \{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ and suppose $\mathbf{q}_1, \mathbf{q}_2$ lie in distinct G -orbits. The primes $\mathbf{q}_i$ are maximal ideals, hence pairwise coprime, so by the CRT there is a ring isomorphism $\frac{B}{\mathbf{q}_1 \dots \mathbf{q}_n} \cong \frac{B}{\mathbf{q}_1} \times \dots \times \frac{B}{\mathbf{q}_n}$ and we may choose $b \in B$ s.t. $b \equiv 0 \pmod{\mathbf{q}_2}, b \equiv 1 \pmod{\sigma^{-1}(\mathbf{q}_1)}, \forall \sigma \in G$. Then $b \in \mathbf{q}_2$ and $N_{L/K}(b) = \prod_{\sigma \in G} \sigma(b) \equiv 1 \pmod{\mathbf{q}_1}$ hence $N_{L/K}(b) \notin A \cap \mathbf{q}_1 = \mathbf{p}$. But $N_{L/K}(b) \in N_{L/K}(\mathbf{q}_2) = \mathbf{p}^{f_{q_2}} \subseteq \mathbf{p}$, a contradiction.

$\rightarrow \forall \sigma \in G = \text{Gal}(L/K)$, it restricts to a ring automorphism of B that fixes every

element of the subring $A = B \cap K$ and thus every element of any prime $\mathfrak{p}$ of A . So it induces an isomorphism of residue field extensions $\bar{\sigma} \in \text{Hom}_{A/\mathfrak{p}}(B/\mathfrak{q}, B/\sigma(\mathfrak{q}))$ defined by $\bar{\sigma}(x + \mathfrak{q}) \equiv \sigma(x) + \sigma(\mathfrak{q})$ for $x \in B$.

Corollary 2.1.23. *Assume AKLBG and let $\mathfrak{p}$ be a prime of A . The residue field degrees $f_{\mathfrak{q}} \equiv [B/\mathfrak{q} : A/\mathfrak{p}]$ are the same for every $\mathfrak{q}|\mathfrak{p}$ as are the ramification indices $e_{\mathfrak{q}} \equiv v_{\mathfrak{q}}(\mathfrak{p}B)$.*

pf. $\forall \sigma \in G$, we have an isomorphism of the residue field $B/\mathfrak{q}$ and $B/\sigma(\mathfrak{q})$ that fixes $A/\mathfrak{p}$, so they clearly have the same degree $f_{\mathfrak{q}} = f_{\sigma(\mathfrak{q})}$ and G acts transitively on $\{\mathfrak{q}|\mathfrak{p}\}$ by the previous corollary, so the function $\mathfrak{q} \mapsto f_{\mathfrak{q}}$ must be constant on $\{\mathfrak{q}|\mathfrak{p}\}$. $\forall \sigma \in G$, $\sigma(\mathfrak{p}) = \mathfrak{p}$, $\sigma(B) = B$, so $\sigma(\mathfrak{p}B) = \mathfrak{p}B$ and for each $\mathfrak{q}|\mathfrak{p}$, $e_{\mathfrak{q}} = v_{\mathfrak{q}}(\mathfrak{p}B) = v_{\mathfrak{q}}(\sigma(\mathfrak{p}B)) = v_{\mathfrak{q}}(\sigma(\Pi_{\tau|\mathfrak{p}} \tau^{e_{\tau}})) = v_{\mathfrak{q}}(\Pi \sigma(\tau)^{e_{\tau}}) = v_{\mathfrak{q}}(\Pi \tau^{e_{\sigma^{-1}(\tau)}}) = e_{\sigma^{-1}(\mathfrak{q})}$. The transitivity of the G -action on $\{\mathfrak{q}|\mathfrak{p}\}$ again implies $\mathfrak{q} \mapsto e_{\mathfrak{q}}$ is constant on $\{\mathfrak{q}|\mathfrak{p}\}$.

Corollary 2.1.24. *Assume AKLBG. For each prime $\mathfrak{p}$ of A , $e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}} = [L : K]$, where $g_{\mathfrak{p}} \equiv \#\{\mathfrak{q}|\mathfrak{p}\}$.*

Assume AKLBG. When $n \equiv [L : K]$ is a prime, there are three ways a prime $\mathfrak{p}$ of A can split in $B \rightarrow L$

$e_{\mathfrak{p}} = n, f_{\mathfrak{p}} = g_{\mathfrak{p}} = 1$	$\mathfrak{p}$ is totally ramified in L
$f_{\mathfrak{p}} = n, e_{\mathfrak{p}} = g_{\mathfrak{p}} = 1$	$\mathfrak{p}$ remains inert in L if $B/\mathfrak{p}B$ is finite etale
$g_{\mathfrak{p}} = n, e_{\mathfrak{p}} = f_{\mathfrak{p}} = 1$	$\mathfrak{p}$ splits completely in L if $B/\mathfrak{p}B$ is finite etale

Definition 2.1.31. *Assume AKLBG. For each prime $\mathfrak{q}$ of B , the decomposition group $D_{\mathfrak{q}}$ denoted $D_{\mathfrak{q}}(L/K)$ is the stabiliser of $\mathfrak{q}$ in G .*

Lemma 2.1.15. *Assume AKLBG and let $\mathfrak{p}$ be a prime of A . The decomposition groups $D_{\mathfrak{q}}$ for $\mathfrak{q}|\mathfrak{p}$ are all conjugates in G , with $\#D_{\mathfrak{q}} = e_{\mathfrak{p}} f_{\mathfrak{p}}$, $[G : D_{\mathfrak{q}}] = g_{\mathfrak{p}}$.*

pf. Points in an orbit of group action have conjugate stabilisers, so the $D_{\mathfrak{q}}$ for $\mathfrak{q}|\mathfrak{p}$ are all conjugate by the corollary 2.2.22. The orbit stabiliser theorem implies $[G : D_{\mathfrak{q}}] = \#\{\mathfrak{q}|\mathfrak{p}\} = g_{\mathfrak{p}}$. So $\#G = [L : K] = e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}}$ by the previous corollary. so $\#D_{\mathfrak{q}} = \#G/[G : D_{\mathfrak{q}}] = e_{\mathfrak{p}} f_{\mathfrak{p}}$.

$\forall \sigma \in G$ induces a residue field isomorphism $\bar{\sigma} \in \text{Hom}_{A/\mathfrak{p}}(B/\mathfrak{q}, B/\sigma(\mathfrak{q}))$. If $\sigma \in D_{\mathfrak{q}}$, $\sigma(\mathfrak{q}) = \mathfrak{q}$, $\bar{\sigma} \in \text{Aut}_{A/\mathfrak{p}}(B/\mathfrak{q})$ and the map $\sigma \mapsto \bar{\sigma}$ defines a group homomorphism $\pi_{\mathfrak{q}} : D_{\mathfrak{q}} \rightarrow \text{Aut}_{A/\mathfrak{p}}(B/\mathfrak{q})$.

Proposition 2.1.18. *Assume AKLBG and let $\mathfrak{q}|\mathfrak{p}$ be a prime of B . The group homomorphism $\pi_{\mathfrak{q}} : D_{\mathfrak{q}} \rightarrow \text{Aut}_{A/\mathfrak{p}}(B/\mathfrak{q})$ is surjective and $B/\mathfrak{q}$ is normal over $A/\mathfrak{p}$.*

pf. Let F be the separable closure of $A/\mathbf{p}$ in $B/\mathbf{q}$ and for $\bar{b} \in F$, $\exists b \in B$ s.t. $b \equiv \bar{b} \pmod{\mathbf{q}}$, $b \equiv 0 \pmod{\sigma^{-1}(\mathbf{q})}$, $\forall \sigma \in G \setminus D_{\mathbf{q}}$; by the CRT, such an b exists since $\forall \sigma \in G \setminus D_{\mathbf{q}}$, the ideal $\mathbf{q}, \sigma(\mathbf{q})$ are distinct and therefore coprime. Now define $g(x) \equiv \prod_{\sigma \in G} (x - \sigma(b)) \in A[x]$ and let $\bar{g}$ denote the image of g in $(A/\mathbf{p})[x]$. Then $\bar{b}$ is the root of a polynomial $\bar{g} \in (A/\mathbf{p})[x]$ that splits completely in $(B/\mathbf{q})[x]$ and since $\bar{b}$ was chosen arbitrarily, this applies to all $\bar{b} \in F^\times$. Thus F is normal extension of $A/\mathbf{p}$ and $\text{Gal}(F/(A/\mathbf{p})) \cong \text{Aut}_{A/\mathbf{p}}(B/\mathbf{q})$, since F is the separable closure of $A/\mathbf{p}$ in $B/\mathbf{q}$. For each $\sigma \in G \setminus D_{\mathbf{q}}$, $\bar{\sigma}(\bar{b}) = 0$ so 0 is a root of $\bar{g}(x)$ with multiplicity at least $m = \#(G \setminus D_{\mathbf{q}})$ and the remaining roots are $\bar{\sigma}(\bar{b})$ for $\sigma \in D_{\mathbf{q}}$, all of which are $\text{Gal}(F/(A/\mathbf{p}))$ -conjugates of $\bar{b}$. Thus $\bar{g}(x)/x^m$ divides a power of the minimal polynomial $f(x)$ of $\bar{b}$, but $f(x)$ is irreducible in $(A/\mathbf{p})[x]$, so $\bar{g}(x)/x^m$ is a power of $f(x)$ and every $\text{Gal}(F/(A/\mathbf{p}))$ -conjugate of $\bar{b}$ has the form $\bar{\sigma}(\bar{b})$ for some $\sigma \in D_{\mathbf{q}}$. Applying this to $\bar{b}$ chosen so that $F = (A/\mathbf{p})(\bar{b})$ shows that the map $\pi_{\mathbf{q}} : D_{\mathbf{q}} \rightarrow \text{Aut}_{A/\mathbf{p}}(B/\mathbf{q}) \cong \text{Gal}(F/(A/\mathbf{p}))$ is surjective.

For each $b \in B$, define $g \in A[x]$ and $\bar{g} \in (A/\mathbf{p})[x]$ as above. Then $\forall b \in B/\mathbf{q}$ is the root of a polynomial in $(A/\mathbf{p})[x]$ that splits completely in $(B/\mathbf{q})[x]$.

Definition 2.1.32. Assume AKLBG and let $\mathbf{q}|\mathbf{p}$ be a prime of B . The Kernel of the surjective homomorphism $\pi_{\mathbf{q}} : D_{\mathbf{q}} \rightarrow \text{Aut}_{A/\mathbf{p}}(B/\mathbf{q})$ is the inertial group $I_{\mathbf{q}}$ of $\mathbf{q}$.

Corollary 2.1.25. Assume AKLBG and let $\mathbf{q}|\mathbf{p}$ be a prime of B . We have an exact sequence

$$0 \rightarrow I_{\mathbf{q}} \rightarrow D_{\mathbf{q}} \rightarrow \text{Aut}_{A/\mathbf{p}}(B/\mathbf{q}) \rightarrow 0$$

and $\#I_{\mathbf{q}} = e_{\mathbf{p}}[B/\mathbf{q} : A/\mathbf{p}]_i$.

Proposition 2.1.19. Assume AKLBG, let $\mathbf{q}|\mathbf{p}$ be a prime of B and suppose $B/\mathbf{q}$ is a separable extension of $A/\mathbf{p}$. We have a tower of field extensions $K \subseteq L^{D_{\mathbf{q}}} \subseteq L^{I_{\mathbf{q}}} \subseteq L$ with $e_{\mathbf{p}} = [L : L^{I_{\mathbf{q}}}] = \#I_{\mathbf{q}}$, $f_{\mathbf{p}} = [L^{I_{\mathbf{q}}} : L^{D_{\mathbf{q}}}] = \#D_{\mathbf{q}}/\#I_{\mathbf{q}}$, $g_{\mathbf{p}} = [L^{D_{\mathbf{q}}} : K] = \#\{\mathbf{q}|\mathbf{p}\}$. The fields $L^{D_{\mathbf{q}}}, L^{I_{\mathbf{q}}}$ are the decomposition field and inertial field associated to $\mathbf{q}$.

pf. The third equality follows by the previous lemma. The second equality follows by the previous proposition and the separability of $(B/\mathbf{q})/(A/\mathbf{p})$, since $D_{\mathbf{q}}/I_{\mathbf{q}} \cong \text{Gal}((B/\mathbf{q})/(A/\mathbf{p}))$ has cardinality $f_{\mathbf{p}} = [B/\mathbf{q} : A/\mathbf{p}]$. Then the first equality follows by the previous two equalities.

Consider an intermediate field E lying between K and L . Fix a prime $\mathbf{q}|\mathbf{p}$ of B and let $\mathbf{q}_E \equiv \mathbf{q} \cap E$ so that $\mathbf{q}_E|\mathbf{p}$. Use $\tilde{G}_{\mathbf{q}}(L/K) \equiv \text{Aut}_{A/\mathbf{p}}(B/\mathbf{q}), \tilde{G}_{\mathbf{q}}(L/E) \equiv \text{Aut}_{(B \cap E)/\mathbf{q}_E}(B/\mathbf{q}), \tilde{G}_{\mathbf{q}_E}(E/K) \equiv \text{Aut}_{A/\mathbf{p}}((B \cap E)/\mathbf{q}_E)$ to denote the automorphism groups of the residue field extensions associated to the tower $K \subseteq E \subseteq L$.

Proposition 2.1.20. Assume AKLBD, and let E be an intermediate field between K and L . Let $\mathbf{q}$ be a prime of B and let $\mathbf{q}_E = \mathbf{q} \cap E, \mathbf{p} = \mathbf{q} \cap K$. Then

$$I_{\mathbf{q}}(L/E) = I_{\mathbf{p}}(L/K) \cap \text{Gal}(L/E), D_{\mathbf{q}}(L/E) = D_{\mathbf{q}}(L/K) \cap \text{Gal}(L/E)$$

If E/K is Galois, we have the commutative diagram of exact sequences

$$\begin{array}{ccccccc}
 & 0 & & 0 & & 0 & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & I_{\mathbf{q}}(L/E) & \longrightarrow & I_{\mathbf{q}}(L/K) & \longrightarrow & I_{\mathbf{q}_E}(E/K) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & D_{\mathbf{q}}(L/E) & \longrightarrow & D_{\mathbf{q}}(L/K) & \longrightarrow & D_{\mathbf{q}_E}(E/K) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \bar{G}_{\mathbf{q}}(L/E) & \longrightarrow & \bar{G}_{\mathbf{q}}(L/K) & \longrightarrow & \bar{G}_{\mathbf{q}_E}(E/K) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

pf. $D_{\mathbf{q}}(L/E) \subseteq \text{Gal}(L/E) \subseteq \text{Gal}(L/K)$. An element $\sigma \in \text{Gal}(L/K)$ lies in $D_{\mathbf{q}}(L/E)$ iff it fixes E hence lies in $\text{Gal}(L/E)$ and satisfies $\sigma(\mathbf{q}) = \mathbf{q}$ hence lies in $D_{\mathbf{q}}(L/K)$. Thus the second equality follows.

The restriction of $\pi_{\mathbf{q}}(L/K) : D_{\mathbf{q}}(L/K) \rightarrow \bar{G}_{\mathbf{q}}(L/K)$ to $D_{\mathbf{q}}(L/E)$ is the map $\pi_{\mathbf{q}} : D_{\mathbf{q}}(L/E) \rightarrow \bar{G}_{\mathbf{q}}(L/E)$ hence the kernels agree after intersecting with $\text{Gal}(L/E)$ and thus the first equality follows.

Exactness of columns follows by the previous corollary. Each row corresponds to an inclusion followed by a restriction in which the inclusion is precisely the kernel of restriction. Thus the exactness of the first two groups in each row follows.

Surjectivity of the restriction maps follows from the bijection used before.

($\text{Hom}_K(L, \Omega) \rightarrow \text{Hom}_E(L, \Omega) \times \text{Hom}_K(E, \Omega)$ whose second factor is restriction and we may view this as a bijection $\phi : \text{Gal}(L/K) \rightarrow \text{Gal}(L/E) \times \text{Gal}(E/K)$. If $\sigma \in \text{Gal}(E/K)$ stabilises $\mathbf{q}_E$, $\phi^{-1}(1, \sigma) \in \text{Gal}(L/K)$ stabilises $\mathbf{q}$ and restricts to σ . Hence the surjectivity of restriction map of the first two rows follows).

The upper left square commute because all the maps are inclusions. The upper right square commutes because inclusion and restriction commute. the lower left square commutes because the horizontal maps are inclusions and the vertical maps coincide on $D_{\mathbf{q}}(L/E)$. The lower right square commutes because the horizontal maps are restrictions and the vertical maps agree after restricting to E .

Corollary 2.1.26. Assume AKLBG, let E be an intermediate field between K and L . Let $\mathbf{q}$ be a prime of B and let $\mathbf{q}_E = \mathbf{q} \cap E, \mathbf{p} = \mathbf{q} \cap K$. Then
$$e_{\mathbf{q}_E/\mathbf{p}} = 1 \quad E \subseteq L^{I_{\mathbf{q}}} \\ e_{\mathbf{q}_E/\mathbf{p}} = f_{\mathbf{q}_E/\mathbf{p}} = 1 \quad E \subseteq L^{D_{\mathbf{q}}}$$
 where $I_{\mathbf{q}}, D_{\mathbf{q}}$ are the inertia and decomposition groups of $\mathbf{q}$.

pf. $I_{\mathbf{q}}(L/E) = I_{\mathbf{q}}(L/K) \cap \text{Gal}(L/E)$ and for $F = L^{I_{\mathbf{q}}}$, $I_{\mathbf{q}}(L/F) = I_{\mathbf{q}}(L/K) = \text{Gal}(L/F)$. Also $\text{Gal}(L/EF) = \text{Gal}(L/E) \cap \text{Gal}(L/F)$ Then we get $I_{\mathbf{q}}(L/E) = I_{\mathbf{q}}(L/K) \cap \text{Gal}(L/E) = \text{Gal}(L/F) \cap \text{Gal}(L/EF) = \text{Gal}(L/EF) = I_{\mathbf{q}}(L/EF)$. Now $e_{\mathbf{q}_E/\mathbf{q}} = \#I_{\mathbf{q}}(L/E) = \#I_{\mathbf{q}}(L/EF) =$

$e_{\mathbf{q}/\mathbf{q}_E}$ is equal to $e_{\mathbf{q}/\mathbf{q}_F} = \#I_{\mathbf{q}}(L/F)$ iff $E \subseteq F$. Thus the first claim follows since $I_{\mathbf{q}}(L/F) = I_{\mathbf{q}}(L/K)$ implies $e_{\mathbf{q}/\mathbf{q}_F} = 1$ since $e_{\mathbf{q}/\mathbf{q}_F} e_{\mathbf{q}_F/\mathbf{p}} = e_{\mathbf{q}/\mathbf{p}}$. The second follows mutatis mutandis.

Corollary 2.1.27. *Assume AKLBG and let $\mathbf{p}$ be a prime of K . The fields $L^{I_{\mathbf{p}}}, L^{D_{\mathbf{p}}}$ are Galois extensions of K and for any intermediate field E , we have $e_{\mathbf{q}_E/\mathbf{p}} = 1$ for all $\mathbf{q}_E | \mathbf{p}$ iff $E \subseteq L^{I_{\mathbf{p}}}$ and $e_{\mathbf{q}_E/\mathbf{p}} = f_{\mathbf{q}_E/\mathbf{p}} = 1$ for all $\mathbf{q}_E | \mathbf{p}$ iff $E \subseteq L^{D_{\mathbf{p}}}$. When $A/\mathbf{p}$ is a perfect field, the inertial field is the largest subfield of L in which $\mathbf{p}$ is unramified, and the decomposition field is the largest subfield in which $\mathbf{p}$ splits completely.*

pf. Since G acts transitively on $\{\mathbf{q} | \mathbf{p}\}$, $I_{\mathbf{p}}$ is generated by a complete set of conjugate subgroups $I_{\mathbf{q}}$ and is therefore stable under conjugation, hence normal, and similarly for $D_{\mathbf{p}}$. Thus the corresponding fixed fields are Galois extensions of K . The rest follows by the previous corollary.

Assume further that the residue fields $A/\mathbf{p}$ are finite for all primes $\mathbf{p}$ of K . (e.g. when K is a global field, such assumption holds). Here $B/\mathbf{q}$ is necessarily a Galois extension of $A/\mathbf{p}$ since finite extensions of finite fields are always Galois. Denote $\mathbb{F}_{\mathbf{q}} \equiv B/\mathbf{q}, \mathbb{F}_{\mathbf{p}} \equiv A/\mathbf{p}$ there are finite fields of p -powers order, where p is the characteristic of $\mathbb{F}_p$.

Let $\mathbf{q} | \mathbf{p}$ be a prime of B . Then we get an exact sequence

$$0 \rightarrow I_{\mathbf{q}} \rightarrow D_{\mathbf{q}} \rightarrow \pi_{\mathbf{q}} \text{Gal}(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}}) \rightarrow 0$$

If $\mathbf{p}$ is unramified, $e_{\mathbf{p}} = e_{\mathbf{q}} = 1$ and $I_{\mathbf{q}}$ is trivial. Then we have an isomorphism $\pi_{\mathbf{q}} : D_{\mathbf{q}} \rightarrow \text{Gal}(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$. Here Galois group $\text{Gal}(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ is the cyclic group of order $f_{\mathbf{p}} = [\mathbb{F}_{\mathbf{q}} : \mathbb{F}_{\mathbf{p}}]$ generated by the Frobenius automorphism $x \mapsto x^{\# \mathbb{F}_{\mathbf{p}}}$.

Definition 2.1.33. *Assume AKLBG with finite residue fields and $\mathbf{q} | \mathbf{p}$ unramified. The inverse image of the Frobenius automorphism of $\text{Gal}(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ under $\pi_{\mathbf{q}} : D_{\mathbf{q}} \rightarrow \text{Gal}(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ is the Frobenius element $\sigma_{\mathbf{q}} \in D_{\mathbf{q}} \subseteq G$.*

Proposition 2.1.21. *Assume AKLBG with finite residue fields and $\mathbf{q} | \mathbf{p}$ unramified. The Frobenius element $\sigma_{\mathbf{q}}$ is the unique $\sigma \in G$ s.t. $\forall x \in B, \sigma(x) \equiv x^{\# \mathbb{F}_{\mathbf{p}}} \pmod{\mathbf{p}}$.*

pf. Clearly $\sigma_{\mathbf{q}}$ has this property. So existence follows. Suppose $\sigma \in G$ also has this property. For any $x \in \mathbf{q}, x \equiv \pmod{\mathbf{q}}, \sigma(x) \equiv x^{\# \mathbb{F}_{\mathbf{p}}} \pmod{\mathbf{q}}$ implies $\sigma(x) \equiv 0 \pmod{\mathbf{q}}$, so $\sigma(x) \in \mathbf{q}$. Thus $\sigma(\mathbf{q}) = \mathbf{q}$ and therefore $\sigma \in D_{\mathbf{q}}$. The isomorphism $\pi_{\mathbf{q}} : D_{\mathbf{q}} \rightarrow \text{Gal}(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ maps both σ and $\sigma_{\mathbf{q}}$ to the Frobenius automorphism $x \mapsto x^{\# \mathbb{F}_{\mathbf{p}}}$ so $\sigma = \sigma_{\mathbf{q}}$.

Proposition 2.1.22. *Assume AKLBG with finite residue fields and $\mathbf{q} | \mathbf{p}$ unramified. For all $\mathbf{q}' | \mathbf{p}$ the Frobenius elements $\sigma_{\mathbf{q}}, \sigma_{\mathbf{q}'}$ are conjugate in G .*

pf, G acts transitively on $\{\mathbf{q}|\mathbf{p}\}$ so let $\tau \in G$ be s.t, $\mathbf{q}' = \tau(\mathbf{q})$. Then for any $x \in B$, we have

$$\begin{aligned} \sigma_{\mathbf{q}}(x) &\equiv x^{\#\mathbb{F}_{\mathbf{p}}} \pmod{\mathbf{q}} \\ \tau(\sigma_{\mathbf{q}}(x)) &\equiv \tau(x^{\#\mathbb{F}_{\mathbf{p}}}) \pmod{\tau(\mathbf{q})} \\ (\tau\sigma_{\mathbf{q}})(x) &\equiv \tau(x)^{\#\mathbb{F}_{\mathbf{p}}} \pmod{\mathbf{q}'} \quad \text{By the previous proposition,} \\ (\tau\sigma_{\mathbf{q}})(\tau^{-1}(x)) &\equiv \tau(\tau^{-1}(x))^{\#\mathbb{F}_{\mathbf{p}}} \pmod{\mathbf{q}'} \\ (\tau\sigma_{\mathbf{q}}\tau^{-1})(x) &\equiv x^{\#\mathbb{F}_{\mathbf{p}}} \pmod{\mathbf{q}'} \end{aligned}$$

the uniqueness of $\sigma_{\mathbf{q}'}$ follows and therefore $\sigma_{\mathbf{q}'} = \tau\sigma_{\mathbf{q}}\tau^{-1}$.

Definition 2.1.34. Assume AKLBG with finite residue fields and $\mathbf{q}|\mathbf{p}$ unramified. The conjugacy class of the Frobenius element $\sigma_{\mathbf{q}} \in G$ is the Frobenius class of $\mathbf{p}$, denoted $\text{Frob}_{\mathbf{p}}$.

Definition 2.1.35. Assume AKLBG with finite residue fields. For each unramified prime $\mathbf{q}$ of L we define the Artin symbol $(\frac{L/K}{\mathbf{q}}) \equiv \sigma_{\mathbf{q}}$.

Proposition 2.1.23. Assume AKLBG with finite residue fields and $\mathbf{q}|\mathbf{p}$ unramified. Then $\mathbf{p}$ splits completely iff $\frac{L/K}{\mathbf{q}} = 1$.

pf. If $\mathbf{p}$ splits, then $e_{\mathbf{p}}f_{\mathbf{p}} = 1$ and $D_{\mathbf{q}} = \langle \sigma_{\mathbf{q}} \rangle = \{1\}$.
Conversely, if $D_{\mathbf{q}} = \langle \sigma_{\mathbf{q}} \rangle = \{1\}$, then $e_{\mathbf{p}}f_{\mathbf{p}} = 1$ and $\mathbf{p}$ splits completely.

Proposition 2.1.24. Assume AKLBG with finite residue fields and let $\mathbf{q}|\mathbf{p}$ be unramified. Let E be an intermediate field between K and L and define $\mathbf{q}_E \equiv \mathbf{q} \cap E$. Then

$$\left(\frac{L/E}{\mathbf{q}}\right) = \left(\frac{L/K}{\mathbf{q}}\right)^{[\mathbb{F}_{\mathbf{q}_E}:\mathbb{F}_{\mathbf{p}}]}$$

and if E/K is Galois, then $(\frac{E/K}{\mathbf{q}_E})$ is the restriction of $(\frac{L/K}{\mathbf{q}})$ to E .

pf. $\#\mathbb{F}_{\mathbf{q}_E} = (\#\mathbb{F}_{\mathbf{p}})^{[\mathbb{F}_{\mathbf{q}_E}:\mathbb{F}_{\mathbf{p}}]}$. So the first claim follows.
By the commutativity of the diagram in the previous proposition 2.2.19, the Frobenius automorphism $x \mapsto x^{\#\mathbb{F}_{\mathbf{p}}}$ of $\text{Gal}(\mathbb{F}_{\mathbf{q}_E}/\mathbb{F}_{\mathbf{p}})$ is the restriction of the Frobenius automorphism $x \mapsto x^{\#\mathbb{F}_{\mathbf{p}}}$ of $\text{Gal}(\mathbb{F}_{\mathbf{q}}/\mathbb{F}_{\mathbf{p}})$ to $\mathbb{F}_{\mathbf{q}_E}$.

2.2 Extensions of Complete DVR

2.2.1 Completions

A metric on a set X is a function $d : X \times X \rightarrow \mathbb{R}_+$ that satisfies

- $d(x, y) = 0$ iff $x = y$
- $d(x, y) = d(y, x)$

- $d(x, z) \leq d(x, y) + d(y, z)$

A metric also satisfies

- $d(x, z) \leq \max(d(x, y), d(y, z))$

is an ultrametric and is said to be nonarchimedean.

→ every absolute value on a ring induces a metric $d(x, y) \equiv |x - y|$.

→ the metric d defines a topology on X generated by open balls $B_{<r}(x) \equiv \{y \in X : d(x, y) < r\}$ where $r \in \mathbb{R}_{++}, x \in X$ and we call X a metric space.

Definition 2.2.1. Let X be a metric space. A sequence (x_n) of elements of X converges to x if there is an $x \in X$ s.t. $\forall \epsilon > 0, \exists N \in \mathbb{Z}_{++}$ s.t. $d(x_n, x) < \epsilon, \forall n \geq N$. The sequence (x_n) is Cauchy if $\forall \epsilon > 0, \exists N \in \mathbb{Z}_{++}$ s.t. $d(x_m, x_n) < \epsilon$ for all $m, n \geq N$. Every convergent sequence is Cauchy but not vice versa. If every Cauchy sequence is convergent, a metric space is said to be complete.

Definition 2.2.2. An abelian group G is a topological group if it is a topological space in which the map $G \times G \rightarrow G$ $(g, h) \mapsto g + h$ and the map $G \rightarrow G$ $g \mapsto -g$ are both continuous. A commutative ring R is a topological ring if it is a topological space in which the maps $R \times R \rightarrow R$ $(r, s) \mapsto r + s$ or rs are both continuous; the additive group of R is a topological group, since $(-1, s) \mapsto -s$ is continuous, but the unit group $R^\times$ need not be a topological group. A field K is a topological field if it is a topological ring whose unit group is a topological group.

If R is a ring with an absolute value, it is a topological ring under the induced topology and its unit group is also a topological group by the triangular inequality and the multiplicative property of absolute value.

Definition 2.2.3. Let X be a metric space. Cauchy sequences $(x_n), (y_n)$ are equivalent if $d(x_n, y_n) \rightarrow 0$ as $n \rightarrow \infty$; this defines an equivalence relation on the set of Cauchy sequences in X and we use $[(x_n)]$ to denote the equivalence class of (x_n) . The completion of X is the metric space $\hat{X}$ whose elements are equivalence classes of Cauchy sequences with the metric $d([(x_n)], [(y_n)]) \equiv \lim d(x_n, y_n)$. We may canonically embed X in its completion $\hat{X}$ via the map $x \mapsto \hat{x} = [(x, x, \dots)]$.

When X is a topological ring, we can extend the ring operation to $\hat{X}$ by defining $[(x_n)] + [(y_n)] \equiv [(x_n + y_n)]$, $[(x_n)][(y_n)] \equiv [(x_n y_n)]$. When the metric on X is induced by an absolute value $|\cdot|$, we extend it to an absolute value on $\hat{X}$ via $[(x_n)] \equiv \lim |x_n|$.

When X is a field with a metric induced by an absolute value, the completion is also a field.

(Given $[(x_n)] \neq 0$, we pick each x_n s.t. $x_n \neq 0$ for all n and by the multiplicative property $(1/x_n)$ is Cauchy. Then $1/[(x_n)] = [(1/x_n)]$ so its completion is a field).

Proposition 2.2.1. *Let K be a field with an absolute value $|\cdot|$ viewed as a topological field under the induced topology, and let $\hat{K}$ be the completion of K . The field $\hat{K}$ is complete, and has the following universal property : every embedding of K as a topological field into a complete field L can be uniquely extended to an embedding of $\hat{K}$ into L which is an isomorphism whenever k is dense in L . Up to a canonical isomorphism, $\hat{K}$ is the unique topological field with this property.*

pf. Assume $\hat{K}$ is not complete. Then there exists a Cauchy sequence (x_n) which is not convergent in $\hat{K}$. But since it is Cauchy, $\forall m, n \geq N, d(x_m, x_n) < \epsilon$

$$\begin{array}{ccc} K & \xrightarrow{\phi} & L \\ & \searrow \iota & \nearrow \exists \\ & \hat{K} & \end{array}$$

Theorem 2.2.1 (Weak Approximation). *Let K be a field and let $|\cdot|_1, \dots, |\cdot|_n$ be pairwise inequivalent nontrivial absolute values on K . Let $a_1, \dots, a_n \in K$ and let $\epsilon_1, \dots, \epsilon_n$ be positive real numbers. Then there exists an $x \in K$ s.t. $|x - a_i|_i < \epsilon_i, 1 \leq i \leq n$.*

pf. **CLAIM : Given inequivalent nontrivial absolute values $|\cdot|_1, \dots, |\cdot|_n$ on k , $\exists x \in k^\times$ s.t. $|x|_1 > 1$ and $|x|_i < 1, 1 < i \leq n$.**

pf. $n = 1$, since k is a field either $a \in k$ or $a^{-1} \in k$ has $|x|_1 > 1$ Pick such an element x .

$n = 2$, if for the previously chosen x , $|x|_2 < 1$, then nothing to be done. Otherwise, pick an element $y \in k$ s.t. $|y|_2 = |x|_2, |y|_1 = 1$ (e.g. $y = x/|x|_1$), which exists due to the pairwise coprimeness. Then xy^{-2} is the desired element.

By repeating this procedure, we get an element x with the desired property.

CLAIM : Given inequivalent nontrivial absolute values $|\cdot|_1, \dots, |\cdot|_n$ on k , there is a sequence $(x_1, x_2, \dots)$ of elements $x_j \in k$ that converges to 1 w.r.t. $|\cdot|_1$ and to 0 w.r.t. $|\cdot|_i, \forall 1 < i \leq n$.

pf. by the previous claim we have x s.t. $|x|_1 > 1, |x|_i < 1, \forall 1 < i \leq n$. Then By setting $x' = \frac{x}{|x|_1}$, we get a sequence (x'^n) with the desired property.

Then $\forall i \in (1, n), \exists x_i$ with $|x_i|_i$ converges to 1 and $|x_i|_j$ converges to 0 for $j \neq i$. Fix $x_n = \sum_i a_i x_i^n$. Then by the triangular inequality the sequence (x_n) is the sequence with the desired property.

Corollary 2.2.1. *Let K be a field with absolute values $|\cdot|_1, |\cdot|_2$. The induced topologies on K coincide iff $|\cdot|_1$ and $|\cdot|_2$ are equivalent.*

pf. If they are not equivalent, by the previous theorem $\exists x_1, x_2 \in K$ s.t. x_1 goes to 1 for $|\cdot|_1$, and 0 for $|\cdot|_2$. Then for an open ball $x_1 \in B_{<\epsilon}^1(x-1)$ but $x_1 \notin B_{<\epsilon'}^2(x-1)$ for any ϵ' . Since open balls induce topology, but they are different for those absolute values and therefore induced topologies are different.

If they are equivalent, $\exists c \in \mathbb{R}_{++}$ s.t. $|\cdot|_1^c = |\cdot|_2$ and we can get a corresponding open balls for each absolute values. Similarly open balls induce the topologies and we get the same open balls. Thus they have the same induced topologies.

PROPERTIES of TOPOLOGY induced by a NONARCHIMEDEAN ABSOLUTE VALUE

- Every point in an open ball is a centre : $B_{<r}(y) = B_{<r}(x) \forall y \in B_{<r}(x)$
 pf. $\forall z \in B_{<r}(y), d(y, z) < r, d(y, x) < r \rightarrow d(z, x) \leq d(z, y) + d(y, x) \leq \min(d(z, y), d(y, x)) < r$.
- Any pair of open balls are either disjoint or concentric.
 pf. If they are concentric, nothing to prove. Otherwise, assume the contrary, so $\exists x' \in B_{<r}(x) \cap B_{<r}(y)$ not concentric. Since they are not concentric, $y \notin B_{<r}(x), x \notin B_{<r}(y)$. But $x' \in B_{<r}(x), B_{<r}(x) = B_{<r}(x')$, and similarly $B_{<r}(x') = B_{<r}(y)$, a contradiction. Thus they are disjoint.
- Every open ball is closed and every closed ball is open.
 pf. $B_{<r}(x) = \{y : d(y, x) < r\}$. Pick a maximum value of $k = d(x, b), b \in B_{<r}(x)$. Then $B_{<r}(x) = B_{\leq k}(x)$ a closed ball.
 Let $B_{\leq k}(x)$ be a closed ball. Then for any $b \in B_{\leq k}(x)^c$, set $r = \min(d(b, x))$. Then $B_{\leq k}(x) = B_{<r}(x)$.
- X is totally disconnected.
 pf. Let x, y be two disjoint points. Let $r = d(x, y)/2$. Then $B_{<r}(x)$ is an open nhoud of x not containing y . Then its complement is the closed ball and by the previous statement, it is an open ball. Since $y \in B_{<r}(x)^c$ by the previous statement it is a centre. Thus we get the desired two open balls.

For an additive topological group G , every translation by h map $g \mapsto g + h$ is a homeomorphism, since it is continuous and has a continuous inverse (translation by $-h$). Thus we can focus on nhouds of the identity to understand the topology of the topological space.

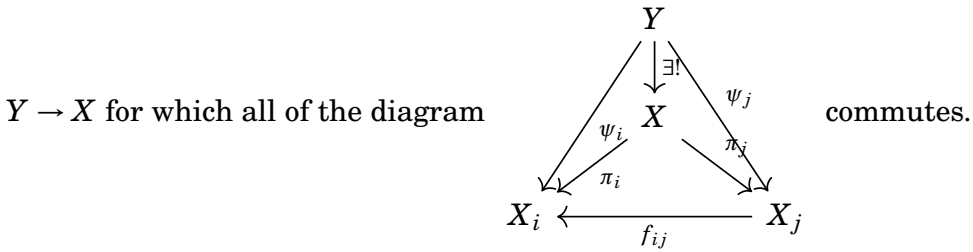
By construction K is dense in $\hat{K}$.

Definition 2.2.4. A directed set is a set I with a relation $\leq$ that is reflexive, antisymmetric($i \leq j \leq i$ implies $i = j$) and transitive, in which every finite subset has an upper bound.

Definition 2.2.5. An inverse system in a category is a family of objects $\{X_i : i \in I\}$ indexed by a directed set I and a family of morphisms $\{f_{ij} : X_i \leftarrow X_j : i \leq j\}$ s.t. each f_{ii} is the identity and $f_{ik} = f_{ij} \circ f_{jk}$ for all $i \leq j \leq k$.

Definition 2.2.6. Let (X_i, f_{ij}) be an inverse system in a concrete category with products. The inverse limit of (X_i, f_{ij}) is the object $X \equiv \lim X_i \equiv \{x \in \prod_{i \in I} X_i : x_i = f_{ij}(x_j), \forall i \leq j\} \subseteq \prod X_i$. The restriction $\pi_i : X \rightarrow X_i$ of the projection $\prod X_i \rightarrow X_i$ satisfy $\pi_i = f_{ij} \circ \pi_j, i \leq j$.

→it has the universal property that if Y is another object with morphisms $\psi_i : Y \rightarrow X_i$ that satisfy $\psi_i = f_{ij} \circ \psi_j, \forall i \leq j$, then there is a unique morphism



Proposition 2.2.2. Let (X_i, f_{ij}) be an inverse system of Hausdorff topological spaces. Then $X \equiv \lim X_i$ is a closed subset of $\prod X_i$ and if the X_i are compact, then X is compact.

pf. The set X is the intersection of the sets $Y_{ij} \equiv \{x \in \prod X_i : x_i = f_{ij}(x_j)\}, i \leq j$, each of which can be written as $Y_{ij} = \prod_{k \neq i, j} X_k \times Z_{ij}$, where Z_{ij} is the preimage of the diagonal $\Delta_i \equiv \{(x_i, x_i) : x_i \in X_i\} \subseteq X_i \times X_i$ under the continuous map $X_i \times X_j \rightarrow X_i \times X_i$ defined by $(x_i, x_j) \mapsto (x_i, f_{ij}(x_j))$. Each Δ_i is closed in $X_i \times X_i$ because X_i is Hausdorff, so each Z_{ij} is closed in $X_i \times X_j$ and each Y_{ij} is closed in $\prod X_i$. Thus X is closed in $\prod X_i$.

By Tychonoff's theorem, if X_i are compact, so is $X = \prod X_i$.

Definition 2.2.7. A local field is a field with a nontrivial absolute value $|\cdot|$ that is locally compact (if every point has a compact nhod) under the topology induced by $|\cdot|$.

Lemma 2.2.1. Let K be a field with a nontrivial absolute value $|\cdot|$. Then K is a local field iff every closed ball in K is compact.

pf. Suppose K is a local field. Then $0 \in K$ lies in a compact nhod that contains a closed ball $B_{\leq s}(0)$, which is compact. Fix $\alpha \in K^\times$ with $|\alpha| > 1$. The map $x \mapsto \alpha x$ is continuous and $|\cdot|$ is multiplicative, so $B_{\leq |\alpha|^n s}(0)$ is compact for every $n \in \mathbb{Z}_{++}$ since the continuous image of a compact set is compact. Thus we have compact balls about 0 of arbitrarily large radii, implying that every closed ball $B_{\leq r}(0)$ is a closed subset of a compact set, hence compact. For every $z \in K$, the translation

map $x \mapsto x + z$ is continuous, so every closed ball $B_{\leq r}(z)$ is compact. By the definition of the local compactness, the converse holds.

Corollary 2.2.2. *Let K be a local field with nontrivial absolute value $|\cdot|$. Then K is complete.*

pf. Suppose the contrary. Then there exists a Cauchy sequence (x_n) in K that converges to a limit $x \in \hat{K} \setminus K$. Pick $N \in \mathbb{Z}_{++}$ s.t. $|x_n - x| \leq 1/2, \forall n \geq N$ and consider the closed ball $S \equiv B_{\leq 1}(x_N)$ in K , which is compact by the previous lemma. The Cauchy sequence $(x_n)_{n \geq N}$ in S has a convergent subsequence whose limit lies in $S \subseteq K$, since S is compact and therefore sequentially compact. But this limit must be equal to $x \notin K$, a contradiction.

Proposition 2.2.3. *Let K be a field with absolute value $|\cdot|_v$ induced by a discrete valuation v with valuation ring A and uniformiser π . Then K is a local field iff K is complete and the residue field $A/\pi A$ is finite.*

pf. If K is a local field, K is complete by the previous corollary and the valuation ring $A = \{x \in K : v(x) \geq 0\} = \{x \in K : |x|_v \leq 1\} = B_{\leq 1}(0)$ is a closed ball, hence by the previous lemma compact. The cosets $x + \pi A$ of the subgroup $\pi A \subseteq A$ are open balls $B_{<1}(x)$ since $y \in x + \pi A$ iff $|x - y|_v \leq |\pi|_v < 1$. The collection $\{x + \pi A : x \in A\}$ of cosets of πA is an open cover of A and therefore it must have a finite subcover given A compact. Thus $A/\pi A$ is finite.

Suppose K is complete and $A/\pi A$ is finite. The valuation ring $A \subseteq K$ is also complete and we have an isomorphism of topological rings $A = \hat{A} \cong \varprojlim \frac{A}{\pi^n A}$. Each quotient $A/\pi^n A$ is finite since $A/\pi A$ is finite and therefore compact. Thus its inverse limit A is also compact. Then by the previous lemma K is a local field, since it contains a compact closed ball $B_{\leq 1}(0) = A$ and $|\cdot|_v$ is nontrivial.

Corollary 2.2.3. *Let L be a global field with a nontrivial absolute value $|\cdot|_v$. Then the completion L_v of L w.r.t. $|\cdot|_v$ is a local field.*

pf. Let L/K be a finite extension with $K = \mathbb{Q}$ or $K = \mathbb{F}_q(t)$ and $A = \mathbb{Z}$ or $\mathbb{F}_q[t]$, so that $K = \text{Frac} A$. Then A is a Dedekind domain, as is its integral closure B in L .

- $|\cdot|_v$ is archimedean
 $K = \mathbb{Q}$, and the completion of L w.r.t. $|\cdot|_v$ must contain the completion of $\mathbb{Q}$ w.r.t. the restriction of $|\cdot|_v$ to $\mathbb{Q}$, which must be isomorphic to $\mathbb{R}$. Thus L_v is a finite extension of $\mathbb{R}$ and must be isomorphic to either $\mathbb{R}$ or $\mathbb{C}$, both of which are local fields.
- $|\cdot|_v$ is nonarchimedean
 Let $C \equiv \{x \in L : |x|_v \leq 1\}$ be the valuation ring of L w.r.t. $|\cdot|_v$ and let $\mathfrak{m} \equiv \{x \in$

$L : |x|_v < 1\}$ be its maximal ideal, which is nonzero because $|\cdot|_v$ is nontrivial. The restriction of $|\cdot|_v$ to K is a nonarchimedean absolute value and from the classification of absolute values on $\mathbb{Q}^{(**)}$ and $\mathbb{F}_q[t]$ we can assume it is induced by a discrete valuation on A ; in particular $|x|_v \leq 1, \forall x \in A$ and therefore $A \subseteq C$. Like all valuation rings, C is integrally closed in its fraction field L , and $A \subseteq C$, and therefore $B \subseteq C$, given B the integral closure of A in L . The ideal $\mathfrak{q} = \mathfrak{m} \cap B$ is maximal and the DVR $B_{\mathfrak{q}}$ lies in C and must equal C , since there are no intermediate rings between a DVR and its fraction field (Since C is not a field $C \neq L$). Thus the absolute value induced by $v_{\mathfrak{q}}$ is equivalent to $|\cdot|_v$ since they have the same valuation rings. By choosing $0 < c < 1$, we can assume $|\cdot|_v = c^{v_{\mathfrak{q}}(\cdot)}$ is induced by $v_{\mathfrak{q}}$. Thus $|\cdot|_v$ is induced by a discrete valuation. The residue field $B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}} \cong B/\mathfrak{q}$ is finite since $B/\mathfrak{q}$ is a finite extension of finite field $A/\mathfrak{q}$ where $\mathfrak{q} = \mathfrak{q} \cap A$. If we consider the completion L_v with valuation ring B_v , we can take any uniformiser π for $\mathfrak{q} \subseteq B \subseteq B_v$ as a uniformiser for B_v , $\frac{B}{\mathfrak{q}} \cong \frac{B_{\mathfrak{q}}}{\mathfrak{q}B_{\mathfrak{q}}} = \frac{B_{\mathfrak{q}}}{\pi B_{\mathfrak{q}}} \cong \frac{B_v}{\pi B_v}$, so $B_v/\pi B_v$ is finite. Thus L_v is a complete field with an absolute value induced by a discrete valuation and finite residue field, and therefore a local field by the previous proposition.

****For each prime $\pi \in \mathbb{F}_q[t]$ and any nonzero $f \in \mathbb{F}_q[t]$, let $v_{\pi}(f)$ be the largest integer for which $\pi^n | f$, equivalently the largest n for which $f \in (\pi^n)$. For each $f/g \in \mathbb{F}_q(t)^{\times}$, define $v_{\pi}(f/g) \equiv v_{\pi}(f) - v_{\pi}(g)$ and let $v_{\pi}(0) \equiv \infty$, and also define $\deg 0 \equiv -\infty, \deg(f/g) \equiv \deg f - \deg g$.**

For each prime $\pi \in \mathbb{F}_q[t]$, define $|r|_{\pi} = (q^{\deg \pi})^{-v_{\pi}(r)}, \forall r \in \mathbb{F}_q(t)$. Then $|\cdot|_{\pi}$ is a nonarchimedean absolute value on $\mathbb{F}_q[t]$.

pf. $|r+s|_{\pi} = (q^{\deg \pi})^{-v_{\pi}(r+s)} = (q^{\deg \pi})^{-v_{\pi}(r)}(q^{\deg \pi})^{-v_{\pi}(s)} = |r|_{\pi}|s|_{\pi} \leq \max(|r|_{\pi}, |s|_{\pi})$ hence nonarchimedean.

Define $|r|_{\infty} \equiv q^{\deg r}, \forall r \in \mathbb{F}_q(t)$. Then $|\cdot|_{\infty}$ is a nonarchimedean absolute value on $\mathbb{F}_q(t)$.

pf. $|r+s|_{\infty} = q^{\deg(r+s)} = q^{\max(\deg r, \deg s)} \leq \max(|r|_{\infty}, |s|_{\infty})$, hence a nonarchimedean absolute value.

The residue field of $\mathbb{F}_q(t)$ w.r.t. $|\cdot|_{\pi}$.

pf. $\mathbb{F}_q(t)/\mathfrak{m}\mathbb{F}_q(t) = \{r + \mathfrak{m} : r \in \mathbb{F}_q(t) : ||r||_{\pi} = 1\}$ a coset of polynomial coprime to π .

Every nontrivial absolute value on $\mathbb{F}_q[t]$ is equivalent to $|\cdot|_{\infty}$ or $|\cdot|_{\pi}$ for some prime $\pi \in \mathbb{F}_q[t]$.

pf. Any element f in $\mathbb{F}_q(t)$ can be expressed as a finite product of irreducible polynomials $f_1, f_2, \dots$. Prime ideals of $\mathbb{F}_q[t]$ consists of principal ideals, generated by an irreducible polynomial. Since $||\cdot||$ is nontrivial, pick any irreducible polynomial s.t. $||f_1|| = a > 1$. Since $||\cdot||$ is multiplicative, $\forall f \in \mathbb{F}_q(t)$ s.t. $f = \prod f_i^{e_i}$ and $e_1 \neq 0, |f| = a^{e_1} ||f/f_i^{e_i}||$. For another $j \neq i$, if $||f_j|| \neq 1$, wlog assume $||f_j|| = b > 1$ (or take an inverse). Then $||f_j||^{\log_b a} = ||f_i|| \leftrightarrow f_i = f_j^{\log_b a}$ a contradiction. Hence

$\forall j \neq i, \|f_j\| = 1$. This implies it is equivalent to $\|\cdot\|_\pi$ where π is a PID generated by the irreducible polynomial s.t. $\|f_i\| > 1$.

If $\|t\| = \alpha > 1$, t is an irreducible polynomial with the above definition, and therefore it is equivalent to $\|\cdot\|_{(t)}$, which is equivalent to $\|\cdot\|_\infty$. If $\|t\| \leq 1, \pi \neq t$ and is equivalent to some prime generated by an irreducible polynomial with no root 0.

Proposition 2.2.4. *A locally compact topological vector space over a nondiscrete locally compact field has finite dimension.*

Theorem 2.2.2. *Let L be a local field. If L is archimedean, then it is isomorphic to $\mathbb{R}$ or $\mathbb{C}$; otherwise L is isomorphic to a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$.*

pf. Let L be a local field with nontrivial absolute value $|\cdot|$; then L is complete by the previous corollary.

- L has characteristic zero
the prime field of L is $\mathbb{Q}$ and L contains the completion of $\mathbb{Q}$ w.r.t. the restriction of $|\cdot|$ to $\mathbb{Q}$. By the Ostrowski's theorem, the restriction of $|\cdot|$ to $\mathbb{Q}$ is equivalent to either the standard archimedean absolute value (the completion is $\mathbb{R}$) or to a p -adic absolute value (the completion is $\mathbb{Q}_p$). Thus L contains a subfield K isomorphic to $\mathbb{R}$ or to $\mathbb{Q}_p$ for some prime p .
- L has positive characteristic p
the prime field of L is $\mathbb{F}_p$ and L must contain a transcendental element s since no algebraic extension of $\mathbb{F}_p$ has a nontrivial absolute value. Thus L contains $\mathbb{F}_p(s)$ and therefore contains the completion of $\mathbb{F}_p(s)$ w.r.t. $|\cdot|$. Every completion of $\mathbb{F}_p(s)$ is isomorphic to $\mathbb{F}_q((t))$ for some q a power of p and t transcendental over $\mathbb{F}_q^{(*)}$. Thus L contains a subfield K isomorphic to $\mathbb{F}_q((t))$.

If K is archimedean, $K = \mathbb{R}$ is a local field and if K is nonarchimedean the $K = \mathbb{Q}_p$ or $K = \mathbb{F}_q((t))$ is a complete field with a discrete valuation and finite residue field, hence a local field by the proposition 2.3.3. Thus K is locally compact and it is nondiscrete because its absolute value is nontrivial. Then by the previous proposition, L has finite degree over K . If K is archimedean, $K = \mathbb{R}$ and L must be $\mathbb{R}$ or $\mathbb{C}$; otherwise L is a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$ as claimed.

**

$\mathbb{F}_q(t)_\pi$ is isomorphic to $\mathbb{F}_\pi((T))$ where $\mathbb{F}_\pi$ denotes the residue field of $\mathbb{F}_q(t)$ w.r.t. $|\cdot|_\pi$ and $T \in \mathbb{F}_q(t)$ is a uniformiser. Let π_1, π_2 be two discrete valuation for which $\mathbb{F}_{\pi_1} \cong \mathbb{F}_{\pi_2} \cong \mathbb{F}_q$ and consider the completions $\mathbb{F}_q(t)_{\pi_1}, \mathbb{F}_q(t)_{\pi_2}$. Then they are isomorphic as fields/topological fields/ $\mathbb{F}_q(t)$ -algebras.

pf For any polynomial $f \in \mathbb{F}_q(t)_\pi$ can be expressed as $\pi \rightarrow f(t) = \cdots + f_{-1}\pi^{-1} + f_0 +$

$f_1\pi + f_2\pi^2 + \dots$. Let $T = \pi$ be a uniformiser, which is an irreducible polynomial. Then we can get a homomorphism $\phi : \mathbb{F}_q(t)_\pi \rightarrow \mathbb{F}_\pi((T))$ by sending any polynomial in the domain to an expansion in terms of $\pi = T$. It is injective, since every different polynomial in the domain has different unique $T = \pi$ expansion. Also it is surjective since for any $f \in \mathbb{F}_\pi((T))$, send T to π and the coefficients unchanged. Then we can similarly say that for any irreducible polynomial π inducing the absolute value with same residue field, have isomorphisms from $\mathbb{F}_\pi((T)) \rightarrow \mathbb{F}_{\pi_2}((T'))$ as a field and $\mathbb{F}_q(t)$ -algebras.

The structure of the abelian group $\mathbb{F}_q(t)_\pi^\times / \mathbb{F}_q(t)_\pi^{\times n}$ where q, n are primes and π is a discrete valuation on $\mathbb{F}_p(t)$.

pf. a free abelian group of dimension n

Definition 2.2.8. Let R be a commutative ring, and let $f(x) = \sum f_i x^i \in R[x]$ be a polynomial. The formal derivative f' of f is the polynomial $f'(x) \equiv \sum i f_i x^{i-1} \in R[x]$.

The formal derivative satisfies the usual identities

$$\left\{ \begin{array}{ll} (af + bg)' = af' + bg' & \text{linearity} \\ (fg)' = f'g + fg' & \text{Leibniz rule} \\ (f \circ g)' = (f' \circ g)g' & \text{chain rule} \end{array} \right.$$

Lemma 2.2.2. Let R be a ring, let $f = \sum f_i x^i \in R[x]$ be a polynomial, and let $a \in R$. Then $f(x) = f(a) + f'(a)(x-a) + g(x)(x-a)^2$ for a unique $g \in R[x]$.

pf. $f(x) = f(a + (x-a)) = \sum_{i \geq 0} f_i (a + (x-a))^i = \sum_{i \geq 0} f_i \sum_{0 \leq j \leq i} \binom{i}{j} a^j (x-a)^{i-j}$
 $= f(a) + \sum_{i \geq 1} f_i \sum_{0 \leq j < i} \binom{i}{j} a^j (x-a)^{i-j} = f(a) + f'(a)(x-a) + \sum_{i \geq 2} f_i \sum_{0 \leq j \leq i-2} \binom{i}{j} a^j (x-a)^{i-j} = f(a) + f'(a)(x-a) + g(x)(x-a)^2$ where $g(x) = \sum_{i \geq 2} f_i \sum_{0 \leq j \leq i-2} \binom{i}{j} a^j (x-a)^{i-j-2} \in R[x]$.

→ the binomial coefficients are integers, hence well-defined elements of R under the canonical homomorphism $\mathbb{Z} \rightarrow R$, even when $j!$ is divisible by the $\text{char} R$.

Corollary 2.2.4. Let R be a ring, $f \in R[x]$ and $a \in R$. Then $f(a) = f'(a) = 0$ iff a is at least a double root of f , that is $f(x) = (x-a)^2 g(x)$ for some $g \in R[x]$.

Definition 2.2.9. Let $f \in R[x]$ be a polynomial over a ring R and let $a \in R$. If $f(a) = 0, f'(a) \neq 0$, then a is a simple root of f .

If R is a ring and I is an R -ideal, a lift of an element $\bar{r}$ of the quotient R/I means a preimage of $\bar{r}$ under the quotient map $R \rightarrow R/I$.

Lemma 2.2.3 (Hensel's Lemma I). Let A be a complete DVR with maximal ideal $\mathfrak{p}$ and residue field $k \equiv A/\mathfrak{p}$. Suppose $f \in A[x]$ is a monic polynomial whose reduction to $k[x]$ has a simple root $\bar{a} \in k$. Then $\bar{a}$ can be lifted to a root of f in A .

pf. Let a_0 be any lift of $\bar{a}$ to A , then it is not necessarily a root of f but it is a root modulo $\mathfrak{p}$. Fix $0 < c < 1$ and define the absolute value $|\cdot| \equiv c^{v_p(\cdot)}$. Since $\bar{a}$ is a simple root, $f(a_0) \in \mathfrak{p}, f'(a_0) \notin \mathfrak{p} \rightarrow |f(a_0)| \leq c < 1, |f'(a_0)| = 1$. Define $\epsilon \equiv \frac{|f(a_0)|}{|f'(a_0)|^2} < 1$.

$|a_n| \leq 1$

CLAIM : $\begin{matrix} |a_n - a_0| \leq \epsilon < 1 \\ |f'(a_n)| = |f'(a_0)| \\ |f(a_n)| \leq \epsilon^{2^n} |f'(a_0)|^2 \end{matrix}$ **for all $n \in \mathbb{Z}_+$.**

pf. when $n = 0$, the claim trivially holds. Assume the claim holds for all n and consider the case $n + 1$.

- $|a_{n+1} - a_n| = |f(a_n)/f'(a_n)| \leq \epsilon^{2^n} |f'(a_0)|^2 / |f'(a_0)| = \epsilon^{2^n} |f'(a_0)| \leq \epsilon^{2^n} \rightarrow |a_{n+1}| = |a_{n+1} - a_n + a_n| \leq \max(|a_{n+1} - a_n|, |a_n|) \leq 1$
- $|a_{n+1} - a_0| \leq \max(|a_{n+1} - a_n|, |a_n - a_0|) \leq \max(\epsilon^{2^n}, \epsilon) = \epsilon$
- $f'(a_{n+1}) = f'(a_n) - f''(a_n) \frac{f(a_n)}{f'(a_n)} + g(a_{n+1}) \left(\frac{f(a_n)}{f'(a_n)}\right)^2$ where $f''(a_n), g(a_{n+1}) \in A$ so $|f''(a_n)|, |g(a_{n+1})| \leq 1, |f(a_n)/f'(a_n)| = |f(a_n)|/|f'(a_0)| \leq \epsilon^{2^n} |f'(a_0)|$ so the absolute values of the last two terms on the RHS are strictly smaller than the first term $|f'(a_n)| = |f'(a_0)| \rightarrow |f'(a_{n+1})| = |f'(a_n)| = |f'(a_0)|$
- $f(a_{n+1}) = f(a_n) - f'(a_n) \frac{f(a_n)}{f'(a_n)} + h(a_{n+1}) \left(\frac{f(a_n)}{f'(a_n)}\right)^2 = h(a_{n+1}) \left(\frac{f(a_n)}{f'(a_n)}\right)^2$ for some $h \in A[x]$. Since $|h(a_{n+1})| \leq 1, |f(a_{n+1})| \leq |f(a_n)|^2 / |f'(a_n)|^2 = |f(a_n)|^2 / |f'(a_0)|^2 \leq \epsilon^{2^{n+1}} |f'(a_0)|^2$

hence the claim follows by the induction.

$|a_{n+1} - a_n| \leq \epsilon^{2^n} \rightarrow 0$ as $n \rightarrow \infty$ and for a nonarchimedean absolute value this implies (a_n) is a Cauchy. Thus $\exists a \equiv \lim a_n \in A$, given A complete. We have $f(a) = \lim f(a_n) = 0$ so a is a root of f and $|a - a_0| = \lim |a_n - a_0| < 1 \rightarrow a \equiv a_0 \pmod{\mathfrak{p}}$ is a lift of $\bar{a}$.

Actually, the above proof proves more general version of the Hensel's Lemma.

Lemma 2.2.4 (Hensel's Lemma II). *Let A be a complete DVR. Let $f \in A[x]$ and suppose $a_0 \in A$ satisfies $|f(a_0)| < |f'(a_0)|^2$ and for $n \geq 0$, define $a_{n+1} \equiv a_n - f(a_n)/f'(a_n)$. The sequence (a_n) is well-defined and converges to the unique root $a \in A$ of f for which $|a - a_0| \leq \epsilon \equiv |f(a_0)|/|f'(a_0)|^2$. Moreover, $|f(a_n)| \leq \epsilon^{2^n} |f'(a_0)|^2, \forall n \geq 0$.*

Both of the Hensel's Lemmas does not require for A to be complete. A local ring A for which the Hensel's lemma I holds is called a henselian ring. One can define the henselisation of a noetherian local ring R as the minimal extension of R that is henselian. Loosely speaking, in henselian rings, Cauchy sequences that converges to the root of a polynomial required to converge, but not every Cauchy sequences need to converge.

Lemma 2.2.5 (Hensel's Lemma III). *Let A be a complete DVR with maximal ideal $\mathfrak{p}$ and residue field k , let $f \in A[x]$ have image $\bar{f} \in k[x]$, and suppose $\bar{f} = \bar{g}\bar{h}$ for some coprime $\bar{g}, \bar{h} \in k[x]$. Then there exists a polynomials $g, h \in A[x]$ for which $f = gh, g \equiv \bar{g} \pmod{\mathfrak{p}}, h \equiv \bar{h} \pmod{\mathfrak{p}}$ s.t. $\deg g = \deg \bar{g}$.*

Lemma 2.2.6 (Hensel-Kurschak Lemma). *Let A be a complete DVR with fraction field K , and let $f \in K[x]$ be an irreducible polynomial whose leading and constant coefficients lie in A . Then $f \in A[x]$.*

pf. Let $\mathfrak{p} = (\pi)$ be the maximal ideal of A , let $k \equiv A/\mathfrak{p}$ and write $f = \sum f_i x^i$, with $f_n \neq 0$. Then $n > 0, f_0 \neq 0$ since f is irreducible. Let $m \equiv \min(v_{\mathfrak{p}}(f_i))$. Suppose the contrary. Then $m < 0$. Let $g \equiv \pi^{-m} f = \sum g_i x^i \in A[x]$, which is an irreducible polynomial in $A[x]$ with $g_0, g_n \in \mathfrak{p}$ since $m < 0$, and $f_0, f_n \in A$ and g_i is a unit for some $0 < i < n$ by the minimality of m . The reduction $\bar{g}$ of g to $k[x]$ has positive degree and constant term 0 and is therefore divisible by x . If we let $\bar{u} \equiv x^d$ be the largest power of x dividing $\bar{g}$, $0 < d \leq \deg \bar{g} < n$ and $\bar{v} \equiv \bar{g}/x^d \in k[x]$ is coprime to $\bar{u}$. By the Hensel's lemma III, $g = uv, u, v \in A[x]$ with $0 < \deg u = \deg \bar{u} < n$. This means g is reducible, a contradiction.

Corollary 2.2.5. *Let A be a complete DVR with fraction field K , and let L/K be a finite extension of degree n . Then $\alpha \in L$ is integral over A iff $N_{L/K}(\alpha) \in A$.*

pf. Let $f = \sum_{i=0}^d f_i x^i \in K[x]$ be the minimal polynomial of α . If α is integral over A , $f \in A[x]$, $N_{L/K}(\alpha) = (-1)^n f(0)^e$ where $e = [L : K(\alpha)]$. Conversely, if $N_{L/K}(\alpha) = (-1)^n f(0)^e \in A$, $f(0) \in A$, since $f(0) \in K$ is a root of $x^e - (-1)^n N_{L/K}(\alpha) \in A[x]$ and A is integrally closed. The constant coefficient of f thus lies in A as does its leading coefficient 1, so $f \in A[x]$ by the previous lemma and α is integral over A .

Theorem 2.2.3. *Assume AKLB and that A is a complete DVR with maximal ideal $\mathfrak{p}$. Then B is a DVR whose maximal ideal $\mathfrak{q}$ is necessarily the unique prime ideal above $\mathfrak{o}$.*

pf. At least one prime $\mathfrak{q}$ of B lies above $\mathfrak{p}$, since the factorisation of $\mathfrak{p}B \subseteq B$ is nontrivial. Suppose the contrary. Then $\exists \mathfrak{q}_1, \mathfrak{q}_2 \in \{\mathfrak{q}|\mathfrak{p}\}$ with $\mathfrak{q}_1 \neq \mathfrak{q}_2$. Choose $b \in \mathfrak{q}_1 \setminus \mathfrak{q}_2$ and consider the ring $A[b] \subseteq B$. The ideals $\mathfrak{q}_1 \cap A[b], \mathfrak{q}_2 \cap A[b]$ are distinct prime ideals of $A[b]$ containing $\mathfrak{p}A[b]$ and both are maximal, since they are nonzero and $\dim A[b] = \dim A = 1$. The quotient ring $A[b]/\mathfrak{p}A[b]$ thus has at least two maximal ideals. Let $f \in A[x]$ be the minimal polynomial of b over K and let $\bar{f} \in (A/\mathfrak{p})[x]$ be its reduction to the residue field $A/\mathfrak{p} \rightarrow \frac{(A/\mathfrak{p})[x]}{(\bar{f})} \cong \frac{A[x]}{(\mathfrak{p}, f)} \cong \frac{A[b]}{\mathfrak{p}A[b]}$ thus the ring $(A/\mathfrak{p})[x]/(\bar{f})$ has at least two maximal ideals, i.e. $\bar{f}$ is divisible by two distinct irreducible polynomials. Thus we can factor $\bar{f} = \bar{g}\bar{h}$, where two factors are coprime. Then by the Hensel's Lemma III, we can lift this to a nontrivial

factorisation of $f = gh \in A[x]$, a contradiction. Hence there is only one prime $\mathbf{q} \in \{\mathbf{q}|\mathbf{p}\}$.

Every maximal ideal of B lies above a maximal ideal of A but A has only one maximal $\mathbf{p}$ and $\#\{\mathbf{q}|\mathbf{p}\} = 1$, and therefore B has a unique maximal ideal $\mathbf{q}$. So B is a local Dedekind domain, hence a local PID that is not a field, hence a DVR.

Definition 2.2.10. Let K be a field with absolute value $|\cdot|$ and let V be a K -vector

$$\|v\| = 0 \text{ iff } v = 0$$

space. A norm on V is a function $\|\cdot\|: V \rightarrow \mathbb{R}_+$ s.t. $\|\lambda v\| = |\lambda| \|v\|, \forall \lambda \in K, v \in V$

$$\|v + w\| \leq \|v\| + \|w\|, \forall w, v \in V$$

Each norm $\|\cdot\|$ induces a topology on V via the distance metric $d(v, w) \equiv \|v - w\|$.

e.g. A sup-norm $\|v\|_\infty \equiv \sup\{|v_i|\}$ is a norm on V where $v = \sum v_i e_i$ where K -basis for V is (e_i) .

Proposition 2.2.5. Let V be a vector space of finite dimension over a complete field K . Every norm on V induces the same topology in which V is a complete metric space.

pf. **CLAIM : Equivalent norms induce the same topology.**

pf. The basis for the topology is a set of open balls $B_{<r}(x)$. For any open ball in a metric 1, $B_{<r}^1(x) = \{\forall y \in V : \|y - x\|_1 < r\} = B_{<r}^2(x) = \{\forall y \in V : \|y - x\|_2 < r\}$ since $\exists c \in \mathbb{R}$ s.t. $\|\cdot\|_1 < c \|\cdot\|_2, \|\cdot\|_2 < c \|\cdot\|_1$. So they have the same basis and therefore their induced topologies are equivalent.

CLAIM : The sup-norm does not depend on the choice of basis.

pf. Let $(e_i), (d_i)$ are two different K -basis for V . Then $\exists M \in \mathbb{R}^{n \times n}$ s.t. $eM = d$, with $\det M = 1$. Then for any $d_i, d_i = \sum_j a_{ij} e_j, \forall v \in V, v = \sum_i v_i e_i = \sum_i a'_i d_i = \sum a'_i \sum_j a_{ij} e_j = \sum_j e_j (\sum_i a'_i a_{ij})$. Then $\|v\|_\infty = \max_j \sum_i a'_i a_{ij} = \max_i v_i$.

CLAIM : For any norm on V is equivalent to the sup-norm.

pf. For any norm $\|\cdot\|$ on $V, \forall v = \sum v_i e_i \in V, \|v\| \leq \sum \|v_i e_i\| = \sum |v_i| \|e_i\| = \sum |v_i| \leq n \max_i |v_i| = n \|v\|_\infty$. So $\exists n \in \mathbb{R} < \infty$ s.t. $\|\cdot\| \leq n \|\cdot\|_\infty$.

Conversely, when $n = 1, \|v\| = \|v_1 e_1\| = |v_1| \|e_1\| = |v_1| \geq 1/c \|v\|_\infty$ for all $c \geq 1$.

When $n > 1$, let c' be the constant we get from the case $n - 1$.

- if $\|v\|_\infty = |v_n|$
 $\|v\|_\infty = |v_n| = r \max_{i \in [1, n-1]} |v_i| \equiv r v'$ with $r \geq 1$. Then $\|v\| = \|v_1 e_1 + \dots + v_n e_n\| \leq \|v_1 e_1 + \dots + v_{n-1} e_{n-1}\| + |v_n| \leftrightarrow \frac{\|v\|}{\|v\|_\infty} \leq \frac{\|v'\|}{\|v\|_\infty} + 1 = \frac{\|v'\|}{r v'} + 1 = \frac{c'}{r} + 1 \leftrightarrow \|v\|_\infty \leq (1 + \frac{c'}{r}) \|v\|$ so we get a new constant.
- if $\|v\|_\infty \neq |v_n|$
 $c' \|v\| = c' \|v_1 e_1 + \dots + v_n e_n\| \geq c' \|v_1 e_1 + \dots + v_{n-1} e_{n-1}\| - c' \|v_n e_n\| \geq \|v_1 e_1 + \dots + v_{n-1} e_{n-1}\|_\infty - c' |v_n|$ by the inductive hypothesis. Then the LHS = $\|v\|_\infty - r \|v\|_\infty = (1 - r) \|v\|_\infty$ where $r \leq 1$. Thus we get a new constant $c = \frac{c'}{1-r} \in \mathbb{R}$

Among the set of constants, we choose a minimum value so that $c\|v\| \geq \|v\|_\infty$. Thus they are equivalent and therefore the proposition holds.

Theorem 2.2.4. *Let A be a complete DVR with fraction field K , maximal ideal $\mathfrak{p}$, discrete valuation $v_{\mathfrak{p}}$, and absolute value $|x|_{\mathfrak{p}} \equiv c^{v_{\mathfrak{p}}(x)}$, with $0 < c < 1$. Let L/K be a finite extension of degree n . The following hold.*

1. *There is a unique absolute value $|x| \equiv |N_{L/K}(x)|_{\mathfrak{p}}^{1/n}$ on L that extends $|\cdot|_{\mathfrak{p}}$*
2. *The field L is complete w.r.t. $|\cdot|$, and its valuation ring $\{x \in L : |x| \leq 1\}$ is equal to the integral closure of B of A in L*
3. *If L/K is separable, then B is a complete DVR whose maximal ideal $\mathfrak{q}$ induces $|x| = |x|_{\mathfrak{q}} \equiv c^{\frac{1}{e_{\mathfrak{q}}}v_{\mathfrak{q}}(x)}$ where $e_{\mathfrak{q}}$ is the ramification index of $\mathfrak{q}$ that is $\mathfrak{p}B = \mathfrak{q}^{e_{\mathfrak{q}}}$.*

pf. **CLAIM : $|\cdot|$ is an absolute value.**

pf. it is clearly multiplicative and $|x| = 0$ iff $x = 0$. So only the triangular inequality remains. Let $|x| \leq 1 \leftrightarrow |N_{L/K}(x)|_{\mathfrak{p}} \leq 1 \leftrightarrow N_{L/K}(x) \in A \leftrightarrow x \in B \leftrightarrow x+1 \in B \leftrightarrow |x+1| \leq 1$ and then for $y, z \in L$, assume wlog $|y| \geq |z| \rightarrow |y+z| = |y|(1 + \frac{z}{y})| \leq |y| + |z| = |y|(\frac{|z|}{|y|} + 1)$ so the triangular inequality holds and it is an absolute value.

$\forall x \in K, |x| = |N_{L/K}(x)|_{\mathfrak{p}}^{1/n} = |x^n|_{\mathfrak{p}}^{1/n} = |x|_{\mathfrak{p}}$ and therefore it extends $|\cdot|_{\mathfrak{p}}$ and is therefore a norm on L . Since it is nontrivial, $\exists x \in K^\times$ s.t. $|x|_{\mathfrak{p}} \neq 1$ and $|x|^a = |x|_{\mathfrak{p}} = |x|$ only for $a = 1$, which implies $|\cdot|$ is the unique absolute value extending $|\cdot|_{\mathfrak{p}}$.

Assume L/K is separable. Then B is a DVR and it is complete because it is a valuation ring of L . Let $\mathfrak{q}$ be the unique maximal ideal of B . The valuation $v_{\mathfrak{q}}$ extends $v_{\mathfrak{p}}$ with index $e_{\mathfrak{q}}$ so $v_{\mathfrak{q}}(x) = v_{\mathfrak{p}}(x)e_{\mathfrak{q}}$ for $x \in K^\times$. Since $0 < c^{1/e_{\mathfrak{q}}} < 1$, $|x|_{\mathfrak{q}} \equiv (c^{1/e_{\mathfrak{q}}})^{v_{\mathfrak{q}}(x)} = c^{\frac{1}{e_{\mathfrak{q}}}e_{\mathfrak{q}}v_{\mathfrak{p}}(x)} = c^{v_{\mathfrak{p}}(x)} = |x|_{\mathfrak{p}}$.

$\rightarrow$ The transitivity of $N_{L/K}$ implies we can uniquely extend the absolute value on the fraction field K of a complete DVR to an algebraic closure $\bar{K}$.

Corollary 2.2.6. *Assume AKLB and A is a complete DVR with maximal ideal $\mathfrak{p}$ and let $\mathfrak{q}|\mathfrak{p}$. Then $v_{\mathfrak{q}}(x) = \frac{1}{f_{\mathfrak{q}}}v_{\mathfrak{p}}(N_{L/K}(x))$ for all $x \in L$.*

pf. $v_{\mathfrak{p}}(N_{L/K}(x)) = v_{\mathfrak{p}}(N_{L/K}(\pi(x))) = v_{\mathfrak{p}}(N_{L/K}(\pi^{f_{\mathfrak{q}}v_{\mathfrak{q}}(x)})) = v_{\mathfrak{p}}(\pi^{f_{\mathfrak{q}}v_{\mathfrak{q}}(x)}) = f_{\mathfrak{q}}v_{\mathfrak{q}}(x)$.

$\rightarrow$ A valuation is a surjective homomorphism $v : K^\times \rightarrow \Gamma$ in which Γ is a totally ordered abelian group and $v(x+y) \geq \min(v(x), v(y))$; we extend v to K by defining $v(0) = \infty$ to be strictly greater than any element of Γ .

e.g. In AKLB with A a complete DVR, a valuation is defined $v(x) = \frac{1}{e_{\mathfrak{q}}}(v_{\mathfrak{q}}(x))$ with image $\frac{1}{e_{\mathfrak{q}}}\mathbb{Z}$ that restricts to the discrete valuation $v_{\mathfrak{p}}$ on K .

Lemma 2.2.7 (Nakayama's Lemma). *Let A be a local ring with maximal ideal $\mathfrak{p}$ and let M be a finitely generated A -module. If the images of $x_1 \dots x_n \in M$ generate $M/\mathfrak{p}M$ as an $(A/\mathfrak{p})$ -vector space, then $x_1, \dots, x_n$ generate M as an A -module.*

Corollary 2.2.7. *Let A be a local noetherian ring with maximal ideal $\mathfrak{p}$, let $g \in A[x]$ and let $B \equiv A[x]/(g)$. Every maximal ideal $\mathfrak{m}$ of B contains the ideal $\mathfrak{p}B$.*

pf. Assume the contrary. Then $\mathfrak{m} + \mathfrak{p}B = B$ for some maximal $\mathfrak{m}$ of B . The ring B is finitely generated over a noetherian ring A , hence a noetherian A -module, so its A -submodules are all finitely generated. Let $z_1, \dots, z_n$ be A -module generator for $\mathfrak{m}$. Every coset of $\mathfrak{p}B$ in B can be written as $z + \mathfrak{p}B$ for some A -linear combination z of $z_1, \dots, z_n$ so the images of them generate $B/\mathfrak{p}B$ as an $(A/\mathfrak{p})$ -vector space. Then by the Nakayama's lemma, $z_1, \dots, z_n$ generate B i.e. $\mathfrak{m} = B$, a contradiction.

Corollary 2.2.8. *Let A be a local noetherian ring with maximal ideal $\mathfrak{p}$, let $g \in A[x]$ be a polynomial with reduction $\bar{g} \in (A/\mathfrak{p})[x]$ and let α be the image of x in the ring $B \equiv A[x]/(g) = A[\alpha]$. The maximal ideals of B are $(\mathfrak{p}, g_i(\alpha))$, where $g_1, \dots, g_m \in A[x]$ are lifts of the distinct irreducible polynomials $\bar{g}_i \in (A/\mathfrak{p})[x]$ that divides $\bar{g}$.*

pf. By the previous corollary, $B \rightarrow B/\mathfrak{p}B$ gives a one-to-one correspondence between maximal ideals of B and maximal ideals of $B/\mathfrak{p}B$ and $\frac{B}{\mathfrak{p}B} \cong \frac{A[x]}{\mathfrak{p}B} \cong \frac{(A/\mathfrak{p})[x]}{(\bar{g})}$. Each maximal of $(A/\mathfrak{p})[x]/(\bar{g})$ is the reduction of an irreducible divisor of $\bar{g}$ hence one of the $\bar{g}_i$. Thus the corollary follows.

Theorem 2.2.5. *Assume AKLB with A, B DVRs with residue fields $k \equiv A/\mathfrak{p}, l \equiv B/\mathfrak{q}$. If l/k is separable, then $B = A[\alpha]$ for some $\alpha \in B$; if L/K is unramified this holds for every lift α of any generator $\bar{\alpha}$ for $l = k(\bar{\alpha})$.*

pf. Let $\mathfrak{p}B = \mathfrak{q}^e$ be the factorisation of $\mathfrak{p}B$ and let $f = [l : k]$ be the residue field degree, so that $ef = n \equiv [L : K]$. The extension l/k is separable, so by the primitive element theorem, $l = k(\bar{\alpha}_0)$ for some $\bar{\alpha}_0 \in l$ whose minimal polynomial $\bar{g}$ is separable of degree f . Let $g \in A[x]$ be a monic lift of $\bar{g}$ and let α_0 be any lift of $\bar{\alpha}_0$ to B . If $v_{\mathfrak{q}}(g(\alpha_0)) = 1$, then $\alpha = \alpha_0$. Otherwise let π_0 be any uniformiser for B and let $\alpha \equiv \alpha_0 + \pi_0 \in B$ so that $v_{\mathfrak{q}}(g(\alpha)) = v_{\mathfrak{q}}(g(\alpha_0 + \pi_0)) = v_{\mathfrak{q}}(g(\alpha_0) + \pi_0 g'(\alpha_0) + \pi_0^2 h(\alpha_0)) = 1$ where $h \in A[x]$, i.e. $\pi \equiv g(\alpha)$ is a uniformiser for B .

CLAIM : $B = A[\alpha]$.

pf. $\mathfrak{p} = \mathfrak{q}^e$ so $\mathfrak{p}B = (\pi^e)$. Represent each element of $B/\mathfrak{p}B$ as a coset $b + \mathfrak{p}B = b_0 + b_1\pi + \dots + b_{e-1}\pi^{e-1} + \mathfrak{p}B$ where b_i s are determined up to equivalence module πB . Let $1, \bar{\alpha}, \dots, \bar{\alpha}^{f-1}$ are a basis for $B/\mathfrak{p}B = B/\mathfrak{q}$ as a k -vector space, and $\pi = g(\alpha)$, so $b + \mathfrak{p}B = (a_0 + a_1\alpha + \dots + a_{f-1}\alpha^{f-1}) + (a_f + a_{f+1}\alpha + \dots + a_{2f-1}\alpha^{f-1})g(\alpha) + \dots + (a_{ef-f+1} + \dots + a_{ef-1}\alpha^{f-1})g(\alpha)^{e-1} + \mathfrak{p}B$. Since $\deg g = f$, $n = ef$, this expresses $b + \mathfrak{p}B$ in the form $b' + \mathfrak{p}B$ with b' in the A -span of $1, \dots, \alpha^{n-1}$ thus the claim follows.

Then by the Nakayama's lemma, $1, \dots, \alpha^{n-1}$ also span B .

If L/K is unramified, l/k is separable and $e = 1, f = n$, in which case there is no need to require $g(\alpha)$ to be a uniformiser and we can just take $\alpha = \alpha_0$ to be any lift of any $\bar{\alpha}_0$ that generate l over k .

2.2.2 Extensions of Complete DVRs-Unramified

Define a category $\mathcal{C}_K^{unr}$ to be the finite unramified extensions L of K with morphisms K -algebra homomorphisms and a category $\mathcal{C}_k^{sep}$ to be the finite separable extensions l of k with morphisms k -algebra homomorphisms.

Theorem 2.2.6. *Let A be a complete DVR with fraction field K and residue field $k \equiv A/\mathfrak{p}$. The categories $\mathcal{C}_K^{unr}, \mathcal{C}_k^{sep}$ are equivalent via functor $\mathcal{F} : \mathcal{C}_K^{unr} \rightarrow \mathcal{C}_k^{sep}$. That sends each unramified extension L of K to its residue field l , and each K -algebra homomorphism $\phi : L_1 \rightarrow L_2$ to the k -algebra homomorphism $\bar{\phi} : l_1 \rightarrow l_2$ defined by $\bar{\phi}(\bar{\alpha}) \equiv \overline{\phi(\alpha)}$, where α is any lift of $\bar{\alpha} \in l_1 \equiv B_1/\mathfrak{q}_1$ to B_1 and $\overline{\phi(\alpha)}$ is the reduction of $\phi(\alpha) \in B_2$ to $l_2 \equiv B_2/\mathfrak{q}_2$; here $\mathfrak{q}_1, \mathfrak{q}_2$ are the maximal ideals of the valuation rings B_1, B_2 of L_1, L_2 respectively. In particular, $\mathcal{F}$ gives a bijection between the isomorphism classes in $\mathcal{C}_K^{unr}$ and $\mathcal{C}_k^{sep}$ and if L_1, L_2 and have residue fields l_1, l_2 , then $\mathcal{F}$ induces a bijection of finite sets $\text{Hom}_K(L_1, L_2) \rightarrow \text{Hom}_k(l_1, l_2)$.*

pf. **CLAIM : $\mathcal{F}$ is well-defined.**

pf. $\mathcal{F}$ maps finite unramified extensions L/K to a finite separable extensions l/k , but we need to check it does not depend on the choice of the lift α of $\bar{\alpha}$ we pick. So let $\phi : L_1 \rightarrow L_2$ be a K -algebra homomorphism and for $\bar{\alpha} \in l_1$, let α, α' be two lifts of $\bar{\alpha}$ to B_1 . Then $\alpha - \alpha' \in \mathfrak{q}_1$ and this implies $\phi(\alpha - \alpha') \in \phi(\mathfrak{q}_1) = \phi(B_1) \cap \mathfrak{q}_2 \subseteq \mathfrak{q}_2$ and therefore $\overline{\phi(\alpha)} = \overline{\phi(\alpha')}$. The identity $\phi(\mathfrak{q}_1) = \phi(B_1) \cap \mathfrak{q}_2 \subseteq \mathfrak{q}_2$ follows from the fact that ϕ restricts to an injective ring homomorphism $B_1 \rightarrow B_2$ and $B_2/\phi(B_1)$ is a finite extension of DVRs in which $\mathfrak{q}_2$ lies over the prime $\phi(\mathfrak{q}_1)$ of $\phi(B_1)$. Then $\mathcal{F}$ sends identity morphisms to the identity morphisms and it is compatible with composition. Thus the claim follows.

CLAIM : $\mathcal{F}$ is essentially surjective, i.e. each separable l/k is isomorphic to the residue field of some unramified L/K

pf. Given a finite separable extension l/k , by the primitive element theorem $l \cong k(\bar{\alpha}) = \frac{k[x]}{(\bar{g})}$ for some $\bar{\alpha} \in l$, whose minimal polynomial $\bar{g} \in k[x]$ is monic irreducible separable and of degree $n \equiv [l : k]$. Let $g \in A[x]$ be any monic lift of $\bar{g}$, then g is also irreducible separable and of degree n . Let $L \equiv \frac{K[x]}{(g)} = K(\alpha)$ where α is the image of x in $K[x]/(g)$. Then L/K is a finite separable extension and by the previous corollary 2.3.6, $(\mathfrak{p}, g(\alpha))$ is the unique maximal ideal of $A[\alpha]$ and $B/\mathfrak{q} \cong \frac{A[\alpha]}{(\mathfrak{p}, g(\alpha))} \cong \frac{A[x]}{(\mathfrak{p}, g)} \cong \frac{(A/\mathfrak{p})[x]}{(\bar{g})} \cong l$. Thus $[L : K] = \deg g = [l : k] = n$ so L/K is an unramified extension of degree n and the ramification index of $\mathfrak{q}$ is necessarily $e = n/f = 1$ and the extension l/k is separable by assumption.

CLAIM : $\mathcal{F}$ is full and faithful; i.e. tie induced map $\text{Hom}_K(L_1, L_2) \rightarrow \text{Hom}_k(l_1, l_2)$

is a bijection.

pf. Given finite unramified extensions L_1, L_2 with valuation rings B_1, B_2 , and residue fields l_1, l_2 we have induced maps $\text{Hom}_K(L_1, L_2) \rightarrow \text{Hom}_A(B_1, B_2) \rightarrow \text{Hom}_k(l_1, l_2)$. The first map is given by the restriction from L_1 to B_1 and since tensoring with K gives an inverse map in the other direction, it is a bijection. The second map sends $\phi : B_1 \rightarrow B_2$ to the k -homomorphism $\bar{\phi}$ that sends $\bar{\alpha} \in l_1 = B_1/\mathfrak{q}_1$ to the reduction of $\phi(\alpha)$ modulo $\mathfrak{q}_2$, where α is any lift of $\bar{\alpha}$. By the primitive element theorem $l_1 = k(\bar{\alpha}) = k[x]/(\bar{g})$ for some $\bar{\alpha} \in l_1$. If we lift $\bar{\alpha}$ to $\alpha \in B_1$, $L_1 = K(\alpha)$ since $[L_1 : K] = [l_1 : k]$ is equal to the degree of the minimal polynomial $\bar{g}$ of $\bar{\alpha}$ which cannot be less than the degree of the minimal polynomial g of α , where both are monic. Moreover $B_1 = A[\alpha]$ since it is true of the valuation ring of every finite unramified extensions in the category. Each A -module homomorphism in $\text{Hom}_A(B_1, B_2) = \text{Hom}_A(\frac{A[x]}{(g)}, B_2)$ is uniquely determined by the image of x in B_2 . This gives a bijection between $\text{Hom}_A(B_1, B_2)$ and the roots of g in B_2 . Similarly each k -algebra homomorphism in $\text{Hom}_k(l_1, l_2) = \text{Hom}_k(\frac{k[x]}{(\bar{g})}, l_2)$ is uniquely determined by the image of x in l_2 and there is a bijection between $\text{Hom}_k(l_1, l_2)$ and the roots of $\bar{g}$ in l_2 . Now $\bar{g}$ is separable, so every root of $\bar{g}$ in $l_2 = B_2/\mathfrak{q}_2$ lifts to a unique root of g in B_2 by the Hensel's Lemma. Thus the second map above is also a bijection and the claim follows.

→ even if L_2/K is not unramified, the map $\text{Hom}_K(L_1, L_2) \rightarrow \text{Hom}_l(l_1, l_2)$ is a bijection.

Corollary 2.2.9. *Assume AKLB with A a complete DVR with residue field k . Then L/K is unramified iff $B = A[\alpha]$ for some $\alpha \in L$ whose minimal polynomial $g \in A[x]$ has separable image $\bar{g}$ in $k[x]$.*

pf. $\bar{g}$ must be irreducible, since otherwise we could use Hensel's lemma to lift a nontrivial factorisation of $\bar{g}$ to a nontrivial factorisation of g so the residue field extension is separable and has the same degree as L/K . So L/K is unramified. The converse follows by the previous theorem.

Corollary 2.2.10. *Let A be a complete DVR with fraction field K and residue field k and let ζ_n be a primitive n th root of unity in some algebraic closure of K with n prime to the characteristic of k . The extension $K(\zeta_n)/K$ is unramified.*

pf. The field $K(\zeta_n)$ is the splitting field of $f(x) = x^n - 1$ over K . The image $\bar{f}$ of f in $k[x]$ is separable when $p \nmid n$ since $\gcd(\bar{f}, f) \neq 1$ only when $\bar{f}' = nx^{n-1}$ is zero, equivalently only when $p|n$. When $\bar{f}$ is separable, so are all of its divisors, including the reduction of the minimal polynomial of ζ_n , which must be irreducible since otherwise we could obtain a contradiction by lifting a nontrivial factorisation via Hensel's lemma. Thus the residue field of $K(\zeta_n)$ is a separable extension of k so $K(\zeta_n)/K$ is unramified.

Theorem 2.2.7. *Assume $AKLB$ with A a complete DVR with the finite residue field. Then L/K is unramified iff $N_{L/K}(B^\times) = A^\times$.*

pf. **CLAIM :** $N_{L/K}(B^\times) \subseteq A^\times$, $N_{l/k}(l^\times) = k^\times$, $T_{l/k}(l) = k$.

pf. $a \in N_{L/K}(B^\times) \leftrightarrow \exists b \in B^\times$ s.t. $N_{L/K}((b)) = a$, $\exists b^{-1} \in B$ s.t. $N_{L/K}((b^{-1})) = \hat{a}$. Since the norm map is a group homomorphism $N_{L/K}((1)) = N_{L/K}((bb^{-1})) = N_{L/K}((b))N_{L/K}((b^{-1})) = a\hat{a}$ implies $a \in A^\times$.

CLAIM : $\forall i \geq 0, U_i \equiv 1 + \mathfrak{p}^i \equiv \{1 + a : a \in \mathfrak{p}^i\}$. Then U_i are distinct closed subgroup of $A^\times$ that form a basis of nhoods $1 \in A^\times$.

pf. $\forall 1 + a, 1 + a' \in U_i, (1 + a)(1 + a') = 1 + (a + a' + aa') \in U_i$ subgroup of $A^\times$.

If $U_i \cap U_j \neq \emptyset, \exists a \in U_i \cap U_j$ i.e. $a = 1 + rp = 1 + sq$ where $r, s \in A^\times, p \in \mathfrak{p}^i, q \in \mathfrak{p}^j, i \neq j$. Wlog assume $i > j$. Then $a = 1 + rp'q = 1 + sq \leftrightarrow rp' = s \in A^\times$ a contradiction. Hence U_i are distinct.

Pick an arbitrary open nhood of 1 in the topological group $A^\times$. Then it contains an open ball $B_{<r}(1) = \{x \in A^\times : |x - 1| < r\}$. For any $r \in \mathbb{R}, \exists i \in \mathbb{Z}_{++}$ s.t. $1/r < i$ and $U_i \subseteq B_{<r}(1)$ implies U_i form a base of nhoods $1 \in A^\times$.

CLAIM : If L/K is totally ramified, the norm of every $b \in B^\times$ lies in a coset of U_1 of the form $u^n U_1$ where $n = [L : K]$.

pf. Since L/K is totally ramified, for an uniformiser of K , π_A , $L = K(\pi_A^{1/n})$, $B = A[\pi_A^{1/n}] = B[\pi]$ where n is defined above. Since $b \in B^\times = A[\pi]^\times, a = N_{L/K}(b) \in A^\times$ i.e. $a \notin \mathfrak{p}$. d

By the previous claim, $\forall n > 1$, the norms of the cosets above do not cover $A^\times$. Thus if $N_{L/K}(B^\times) = A^\times, L/K$ must be unramified.

CLAIM : Assume L/K is unramified. Then for every $u \in A^\times, \exists \alpha_0 \in B^\times$ with $N_{L/K}(\alpha_0) \equiv u \pmod{\mathfrak{p}}$.

pf. Since L/K is unramified, there is a corresponding finite separable extension l/k with $N_{l/k}(l^\times) = k^\times$. So $\bar{u} \in k^\times, \exists \bar{v} \in l^\times$ s.t. $N_{l/k}(\bar{v}) = \bar{u}$. Then we can get a lift for $\bar{v}$, say v s.t. $N_{L/K}(v) \equiv u \pmod{\mathfrak{p}}$. $\exists \alpha_1 \in B^\times$ with $N_{L/K}(\alpha_0 \alpha_1) \equiv u \pmod{\mathfrak{p}^2}$. Continuing by this fashion, we get $\alpha \in B^\times$ s.t. $N_{L/K}(\alpha) = u$.

Definition 2.2.11. Let L/K be a separable extension. The maximal unramified extension of K in L is the subfield $\cup_{L \subseteq E \subseteq L, E/L \text{ fin. umr}} E \subseteq L$ where the union is over finite unramified extensions E/K . When $L = K^{sep}$ is the separable closure of K , this is the maximal unramified extension of K , denoted K^{unr} .

e.g. $\mathbb{Q}_p^{unr}$ is an infinite extension of $\mathbb{Q}_p$ with Galois group $Gal(\mathbb{Q}_p^{unr}/\mathbb{Q}_p) \cong Gal(\bar{\mathbb{F}}_p/\mathbb{F}_p) = \lim Gal(\mathbb{F}_{p^n}/\mathbb{F}_p) \cong \lim \mathbb{Z}/n\mathbb{Z} \cong \hat{\mathbb{Z}}$, which is the profinite completion of $\mathbb{Z}$.

2.2.3 Extensions of Complete DVRs-Totally Ramified

Definition 2.2.12. Let A be a DVR with maximal ideal $\mathfrak{p}$. A monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in A[x]$ is Eisenstein if $a_i \in \mathfrak{p}, \forall 0 \leq i < n$ and $a_0 \notin \mathfrak{p}^2$; equivalently, $v_{\mathfrak{p}}(a_i) \geq 1, \forall 0 \leq i < n$ and $v_{\mathfrak{p}}(a_0) = 1$.

→ the definition implies that a_0 is a uniformiser.

Lemma 2.2.8 (Eisenstein Irreducibility). *Let A be a DVR with fraction field K and let $f \in A[x]$ be Eisenstein. Then f is irreducible in both $A[x]$ and $K[x]$.*

pf. Suppose the contrary. Then $f = gh$ has degree $n \geq 2$ for some nonconstant monic $g, h \in A[x]$. Put $f(x) = \sum f_i x^i, g = \sum g_i x^i, h = \sum h_i x^i$. Then $v_{\mathbf{p}}(f_0) = v_{\mathbf{p}}(g_0 h_0) = v_{\mathbf{p}}(g_0) + v_{\mathbf{p}}(h_0) = 1$ where $\mathbf{p}$ is a maximal ideal of A . Wlog assume $v_{\mathbf{p}}(h_0) = 1$. Let $i > 0$ be the least i for which $v_{\mathbf{p}}(h_i) = 0$; such an $i < n$ exists since h is monic and $\deg h < n$. Then we have $f_i = g_0 h_i + \dots + g_i h_0$ with $v_{\mathbf{p}}(f_i) \geq 1$, given f Eisenstein and $i < n$. But the valuation of the RHS is zero since $v_{\mathbf{p}}(g_0 h_i) = 0, v_{\mathbf{p}}(g_i h_j) \geq 1, \forall 0 \leq j < i$ by the minimality of i , a contradiction. Thus f is irreducible in $A[x]$ and since A is a DVR, and therefore a PID and UFD. Thus by Gauss' lemma f is irreducible in $K[x]$.

→ **EISENSTEIN'S CRITERION** : Any polynomial $f(x)$ over a Dedekind domain A that is Eisenstein over a localisation $A_{\mathbf{p}}$, f is irreducible in $K[x]$, hence in $A[x]$, given $A_{\mathbf{p}}, A$ have the same fraction field K .

Lemma 2.2.9. *Let A be a DVR and let $f \in A[x]$ be an Eisenstein polynomial. Then $B = A[\pi] \equiv A[x]/(f)$ is a DVR with uniformiser π , where π is the image of x in $A[x]/(f)$.*

pf. Let $\mathbf{p}$ be the maximal of A . Then $f \equiv x^n \pmod{\mathbf{p}}$ so by the Dedekind-Kummer theorem $\mathbf{q} = (\mathbf{p}, x) = (\mathbf{p}, \pi)$ is the only maximal ideal of B . Let $f = \sum f_i x^i$, then $\mathbf{p} = (f_0)$ and $\mathbf{q} = (f_0, \pi)$ and $f_0 = -\pi f_1 - \pi^2 f_2 - \dots - \pi^n \in (\pi)$, so $\mathbf{q} = (\pi)$. The unique maximal (π) of B is nonzero and principal, so B is a DVR with uniformiser π .

Theorem 2.2.8. *Assume AKLB with A a complete DVR and π a uniformiser for B . The extension L/K is totally ramified iff $B = A[\pi]$ and the minimal polynomial of π is Eisenstein.*

pf. Let $n = [L : K]$, $\mathbf{p}$ be the maximal ideal of A , $\mathbf{q}$ be the maximal ideal of B and let π be a uniformiser for B with minimal polynomial f . If $B = A[\pi]$ and f is Eisenstein, then by the previous lemma, we have $\mathbf{p} = \mathbf{q}^n$, so $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$ with index $e_{\mathbf{q}} = n$ and L/K is totally ramified.

Suppose L/K is totally ramified. Then $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$ with index n , which implies $v_{\mathbf{q}}(K) = n\mathbb{Z}$. The set $\{\pi^0, \dots, \pi^{n-1}\}$ is linearly independent over K , since the valuation of $\pi^0, \dots, \pi^{n-1}$ are distinct modulo $v_{\mathbf{q}}(K) = n\mathbb{Z}$. Thus $L = K(\pi)$. Let $f = \sum a_i x^i$ be the minimal polynomial of π . We have $v_{\mathbf{q}}(f(\pi)) = \infty$ and $v_{\mathbf{q}}(a_i \pi^i) \equiv i \pmod{n}, \forall 0 \leq i < n$. This is possible only if $v_{\mathbf{q}}(a_0) = v_{\mathbf{q}}(a_0 \pi^0) = v_{\mathbf{q}}(a_n \pi^n) = v_{\mathbf{q}}(\pi^n) = n, v_{\mathbf{q}}(a_i) \geq n, \forall 0 \leq i < n$. This implies $v_{\mathbf{p}}(a_0) = 1$ since $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$ with index

n and $v_{\mathbf{p}}(a_i) \geq 1, \forall 1 \leq i < n$. Thus f is Eisenstein and by the previous lemma $A[\pi] \subseteq B$ is a DVR and hence a maximal, so $B = A[\pi]$.

e.g. $K = \mathbb{Q}_3$ has distinct three quadratic extensions :

$\mathbb{Q}_3(\sqrt{2})$	unramified quadratic extension
$\mathbb{Q}_3(\sqrt{3})$	ramified extension
$\mathbb{Q}_3(\sqrt{6})$	ramified extension

Definition 2.2.13. Assume AKLB with A a complete DVR and separable residue field extension of characteristic $p \geq 0$. The extension L/K is tamely ramified if $p \nmid e_{L/K}$. Otherwise L/K is wildly ramified if $p \mid e_{L/K}$. A totally ramified extension L/K is totally tamely ramified if $p \nmid e_{L/K}$ and it is totally wildly ramified if $e_{L/K}$ is a power of p .

Proposition 2.2.6. The properties of being unramified, tamely ramified, wildly ramified, totally ramified, totally tamely ramified and totally wildly ramified are all transitive in towers of extensions fraction fields of complete DVRs with separable residue field extensions.

pf. follows by the transitivity of separability and the multiplicativity of ramification indices and degrees in towers.

A compositum of totally ramified extensions need not be totally ramified. (A compositum of $\mathbb{Q}_3(\sqrt{3}), \mathbb{Q}_3(\sqrt{6})$ of $\mathbb{Q}_3$ contains the unramified quadratic extension $\mathbb{Q}_3(\sqrt{2})$).

Theorem 2.2.9. Assume AKLB with A a complete DVR and separable residue field extension of characteristic $p \geq 0$ not dividing $n \equiv [L : K]$. The extension L/K is totally ramified iff $L = K(\pi_A^{1/n})$ for some uniformiser π_A of A .

pf. If $L = K(\pi_A^{1/n})$, then $\pi = \pi_A^{1/n}$ has minimal polynomial $x^n - \pi_A$, which is Eisenstein, so $A[\pi]$ is a DVR by the first lemma, and therefore $B = A[\pi]$, since DVRs are maximal and by the previous theorem L/K is totally ramified, given $p \nmid n$.

Conversely, assume L/K is totally tamely ramified, and let $\mathbf{p}, \mathbf{q}$ be the maximal ideals of A, B with uniformisers π_A, π_B respectively. Then $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$ with index $e_{\mathbf{q}} = n$ and $v_{\mathbf{q}}(\pi_B^n) = n = v_{\mathbf{q}}(\pi_A) \rightarrow \pi_B^n = u\pi_A$ for some unit $u \in B^\times$. We have $f_{\mathbf{q}} = 1$ so B, A have the same residue field and if we lift the image of u in $B/\mathbf{q} \cong A/\mathbf{p}$ to a unit u_A in A and replace π_A with $u_A^{-1}\pi_A$, we can assume $u \equiv 1 \pmod{\mathbf{q}}$. Define $g(x) \equiv x^n - u \in B[x]$ with reduction $\bar{g} = x^n - 1$ in $(B/\mathbf{q})[x]$. Then $\bar{g}'(1) \neq 0$ so by the Hensel's Lemma I, we can lift the root 1 of $\bar{g}(x)$ in $B/\mathbf{q}$ to be a root r of $g(x)$ in B . Let $\pi \equiv \pi_B/r$. Then π is a uniformiser for B and $B = A[\pi]$ by the previous theorem, so $L = K(\pi)$ and $\pi^n = \pi_B^n/r^n = \pi_B^n/u = \pi_A$ so $L = K(\pi_A^{1/n})$ as desired.

Proposition 2.2.7. *Let L be a totally ramified extension of the fraction field K of a complete DVR. There is a unique intermediate field E s.t. E/K is totally tamely ramified and L/E is totally wildly ramified.*

pf. Let $e \equiv e_{L/K}$ be the ramification index and let $p \geq 0$ be the characteristic of the residue field. If $p \nmid e$, the proposition holds with $E = L$, so we assume $p \mid e$ and put $e = mp^a$, where $p \nmid m$. Let A be the valuation ring of K with maximal ideal $\mathfrak{p}$ and let B be the valuation ring of L , which is also a complete DVR, with maximal ideal $\mathfrak{q}$. Then by the previous theorem, we can choose the uniformiser π_A, π_B of A, B respectively, s.t. $\pi_B^n = u\pi_A$ with $u \in B^\times$, $u \equiv 1 \pmod{\mathfrak{q}}$. Let $g(x) = x^m - u \in B[x]$, then we can construct a root $r \in B$ of $g(x)$ by the Hensel's lemma, lifting the root 1 of $\bar{g} \in (B/\mathfrak{q})[x]$. Consider the field extension $E \equiv K(\pi)$ where $\pi \equiv \pi_B^{o_a}/r$. Then we have $\pi^m = \pi_B^n/r^m = \pi_B^n/u = \pi_A \rightarrow E = K(\pi_A^{1/m})$ with $p \nmid m$. The polynomial $x^m - \pi_A$ is Eisenstein, hence irreducible and has π as a root, so E/K has degree m . Then by the previous theorem E/K is totally tamely ramified, and L/E has degree p^a and is thus totally wildly ramified.

Suppose $E' \subseteq L$ is another totally tamely ramified extensions of K s.t. L/E' is totally wildly ramified. Then E' must be of the form $E' = K(\pi_A^{1/m})$ by the previous theorem. In other words, E, E' are both generated by the roots of $x^m - \pi_A$. The ratio of these roots is a m th root of unity $\zeta \in L$ that must lie in K because L/K is totally ramified and the extension $K(\zeta)/K$ is unramified. Thus $E = E'$.

Corollary 2.2.11. *Let L be a finite separable extension of the fraction field K of a complete DVR with separable field extension. There is a unique intermediate field E s.t. E/K is tamely ramified and L/E is totally wildly ramified.*

pf. Let F be the maximal unramified extension of K in L . Then we can assume $F = K(\alpha)$ where α is integral, whose minimal polynomial g has separable image in $k[x]$ where k is the residue field of K . Then by the previous proposition to L/F , we get a tamely ramified extension E/F with L/E totally wildly ramified. Unramified extensions are tamely ramified, so $E/F/K$ is a tower of tamely ramified extensions, hence tamely ramified.

Any field E' with L/E' totally wildly ramified must contain α , otherwise $E'(\alpha)$ would be a nontrivial unramified subextension L/E' . By the previous proposition, $E' = E$.

Let K be the fraction field of a complete DVR with absolute value $|\cdot|$. Then we can uniquely extend $|\cdot|$ to any finite extension L/K by defining $|x| \equiv |N_{L/K}(x)|^{1/n}$ where $n = [L : K]$. This induces a unique absolute value on $\bar{K}$ that restricts to the absolute value of K .

Lemma 2.2.10. *Let K be the fraction field of a complete DVR with algebraic closure $\bar{K}$ and absolute value $|\cdot|$ extended to $\bar{K}$. For all $\alpha \in \bar{K}$ and $\sigma \in \text{Aut}_K(\bar{K})$ we*

have $|\sigma(\alpha)| = |\alpha|$.

pf. The elements $\alpha, \sigma(\alpha)$ must have the same minimal polynomial $f \in K[x]$ since $f(\sigma(\alpha)) = \sigma(f(\alpha)) = \sigma(0) = 0$, so $N_{L/K}(\alpha) = f(0) = N_{K(\sigma(\alpha))/K}(\sigma(\alpha))$. It follows that $|\sigma(\alpha)| = |N_{K(\sigma(\alpha))/K}(\sigma(\alpha))|^{1/n} = |N_{K(\alpha)/K}(\alpha)|^{1/n} = |\alpha|$ where $n = \deg f$.

Definition 2.2.14. Let K be the fraction field of a complete DVR with absolute value $|\cdot|$ extended to an algebraic closure $\bar{K}$. For $\alpha, \beta \in \bar{K}$, we say β belongs to α is $|\beta - \alpha| < |\beta - \sigma(\alpha)|, \forall \sigma \in \text{Aut}_K(\bar{K})$ with $\sigma(\alpha) \neq \alpha$, that is β is strictly closer to α than to any of its conjugates. This is equivalent to requiring that $|\beta - \alpha| < |\alpha - \sigma(\alpha)|, \forall \sigma(\alpha) \neq \alpha$, since every nonarchimedean triangle is isosceles.

Lemma 2.2.11 (Krasner's Lemma). Let K be the fraction field of a complete DVR and let $\alpha, \beta \in \bar{K}$ with α separable over K . If β belongs to α , then $K(\alpha) \subseteq K(\beta)$.

pf. Suppose the contrary. Then β belongs to α , but $\alpha \notin K(\beta)$. The extension $K(\alpha, \beta)/K(\beta)$ is separable and nontrivial, so there is an automorphism $\sigma \in \text{Aut}_{K(\beta)}(\bar{K}/K(\beta))$ for which $\sigma(\alpha) \neq \alpha$. By the previous lemma to $\beta - \alpha \in \bar{K}, |\beta - \alpha| = |\sigma(\beta - \alpha)| = |\sigma(\beta) - \sigma(\alpha)| = |\beta - \sigma(\alpha)|$, a contradiction to the fact that β belongs to α .

Definition 2.2.15. For a field K with absolute value $|\cdot|$, the L^1 -norm of $f = \sum f_i x^i \in K[x]$ is $\|f\|_1 \equiv \sum_i |f_i|$.

Lemma 2.2.12. Let K be a field with absolute value $|\cdot|$ and let $f \equiv \prod_i (x - \alpha_i) \in K[x]$ be a monic polynomial with roots $\alpha_1, \dots, \alpha_n \in L$, where L/K is a field with an absolute value that extends $|\cdot|$. Then $|\alpha| \leq \|f\|_1$ for every root α of f .

pf. When $n = 0, 1$, the lemma is clear, so assume $n \geq 2$. If $\|f\|_1 = 1, f = x^n, \alpha = 0$ so $|\alpha| = 0 < 1 = \|f\|_1$ the lemma holds. Otherwise, $\|f\|_1 > 1$, and if $|\alpha| \leq 1$, also the lemma trivially holds, so assume $|\alpha| > 1$. Then we have $0 = |f(\alpha)| = |\alpha^n + \sum_{i=0}^{n-1} f_i \alpha^i| \geq |\alpha|^n - \sum_{i=0}^{n-1} |f_i| |\alpha|^i \geq |\alpha|^n - |\alpha|^{n-1} \sum_{i=0}^{n-1} |f_i| \geq |\alpha| - (\|f\|_1 - 1)$. Thus $\|f\|_1 - 1 \geq |\alpha| \leftrightarrow \|f\|_1 \geq |\alpha| + 1 > |\alpha|$.

Theorem 2.2.10 (Continuity of Roots). Let K be the fraction field of a complete DVR and $f \in K[x]$, a monic irreducible separable polynomial. There exists $\delta = \delta(f) \in \mathbb{R}_{++}$ s.t. for every monic polynomial $g \in K[x]$ with $\|f - g\|_1 < \delta$, the following holds:

Every root β of g belongs to a root α of f for which $K(\beta) = K(\alpha)$.
In particular, every such g is separable, irreducible, and has the same splitting field as f .

pf. We can always pick $\delta < 1$, in which case any monic $g \in K[x]$ with $\|f - g\|_1 < \delta$ must have the same degree as f , so we can assume $\deg g = \deg f$. Fix an algebraic closure $\bar{K}$ of K with absolute value $|\cdot|$ extending the absolute value on K .

Let $\alpha_1, \dots, \alpha_n$ be the roots of f in $\bar{K}$ and write $f(x) = \prod (x - \alpha_i) = \sum_{i=0}^n f_i x^i$. Let $\epsilon = \min(1, \min_{i \neq j} |\alpha_i - \alpha_j|)$. Define $\delta \equiv \delta(f) = (\frac{\epsilon}{2(\|f\|_1 + 1)})^n > 0$ s.t. $\delta < 1$, given $\|f\|_1 \geq 1, \epsilon \leq 1$. Let $g(x) = \sum g_i x^i$ be a monic polynomial of degree n with $\|f - g\|_1 < \delta$. Then $\|g\|_1 \leq \|f\|_1 + \|g - f\|_1 = \|f\|_1 + \|f - g\|_1 < \|f\|_1 + \delta$ and $\forall \beta \in \bar{K}$ any root of g , we have

$|f(\beta)| = |f(\beta) - g(\beta)| = |(f - g)(\beta)| = |\sum (f_i - g_i)\beta^i| \leq \sum |f_i - g_i| |\beta|^i$. By the previous lemma

$|\beta| < \|g\|_1, \|g\|_1 \geq 1$, so $|\beta|^i < \|g\|_1^i \leq \|g\|_1^n$ and therefore $|f(\beta)| < \|f - g\|_1 \cdot \|g\|_1^n < \delta(\|f\|_1 + \delta)^n < \delta(\|f\|_1 + 1)^n \leq (\epsilon/2)^n \rightarrow \prod |\beta - \alpha_i| = |f(\beta)| < (\epsilon/2)^n$ and at least one $i, |\beta - \alpha_i| < \epsilon/2$. Also by the triangular inequality α_i is unique, given $|\alpha_i - \alpha_j| \geq \epsilon, \forall j \neq i$. This implies β belongs to $\alpha \equiv \alpha_i$.

By the Krasner's lemma, $K(\alpha) \subseteq K(\beta)$ and $n = [K(\alpha) : K] \leq [K(\beta) : K] \leq n$ so $K(\alpha) = K(\beta)$. Thus g is the minimal polynomial of β since $\deg g = [K(\beta) : K]$. Thus g is irreducible and it is also separable, since $\beta \in K(\beta) = K(\alpha)$ lies in a separable extension of K . If a root β of g belongs to a root α of f , then for any $\tau \in \text{Aut}_K(\bar{K}), \forall \sigma \in \text{Aut}_K(\bar{K})$, s.t. $\sigma(\alpha) \neq \alpha, |\tau(\beta) - \tau(\alpha)| = |\tau(\beta - \alpha)| = |\beta - \alpha| < |\alpha - \sigma(\alpha)| = |\tau(\alpha - \sigma(\alpha))| = |\tau(\alpha) - \tau(\sigma(\alpha))|$ so $\tau(\beta)$ belongs to $\tau(\alpha)$. Since $\text{Aut}_K(\bar{K})$ acts transitively on the roots of f, g , so every root β of g belongs to a distinct root α of f for which $K(\beta) = K(\alpha)$. Therefore g has the same splitting field as f .

Theorem 2.2.11. *Let K be a global field with a nontrivial absolute value $|\cdot|$ and let $\hat{K}$ be the completion of K w.r.t. $|\cdot|$. Every finite separable extension $\hat{L}$ of $\hat{K}$ is the completion of a finite separable extension L of K w.r.t. an absolute value that restricts to $|\cdot|$. One can choose L so that $[L : K] = [\hat{L} : \hat{K}]$, in which case $\hat{L} = \hat{K} \cdot L$.*

pf. Let $\hat{L}/\hat{K}$ be a separable extension of degree n .

- $|\cdot|$ is archimedean, K is a number field and $\hat{K}$ is either $\mathbb{R}$ or $\mathbb{C}$; the only nontrivial case is $\hat{K} \cong \mathbb{R}, n = 2$ and we may assume $\hat{L} = \hat{K}(\sqrt{d}), d \in \mathbb{Z}_{<0}$ is any nonsquare in K . Wlog assume $|\cdot|$ is the Euclidean absolute value on $\hat{K} \cong \mathbb{R}$ and uniquely extend $|\cdot|$ to $L \equiv K(\sqrt{d})$ by requiring $|\sqrt{d}| = \sqrt{-d}$. Then $\hat{L}$ is the completion of L w.r.t. $|\cdot|$ and $[L : K] = [\hat{L} : \hat{K}] = 2$ and $\hat{L} = \hat{K} \cdot L$.
- $|\cdot|$ is nonarchimedean, the valuation ring of $\hat{K}$ is a complete DVR and $|\cdot|$ is induced by its discrete valuation. By the primitive element theorem, we may assume $\hat{L} = \hat{K}[x]/(f)$ where $f \in \hat{K}[x]$ is monic, irreducible and separable. The field K is dense in its completion $\hat{K}$, so we can find a monic $g \in K[x] \subseteq \hat{K}[x]$ s.t. $\|g - f\|_1 \leq \delta, \forall \delta > 0$. Then by the previous theorem $\hat{L} = \hat{K}[x]/(f)$ where g is separable. The field $\hat{L}$ is a finite separable extension of the fraction field of a complete DVR, so it is itself the fraction field of a complete DVR and has a unique absolute value that extends the absolute value $|\cdot|$ on $\hat{K}$. Let $L \equiv K[x]/(g)$. Then g is irreducible in $\hat{K}[x]$ hence in $K[x]$, so $[L : K] =$

$[\hat{L} : \hat{K}] = \deg g$. The field $\hat{L}$ contains both $\hat{K}, L$ and it is clearly the smallest field that does, so $\hat{L}$ is the compositum of $\hat{K}$ and L . The absolute value on $\hat{L}$ restricts to an absolute value on L , extending the absolute value on K , and $\hat{L}$ is complete, so $\hat{L}$ contains the completion of L w.r.t. $|\cdot|$. On the other hand, the completion of L w.r.t. $|\cdot|$ contains $L, \hat{K}$, so it must be $\hat{L}$.

Corollary 2.2.12. *For every finite Galois extension $\hat{L}/\hat{K}$ of local fields, there is a finite Galois extension of global fields L/K and an absolute value $|\cdot|$ on L s.t. $\hat{L}$ is the completion of L w.r.t. $|\cdot|$, $\hat{K}$ is the completion of K w.r.t. the restriction of $|\cdot|$ to K and $\text{Gal}(L/K) \cong \text{Gal}(\hat{L}/\hat{K})$.*

pf. The archimedean case follows by the previous theorem with $K = \mathbb{Q}$. So assume $\hat{L}$ is nonarchimedean and we take $|\cdot|$ to be the absolute value on $\hat{K}, \hat{L}$. The field $\hat{K}$ is an extension of either $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$, and by the previous theorem, we may assume $\hat{K}$ is the completion of a global field K w.r.t. the restriction of $|\cdot|$. Let $g \in K[x]$ be a monic separable polynomial irreducible in $\hat{K}[x]$ s.t. $\hat{L}[x] = \hat{K}[x]/(g)$ and define $L \equiv K[x]/(g)$ so that $\hat{L}$ is the compositum of $\hat{K}$ and L . Let M be the splitting field of g over K , the minimal extension of K that contains all the roots of g . The field $\hat{L}$ also contains these roots and $\hat{L}$ contains K , so $\hat{K}$ also contains a subextension of K isomorphic to M , which we now identify with M . Then $\hat{L}$ is also the completion of M w.r.t. the restriction of $|\cdot|$ to M . We have a group homomorphism $\phi : \text{Gal}(\hat{L}/\hat{K}) \rightarrow \text{Gal}(M/K)$ induced by restriction and ϕ is injective since $\forall \sigma \in \text{Gal}(\hat{L}/\hat{K})$ is determined by its action on any root of g in M . If we now replace K by the fixed field of the image of ϕ and replace L with M , the completion of K w.r.t. the restriction of $|\cdot|$ is still equal to $\hat{K}$ and similarly for L and $\hat{L}$. Now the corollary follows.

Under AKLB assumption, if $\mathfrak{p}$ is a prime of K , each prime $\mathfrak{q}|\mathfrak{p}$ induces a valuation $v_{\mathfrak{q}}$ of L that extends the valuation $v_{\mathfrak{p}}$ of K with index $e_{\mathfrak{q}}$, meaning $v_{\mathfrak{q}}|_K = e_{\mathfrak{q}}v_{\mathfrak{p}}$. The embedding $K \rightarrow L$ induces an embedding $K_{\mathfrak{p}} \rightarrow L_{\mathfrak{q}}$ of local fields for each $\mathfrak{q}|\mathfrak{p}$.

Theorem 2.2.12. *Assume AKLB, let $\mathfrak{p}$ be a prime of A , and let $\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}}}$ be the factorisation of $\mathfrak{p}B$ in B . Let $K_{\mathfrak{p}}$ be the completion of K w.r.t. $|\cdot|_{\mathfrak{p}}$ and let $\hat{\mathfrak{p}}$ be the maximal ideal of its valuation ring. For each $\mathfrak{q}|\mathfrak{p}$, let $L_{\mathfrak{q}}$ denote the completion of L w.r.t. $|\cdot|_{\mathfrak{q}}$, and $\hat{\mathfrak{q}}$ the maximal ideal of its valuation ring. The following hold:*

1. Each $L_{\mathfrak{q}}$ is a finite separable extension of $K_{\mathfrak{p}}$ with $[L_{\mathfrak{q}} : K_{\mathfrak{p}}] \leq [L : K]$
2. Each $\hat{\mathfrak{q}}$ is the unique prime of $L_{\mathfrak{q}}$ lying over $\hat{\mathfrak{p}}$
3. Each $\hat{\mathfrak{q}}$ has ramification index $e_{\hat{\mathfrak{q}}} = e_{\mathfrak{q}}$ and residue field degree $f_{\hat{\mathfrak{q}}} = f_{\mathfrak{q}}$
4. $[L_{\mathfrak{q}} : K_{\mathfrak{p}}] = e_{\mathfrak{q}}f_{\mathfrak{q}}$

5. the map
$$\begin{array}{ccc} L \otimes_K K_{\mathbf{p}} & \rightarrow & \Pi_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}} \\ l \otimes x & \mapsto & (lx, lx, \dots, lx) \end{array}$$
 is an isomorphism of finite etale $K_{\mathbf{p}}$ -algebras

6. If L/K is Galois, then each $L_{\mathbf{q}}/K_{\mathbf{p}}$ is Galois and we have isomorphisms of decomposition groups $D_{\mathbf{q}} \cong D_{\hat{\mathbf{q}}} = \text{Gal}(L_{\mathbf{q}}/K_{\mathbf{p}})$ and inertial groups $I_{\mathbf{q}} \cong I_{\hat{\mathbf{q}}}$.

pf.

1. For each $\mathbf{q}|\mathbf{p}$, the embedding $K \rightarrow L$ induces an embedding $K_{\mathbf{p}} \rightarrow L_{\mathbf{q}}$ defined by $[(x_n)] \mapsto [(x_n)]$ on equivalence classes of Cauchy sequences; a sequence (x_n) that is Cauchy in K w.r.t. $|\cdot|_{\mathbf{p}}$ is also Cauchy in L w.r.t. $|\cdot|_{\mathbf{q}}$ because $v_{\mathbf{q}}$ extends $v_{\mathbf{p}}$. Thus we may view $K_{\mathbf{p}}$ as a topological subfield of $L_{\mathbf{q}}$ and it is clear that $[L_{\mathbf{q}} : K_{\mathbf{p}}] \leq [L : K]$ since any K -basis $b_1, \dots, b_m$ for $L \subseteq L_{\mathbf{q}}$ spans $L_{\mathbf{q}}$ as a $K_{\mathbf{p}}$ -vector space; given a Cauchy sequence $y \equiv (y_n)$ of elements of L , if we write each y_n as $\sum_i x_{in} b_i, x_{in} \in K$, we obtain Cauchy sequence $x_1 \equiv (x_{1n}), \dots, x_m \equiv (x_{mn})$ of elements in K and we can write $[y] = [x_1]b_1 + \dots + [x_m]b_m$ as a $K_{\mathbf{p}}$ -linear combination of b_i . The field L is a finite etale K -algebra, since L/K is a separable extension so its base change $L \otimes_K K_{\mathbf{p}}$ to $K_{\mathbf{p}}$ is a finite etale $K_{\mathbf{p}}$ -algebra. Consider the $K_{\mathbf{p}}$ -algebra homomorphism
$$\begin{array}{ccc} \phi_{\mathbf{q}} : L \otimes_K K_{\mathbf{p}} & \rightarrow & L_{\mathbf{q}} \\ l \otimes x & \mapsto & lx \end{array}$$
. Then $\phi_{\mathbf{q}}(b_i \otimes 1) = b_i$ for each of the K -basis elements $b_i \in L$, and b_i s span $L_{\mathbf{q}}$ as $K_{\mathbf{p}}$ -vector space, thus $\phi_{\mathbf{q}}$ is surjective. As a finite etale $K_{\mathbf{p}}$ -algebra, $L \otimes_K K_{\mathbf{p}}$ is isomorphic to a finite product of finite separable extensions of $K_{\mathbf{p}}$; $L_{\mathbf{q}}$ is isomorphic to a subproduct and thus also a finite etale $K_{\mathbf{p}}$ -algebra. In particular $L_{\mathbf{q}}/K_{\mathbf{p}}$ is separable.
2. The valuation rings of $K_{\mathbf{p}}, L_{\mathbf{q}}$ are complete DVRs, so this follows by the previous theorem.
3. The valuation ring $v_{\hat{\mathbf{q}}}$ extends $v_{\mathbf{q}}$ with index 1, which in turn extends $v_{\mathbf{p}}$ with index $e_{\mathbf{q}}$. The valuation $v_{\hat{\mathbf{p}}}$ extends $v_{\mathbf{p}}$ with index 1, and therefore $v_{\hat{\mathbf{q}}}$ extends $v_{\hat{\mathbf{p}}}$ with index $e_{\hat{\mathbf{q}}}$ and therefore $e_{\hat{\mathbf{q}}} = e_{\mathbf{q}}$. The residue field of $\hat{\mathbf{p}}$ is the same as that of $\mathbf{p}$: for any Cauchy sequence (a_n) over K , a_n eventually all have the same image in the residue field at $\mathbf{p}$ since $v_{\mathbf{p}}(a_n - a_m) > 0$ for all sufficiently large m, n . Similar statements apply to all $\hat{\mathbf{q}}$ and $\mathbf{q}$ and thus $f_{\hat{\mathbf{q}}} = f_{\mathbf{q}}$.
4. By 2, $[L_{\mathbf{q}} : K_{\mathbf{p}}] = e_{\hat{\mathbf{q}}} f_{\hat{\mathbf{q}}}$ since $\hat{\mathbf{q}}$ is the only prime above $\hat{\mathbf{p}}$ and by 3, this implies $[L_{\mathbf{q}} : K_{\mathbf{p}}] = e_{\mathbf{q}} f_{\mathbf{q}}$.
5. Let $\phi \equiv \Pi_{\mathbf{q}|\mathbf{p}} \phi_{\mathbf{q}}$ where $\phi_{\mathbf{q}} : L \otimes_K K_{\mathbf{p}} \rightarrow \Pi_{\mathbf{q}|\mathbf{p}} L \otimes_K K_{\mathbf{p}} \rightarrow L_{\mathbf{q}}$ is surjective $K_{\mathbf{p}}$ -algebra homomorphism, defined in the proof of 1. Then $\phi : L \otimes_K K_{\mathbf{p}} \rightarrow$

$\Pi_{\mathbf{q}|\mathbf{p}}L_{\mathbf{q}}$ is a $K_{\mathbf{p}}$ -algebra homomorphism. By 4, and by the fact that taking the base change of a finite etale algebra does not change its dimension, we have $\dim_{K_{\mathbf{p}}}(L \otimes_K K_{\mathbf{p}}) = \dim_K L = [L : K] = \sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{q}}f_{\mathbf{q}} = \sum_{\mathbf{q}|\mathbf{p}} [L_{\mathbf{q}} : K_{\mathbf{p}}] = \dim_{K_{\mathbf{p}}} \Pi_{\mathbf{q}|\mathbf{p}}L_{\mathbf{q}}$. Pick a $K_{\mathbf{p}}$ -basis $\{\beta_i\}$ for $\Pi_{\mathbf{q}|\mathbf{p}}L_{\mathbf{q}}$, fix $\epsilon > 0$, and for each basis element $\beta_i = (\beta_{i\mathbf{q}})_{\mathbf{q}|\mathbf{p}}$ use the weak approximation theorem to construct $\alpha_i \in L$ s.t. $|\alpha_i - \beta_{i\mathbf{q}}|_{\mathbf{q}} < \epsilon$, $\forall \mathbf{q}|\mathbf{p}$. In the metric space $\Pi_{\mathbf{q}|\mathbf{p}}L_{\mathbf{q}}$ with the sup-norm, each $\phi(\alpha_i \otimes 1)$ is close to β_i . The $K_{\mathbf{p}}$ -matrix whose j th column expresses $\phi(\alpha_j \otimes 1)$ in terms of the basis $\{\beta_i\}$ is then close to the identity matrix w.r.t. $|\cdot|_{\mathbf{p}}$ and the determinant D of this matrix is close to 1. For sufficiently small ϵ , we must have $D \neq 0$ and then $\{\phi(\alpha_i \otimes 1)\}$ is a basis for $\Pi_{\mathbf{q}|\mathbf{p}}L_{\mathbf{q}}$. Thus ϕ is surjective and therefore an isomorphism, since its domain and codomain have the same dimension.

6. Assume L/K is Galois. Then $\forall \sigma \in D_{\mathbf{q}}$ acts on L and respects the valuation $v_{\mathbf{q}}$, since it fixes $\mathbf{q}(x \in \mathbf{q}^n \rightarrow \sigma(x) \in \sigma(\mathbf{q}^n) = \sigma(\mathbf{q})^n = \mathbf{q}^n)$ and therefore if (x_n) is a Cauchy in L , so is $(\sigma(x_n))$, thus σ is an automorphism of $L_{\mathbf{q}}$ and it fixes $K_{\mathbf{p}}$. Thus $\exists \phi : D_{\mathbf{q}} \rightarrow \text{Aut}_{K_{\mathbf{p}}}(L_{\mathbf{q}})$. If $\sigma \in D_{\mathbf{q}}$ acts trivially on $L_{\mathbf{q}}$, then it acts trivially on $L \subseteq L_{\mathbf{q}}$, so $\ker \phi$ is trivial. Also $e_{\mathbf{q}}f_{\mathbf{q}} = |D_{\mathbf{q}}| \leq \#\text{Aut}_{K_{\mathbf{p}}}(L_{\mathbf{q}}) \leq [L_{\mathbf{q}} : K_{\mathbf{p}}] = e_{\mathbf{q}}f_{\mathbf{q}}$ so $\#\text{Aut}_{K_{\mathbf{p}}}(L_{\mathbf{q}}) = [L_{\mathbf{q}} : K_{\mathbf{p}}]$ and $L_{\mathbf{q}}/K_{\mathbf{p}}$ is Galois, and this also shows ϕ is surjective and therefore an isomorphism. There is only one prime $\hat{\mathbf{q}}$ of $L_{\mathbf{q}}$, and it necessarily fixed by $\forall \sigma \in \text{Gal}(L_{\mathbf{q}}/K_{\mathbf{p}})$, $\text{Gal}(L_{\mathbf{q}}/K_{\mathbf{p}}) = D_{\hat{\mathbf{q}}}$. The inertial groups $I_{\mathbf{q}}$ and $I_{\hat{\mathbf{q}}}$ have order $e_{\mathbf{q}} = e_{\hat{\mathbf{q}}}$ and ϕ restricts to a homomorphism $I_{\mathbf{q}} \rightarrow I_{\hat{\mathbf{q}}}$, so the inertial groups are also isomorphic.

Corollary 2.2.13. *Assume AKLB and let $\mathbf{p}$ be a prime of A . For every $\alpha \in L$, $N_{L/K}(\alpha) = \Pi_{\mathbf{q}|\mathbf{p}} N_{L_{\mathbf{q}}/K_{\mathbf{p}}}(\alpha)$, $T_{L/K}(\alpha) = \sum_{\mathbf{q}|\mathbf{p}} T_{L_{\mathbf{q}}/K_{\mathbf{p}}}(\alpha)$ where we view α as an element of $K_{\mathbf{q}}$ and $N_{L/K}(\alpha)$ as an element of $K_{\mathbf{p}}$ via the canonical embeddings $L \rightarrow L_{\mathbf{q}}, K \rightarrow K_{\mathbf{p}}$.*

pf. The norm and trace are defined as the determinant and trace of K -linear maps $L \rightarrow_{\times \alpha} L$ that are unchanged upon tensoring with $K_{\mathbf{p}}$, and therefore by the previous theorem 5's isomorphism the corollary follows.

Corollary 2.2.14. *Assume AKLB and let $\mathbf{p}$ be a prime of A . Let $\mathbf{p}B = \Pi_{\mathbf{q}|\mathbf{p}} \mathbf{q}^{e_{\mathbf{q}}}$ be the factorisation of $\mathbf{p}B$ in B . Let $\hat{A}_{\mathbf{p}}$ be the completion of A w.r.t. $|\cdot|_{\mathbf{p}}$ and for each $\mathbf{q}|\mathbf{p}$, let $\hat{B}_{\mathbf{q}}$ denote the completion of B w.r.t. $|\cdot|_{\mathbf{q}}$. Then $B \otimes_A \hat{A}_{\mathbf{p}} \cong \Pi_{\mathbf{q}|\mathbf{p}} \hat{B}_{\mathbf{q}}$ as $\hat{A}_{\mathbf{p}}$ -algebras.*

pf. After localising, we can replace A with $A_{\mathbf{p}}$, B with $B_{\mathbf{p}}$, and we may assume A is a DVR, B/A as a free A -module of rank $n \equiv [L : K] = \sum_{\mathbf{q}|\mathbf{p}} e_{\mathbf{q}}f_{\mathbf{q}}$. Then $B \otimes_A \hat{A}_{\mathbf{p}}$ is a free $\hat{A}_{\mathbf{p}}$ -module of rank n . Viewing $\hat{A}_{\mathbf{p}}$ and the $\hat{B}_{\mathbf{q}}$ as valuation rings of $K_{\mathbf{p}}, L_{\mathbf{q}}$, by the previous theorem 4, $\Pi \hat{B}_{\mathbf{q}}$ is a free $\hat{A}_{\mathbf{p}}$ -module of rank

$\sum_{\mathbf{q}|\mathbf{p}}[L_{\mathbf{q}} : K_{\mathbf{p}}] = \sum e_{\mathbf{q}} f_{\mathbf{q}} = n$. These isomorphic $\hat{A}_{\mathbf{p}}$ -modules lie in isomorphic finite etale $K_{\mathbf{p}}$ -algebras $L \otimes_K K_{\mathbf{p}} \cong \prod L_{\mathbf{q}}$ by the previous theorem 6, and this $K_{\mathbf{p}}$ -algebra isomorphism restricts to an $\hat{A}_{\mathbf{p}}$ -algebra isomorphisms.

2.2.4 Ramification of the Extensions of the Complete DVRs

The trace mapping $\begin{matrix} L \times L & \rightarrow & K \\ (x, y) & \mapsto & T_{L/K}(xy) \end{matrix}$ is a perfect pairing under the AKLB set-up. An A -lattice M in L is a finitely generated A -module that spans L as a K -vector space. Every A -lattice M in L has a dual lattice $M^* \equiv \{x \in L : T_{L/K}(xm) \in A, \forall m \in M\}$ which is an A -lattice in L isomorphic to the dual A -module $M^\vee \equiv \text{Hom}_A(M, A)$. Every fractional ideal I of B is finitely generated as a B -module and therefore finitely generated as an A -module. If I is nonzero, it necessarily spans L , since B does. Thus every element of $\mathcal{I}_B$ of nonzero fractional ideals of B is an A -lattice in L .

Lemma 2.2.13. *Assume AKLB. If $I \in \mathcal{I}_B$, $I^* \in \mathcal{I}_B$.*

pf. The dual lattice I^* is a finitely generated A -module. Consider $b \in B, x \in I^*, m \in I$. Then we have $T_{L/K}((bx)m) = T_{L/K}(x(bm)) \in A$ since $x \in I^*, bm \in I$, so $bx \in I^*$ and it is closed by the multiplication by the element $b \in B$, hence a finitely generated B -module. Thus it is a fractional ideal of B .

Definition 2.2.16. *Assume AKLB. The different $\mathcal{D}_{L/K}$ of L/K is the inverse of B^* in $\mathcal{I}_B$. Explicitly $B^* \equiv \{x \in L : T_{L/K}(xb) \in A, \forall b \in B\}$ and we define*

$$\mathcal{D}_{L/K} \equiv \mathcal{D}_{B/A} = (B^*)^{-1} = (B : B^*) = \{x \in L : xB^* \subseteq B\}$$

$B \subseteq B^*$ since $T_{L/K}(ab) \in A, \forall a, b \in B \rightarrow \mathcal{D}_{B/A} = (B^*)^{-1} \subseteq B^{-1} = B$ so the different is an ideal.

Proposition 2.2.8. *Assume AKLB and let S be a multiplicative subset of A . Then $S^{-1}\mathcal{D}_{B/A} = \mathcal{D}_{S^{-1}B/S^{-1}A}$.*

pf. by the fact that inverses and duals are both compatible with localisation

Proposition 2.2.9. *Assume AKLB and let $\mathbf{q}|\mathbf{p}$ be a prime of B . Then $\mathcal{D}_{\hat{B}_{\mathbf{q}}/\hat{A}_{\mathbf{p}}} = \mathcal{D}_{B/A}\hat{B}_{\mathbf{q}}$, where $\hat{A}_{\mathbf{p}}, \hat{B}_{\mathbf{q}}$ are the completions of A, B at $\mathbf{p}, \mathbf{q}$ respectively.*

pf. Let $\hat{L} \equiv L \otimes_K K_{\mathbf{p}}$ be the basechange of the finite etale K -algebra L from K to $K_{\mathbf{p}}$. Then $\hat{L} \cong \prod_{\mathbf{q}|\mathbf{p}} L_{\mathbf{q}}$ by the theorem 2.3.12. (5). Even though $\hat{L}$ needs not be a field, it is a free $K_{\mathbf{p}}$ -module of finite rank and thus is equipped with a trace map that necessarily satisfies $T_{\hat{L}/K_{\mathbf{p}}}(x) = \sum_{\mathbf{q}|\mathbf{p}} T_{L_{\mathbf{q}}/K_{\mathbf{p}}}(x)$ that defines a trace

pairing on $\hat{L}$. Let $\hat{B} \equiv B \otimes \hat{A}_{\mathfrak{p}}$, which is an A -lattice in the $K_{\mathfrak{p}}$ -vector space $\hat{L}$. Then $\hat{B} \cong \Pi_{\mathfrak{q}|\mathfrak{p}} \hat{B}_{\mathfrak{q}} \cong \oplus_{\mathfrak{q}|\mathfrak{p}} \hat{B}_{\mathfrak{q}}$ and therefore $\hat{B}^* \cong \oplus_{\mathfrak{q}|\mathfrak{p}} \hat{B}_{\mathfrak{q}}^*$. Then $\hat{B}^* \cong B^* \otimes_A \hat{A}_{\mathfrak{p}}$. In particular, B^* generates each fractional ideal $\hat{B}_{\mathfrak{q}}^* \in \mathcal{I}_{\hat{B}_{\mathfrak{q}}}$. Taking inverses, $\mathcal{D}_{B/A} = (B^*)^{-1}$ generates the $\hat{B}_{\mathfrak{q}}$ -ideal $(\hat{B}_{\mathfrak{q}}^*)^{-1} = \mathcal{D}_{\hat{B}_{\mathfrak{q}}/\hat{A}_{\mathfrak{p}}}$.

Definition 2.2.17. Let S/R be a ring extension in which S is a free R -module of rank n . For any $x_1, \dots, x_n \in S$, we define the discriminant $\text{disc}(x_1, \dots, x_n) \equiv \text{disc}_{S/R}(x_1, \dots, x_n) \equiv \det[T_{S/R}(x_i x_j)]_{i,j} \in R$.

→ if x_i s are not R -linearly independent, the discriminant will be zero by the linearity of the trace.

Under AKLB assumption where $(e_i) \in B$ is a basis for L as a K -vector space, $\text{disc}(e_1, \dots, e_n) = \det[T_{L/K}(e_i e_j)]_{i,j} \in A$.

Proposition 2.2.10. Let L/K be a finite separable extension of degree n and let Ω/K be a field extension for which there are distinct $\sigma_1, \dots, \sigma_n \in \text{Hom}_K(L, \Omega)$. For any $e_1, \dots, e_n \in L$, $\text{disc}(e_1, \dots, e_n) = \det[\sigma_i(e_j)]_{i,j}^2$ and for any $x \in L$ $\text{disc}(1, x, \dots, x^{n-1}) = \Pi_{i < j} (\sigma_i(x) - \sigma_j(x))^2$. Such a field extension Ω/K always exists, since L/K is separable.

pf. For $1 \leq i, j \leq n$, $T_{L/K}(e_i e_j) = \sum_{k=1}^n \sigma_k(e_i e_j)$ and therefore $\text{disc}(e_1, \dots, e_n) = \det[T_{L/K}(e_i e_j)]_{i,j} = \det([\sigma_k(e_i)]_{i,k} [\sigma_k(e_j)]_{k,j}) = \det([\sigma_k(e_i)]_{i,l} [\sigma_k(e_j)]_{l,j}^y) = \det[\sigma_i(e_j)]_{i,j}^2$, since the determinant is multiplicative and $\det M = \det M^t$ for any matrix M . Let $x \in L$, and put $e_i \equiv x^{i-1}$, $\forall 1 \leq i \leq n$. Then $\text{disc}(1, x, \dots, x^{n-1}) = \det[\sigma_i(x^{j-1})]_{i,j}^2 = \det[\sigma_i(x)^{j-1}]_{i,j}^2 = \Pi_{i < j} (\sigma_i(x) - \sigma_j(x))^2$ since $[\sigma_i(x)^{j-1}]_{i,j}$ is a Vandermonde matrix.

Definition 2.2.18. For a polynomial $f(x) = \Pi_i (x - \alpha_i)$, the discriminant of f is $\text{disc}(f) \equiv \Pi_{i < j} (\alpha_i - \alpha_j)^2$, equivalently if A is a Dedekind domain, $f \in A[x]$ is a monic polynomial, and α is the image of x in $A[x]/(f)$, then $\text{disc}(f) = \text{disc}(1, \alpha, \dots, \alpha^{n-1}) \in A$.

$$\text{e.g. } \text{disc}(x^2 + bx + c) = b^2 - 4c, \text{disc}(x^3 + ax + b) = -4a^4 - 27b^2.$$

Assume AKLB and let M be an A -lattice in L . Then M is a finitely generated A -module that contains a K -basis for L .

Consider the case where M is a free A -lattice. If $e_1, \dots, e_n \in M \subseteq L$ and $e'_1, \dots, e'_n \in M \subseteq L$ are two A -basis for M , then $\text{disc}(e'_1, \dots, e'_n) = u^2 \text{disc}(e_1, \dots, e_n)$ where $u \in A^\times$.

pf. $\exists P \in A^{n \times n}$ invertible and $\det P = u$ s.t. it changes the base matrix from e to e' , i.e. $e' = eP$, where $e = (e_1, \dots, e_n)$, $e' = (e'_1, \dots, e'_n)$. Then $\text{disc}(e') = \det[T_{L/K}(e'_i e'_j)]_{i,j} =$

$\det[T_{L/K}((eP)_i(eP)_j)]_{ij} = \det[P^t(T_{L/K}(e_i e_j))_{ij}P] = (\det P^t) \operatorname{disc}(e)(\det P) = (\det P)^2 \operatorname{disc}(e)$.
 $\rightarrow A = \mathbb{Z}$, B is always a free $\mathbb{Z}$ -lattice and the only units in $\mathbb{Z}$ is $u = \pm 1$, so $u^2 = 1$.

Definition 2.2.19. Assume AKLB, let M be an A -lattice in L and let $n \equiv [L : K]$. Then discriminant $D(M)$ of M is the A -module generated by $\{\operatorname{disc}(x_1, \dots, x_n), x_i \in M\}$.

Lemma 2.2.14. Assume AKLB and let $M' \subseteq M$ be free A -lattices in L . The discriminants $D(M') \subseteq D(M)$ are nonzero principal fractional ideals. If $D(M') = D(M)$, $M' = M$.

pf. Let $e \equiv (e_1, \dots, e_n)$ be an A -basis for M . Then $\operatorname{disc}(e) \in D(M)$ and for any row vector $x \equiv (x_1, \dots, x_n)$ with entries in M there is a matrix $P \in A^{n \times n}$ for which $x = eP$ and we then have $\operatorname{disc}(x) = (\det P)^2 \operatorname{disc}(e)$ as above. Then $D(M) = (\operatorname{disc}(e))$ is principal, and it is nonzero because e is a basis for L and the trace pairing is nondegenerate. If we let $e' \equiv (e'_1, \dots, e'_n)$ be an A -basis for M' , $D(M') = (\operatorname{disc}(e'))$ is also nonzero and principal. $M' \subseteq M$ implies $e' = eP$ for some matrix $P \in A^{n \times n}$ and $\operatorname{disc}(e') = (\det P)^2 \operatorname{disc}(e)$. If $D(M') = D(M)$, $\det P$ must be a unit, i.e. P is invertible and $e = e'P^{-1}$, which implies $M \subseteq M' \rightarrow M = M'$.

Proposition 2.2.11. Assume AKLB and let M be an A -lattice in L . Then $D(M) \in \mathcal{I}_A$.

pf. The A -module $D(M) \subseteq K$ is nonzero because M contains a K -basis $e = (e_1, \dots, e_n)$ for L and $\operatorname{disc}(e) \neq 0$ because the trace pairing is nondegenerate. Let N be the free A -lattice in L generated by e and then pick a nonzero $a \in A$ s.t. $M \subseteq a^{-1}N$. Then $D(M) \subseteq D(a^{-1}N)$, which is a principal fractional ideal of A , hence a noetherian A -module. Thus its submodule $D(M)$ is finitely generated, and therefore the proposition follows.

Definition 2.2.20. Assume AKLB. The discriminant $D_{L/K}$ of L/K is the discriminant of B as an A -module:

$$D_{L/K} \equiv D_{B/A} \equiv D(b) \in \mathcal{I}_A$$

which is an A -ideal, since $\operatorname{disc}(x_1, \dots, x_n) = \det[T_{B/A}(x_i x_j)]_{ij} \in A, \forall x_i \in B$.

e.g. $A = \mathbb{Z}, K = \mathbb{Q}, L = \mathbb{Q}(i), B = \mathbb{Z} = [i]$ where B is a free A -lattice with basis $(1, i)$. Then

$$\begin{aligned} \operatorname{disc}(1, i) &= \det \begin{bmatrix} T_{L/K}(1 \cdot 1) & T_{L/K}(1 \cdot i) \\ T_{L/K}(i \cdot 1) & T_{L/K}(i \cdot i) \end{bmatrix} = -4 \\ D_{L/K} &= \operatorname{disc}(1, i) = (\det \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix})^2 = -4 \\ B = \mathbb{Z}[i] &= \mathbb{Z}[x]/(x^2 + 1) \rightarrow \operatorname{disc}(x^2 + 1) = -4 \end{aligned}$$

Proposition 2.2.12. *Assume AKLB and let S be a multiplicative subset of A . Then $S^{-1}D_{B/A} = D_{S^{-1}B/S^{-1}A}$.*

pf. Let $x = s^{-1}disc(e_1, \dots, e_n) \in S^{-1}D_{B/A}$ for some $s \in S, e_i \in B$. Then $x = s^{2n-1}disc(s^{-1}e_1, \dots, s^{-1}e_n)$ lies in $D_{S^{-1}B/S^{-1}A}$.

For any $e_1, \dots, e_n \in S^{-1}B$, $\exists s \in S \subseteq A$ s.t. $se_i \in B \forall i$. Then $disc(e_1, \dots, e_n) = s^{-2n}disc(se_1, \dots, se_n) \in S^{-1}D_{B/A}$.

Proposition 2.2.13. *Assume AKLB and let $\mathfrak{p}$ be a prime of A . Then $D_{B/A}\hat{A}_{\mathfrak{p}} = \Pi_{\mathfrak{q}|\mathfrak{p}}D_{\hat{B}_{\mathfrak{q}}/\hat{A}_{\mathfrak{p}}}$ where $\hat{A}_{\mathfrak{p}}, \hat{B}_{\mathfrak{q}}$ are the completions of A, B at $\mathfrak{p}, \mathfrak{q}$ respectively.*

pf. After localising at $\mathfrak{p}$, assume A is a DVR, B is a free A -module of rank n . Then we have a trace pairing on the finite etale $K_{\mathfrak{p}}$ -algebra $\hat{L} \equiv L \otimes K_{\mathfrak{p}}, \hat{B} \equiv B \otimes \hat{A}_{\mathfrak{p}} \cong \oplus_{\mathfrak{q}|\mathfrak{p}}\hat{B}_{\mathfrak{q}}$ is an $\hat{A}_{\mathfrak{p}}$ -lattice in the $K_{\mathfrak{p}}$ -vector space $\hat{L}$ that is a direct sum of free $\hat{A}_{\mathfrak{p}}$ -module, and thus a free $\hat{A}_{\mathfrak{p}}$ -module of rank $n = \sum e_{\mathfrak{q}}f_{\mathfrak{q}}$. We can choose $\hat{A}_{\mathfrak{p}}$ basis for each $\hat{B}_{\mathfrak{q}}$ using elements in B by the weak approximation and the fact that B is dense in $\hat{B}_{\mathfrak{q}}$. Then we can construct an $\hat{A}_{\mathfrak{p}}$ -basis $\hat{e}$ for the direct sum $\oplus \hat{B}_{\mathfrak{q}} \cong \hat{B}$ whose elements have nonzero projections to exactly one of the $\hat{B}_{\mathfrak{q}}$, along with a corresponding A -basis e for B obtained from $\hat{e}$ as the union of these projections. The matrix $[T_{\hat{L}/K_{\mathfrak{p}}}(\hat{e}_i\hat{e}_j)]$ is a block diagonal; each block corresponds to a matrix whose determinant is the discriminant of the $\hat{A}_{\mathfrak{p}}$ -basis we chose for each $\hat{B}_{\mathfrak{q}}$. Thus $D_{\hat{B}/\hat{A}_{\mathfrak{p}}} = \Pi_{\mathfrak{q}|\mathfrak{p}}D_{\hat{B}_{\mathfrak{q}}/\hat{A}_{\mathfrak{p}}}$. Then $disc_{B/A}(e_1, \dots, e_n) = disc_{(B \otimes \hat{A}_{\mathfrak{p}})/\hat{A}_{\mathfrak{p}}}(e_1 \otimes 1, \dots, e_n \otimes 1)$ generated $D_{B/A}$ as an A -ideal and also generate $D_{\hat{B}/\hat{A}_{\mathfrak{p}}}$ as an $\hat{A}_{\mathfrak{p}}$ -ideal. Thus $D_{B/A}\hat{A}_{\mathfrak{p}} = D_{\hat{B}/\hat{A}_{\mathfrak{p}}} = \Pi_{\mathfrak{q}|\mathfrak{p}}D_{\hat{B}_{\mathfrak{q}}/\hat{A}_{\mathfrak{p}}}$.

Theorem 2.2.13. *Assume AKLB. Then $D_{B/A} = N_{B/A}(\mathcal{D}_{B/A})$.*

pf. The different and discriminant are both compatible with localisation by the previous propositions, and the A -module $D_{B/A}, N_{B/A}(\mathcal{D}_{B/A})$ of A are both determined by the intersection of their localisations at the maximal ideals, so we can reduce to the case where A is its localisation at a prime of A . Then A is a DVR, B is a free A -lattice in L ; fix an A -basis $(e_1, \dots, e_n)$ for B . The dual A -lattice $B^* = \{x \in L : T_{L/K}(xb) \in A, \forall b \in B\} \in \mathcal{J}_B$ is also a free A -lattice in L with basis $(e_1^*, \dots, e_n^*)$ uniquely determined by $T_{L/K}(e_i^*e_j) = \delta_{ij}$. If we write $e_i = \sum_j a_{ij}e_j^*$ in terms of the K -basis $(e_1^*, \dots, e_n^*)$ for L , then $T_{L/K}(e_ie_j) = T_{L/K}(\sum_k a_{ik}e_k^*e_j) = \sum_k a_{ik}T_{L/K}(e_k^*e_j) = \sum_k a_{ik}\delta_{kj} = a_{ij}$. Thus $P \equiv [T_{L/K}(e_ie_j)]_{ij}$ is the change of basis matrix from $e^* \equiv (e_1^*, \dots, e_n^*)$ to $e \equiv (e_1, \dots, e_n)$. If we let ϕ denote the K -linear transformation with matrix P , it is an isomorphism with matrix P , then ϕ is an isomorphism of free A -modules and $D_{B/A}(\det[T_{L/K}(e_ie_j)]_{ij}) = (\det \phi) = [B^* : B]_A$ which is a module index. Then $D_{B/A} = [B^* : B]_A = N_{B/A}((B : B^*)) = N_{B/A}((B^*)^{-1}) = N_{B/A}(\mathcal{D}_{B/A})$.

Under AKLB assumption, if $\mathfrak{p}$ is a prime of A , $\mathfrak{p}B = \mathfrak{q}_1^{e_1} \dots \mathfrak{q}_r^{e_r}$ is the factorisation of $\mathfrak{p}B$. By the CRT, $B/\mathfrak{p}B \cong B/\mathfrak{q}_1^{e_1} \times \dots \times B/\mathfrak{q}_r^{e_r}$, which is a commutative $A/\mathfrak{p}$ -algebra of dimension $\sum e_i f_i$ where $f_i = [B/\mathfrak{q}_i : A/\mathfrak{p}]$ is the residue degree. It is a product of fields iff $e_i = 1, \forall i$ and it is a finite etale algebra iff it is a product of fields that are separable extensions of $A/\mathfrak{p}$.

Lemma 2.2.15. *Let k be a field and let R be a commutative k -algebra with k -basis $r_1, \dots, r_n$. Then R is a finite etale k -algebra iff $\text{disc}(r_1, \dots, r_n) \neq 0$.*

pf. R is a finite etale k -algebra iff the trace pairing on R is a perfect pairing, which is equivalent to being nondegenerate, given k a field. If the trace pairing is degenerate, then for some nonzero $x \in R$, $T_{R/k}(xy) = 0, \forall y \in R$. If we write $x = \sum x_i r_i, \forall x_i \in k, T_{R/k}(x r_j) = \sum_i x_i T_{R/k}(r_i r_j) = 0,$

$\forall r_j$ and this implies that the column of the matrix $[T_{R/k}(r_i r_j)]_{ij}$ are linearly dependent and $\text{disc}(r_1, \dots, r_n) = \det[T_{R/k}(r_i r_j)]_{ij} = 0$.

Conversely, if $\text{disc}(r_1, \dots, r_n) = 0$, the columns of $[T_{R/k}(r_i r_j)]_{ij}$ are linearly dependent and for some $x_i \in k$, not identically zero, $\sum_i x_i T_{R/k}(r_i r_j) = 0 \forall j$. For $x \equiv \sum x_i r_i$, and for any $y = \sum y_j r_j \in R, T_{R/k}(xy) = \sum_j y_j \sum_i x_i T_{R/k}(r_i r_j) = 0$ and the trace pairing is nondegenerate.

Theorem 2.2.14. *Assume AKLB, let $\mathfrak{q}$ be a prime of B lying above a prime $\mathfrak{p}$ of A s.t. $B/\mathfrak{q}$ is a separable extension of $A/\mathfrak{p}$. The extension L/K is unramified at $\mathfrak{q}$ iff $\mathfrak{q}$ does not divide $\mathcal{D}_{B/A}$, and it is unramified at $\mathfrak{p}$ iff $\mathfrak{p}$ does not divide $D_{B/A}$.*

pf. By the previous proposition, the different is compatible with completion, so it reduces to the case where A, B are complete DVRs. Then $[L : K] = e_{\mathfrak{q}} f_{\mathfrak{q}}$, where $\mathfrak{p}B = \mathfrak{q}^{e_{\mathfrak{q}}}$. Since B is a DVR with maximal $\mathfrak{q}$, we must have $\mathcal{D}_{B/A} = \mathfrak{q}^m, m \geq 0$. By the previous theorem, $D_{B/A} = N_{B/A}(\mathcal{D}_{B/A}) = N_{B/A}(\mathfrak{q}^m) = \mathfrak{p}^{f_{\mathfrak{q}} m}$. Thus $\mathfrak{q} | \mathcal{D}_{B/A}$ iff $\mathfrak{p} | D_{B/A}$. Since A is a PID, B is a free A -module and we may choose an A -module bases $e_1, \dots, e_n$ for B that is also a K -basis for L . Let $k \equiv A/\mathfrak{p}$, and $\bar{e}_i$ be the reduction of e_i to the k -algebra $R \equiv B/\mathfrak{p}B$. Then $(\bar{e}_1, \dots, \bar{e}_n)$ is a k -basis for R , and $[R : k] = [B/\mathfrak{q}^{e_{\mathfrak{q}}} : A/\mathfrak{p}] = e_{\mathfrak{q}} f_{\mathfrak{q}} = [L : K] = n$. Since B has an A -module basis, we may compute its discriminant as $D_{B/A} = \text{disc}(e_1, \dots, e_n)$. Thus $\mathfrak{p} | D_{B/A}$ iff $\text{disc}(e_1, \dots, e_n) \in \mathfrak{p}$, i.e. $\text{disc}(\bar{e}_1, \dots, \bar{e}_n) = 0$. By the previous lemma, $\text{disc}(\bar{e}_1, \dots, \bar{e}_n) = 0$ iff the k -algebra $B/\mathfrak{p}B$ is not finite etale, i.e. iff $\mathfrak{p}$ is ramified. Thus $\mathfrak{p} | D_{B/A}$ iff $\mathfrak{p}$ is ramified. There is only one prime over $\mathfrak{q}$ above $\mathfrak{p}$, $\mathfrak{q} | \mathcal{D}_{B/A}$ iff $\mathfrak{q}$ is ramified.

Corollary 2.2.15. *Assume AKLB. Only finitely many primes of A ramify.*

pf. A, B are both Dedekind domains, so the ideals $D_{B/A}, \mathcal{D}_{B/A}$ have unique factorisation into prime ideals, in which only finitely many primes appear.

If $\mathcal{O}$ is an order of the form $A[\alpha]$, where $\alpha \in B$ generates $L = K(\alpha)$ with minimal polynomial $f \in A[x]$, $\mathcal{O}$ is a free A -lattice with basis $1, \alpha, \dots, \alpha^{n-1}$, where $n = \deg f$ and $D_{\mathcal{O}/A} = (\text{disc}(1, \alpha, \dots, \alpha^{n-1})) = (\text{disc}(f))$, which is a principal A -ideal contained in $D_{B/A}$. If B is also a free A -lattice, $D_{\mathcal{O}/A} = (\det P)^2 D_{B/A} = [B : \mathcal{O}]_A^2 D_{B/A}$ where P is the matrix of the A -linear map $\phi : B \rightarrow \mathcal{O}$ that sends an A -basis for B to an A -basis for $\mathcal{O}$ and $[B : \mathcal{O}]_A$ is the module index.

Theorem 2.2.15. *Assume AKLB and let $\mathcal{O}$ be an order with integral closure B and conductor $\mathfrak{f}$. Then $D_{\mathcal{O}/A} = N_{B/A}(\mathfrak{f})D_{B/A}$.*

pf. **CLAIM :** $\mathfrak{c} = (B^* : C^*) \equiv \{x \in L : xC^* \subseteq B^*\}$ **where $C = A[\alpha]$ is an order.**

pf. Let A -basis for B, C be $(b_1, \dots, b_n), (c_1, \dots, c_n)$ s.t. $\exists(b_1^*, \dots, b_n^*), (c_1^*, \dots, c_n^*)$ with $b_i^*(b_j) = \delta_{ij} = c_i^*(c_j)$ respectively and since $C \subseteq B$, $B^* \subseteq C^*$, $c_i = \sum_j l_{ij} b_j$. If $c \in \mathfrak{c}$ is a conductor, $cb_i \in C$. Then $(c \sum_i x_i c_i^*, \sum_j y_j b_j) = \sum_{i,j} c x_i y_j c_i^*(b_j)$. Since $c \in \mathfrak{c}$, $\sum_{ij} c c_i^*(b_j)$ is well-defined and therefore it is in $(B^* : C^*)$.

If $\exists c$ s.t. $(c \sum_i x_i c_i^*, \sum_j y_j b_j) = \sum_{i,j} c x_i y_j c_i^*(b_j)$, $\sum_{ij} c c_i^*(b_j)$ is a well-defined pairing. This implies $c \in \mathfrak{c}$.

Then if we define $\mathcal{D}_{C/A} \equiv (B : C^*), D_{C/A} \equiv D(C)$,

$$\begin{cases} \mathcal{D}_{C/A} = (B : C^*) = (B^* : C^*)(B : B^*) = \mathfrak{c} \mathcal{D}_{B/A}, \\ D_{C/A} = D(C) = N_{B/A}(\mathfrak{c} \mathcal{D}_{B/A}) = N_{B/A}(\mathfrak{c}) D_{B/A} \end{cases} \rightarrow D_{C/A} = N_{B/A}(\mathcal{D}_{C/A}).$$

Theorem 2.2.16 (Dedekind). *Let $K = \mathbb{Q}(\alpha)$ be a number field, let f be the minimal polynomial of α , let p be a prime, and let $\bar{f} = \prod_{i=1}^r \bar{f}_i^{e_i}$ be the factorisation of $\bar{f} \equiv f \pmod{p}$ in $\mathbb{F}_p[x]$. Let f_i be any monic lift of $\bar{f}_i$ to $\mathbb{Z}[x]$, let $u \equiv \prod_i f_i$, let v be any monic lift of $\bar{f}/\bar{f}$ to $\mathbb{Z}[x]$ and let $w = (uv - f)/p \in \mathbb{Z}[x]$. Then p divides $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ iff $\bar{u}, \bar{v}, \bar{w}$ have no common factor in $\mathbb{F}_p[x]$.*

Proposition 2.2.14. *Assume AKLB. If $B = A[\alpha]$ for some $\alpha \in L$, $f \in A[x]$ is the minimal polynomial of α , then $\mathcal{D}_{B/A} = (f'(\alpha))$ is the B -ideal generated by $f'(\alpha)$.*

pf. **CLAIM :** $B^* = \{x \in L : T_{L/K}(xb) \in A \forall b \in B\}$ **is the principal B -ideal $(1/f'(\alpha))$.**

$$\text{pf. } T_{L/K}\left(\frac{\alpha^i}{f'(\alpha)}\right) = \begin{cases} 0 & 0 \leq i \leq n-2 \\ 1 & i = n-1 \\ \in A & i \geq n \end{cases} \text{ and therefore } \left(\frac{1}{f'(\alpha)}\right) \subseteq B^*.$$

Assume there exists $a \in B^*$. Then $T_{L/K}(ab) \in A, \forall b \in B$. Since $f'(\alpha) \in B$, $T_{L/K}(af'(\alpha)) \in A \rightarrow a \in (1/f'(\alpha))$.

Thus $\mathcal{D}_{B/A} = (B^*)^{-1} = (f'(\alpha))$

Definition 2.2.21. *Assume AKLB and let $\alpha \in B$ have minimal polynomial $f \in A[x]$.*

The different of α is defined by $\delta_{B/A}(\alpha) \equiv \begin{cases} f'(\alpha) & L = K(\alpha) \\ 0 & \text{otherwise} \end{cases}$

Proposition 2.2.15. *Assume AKLB. Then $\mathcal{D}_{B/A} = (\delta_{B/A}(\alpha) : \alpha \in B)$.*

$$\rightarrow D_{B/A} = \{N_{L/K}(\delta_{B/A}(\beta)) : \beta \in B\}$$

pf.

Theorem 2.2.17. *Assume AKLB and let $\mathfrak{q}$ be a prime of L lying above $\mathfrak{p} = \mathfrak{q} \cap A$ for which the residue field extension $(B/\mathfrak{q})/(A/\mathfrak{p})$ is separable. Then*

$$e_{\mathfrak{q}} - 1 \leq v_{\mathfrak{q}}(\mathcal{D}_{B/A}) \leq e_{\mathfrak{q}} - 1 + v_{\mathfrak{q}}(e_{\mathfrak{q}})$$

and the lower bound is an equality iff $\mathfrak{q}$ is tamely ramified.

pf. $e_{\mathfrak{q}} = v_{\mathfrak{q}}(\mathcal{D}_{C/A}) = v_{\mathfrak{q}}(\mathfrak{c}\mathcal{C}_{B/A}) = v_{\mathfrak{q}}(\mathfrak{c}) + v_{\mathfrak{q}}(\mathcal{D}_{B/A}) \leftrightarrow e_{\mathfrak{q}} - 1 \leq e_{\mathfrak{q}} - v_{\mathfrak{q}}(\mathfrak{c}) = v_{\mathfrak{q}}(\mathcal{D}_{B/A})$, given $0 \leq v_{\mathfrak{q}}(\mathfrak{c}) \leq 1$. So the first inequality follows. Also $\min v_{\mathfrak{q}}(\mathfrak{c}) = 1 - v_{\mathfrak{q}}(e_{\mathfrak{q}})$ so the second inequality and the last statement follows.

Proposition 2.2.16. *Assume AKLB and let M/L be a finite separable extension and let C be the integral closure of A in M . Then $\mathcal{D}_{C/A} = \mathcal{D}_{C/B} \cdot \mathcal{D}_{B/A}$ and $D_{C/A} = (D_{B/A})^{[M:L]} N_{B/A}(D_{C/B})$.*

Chapter 3

Global Fields

3.1 Global Fields and Primes

3.1.1 Global Fields

Definition 3.1.1. *Let K be a field. A place of K is an equivalence class of nontrivial absolute values on K . There is an one-to-one correspondence between places of K and completions of K , given the completion of K at an absolute value depends only on its equivalence class. M_K denotes the set of places of K , and for each place v , we use $|\cdot|_v$ to denote any representative absolute value and K_v to denote the completion of K w.r.t. $|\cdot|_v$.*

Let K be a global field. Then for any place v of K , the completion K_v is a local field.

- If $K_v \cong \mathbb{R}$, v is a real place
- If $K_v \cong \mathbb{C}$, v is a complex place
- If $|\cdot|_v$ is induced by a DVR v_p corresponding to a prime $\mathfrak{p}$ of K , then v is a finite place; otherwise v is an infinite place

e.g.

- $M_{\mathbb{Q}}$ consists of finite places p corresponding to p -adic absolute values $|\cdot|_p$ and a single archimedean infinite place ∞ corresponding to the Euclidean absolute value $|\cdot|_{\infty}$.
- $M_{\mathbb{F}_q(t)}$ consists of finite places corresponding to irreducible polynomials in $\mathbb{F}_q[t]$ and a single nonarchimedean infinite place ∞ corresponding to the absolute value $|\cdot|_{\infty} \equiv q^{\deg(\cdot)}$

Definition 3.1.2. If L/K is an extension of global fields, for every place w of L , any absolute value $|\cdot|_w$ that represents the equivalence class w restricts to an absolute value on K that represents a place v of K that is independent of the choice of $|\cdot|_w$. This is said that w extends v or w lies above v , denoted $w|v$.

Theorem 3.1.1. Let L/K be a finite separable extension of global fields and let v be a place of K . We have an isomorphism of finite etale K_v -algebras

$$\begin{aligned} L \otimes_K K_v &\xrightarrow{\sim} \prod_{w|v} L_w \\ l \otimes x &\mapsto (lx, \dots, lx) \end{aligned}$$

pf. The separable extension L/K is a finite etale K -algebra, so the base change $L \otimes_K K_v$ is a finite etale K_v -algebra, and is therefore isomorphic to a finite product $\prod_{i \in I} L_i$ of finite separable extensions L_i of K_v , each of which is a local field, given the fact that any finite extension of a local field is a local field. Fix an absolute value $|\cdot|_v$ on K_v representing the place v . Each L_i is a local field extending K_v , and therefore has a unique absolute value $|\cdot|_w$ that restricts to $|\cdot|_v$, by the theorem for v nonarchimedean, and by the fact that $K_v \cong L_w$ or $K_v \cong \mathbb{R} \subseteq \mathbb{C} \cong L_w$ and the Euclidean absolute value on $\mathbb{R}$ is the restriction of the Euclidean absolute value on $\mathbb{C}$, for v archimedean. The map $L \rightarrow L \otimes_K K_v \cong \prod_i L_i \rightarrow L_i$ allows to view L as a subfield of each L_i , so the absolute value $|\cdot|_w$ on L_i restricts to an absolute value on L that uniquely determines a place $w|v$. This defines a map $\phi : \{L_i : i \in I\} \rightarrow \{L_w : w|v\}$.

CLAIM : ϕ above is a bijection, satisfying $\phi(L_i) \cong L_i$.

pf. $L \otimes_K K_v \cong \prod_i L_i$ can be viewed as an isomorphism of topological rings, since both sides are finite dimensional vector spaces over the complete field K_v . Then there exists a unique topology induced by the sup norm and this topology agrees with the product topology on $\prod_i L_i$. The image of the canonical embedding

$$\begin{array}{ccc} L & \rightarrow & L \otimes_K K_v \\ l & \mapsto & l \otimes 1 \end{array}$$

is dense because $K \subseteq L$ is dense in K_v . The image of L is therefore dense in $\prod_i L_i$ and has dense image under the projections $\prod_i L_i \rightarrow L_i$ and $\prod_i L_i \rightarrow L_i \times L_j (i \neq j)$. If $\phi(L_i) = L_w, L_i \cong L_w$, since L is dense in the completion field L_i and L_w is the completion of L w.r.t. the restriction of the absolute value on L_i to L , by the universal property of the completions.

If $\phi(L_i) = \phi(L_j) = L_w, i \neq j$, because the image of the diagonal embedding $L \rightarrow L_w \times L_w$ is not dense in $L_w \times L_w$ but the image of L is dense in $L_i \times L_j$ we get the contradiction. So injectivity follows.

For each $w|v$, define a continuous homomorphism of finite etale K_v -algebras and topological rings

$$\begin{array}{ccc} \phi_w : L \otimes_K K_v & \rightarrow & L_w \\ l \otimes x & \mapsto & lx \end{array}$$

The map ϕ_w is surjective because its image contains L and is complete, and L_w is the completion of L . Thus ϕ_w factors through the projection of $L \otimes_K K_v \cong \prod_i L_i$ onto one of its factors L_i , and induces a homeomorphism from L_i to L_w . So $\phi(L_i) = L_w$ and the map is surjective.

Corollary 3.1.1. *Let L/K be a finite separable extension of global fields, v be a place of K and $f \in K[x]$ be a monic irreducible polynomial s.t. $L \cong K[x]/(f)$. There is an one-to-one correspondence between the irreducible factors of f in $K_v[x]$ and the places of L lying above v . If $f = f_1 \cdots f_r$ is the factorisation of f in $K_v[x]$, then we can order the set $\{w|v\} = \{w_1, \dots, w_r\}$ so that $L_{w_i} \cong K_v[x]/(f_i)$ for $i \in [1, r]$.*

pf. f_i are distinct since f is separable over K and therefore separable over every extension of K . Then corollary follows.

Definition 3.1.3. *Let X be a locally compact Hausdorff space. The σ -algebra Σ of X is the collection of subsets of X generated by the open and closed sets under countable unions and countable intersections. Its elements are Borel sets or measurable sets. A Borel measure on X is a countably additive function $\mu : \Sigma \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$. A Radon measure on X is a Borel measure on X that additionally satisfies*

1. $\mu(X) < \infty$ if S is compact
2. $\mu(X) = \inf\{\mu(U) : S \subseteq U, U \text{ is open}\}$
3. $\mu(S) = \sup\{\mu(C) : C \subseteq S, C \text{ is compact}\}$

for all Borel sets S .

Definition 3.1.4. *A locally compact group is a topological group that is Hausdorff and locally compact. A Haar measure μ on a locally compact group G is a nonzero Radon measure that is translation invariant, meaning $\mu(S) = \mu(x + S)$, $\forall x \in G$ and measurable $S \subseteq G$. A compact group is a locally compact group that is compact.*

Theorem 3.1.2 (Weil). *Every locally compact group G has a Haar measure. If μ, μ' are two Haar measure on G , then there is a positive real number λ s.t. $\mu'(S) = \lambda\mu(S)$ for all measurable sets S .*

e.g. The standard Lebesgue measure on $\mathbb{R}^n$ with $\mu(\prod_i |a_i, b_i|) = \prod_i |b_i - a_i|$ is the unique Haar measure on $\mathbb{R}^n$ for which the unit cube has measure 1.

The additive group of a local field K is a locally compact group since it is a metric space and hence Hausdorff. For compact group G , normalise the Haar measure so that $\mu(G) = 1$, but local fields are never compact and therefore $\mu(K) = \infty$. For a nonarchimedean local fields, the valuation ring $A = B_{\leq 1}(0)$ is a compact group so it is natural to normalise the Haar measure on K so that $\mu(A) = 1$.

Proposition 3.1.1. *Let K is a local field with discrete valuation v , residue field k and absolute value $|\cdot|_v \equiv (\#k)^{-v(\cdot)}$, and let μ be a Haar measure on K . $\forall x \in K$ and measurable $S \subseteq K$, we have $\mu(xS) = |x|_v \mu(S)$. Moreover the absolute value $|\cdot|_v$ is the unique absolute value compatible with the topology on K for which this is true.*

pf. Let A be a valuation ring of K with maximal $\mathfrak{p}$. When $x = 0$, the proposition holds trivially. Assume $x \neq 0$, and the map $\phi_x : y \mapsto xy$ is an automorphism of the additive group of K and it follows that the composition $\mu_x = \mu \circ \phi_x$ is a Haar measure on K and hence a multiple of μ i.e. $\mu_x = \lambda_x \mu$ for some $\lambda_x \in \mathbb{R}_{>0}$. Define a function $\chi : K^\times \rightarrow \mathbb{R}_{>0}$ by $\chi(x) = \mu_x(A)/\mu(A) = \lambda_x$. Then $\mu_x = \chi(x)\mu$ and for all $x, y \in K^\times$, $\chi(xy) = \frac{\mu_{xy}(A)}{\mu(A)} = \frac{\mu_x(yA)}{\mu(A)} = \frac{\chi(x)\chi(y)\mu(A)}{\mu(A)} = \chi(x)\chi(y)$ which implies χ is multiplicative. **CLAIM :** $\chi(x) = |x|_v$ for all $x \in K^\times$

pf. Since both are multiplicative, it suffices to consider $x \in A \setminus \{0\}$. For any such x , the ideal xA is equal to $\mathfrak{p}^{v(x)}$, since A is a DVR. The residue field $k \equiv A/\mathfrak{p}$ is finite, hence A/xA is also finite. Writing A as a finite disjoint union of cosets of xA , we have $\mu(A) = [A : xA]\mu(xA) = (\#k)^{v(x)}\chi(x)\mu(A)$, which implies $\chi(x) = (\#k)^{-v(x)} = |x|_v$. Thus $\mu(xS) = \mu_x(S) = \chi(x)\mu(S) = |x|_v\mu(S)$, $\forall x \in K, S \subseteq K$, measurable.

If $|\cdot|$ is an absolute value on K that induces the same topology as $|\cdot|_v$, then for some $0 < c < 1$, $|x| = |x|_v^c$ for all $x \in K^\times$. Fix $x \in K^\times$ with $|x|_v \neq 1$. If $|\cdot|$ satisfies $\mu(xS) = |x|\mu(S)$, $\frac{\mu(xA)}{\mu(A)} = |x| = |x|_v^c = (\frac{\mu(xA)}{\mu(A)})^c$ which implies $c = 1$, $|\cdot| = |\cdot|_v$. Hence it is unique.

Definition 3.1.5. Let K be a global field. For each place v of K , the normalised absolute value $\|\cdot\|_v : K_v \rightarrow \mathbb{R}_{\geq 0}$ on the completion of K at v is defined by $\|x\|_v \equiv \frac{\mu(xS)}{\mu(S)}$, where μ is a Haar measure on K_v and $S \subseteq K_v$ is a measurable set with finite nonzero measure.

Then by the previous theorem, it does not depend on the choice of μ and S .

- If v is nonarchimedean, $\|\cdot\|_v$ is the absolute value $|\cdot|_v$ defined by the previous proposition
- If v is a real place, $\|\cdot\|_v$ is the Euclidean absolute value $|\cdot|_{\mathbb{R}}$
- If v is a complex place, $\|\cdot\|_v$ is the square of the Euclidean absolute value $|\cdot|_{\mathbb{C}}$

Lemma 3.1.1. Let L/K be a finite separable extension of global fields, let v be a place of K and let $w|v$ be a place of L . Then $\|x\|_w = \|N_{L_w/K_v}(x)\|_v$.

pf. If $[L_w : K_v] = 1$, the lemma trivially holds.

Otherwise, if v is archimedean, $L_w \cong \mathbb{C}$, $K_v \cong \mathbb{R}$, in which case $\forall x \in L_w$, $\|x\|_w = \mu(xS)/\mu(S) = |x|_{\mathbb{C}}^2 = |x\bar{x}|_{\mathbb{R}} = |N_{\mathbb{C}/\mathbb{R}}(x)|_{\mathbb{R}} = \|N_{L_w/K_v}(x)\|_v$. If v is nonarchimedean, let π_v, π_w be the uniformisers for the local fields K_v, L_w respectively, and let f be the degree of the corresponding residue field extension k_w/k_v . Wlog, assume $x = \pi_w^{w(x)}$ since $\|x\|_v = |x|_v$ depends only on $w(x)$. Then $\|N_{L_w/K_v}(\pi_w)\|_v = \|\pi_v^f\|_v = (\#k)^{-f}$, so $\|N_{L_w/K_v}(x)\|_v = (\#k)^{-fw(x)}$. By the previous proposition $\|x\|_w = (\#k)^{-w(x)} = (\#k_v)^{-fw(x)} = \|N_{L_w/K_v}(x)\|_v$.

Theorem 3.1.3 (Product Formula). *Let L be a global field. $\forall x \in L^\times$, we have $\prod_{v \in M_L} \|x\|_v = 1$ where $\|\cdot\|_v$ denotes the normalised absolute value for each place $v \in M_L$.*

pf. The global field L is a finite separable extension of $K = \mathbb{Q}$ or $K = \mathbb{F}_q(t)$. Let p be a place of K . Then by the theorem, any basis for L as a K -vector space is also a basis for $L \otimes_K K_p \cong \prod_{v|p} K_v$ as a K_v -vector space. Thus $N_{L/K}(x) = N_{(L \otimes_K K_p)/K_p}(x) = \prod_{v|p} N_{L_v/K_p}(x)$. Taking normalised absolute values on both sides yields $\|N_{L/K}(x)\|_p = \prod_{v|p} \|N_{L_v/K_p}(x)\|_p = \prod_{v|p} \|x\|_v$. By taking the product of both sides over all places $p \in M_K$, $1 = \prod_{p \in M_K} \|N_{L/K}(x)\|_p = \prod_{p \in M_K} \prod_{v|p} \|x\|_v = \prod_{v \in M_L} \|x\|_v$.

Definition 3.1.6. *A global field is a field K with at least one place whose completion at each of its place $v \in M_K$ is a local field K_v and which has a product formula of the form*

$$\prod_{v \in M_K} \|x\|_v = 1$$

where each normalised absolute value $\|\cdot\|_v : K_v \rightarrow \mathbb{R}_{\geq 0}$ satisfies $\|\cdot\|_v = |\cdot|_v^{m_v}$ for some absolute value $|\cdot|_v$ representing v and some fixed $m_v \in \mathbb{R}_{>0}$.

By the above result, we can define a global field K if it has a product formula and the completion K_v of K at each place $v \in M_K$ is a local field.

Definition 3.1.7. *Define an M_K -divisor to be a sequence of positive real numbers $c = (c_v)$ indexed by $v \in M_K$ with all but finitely many $c_v = 1$ s.t. $\forall v \in M_K, \exists x \in K_v^\times$ for which $c_v = \|x\|_v$.*

Theorem 3.1.4 (Artin-Whaples). *Every global field is a finite extension of $\mathbb{Q}$ or $\mathbb{F}_q(t)$.*

pf. **CLAIM : Let E/F be a finite extension. Then E is a global field iff F is.**

pf. Given E/F is Galois, if F is a global field, its finite Galois extension has absolute value extending the absolute value induced by the place $v \in M_F \rightarrow \prod_{w \in M_E} \|x\|_w = (\prod_{v \in M_F} \|x\|_v)^{m_v} = 1$ so the product formula holds. Also E_w is a finite etale F_v -algebra and therefore it is also a local field. Thus E is a global field. Conversely, if E is a global field, for any place w has $v \in M_F$ and therefore $1 = \prod_w \|x\|_w = (\prod_v \|x\|_v)^{m_v}$ so the product formula holds. Also for any local field L_w it is reduced to K_v which is also a local field since its closed ball is compact. Thus F is also a global field.

If E/F is not Galois, pick an intermediate Galois field extension $E/\bar{F}/F$. Then by the previous statement, we can reduce to the case $E/\bar{F}$, which is totally ramified. By the similar argument, we get the equality.

CLAIM : M_K is infinite but contains only finitely many archimedean places.

pf. Assume the contrary. Then there are finite discrete value induced places and finite archimedean places. Since the product formula holds, $1 = \prod_v \|x\|_v = \prod_{v|\infty} \|x\|_v \prod_{v|\infty} \|x\|_v \leftrightarrow (\prod_{v|\infty} \|x\|_v)^{-1} = (\prod_{v|\infty} \|x\|_v)^{-1} = \prod_{v|\infty} \|x\|_v$. Then for $x \in K$ s.t. every discrete valuation induced absolute value $v(x) = 1$. Then $\prod_{v|\infty} \|x\|_v = \prod_{v|\infty} \|x\|_v$. Here LHS is nonarchimedean but RHS is archimedean, so contradiction. Hence M_K is infinite.

Assume the contrary. Set c s.t. $\forall v|\infty, c_v = 1 \rightarrow 1 = \prod_{v|\infty} \|c\|_v \prod_{v|\infty} \|c\|_v = \prod_{v|\infty} \|c\|_v = I_c$ a contradiction since 1 is a unit. Hence the archimedean places are finitely many.

CLAIM : $L(c) = \{x \in K : \|x\|_v \leq c_v, \forall v \in M_K\}$ is finite for every M_K -divisor c .

pf. for v , archimedean, $L(c) = \|x\|_v \leq c_v$ for $c_v = 1$ for all nonarchimedean places. $\{x \in K_v : \|x\|_v \leq c_v\}$ is a compact set and given K dense in K_v , $K \cap \{x \in K_v : \|x\|_v \leq c_v\}$ is discrete and compact and therefore finite.

It there is no archimedean place, pick an arbitrary discrete valuation induced place v and for other places, set $c_{v'} = 1$. Then $L(c) = \{x \in K : \|x\|_v \leq c_v\}$ which is dense in K_v . $L(c)$ is a compact ball since K_v is a local field and $L(c)$ is again an intersection of discrete set and compact set and therefore finite.

CLAIM : If M_K contains an Archimedean place, then K is a finite extension of $\mathbb{R}$.

pf. For any $a/b \in \mathbb{Q}$, $a, b \in \mathbb{Z}$ s.t. $(a, b) = 1$. Then $\prod_v \|a/b\|_v = \prod_v \|a\|_v \prod_v \|b\|_v^{-1} = 1$ hence $a/b \in K$. Pick c s.t. other than place induced by $\mathbb{Q}$, $c_v = 1$. Then $L(c) = \{x \in K : \|x\|_v \leq c_v\} \leq \infty$ for all $c_v \in \mathbb{R}$. Thus K is a finite extension of $\mathbb{Q}$.

CLAIM : If M_K does not have an archimedean place, then K is a finite extension of $\mathbb{F}_q(t)$ for some finite field $\mathbb{F}_q$.

pf. Pick c s.t. for all places v , $c_v = 1$. Then $L(c) < \infty$. For $x \in L(c)$, $\|x^2\|_v = \#k_v^{-2v(x)} = (\#k_v^{-v(x)})^2 \leq \#k_v^{-v(x)} \leq c_v$ hence $x^2 \in L(c)$. But it is finite and therefore $\exists n \in \mathbb{Z}_{++}$ s.t. $x^n = 1$ and therefore $L(c)$ is a field. Let $k = L(c)$. $\forall t \in K \setminus k$, assume it is not transcendental. Then $\exists f(x) \in k[x]$ s.t. $f(t) = 0 = t^n + a_{n-1}t^{n-1} + \dots + a_0 \leftrightarrow a_0 = t(t^{n-1} + \dots + a_1) \in k$ a contradiction. Hence $\forall t \in K \setminus k$, it is transcendental. Also $k(t) \subseteq K$, and K is an extension of $k(t)$. Pick c s.t. for v induced by $(k(t))$, $c_v \neq 1$ and 1 everywhere else. Then $L(c) = \{x \in K : \|x\|_v \leq c_v\} < \infty$. For any c_v it is finite and therefore K is a finite extension.

3.1.2 Number Fields

Definition 3.1.8. A subgroup H of a topological group G is discrete if the subspace topology on H is the discrete topology and cocompact if H is a normal subgroup of G and the quotient G/H is compact.

Lemma 3.1.2. A subgroup $G \subseteq V \cong \mathbb{R}^n$ is finitely generated iff it is discrete, in which case $G \cong \mathbb{Z}^m$ for some $m \leq n$, and G is then cocompact iff $m = n$.

pf. If G is finitely generated, it is a free $\mathbb{Z}$ -module, since it lies in the torsion free abelian group V and the rank of G must be equal to the dimension $m \leq n$ of the subspace it spans. A subgroup $G \cong \mathbb{Z}^m$ of $V \cong \mathbb{R}^n$ is discrete.

Let $v_1, \dots, v_n$ be a basis for the subspace W of V spanned by G and let $\Lambda \cong \mathbb{Z}^m$ be the subgroup of G generated by this basis. The quotient $W/\Lambda \cong \mathbb{R}^m/\mathbb{Z}^m$ is compact, so the image of the discrete group G under the quotient map $\pi : W \rightarrow W/\Lambda$ must be finite. Thus we can choose a finite $S \subseteq G$ s.t. $\pi(S) = \pi(G)$ so G is generated by $v_1, \dots, v_m$ and S .

Definition 3.1.9. A lattice in $V \cong \mathbb{R}^n$ is a $\mathbb{Z}$ -submodule generated by an $\mathbb{R}$ -basis, equivalently a discrete cocompact subgroup.

A discrete subgroup of a Hausdorff topological group is always closed and therefore the quotient of a Hausdorff topological group by a normal discrete subgroup is Hausdorff. — The quotient of a Hausdorff topological group by a discrete cocompact subgroup is a compact group.

Any basis $v_1, \dots, v_n$ for V determines a parallelepiped $F(v_1, \dots, v_n) \equiv \{\sum_i t_i v_i : \forall t_i \in [0, 1)\}$ which can be viewed as the unit cube by fixing an isomorphism $\phi : V \rightarrow \mathbb{R}^n$ that maps $(v_1, \dots, v_n)$ to the standard basis of unit vectors of $\mathbb{R}^n$. Normalise the Haar measure μ so that $\mu(F(v_1, \dots, v_n)) = 1$ so that we have $\mu(S) = \mu_{\mathbb{R}^n}(\phi(S))$ for every measurable set $S \subseteq V$, where $\mu_{\mathbb{R}^n}$ is the standard Lebesgue measure on $\mathbb{R}^n$. For any other basis $e_1, \dots, e_n$ of V , if we let $E = [e_{ij}]$ be the matrix whose j th column expresses $e_j = \sum_i e_{ij} v_i$ in terms of the normalised basis $v_1, \dots, v_n$, then $\mu(F(e_1, \dots, e_n)) = |\det E| = \sqrt{\det E' \det E} = \sqrt{\det(E'E)} = \sqrt{\det(\langle e_i, e_j \rangle)}$ where $\langle e_i, e_j \rangle$ is the canonical inner product on $\mathbb{R}^n$.

Proposition 3.1.2. Let $T : V \rightarrow V$ be a linear transformation of $V \cong \mathbb{R}^n$. For any Haar measure μ on V , and every measurable set $S \subseteq V$, $\mu(T(S)) = |\det T| \mu(S)$.

Definition 3.1.10. Let Λ be a lattice in $V \cong \mathbb{R}^n$. A fundamental domain for Λ is a measurable set $F \subseteq V$ s.t. $V = \coprod_{\lambda \in \Lambda} (F + \lambda)$, i.e. F is a measurable set of coset representatives for V/Λ .

Proposition 3.1.3. Let Λ be a lattice in $V \cong \mathbb{R}^n$ and let μ be a Haar measure on V . Every fundamental domain for Λ has the same measure, and this measure is finite and nonzero.

pf. Let F, G be two fundamental domains for Λ . By the translation invariance and countable additivity of μ , $\mu(F) = \mu(F \cap V) = \mu(F \cap \coprod_{\lambda \in \Lambda} (G + \lambda)) = \mu(\coprod_{\lambda \in \Lambda} (F \cap (G + \lambda))) = \sum_{\lambda \in \Lambda} \mu(F \cap (G + \lambda)) = \sum_{\lambda \in \Lambda} \mu((F - \lambda) \cap G) = \sum \mu(G \cap (F + \lambda)) = \mu(G)$. If we fix a $\mathbb{Z}$ -basis $e_1, \dots, e_n$ for Λ , the parallelepiped $F(e_1, \dots, e_n)$ is a fundamental domain for Λ and its closure is compact, so $\mu(F(e_1, \dots, e_n))$ is finite and it is nonzero since

there is an isomorphism $V \cong \mathbb{R}^n$ that maps the closure of $F(e_1, \dots, e_n)$ to the unit cube in $\mathbb{R}^n$ whose Lebesgue measure is nonzero.

Definition 3.1.11. Let Λ be a lattice in $V \cong \mathbb{R}^n$ and fix a Haar measure μ on V . Then covolume $\text{covol}(\Lambda) \in \mathbb{R}_{>0}$ of Λ is the measure $\mu(F)$ of any fundamental domain F for Λ .

Proposition 3.1.4. If $\Lambda' \subseteq \Lambda$ are lattices in $V \cong \mathbb{R}^n$, then $\text{covol}(\Lambda') = [\Lambda : \Lambda'] \text{covol}(\Lambda)$.

pf. Fix a fundamental domain F for Λ and a set of coset representatives S for Λ/Λ' . Then $F' \equiv \coprod_{\lambda \in S} (F + \lambda)$ is a fundamental domain for Λ' and $\#S = [\Lambda : \Lambda'] = \mu(F')/\mu(F)$ is finite. Then $\text{covol}(\Lambda') = \mu(F') = \sum_{\lambda \in S} \mu(F + \lambda) = (\#S)\mu(F) = [\Lambda : \Lambda'] \text{covol}(\Lambda)$ since every translation $F + \lambda$ is a fundamental domain for Λ .

Definition 3.1.12. Let S be a subset of a real vector space. The set S is symmetric if it is closed under negation, and convex if $\forall x, y \in S, \{tx + (1-t)y : t \in [0, 1]\} \subseteq S$.

Theorem 3.1.5 (Minkowski's Lattice Point Theorem). Let Λ be a lattice in $V \cong \mathbb{R}^n$ and μ a Haar measure on V . If $S \subseteq V$ is a symmetric convex measurable set that satisfies $\mu(S) > 2^n \text{covol}(\Lambda)$, then S contains a nonzero element of Λ .

pf. **CLAIM : For any measurable $S \subseteq \mathbb{R}^n$ with measure $\mu(S) > 1$, $\exists$ distinct $s, t \in S$ s.t. $s - t \in \mathbb{Z}^n$.**

pf. $\mu(S) = \prod_{i=1}^n |a_i - b_i| > 1 \leftrightarrow \exists i \in [1, n]$ s.t. $|a_i - b_i| > 1$. Then we draw a line from a_i to b_i with the coordinate $j \neq i$ fixed. Then on the line $\exists b'_i$ s.t. $|a_i - b'_i| = 1$. Then we set those two points a_i, b'_i as s, t and the claim follows.

Since S is symmetric, we set $\mu(S) > 2$ and apply the above claim $\forall i \in [1, n]$ to get $a_i, b'_i, \forall i$ and the theorem follows.

→ the theorem is tight in the sense that convexity (consider two circles around zero with $r_1 = 1.9, r_2 = 1$. Then between them area is symmetric and $\mu(S) > 2^2$, but not convex and it does not contain the point in $\mathbb{Z}^2$ and symmetry and $\mu(S) > 2^n$ (consider two circle with the same radius r ; $(x-r)^2 + y^2 = r^2, (x+r)^2 + y^2 = r^2$ s.t. $\exists \theta r^2 \theta / 2\pi - 1/2r(r-1)\cos\theta = 1$. Set $S' = \text{two circles} \cap ((-1, 1) \times (-\infty, \infty))$ and set $S = \bar{S}' \setminus \partial(S')$. Then a symmetric and convex with $\mu(S) = 4$, S does not contain a point in $\mathbb{Z}^n$) are all necessary.

Let $K/\mathbb{Q}$ be a number field of degree n with r real places and s complex places. Then $n = r + 2s$ is the total degree. By the base changes of K , $K_{\mathbb{R}} \equiv K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^r \times \mathbb{C}^s$.
 $K_{\mathbb{C}} \equiv K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C}^n$.

where the second isomorphism follows from the fact that $\mathbb{C}$ is separably closed. Note $K_{\mathbb{R}}$ is an $\mathbb{R}$ -vector space of dimension n , thus $K_{\mathbb{R}} \cong \mathbb{R}^n$. Then we have a sequence of injective homomorphisms of topological rings

$$\mathcal{O}_K \rightarrow K \rightarrow K_{\mathbb{R}} \rightarrow K_{\mathbb{C}} \quad (3.1)$$

where $\mathcal{O}_K \rightarrow K$ inclusion
 $K \rightarrow K_{\mathbb{R}} = K \otimes_{\mathbb{Q}} \mathbb{R}$ $\alpha \mapsto \alpha \otimes 1$. Each $\mathbb{C}$ in $\mathbb{C}^s$ arises as $\mathbb{R}[\alpha] = K_{\mathbb{R}} \cong \mathbb{R}^r \times \mathbb{C}^s \rightarrow \mathbb{C}^r \times \mathbb{C}^{2s} \cong K_{\mathbb{C}}$ $(x, z) \mapsto (x, z, \bar{z})$
 $\mathbb{R}[x].(f) \cong \mathbb{C}$ for some monic irreducible polynomial $f \in \mathbb{R}[x]$ with degree 2, but when we base-change to $\mathbb{C}$, the field $\mathbb{R}[\alpha]$ splits into the etale algebra $\mathbb{C}[x]/(x - \alpha) \times \mathbb{C}[x]/(x - \bar{\alpha}) \cong \mathbb{C} \times \mathbb{C}$. So the composition $K \rightarrow K_{\mathbb{R}} \rightarrow K_{\mathbb{C}}$ is given by the map $x \mapsto (\sigma_1(x), \dots, \sigma_n(x))$ where $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C}) = \{\sigma_1, \dots, \sigma_n\}$. If we put $K = \mathbb{Q}(\alpha) \equiv K[x]/(f)$ and let $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ be the roots of f in $\mathbb{C}$, each σ_i is the $\mathbb{Q}$ -algebra homomorphism $K \rightarrow \mathbb{C}$ $\alpha \mapsto \alpha_i$. The ring of integers $\mathcal{O}_K$ is a lattice in the real vector space $K_{\mathbb{R}} \cong \mathbb{R}^n$, which inherits an inner product from the canonical Hermitian inner product on $K_{\mathbb{C}} \cong \mathbb{C}^n$ defined by $\langle z, z' \rangle \equiv \sum z_i \bar{z}'_i \in \mathbb{C}$. For elements $x, y \in K \rightarrow K_{\mathbb{R}} \rightarrow K_{\mathbb{C}}$, the Hermitian inner product can be computed as $\langle x, y \rangle \equiv \sum_{\sigma \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})} \sigma(x) \bar{\sigma}(y) \in \mathbb{R}$, which is a real number because the non-real embeddings in $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ come in complex conjugate pairs. Also the inner product defined above agrees with the restriction of the Hermitian inner product on $K_{\mathbb{R}} \rightarrow K_{\mathbb{C}}$.

If $z \in K_{\mathbb{C}} \cong \mathbb{C}^n$ is written as vectors (z_{σ}) indexed by the set $\sigma \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$, $K_{\mathbb{R}}$ as the image in $K_{\mathbb{C}}$ is the set $K_{\mathbb{R}} = \{z \in K_{\mathbb{C}} : \bar{z}_{\sigma} = z_{\bar{\sigma}}, \forall \sigma \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})\}$. Then each $x \in K_{\mathbb{R}}$ is uniquely written in the form $(w_1, \dots, w_r, x_1 + iy_1, x_1 - iy_1, \dots, x_s + iy_s, x_s - iy_s)$, $w_i, x_j, y_j \in \mathbb{R}$, where each w_i corresponds to a z_{σ} with $\sigma = \bar{\sigma}$ and each $(x_j - iy_j, x_j + iy_j)$ corresponds to a complex conjugate pair $(z_{\sigma}, z_{\bar{\sigma}})$ with $\sigma \neq \bar{\sigma}$. Then the canonical inner product on $K_{\mathbb{R}}$ is $\langle z, z' \rangle = \sum_i w_i w'_i + 2 \sum_j (x_j x'_j + y_j y'_j)$. Thus if we take w_i, x_j, y_j as coordinates for $K_{\mathbb{R}} \cong \mathbb{R}^n$, in order to normalise the Haar measure μ on $K_{\mathbb{R}}$, so that it is consistent with the Lebesgue measure $\mu_{\mathbb{R}^n}$, define $\mu(S) \equiv 2^s \mu_{\mathbb{R}^n}(S)$ for any measurable set $S \subseteq K_{\mathbb{R}}$. Given this normalisation, $\mu(F(e_1, \dots, e_n)) = \sqrt{|\det \langle \sigma_i, \sigma_j \rangle|}$ with e_i for any $\mathbb{R}$ -basis of $K_{\mathbb{R}}$. Using the Hermitian inner product on $K_{\mathbb{R}} \subseteq K_{\mathbb{C}}$, the above equality still holds.

Recall the discriminant of a number field K is $D_K \equiv \text{disc} \mathcal{O}_K \equiv \text{disc}(e_1, \dots, e_n) \in \mathbb{Z}$

Proposition 3.1.5. *Let K be a number field. Using the normalised Haar measure on $K_{\mathbb{R}}$ defined previously,*

$$\text{covol}(\mathcal{O}_K) = \sqrt{|D_K|}$$

pf. Let $e_1, \dots, e_n \in \mathcal{O}_K$ be a $\mathbb{Z}$ -basis for $\mathcal{O}_K$, let $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C}) = \{\sigma_1, \dots, \sigma_n\}$ and define $A \equiv [\sigma_i(e_j)] \in \mathbb{C}^{n \times n}$. Then $D_K = \text{disc}(e_1, \dots, e_n) = (\det A)^2$. Viewing $\mathcal{O}_K \rightarrow K_{\mathbb{R}}$ as a lattice in $K_{\mathbb{R}}$ with basis e_i , $\text{covol}(\mathcal{O}_K) = \mu(F(e_1, \dots, e_n)) = \sqrt{|\det \langle e_i, e_j \rangle|} = \sqrt{|\det [\sum_j \sigma_k(e_i) \bar{\sigma}_k(e_j)]|} = \sqrt{|\det(A' \bar{A})|} = \sqrt{|\det(A)| |\det(\bar{A})|}$. Since $\det A$ is the square root of an integer, $\text{covol}(\mathcal{O}_K)^2 = |(\det A)^2| = |D_K|$.

For a number field K , we view the absolute norm $N : \mathcal{I}_{\mathcal{O}} \rightarrow \mathcal{I}_{\mathbb{Z}}$ as $I \mapsto [\mathcal{O}_K : I]_{\mathbb{Z}}$ having image in $\mathbb{Q}_{++}$ by identifying $N(I) \in \mathcal{I}_{\mathbb{Z}}$ with a positive generator for $N(I)$, given $\mathbb{Z}$ is a PID. Also $[\mathcal{O}_K : I]_{\mathbb{Z}}$ is a module index of $\mathbb{Z}$ -lattices in the $\mathbb{Q}$ -vector space K , and for ideal $I \subseteq \mathcal{O}_K$, it is the positive integer $[\mathcal{O}_K : I]_{\mathbb{Z}} = [\mathcal{O}_K : I]$. When $I = (a)$, $N(x) \equiv N((a)) = |N_{K/\mathbb{Q}}(a)|$.

Corollary 3.1.2. *Let K be a number field and let I be a nonzero fractional ideal of $\mathcal{O}_K$. Then $\text{covol}(I) = N(I)\sqrt{|D_K|}$.*

pf. Let $n = [K : \mathbb{Q}]$. Since $\text{covol}(bI) = b^n \text{covol}(I)$ and $N(bI) = b^n N(I)$ for any $b \in \mathbb{Z}_{++}$, wlog we may assume $I \subseteq \mathcal{O}_K$. Then $\text{covol}(I) = [\mathcal{O}_K : I] \text{covol}(\mathcal{O}_K) = N(I) \text{covol}(\mathcal{O}_K) = N(I)\sqrt{|D_K|}$.

Lemma 3.1.3. *Let K be a number field of degree n with r real and s complex places. For each $t \in \mathbb{R}_{++}$, the measure of the convex symmetric set $S_t \equiv \{(z_\sigma) \in K_{\mathbb{R}} : \sum |z_\sigma| \leq t\} \subseteq K_{\mathbb{R}}$ w.r.t. the normalised Haar measure μ on $K_{\mathbb{R}}$ is $\mu(S_t) = 2^r \pi^s \frac{t^n}{n!}$.*

pf. Write each $z = (z_\sigma) \in K_{\mathbb{R}}$ in the form $(w_1, \dots, w_r, x_1 + iy_1, \dots, x_s - iy_s)$ with $w_i, x_j, y_j \in \mathbb{R}$. Then $\sum_\sigma |z_\sigma| \leq t$ iff $\sum_i |w_i| + \sum_j 2\sqrt{|x_j|^2 + |y_j|^2} \leq t$. Compute the volume of this region in $\mathbb{R}^n$ by relating it to the volume of the simplex $U_t \equiv \{(u_1, \dots, u_n) \in \mathbb{R}_+^n : \sum u_i \leq t\} \subseteq \mathbb{R}^n$, which is $\mu_{\mathbb{R}^n} t^n / n!$. If we view all w_i, x_j, y_j as fixed except the last pair (x_s, y_s) , the (x_s, y_s) ranges over a disk of some radius $d \in [0, t/2]$ determined by the previous inequality. If we replace (x_s, y_s) with (u_{n-1}, u_n) ranging over the triangular region bounded by $u_{n-1} + u_n \leq 2d, u_{n-1}, u_n \in \mathbb{R}_+$, we need to incorporate a factor of $\pi/2$ to account for the difference between $(2d)^2/2 = 2d^2$ and πd^2 . Repeat this s times. Similarly, if we hold everything but w_r fixed, w_r ranges over $[-d, d]$ and we can replace it with $\exists d \in [0, t]$ with u_r ranging over $[0, d]$ by incorporating a factor of 2 to account for the change of variables. Repeat this r times. Then $\mu(S_t) = 2^s \mu_{\mathbb{R}^n}(S_t) = 2^s (\frac{\pi}{2})^s 2^r \mu_{\mathbb{R}^n}(U) = 2^r \pi^s \frac{t^n}{n!}$.

Theorem 3.1.6 (Minkowski Bound). *Let K be a number field of degree n with s complex places. Define the Minkowski constant m_K for K as the positive real number $m_K \equiv \frac{n!}{n^n} (\frac{4}{\pi})^s \sqrt{|D_K|}$. For every nonzero fractional ideal I of $\mathcal{O}_K$, there is a nonzero $a \in I$ for which $N(a) \leq m_K N(I)$.*

pf. Let I be a nonzero fractional ideal of $\mathcal{O}_K$. By the Minkowski lattice point theorem, if we choose t so that $\mu(S_t) > 2^n \text{covol}(I)$, then S_t will contain a nonzero $a \in I$. By the previous lemma, and previous corollary, it suffices to choose t so that $(\frac{t}{n})^n = \frac{n! \mu(S_t)}{n^n 2^r \pi^s} > \frac{n! 2^n}{n^n 2^r \pi^s} \text{covol}(I) = \frac{n!}{n^n} (\frac{4}{\pi})^s \sqrt{|D_K|} N(I) = m_K N(I)$. Pick t so that $(\frac{t}{n})^n > m_K N(I)$. Then S_t contains $a \in I$ with $\sum_\sigma |\sigma(a)| \leq t$. By the geometric-arithmetic inequality, $N(a) = (N(a)^{1/n})^n = (\pi_\sigma |\sigma(a)|^{1/n})^n \leq (\frac{q}{n} \sum_\sigma |\sigma(a)|)^n \leq (\frac{t}{n})^n$. By taking $(\frac{t}{n})^n \rightarrow M_K N(I)$, $N(a) \leq m_K N(I)$.

$cl\mathcal{O}_K$ is the quotient of the ideal group $\mathcal{I}_K$ of $\mathcal{O}_K$ by its subgroup of principal fractional ideals.

Theorem 3.1.7. *Let K be a number field. Every ideal class in $cl\mathcal{O}_K$ contains an ideal $I \subseteq \mathcal{O}_K$ of absolute norm $N(I) \leq m_K$, where m_K is the Minkowski constant for K .*

pf. Let $[J]$ be an ideal class of $\mathcal{O}_K$ represented by the nonzero fractional ideal J . By the Minkowski bound, the fractional ideal J^{-1} contains a nonzero element a for which $N(a) \leq m_K N(J^{-1}) = m_K N(J)^{-1}$. Thus $N(aJ) = N(a)N(J) \leq m_K$. We have $a \in J^{-1} \rightarrow aJ \subseteq J^{-1}J = \mathcal{O}_K$ so $I = aJ$ is an $\mathcal{O}_K$ -ideal in the ideal class $[J]$ with $N(I) \leq m_K$.

Lemma 3.1.4. *Let K be a number field of degree n and let $M \in \mathbb{R}_{++}$. The number of $\mathcal{O}_K$ -ideals of norm $N(I) \leq M$ is bounded by $(nM)^{\log_2 M}$.*

pf. Let I be an ideal of absolute norm $N(I) \leq M$, and let $I = \mathbf{p}_1 \dots \mathbf{p}_k$ be its factorisation into prime ideals. Then $N(\mathbf{p}_i) \geq 2$ for each $\mathbf{p}_i$, so $k \leq \log_2 M$. There are less than M primes $p \leq M$ and at most n primes $\mathbf{p}$ of $\mathcal{O}_K$ above each $p \leq M$. Thus there are less than $(nM)^{\log_2 M}$ $\mathcal{O}_K$ -ideals with norm $N(I) \leq M$.

Corollary 3.1.3. *Let K be a number field. The ideal class group of $\mathcal{O}_K$ is finite.*

pf. By the previous theorem, each ideal class is represented by an ideal of norm at most m_K , and by the previous lemma, the number of such ideals is finite.

Theorem 3.1.8. *Let A be the ring $\mathbb{Z}$ or $\mathbb{F}_q[t]$ and let B the integral closure of A in a finite separable extension of its fraction field. The ideal class group of B is finite.*

pf. $\rightarrow K = \mathbb{Z}(\sqrt{d})$ has class number $h_K \equiv \#cl\mathcal{O}_K \rightarrow \infty$ as $d \rightarrow \infty$. This implies there are only a finite number of imaginary quadratic fields with any particular class number.

Corollary 3.1.4. *Let K be a number field of degree n with s complex places. Then*

$$|D_K| \geq \left(\frac{n^n}{n!}\right)^2 \left(\frac{\pi}{4}\right)^{2s} > \frac{1}{e^2 n} \left(\frac{\pi e^2}{4}\right)^n$$

pf. If I is an ideal and $a \in I, a \neq 0$, $N(a) \geq N(I)$ so by the previous theorem $m_K = \frac{n!}{n^n} \left(\frac{4}{\pi}\right)^2 \sqrt{|D_K|} \geq 1$ and the first inequality follows. By the Stirling's approximation $n! \leq e\sqrt{n} \left(\frac{n}{e}\right)^n$. Also $2s \leq n$, the second inequality follows.

Corollary 3.1.5. *If K is a number field other than $\mathbb{Q}$, then $|D_K| > 1$, equivalently there are no nontrivial unramified extensions of $\mathbb{Q}$.*

A commutative ring R with finite quotients by all nonzero ideals is a finite quotient domain. Assume R is not a field. For such R , define the absolute ideal norm $N_R \equiv \#R/I$ for each nonzero R -ideal I , let $N_R((0)) \equiv 0$, and put $N_R(r) \equiv N_R((r)), \forall r \in R$. Assume AKLB. If A is a finite quotient domain, then so is B and we have the identity $N_B(\alpha) = N_A(N_{L/K}(\alpha))$ for all $\alpha \in B$.
pf.

Definition 3.1.13. *A PID is basic if it is a finite quotient domain and $\exists c_1, c_2 \in \mathbb{Z}_{++}$ s.t.*

$$\begin{aligned} \#\{x \in A : N_A(x) \leq c_1 m\} &\geq m, \forall m \in \mathbb{Z} \\ N_A(x+y) &\leq c_2(N_A(x) + N_A(y)), \forall x, y \in A \end{aligned}$$

CLAIM : $\mathbb{Z}$ is a basic PID but $\mathbb{Z}[\sqrt{5}]$ is not. Also $\mathbb{F}_q[t]$ is a basic PID.

pf. $\#\mathbb{Z}/(x) = x \rightarrow \forall c > 1, \#\{x \in \mathbb{Z} : N_{\mathbb{Z}}(x) \leq cm\} \geq m$. Also $\#\mathbb{Z}/(x+y) = x+y, \mathbb{Z}/(x) = x, \mathbb{Z}/(y) = y \rightarrow \forall c \geq 1, N(x+y) \leq c(N(x) + N(y))$. Hence $\mathbb{Z}$ is a basic PID.

Let f be the monic irreducible polynomial in $\mathbb{F}_q[t]$. $\#\mathbb{F}_q[t]/(f) = \begin{matrix} \alpha \leq q & \deg f = 1 \\ n \leq q^2 & \deg f = 2 \end{matrix}$ and so on. This implies $\#\{N_{\mathbb{F}_q[t]}(f) \leq cm\} \geq m$ where $c = q^n < \infty$. Also $N(f+g) = \#\mathbb{F}_q[t]/(f+g) \leq \#\mathbb{F}_q[t]/(f) + \#\mathbb{F}_q[t]/(g) \rightarrow \forall c \geq 1$, inequality holds and therefore it is a basic PID.

Definition 3.1.14. *A global Dedekind domain over A is a Dedekind domain that is free of finite rank as a module over a subring A that is a basic PID.*

CLAIM : Given AKLB, if A is a basic PID, then B is a global Dedekind domain.

pf. Given AKLB, B is a Dedekind domain with finite rank. $\forall a \in A, \forall \phi \in \text{Aut}(B), \phi(ab) = a\phi(b)$ implies it is an A -module. Hence the claim follows.

CLAIM : Let B be a global Dedekind domain over a basic PID A and let e_i be an A -basis for B . Then there exists a homogeneous $f \in A[x_1, \dots, x_n]$ of degree n s.t. $\forall \alpha \in B, N_{L/K}(\alpha) = f(a_1, \dots, a_n)$, where $\alpha = \sum a_i e_i$ with $a_i \in A$. Also $\exists c \in \mathbb{Z}_{++}$ s.t.

$N_B(\alpha) \leq c \max(N_A(a_1), \dots, N_A(a_n))^n$ for all $\alpha = \sum a_i e_i \in B$.

pf. The determinant for the matrix of base of change gives such f , which is homogeneous of degree n . Then $N_{L/K}(\alpha) = \det(L \rightarrow_{\times \alpha} L) = f(a_1, \dots, a_n) \rightarrow N_B(\alpha) = \frac{B}{(\alpha)} = \prod_{i=1}^n A/(a_i) \leq c \max(N_A(a_1), \dots, N_A(a_n))^n$ with $c \geq 1$.

CLAIM : Let B be a global Dedekind domain. Then there is a real number m s.t. $\forall B$ -ideal I , nonzero, there is a nonzero $\alpha \in I$ for which $N_B(\alpha) \leq mN_B(I)$.

pf. $\mathcal{O}_K \subseteq B$ and by the Minkowski bound, for every nonzero fractional ideal I of

$\mathcal{O}_K$, $\exists a \in I$ for which $N(a) \leq m_L N(I)$. Also $(\text{disc}(f')) = D_{\mathcal{O}_K/A} = N_{B/A}(\mathbf{c}) D_{B/A} = (\mathbf{c})(\text{disc}(f))$. For any B -ideal I containing $\mathbf{c}$,

Theorem 3.1.9. *Let A be the ring $\mathbb{Z}$ or $\mathbb{F}_q[t]$ and let B be the integral closure of A in a finite separable extension of its fraction field. The ideal class group of B is finite.*

pf. Since $\mathbb{Z}, \mathbb{F}_q[t]$ are basic PID, B is a global Dedekind domain.
 $\rightarrow K = \mathbb{Z}(\sqrt{d})$ has class number $h_K \equiv \#cl\mathcal{O}_K \rightarrow \infty$ as $d \rightarrow \infty$. This implies there are only a finite number of imaginary quadratic fields with any particular class number.

Corollary 3.1.6. *Let K be a number field of degree n with s complex places. Then $|D_K| \geq \left(\frac{n^n}{n!}\right)^2 \left(\frac{\pi}{4}\right)^{2s} > \frac{1}{e^{2n}} \left(\frac{\pi e^2}{4}\right)^n$.*

pf. If I is an ideal and $a \in I$ is nonzero, $N(a) \geq N(1)$ so $m_K = \frac{n!}{n^n} \left(\frac{4}{\pi}\right)^s \sqrt{|D_K|} \geq 1$. By Stirling's approximation, $n! \leq e\sqrt{n} \left(\frac{n}{e}\right)^n$ and $2s \leq n$ gives the second inequality.

Corollary 3.1.7. *If K is a number field other than $\mathbb{Q}$, then $|D_K| > 1$, equivalently there are no nontrivial unramified extensions of $\mathbb{Q}$.*

Let K be a global field, M_K be the set of places of K . For each place $v \in M_K$, use K_v to denote the completion of K w.r.t. v and we have a normalised absolute

value $\|\cdot\| : K_v \rightarrow \mathbb{R}_+$
 $x \mapsto \frac{\mu(xS)}{\mu(S)}$, where μ is a Haar measure and S is any measurable set of positive finite measure. Denote k_v to denote the residue field which is a finite field.

$$\|x\| : |x|_v = (\#k_v)^{-v(x)} \quad v \text{ nonarchimedean}$$

$$\begin{array}{ll} |x|_{\mathbb{R}} & K_v \cong \mathbb{R} \\ |x|_{\mathbb{C}}^2 & K_v \cong \mathbb{C} \end{array}$$

Definition 3.1.15. *Let K be a global field. A multiplicative Arakelov divisor is a sequence of positive real numbers $c = (c_v)$ indexed by $v \in M_K$ with all but finitely many $c_v = 1$ and $c_v \in \|K_v^\times\| \equiv \{\|x\|_v : x \in K_v^\times\}$. The set of Arakelov divisors $\text{Div}K$ forms an abelian group under pointwise multiplication $(c_v)(d_v) \equiv (c_v d_v)$. The multiplicative group $K^\times$ is canonically embedded in $\text{Div}K$ via the map $x \mapsto (\|x\|_v)$, where it forms the subgroup $\text{Princ}K$ of principal Arakelov divisors.*

Definition 3.1.16. *Let K be a global field. The size of an Arakelov divisor c is*

$\|c\| \equiv \prod_{v \in M_K} c_v \in \mathbb{R}_{++}$. The map $\begin{array}{ll} \text{Div}K & \rightarrow \mathbb{R}_{++}^\times \\ c & \mapsto \|c\| \end{array}$ is a group homomorphism whose

kernel contains $\text{Princ}K$. Corresponding to each Arakelov divisor c is a subset $L(c)$ of K defined by $L(c) \equiv \{x \in K : \|x\|_v \leq c_v \forall v \in M_K\}$ and a nonzero fractional ideal of $\mathcal{O}_K$ defined by $I_c \equiv \prod_{v \in M_K} \mathfrak{q}_v^{v(c)}$ where $\mathfrak{q}_v \equiv \{a \in \mathcal{O}_K : v(a) > 0\}$ is the prime ideal corresponding to the discrete valuation v that induces $\|\cdot\|_v$ and $v(c) \equiv -\log_{\#k_v}(c_v) \in \mathbb{Z}$.

$\rightarrow L(c) \subseteq I_c \subseteq K$ and the map $c \mapsto I_c$ defines a group homomorphism $DivK \rightarrow \mathcal{J}_K$. The quotient of $DivK$ by the subgroup $PrincK$ is denoted $PicK$. Then we

have a commutative diagram

$$\begin{array}{ccc} DivK & \longrightarrow & \mathcal{J}_K \\ \downarrow & & \downarrow \\ PicK & \longrightarrow & cl\mathcal{O}_K \end{array} .$$

The Arakelov divisors of size 1

form a subgroup of $DivK$, denoted Div^0K that contains $PrincK$ and surjects onto $\mathcal{J}_K$ via the map $DivK \rightarrow \mathcal{J}_K$. The quotient of Div^0K by $PrincK$ is the Arakelov class group Pic^0K , which also admits the ideal class group $cl\mathcal{O}_K$ as a quotient.

Let K be a number field. Then the absolute norm $N(I)$ of a fractional ideal $\mathcal{O}_K$ is the unique $t \in \mathbb{Q}_{++}$ for which $N_{\mathcal{O}_K/\mathbb{Z}}(I) = (t)$. Then $N(I_c) = \prod_{v|\infty} N(\mathbf{q}_v)^{v(c)} = \prod_{v|\infty} (\#k_v)^{v(c)} = \prod_{v|\infty} c_v^{-1} \rightarrow \|c\| = N(I_c)^{-1} \prod_{v|\infty} c_v$. Define $R_c \equiv \{x \in K_{\mathbb{R}} : |x|_v \leq c_v, \forall v|\infty\}$, which is a compact symmetric subset of the real vector space $K_{\mathbb{R}} \equiv K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^r \times \mathbb{C}^s$, where r is the number of real places of K and s is the number of complex places. If we view $I_c, L(c)$ as subsets of $K_{\mathbb{R}}$ via the canonical embedding $K \rightarrow K_{\mathbb{R}}$, $L(c) = I_c \cap R_c$.

Lemma 3.1.5. *Let c be an Arakelov divisor of a global field K . The set $L(c)$ is finite.*

pf. Assume K is a number field. The fractional ideal I_c is a lattice in $K_{\mathbb{R}}$ and is thus a closed discrete subset of $K_{\mathbb{R}}$. In $K_{\mathbb{R}}$ we may view $L(c) = I_c \cap R_c$ as the intersection of a discrete closed set with a compact set, which is a compact discrete set and therefore finite.

Corollary 3.1.8. *Let K be a global field, and let μ_K denote the torsion subgroup of $K^{\times}$ (equivalently, the roots of unity in K). The group μ_K is finite and equal to the kernel of the map*

$$\begin{array}{ccc} K^{\times} & \rightarrow & DivK \\ x & \mapsto & (\|x\|_v) \end{array} .$$

It is also the torsion subgroup of $\mathcal{O}_K^{\times}$.

pf. Each $\zeta \in \mu_K$ satisfies $\zeta^n = 1$ for some positive integer n . For every place $v \in M_K$, $\|\zeta^n\|_v = \|\zeta\|_v^n = 1$ and therefore $\|\zeta\|_v = 1$. Thus $\mu_K \subseteq \ker(K^{\times} \rightarrow DivK)$. Let c be the Arakelov divisor with $c_v = 1$ for all $v \in M_K$. Then $\ker(K^{\times} \rightarrow DivK) \subseteq L(c)$ is a finite subgroup of $K^{\times}$ and is therefore contained in the torsion subgroup μ_K . Every element of μ_K is an algebraic integer (since they are roots of $x^n - 1$), so $\mu_K \subseteq \mathcal{O}_K^{\times}$.

$\rightarrow 0 \longrightarrow \mu_K \longrightarrow K^{\times} \longrightarrow DivK \longrightarrow PicK \longrightarrow 0$ an exact sequence of abelian groups.

Proposition 3.1.6. *Let K be a number field with s complex places, define $B_K \equiv (\frac{2}{\pi})^s \sqrt{|D_K|}$. If c is an Arakelov divisor of size $\|c\| > B_K$, then $L(c)$ contains an element of $K^{\times}$.*

pf. Normalise the Haar measure μ on the locally compact group $K_{\mathbb{R}} \cong \mathbb{R}^r \times \mathbb{C}^s \cong \mathbb{R}^n$ so that $\mu(S) = 2^s \mu_{\mathbb{R}^n}(S)$ for measurable $S \subseteq K_{\mathbb{R}}$. For each real place v , the constraint $\|x\|_v = |x|_{\mathbb{R}} \leq c_v$ contributes a factor of $2c_v$ to $\mu(R_c)$ and for each complex place v the constraint $\|x\|_v = |x|_{\mathbb{C}}^2 \leq c_v$ contributes a factor of πc_v . Then
$$\frac{\mu(R_c)}{\text{covol}(I_c)} = \frac{2^n \mu_{\mathbb{R}^n}(R_c)}{\text{covol}(I_c)} = \frac{2^s (\prod_{v \text{ real}} 2c_v) (\prod_{v \text{ complex}} \pi c_v)}{\text{covol}(I_c)} = \frac{2^r (2\pi)^s \prod_{v|\infty} c_v}{\sqrt{|D_K| N(I_c)}} = \frac{2^{r+s} \pi^s}{\sqrt{|D_K|}} \|c\| = \frac{\|c\|}{B_K} 2^n > 2^n.$$
 Then by the Minkowski lattice point theorem, $L(c) = R_c \cap I_c$, a convex discrete set, contains an element $K^{\times}$.

→ the above bound can be an asymptotic one, $\forall c \in \text{Div } K, \|c\| \rightarrow \infty, \#L(x) = (\frac{2^r (2\pi)^s}{\sqrt{|D_K|}} + o(1)) \|c\|$.

Let K be a number field with ring of integers $\mathcal{O}_K$. The multiplicative group $\mathcal{O}_K^{\times}$ is the unit group of $\mathcal{O}_K$. As a ring, the finite etale $\mathbb{R}$ -algebra $K_{\mathbb{R}} = K \otimes_{\mathbb{Q}} \mathbb{R}$ also has a unit group and we have an isomorphism of topological groups $K_{\mathbb{R}}^{\times} \cong \prod_{v|\infty} K_v^{\times} \cong \prod_{v|\infty \text{ real}} \mathbb{R}^{\times} \prod_{v|\infty \text{ complex}} \mathbb{C}^{\times} = (\mathbb{R}^{\times})^r \times (\mathbb{C}^{\times})^s$. Writing elements of $K_{\mathbb{R}}^{\times}$ as vectors $x = (x_v)$ indexed by the infinite places v of K , define a surjective homomorphism of locally compact groups
$$\text{Log} : K_{\mathbb{R}}^{\times} \rightarrow \mathbb{R}^{r+s} \\ (x_v) \mapsto (\log \|x_v\|_v).$$
 It is surjective and continuous because each of the maps $x_v \mapsto \log \|x_v\|_v$ is, and it is a group homomorphism, because $\text{Log}(xy) = (\log \|x_v y_v\|_v) = (\log \|x_v\|_v + \log \|y_v\|_v) = (\log \|x_v\|_v) + (\log \|y_v\|_v) = \text{Log } x + \text{Log } y$. There is an one-to-one correspondence between the infinite places of K and the $\text{Gal}(\mathbb{C}/\mathbb{R})$ -orbits of $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$. For each $v|\infty$, pick a representative σ_v of its corresponding $\text{Gal}(\mathbb{C}/\mathbb{R})$ -orbit in $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$;

σ_v $v \text{ real}$ $\|x\|_v = |\sigma_v(x)|_{\mathbb{R}}$ $v|\infty \text{ real}$
 $\sigma_v, \bar{\sigma}_v$ $v \text{ complex}$ $|\sigma_v(x) \bar{\sigma}_v(x)|_{\mathbb{R}}$ $v|\infty \text{ complex}$. Then we have

The absolute norm $N : K^{\times} \rightarrow \mathbb{Q}_{++}^{\times}$ extends naturally to a continuous homomorphism of locally compact groups
$$N : K_{\mathbb{R}}^{\times} \rightarrow \mathbb{R}_{++}^{\times} \\ (x_v) \mapsto \prod_{v|\infty} \|x_v\|_v,$$
 which is compatible with the canonical embedding $K^{\times} \rightarrow K_{\mathbb{R}}^{\times}$. Then we have $N(x) = |N_{K/\mathbb{Q}}(x)| =$

$$\begin{array}{ccccc} K^{\times} & \longrightarrow & K_{\mathbb{R}}^{\times} & \xrightarrow{\text{Log}} & \mathbb{R}^{r+s} \\ \downarrow N & & \downarrow N & & \downarrow T \\ \mathbb{Q}_{++}^{\times} & \longrightarrow & \mathbb{R}_{++}^{\times} & \xrightarrow{\log} & \mathbb{R} \end{array}$$

$|\Pi_{\sigma} \sigma(x)|_{\mathbb{R}} = \prod_{v|\infty} \|x\|_v$ and we have a commutative diagram

where
$$T : \mathbb{R}^{r+s} \rightarrow \mathbb{R} \\ x \mapsto \sum_i x_i.$$
 The commutativity can be expressed as $T(\text{Log } x) = \log(N(x))$, which holds for all $x \in K^{\times}, \forall x \in K_{\mathbb{R}}^{\times}$. The norm of a unit in $\mathcal{O}_K$ must be a unit in $\mathbb{Z}$, hence have absolute value 1. Thus $\mathcal{O}_K^{\times}$ lies in the kernel of the map $x \mapsto \log N(x)$ and therefore also lies in the kernel of the map $x \mapsto T(\text{Log } x)$. Thus $\text{Log}(\mathcal{O}_K^{\times})$ is a subgroup of the trace zero hyperplane $\mathbb{R}_0^{r+s} \equiv \{x \in \mathbb{R}^{r+s} : T(x) = 0\} \subseteq \mathbb{R}^{r+s}$ a subgroup and an $\mathbb{R}$ -vector subspace of dimension $r + s - 1$.

Proposition 3.1.7. *Let K be a number field with r real and s complex places and let Λ_K be the image of the unit group $\mathcal{O}_K^\times$ in $\mathbb{R}_0^{r+s}$ under the Log map. The following holds:*

1. *We have a split exact sequence of finitely generated abelian groups*

$$0 \rightarrow \mu_K \rightarrow \mathcal{O}_K^\times \rightarrow \Lambda_K \rightarrow 0$$

2. *Λ_K is a lattice in the trace zero hyperplane $\mathbb{R}_0^{r+s}$.*

Here μ_K is not a Haar measure, but the group of roots of unity in K , all of which are clearly torsion elements of $\mathcal{O}_K^\times$.

pf.

1. Let Z be the kernel of $\mathcal{O}_K^\times \rightarrow \Lambda_K$. Then $\mu_K \subseteq Z$ since $\Lambda_K \subseteq \mathbb{R}_0^{r+s}$ is torsion free. Let x be the Arakelov divisor with $I_c = \mathcal{O}_K$ and $c_v = 2$ for $v|\infty$ so that $L(c) = \{x \in \mathcal{O}_K : \|x\|_v \leq 2 \forall v|\infty\}$. For $x \in \mathcal{O}_K^\times$, $x \in L(c) \leftrightarrow \text{Log}(x) \in \text{Log}R_c = \{z \in \mathbb{R}^{r+s} : z_i \leq \log 2\}$. The set on the RHS includes the zero vector, thus $Z \subseteq L(c)$, which is a finite set by the previous lemma. As a finite subgroup of $\mathcal{O}_K^\times$, $Z \subseteq \mu_K$ thus the sequence is exact at $\mathcal{O}_K^{\text{times}}$. By the definition Log map is surjective and the exactness follows.

$\Lambda_K \cap \text{Log}(R_c) = \text{Log}(\mathcal{O}_K^\times \cap L(c)) < \infty$ since $L(c) < \infty$. Thus 0 is an isolated point of Λ_K in $\mathbb{R}^{r+s}$ and in $\mathbb{R}_0^{r+s}$, so Λ_K is a discrete subgroup of $\mathbb{R}_0^{r+s} \cong \mathbb{R}^{r+s-1}$ and must be finitely generated. It follows that $\mathcal{O}_K^\times$ is finitely generated, since it lies in a s.e.s. whose left and right terms are finitely generated. By the structure theorem for finitely generated abelian groups, the sequence must split, since μ_K is the torsion subgroup of $\mathcal{O}_K^\times$.

2. By the previous statement, we know Λ_K is a finitely generated $\mathbb{R}_0^{r+s}$ -module. So it is a lattice if it spans $\mathbb{R}_0^{r+s}$. Let V be the subspace of $\mathbb{R}_0^{r+s}$ spanned by Λ_K and suppose the contrary; $\dim V < \dim \mathbb{R}_0^{r+s}$. The orthogonal subspace $V^\perp$ then contains a unit vector u and $\forall \lambda \in \mathbb{R}_{++}$, the open ball $B_{<\lambda}(\lambda u)$ does not intersect Λ_K . Thus $\mathbb{R}_0^{s+r}$ contains points arbitrarily far away from every point in Λ_K . Fix a real number $B > B_K$, where B_K is as in the previous proposition, so that for every $c \in \text{Div}K$ with $\|c\| \geq B$, $L(c)$ contains a nonzero element and fix a vector $b \in \mathbb{R}^{r+s}$ with nonnegative components b_i s.t. $T(b) = \sum b_i = \log B$. Let $(\alpha_1), \dots, (\alpha_m)$ be the list of all nonzero principal ideals with $N(\alpha_j) \leq B$, which is finite. Let M be the twice the maximum of $(r+s)B$ and $\max_j \|\text{Log}(\alpha_j)\|$. Let $h \in \mathbb{R}_0^{r+s}$ and define $c \in \text{Div}K$ by $I_c \equiv \mathcal{O}_K, c_v \equiv \exp(h_i + b_i)$ for $v|\infty$ where i is the coordinate in $\mathbb{R}^{r+s}$ corresponding to v under the Log map. Then

$$\|c\| = \Pi \|c_v\| = \exp(\sum (h_i + b_i)) = \exp T(h+b) = \exp(T(h)+T(b)) = \exp(T(b)) = B > B_K$$

thus $L(c)$ contains a nonzero $\gamma \in I_c \cap K = \mathcal{O}_K$ and $g = \text{Log}(\gamma)$ satisfies $g_i \leq \log c_v = h_i + b_i$. Also $T(g) = T(\text{Log} \gamma) = \log N(\gamma) \geq 0$ since $N(\gamma) \geq 1$, $\forall \gamma \in \mathcal{O}_K$ nonzero. The vector $w \equiv g - h \in \mathbb{R}^{r+s}$ satisfies $\sum w_i = T(v) = T(g) - T(h) = T(g) \geq 0$, $w_i \leq b_i \leq B$ with together implies $|w_i| \leq (r+s)B$, $\|g - h\| = \|w\| \leq M/2$. Additionally, $\log N(\gamma) = T(\log \gamma) \leq T(h + b) = T(b) = \log B \rightarrow N(\gamma) \leq B$, $(\gamma) = (\alpha_j)$ for one of the α_j fixed above. Thus $\gamma/\alpha_j \in \mathcal{O}_K^\times$ is a unit and $l \equiv \text{Log}(\gamma/\alpha_j) = \text{Log}(\gamma) - \text{Log}(\alpha_j) \in \Lambda_K$ satisfies $\|g - l\| = \|\text{Log}(\alpha_j)\| \leq M/2$. Thus $\|h - l\| \leq \|h - g\| + \|g - l\| \leq M$, which contradicts that $\mathbb{R}_0^{r+s}$ contains a points arbitrarily far away from every point in Λ_K so $\dim V = \dim \mathbb{R}_0^{r+s}$.

Theorem 3.1.10 (Dirichlet's Unit Theorem). *Let K be a number field with r real and s complex places. Then $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{r+s-1}$ is a finitely generated abelian group.*

pf. The image of the torsion-free part of the unit group $\mathcal{O}_K^\times$ under the Log map is the lattice Λ_K in the trace-zero hyperplane $\mathbb{R}_0^{r+s}$, which has dimension $r + s - 1$.

Theorem 3.1.11 (Unit Theorem for Global Fields). *Let K/F be a finite separable extension, with $F = \mathbb{Q}$ or $F = \mathbb{F}_q(t)$, let $S \subseteq M_K$ be the set of places of K lying above the unique infinite place of F and define $\mathcal{O}_K^\times \equiv \{x \in K^\times : v(x) = 0, \forall v \in M_K \setminus S\}$. Then $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{\#S-1}$ is a finitely generated abelian group.*

pf. When $F = \mathbb{Q}$, $\#S = r + s$ and this is Dirichlet's unit theorem.

Chapter 4

Local Class Field Theory

4.1 Artin Map and Local Class Field Theory

4.1.1 The Prime Number Theorem

Definition 4.1.1. *The Riemann zeta function is the complex function defined by the series*

$$\zeta(s) \equiv \sum_{n \geq 1} n^{-s}$$

for $\text{Re}(s) > 1$, where n varies over positive integers.

→ it converges absolutely and locally uniformly on $\text{Re}(s) > 1$.

Theorem 4.1.1 (Euler Product). *For $\text{Re}(s) > 1$, $\zeta(s) = \sum_{n \geq 1} n^{-s} = \prod_p (1 - p^{-s})^{-1}$ where the product converges absolutely. In particular, $\zeta(s) \neq 0$, $\text{Re}(s) > 1$.*

pf. $\sum n^{-s} = \sum \prod p^{-v_p(n)s}$. Define the partial zeta function $\zeta_m(s) \equiv \sum_{n \in S_m} n^{-s} = \sum_{e_1, \dots, e_k \geq 0} (p_1^{e_1} \dots p_k^{e_k})^{-s} = \prod_{1 \leq i \leq k} \sum_{e_i \geq 0} (p_i^{e_i})^{-s} = \prod_{p \leq m} (1 - p^{-s})^{-1}$ where S_m is the set of m -smooth integers (those with no prime factors $p > m$) and p_i are the primes upto m . For any $\delta > 0$, $\zeta_m(s)$ converges uniformly on $\text{Re}(s) > 1 + \delta$ to $\zeta(s)$ → $\forall \epsilon > 0$, $|\zeta_m(s) - \zeta(s)| \leq |\sum_{n \geq m} n^{-s}| \leq \sum |n^{-s}| = \sum n^{-\text{Re}(s)} \leq \int_m^\infty x^{-1-\delta} dx \leq \frac{1}{\delta} m^{-\delta} < \epsilon$ for all sufficiently large m . Thus the sequence $\zeta_m(s)$ converges locally uniformly to $\zeta(s)$ on $\text{Re}(s) > 1$. The sequence of functions $P_m \equiv \prod_{p \leq m} (1 - p^{-s})^{-1}$ also converges locally uniformly to $\prod (1 - p^{-s})^{-1}$ on any region in which the latter function is absolutely convergent. For any s in $\text{Re}(s) > 1$,

$$\sum_p \log(1 - p^{-s})^{-1} = \sum_p \left| \sum_{e \geq 1} \frac{1}{e} p^{-es} \right| \quad \text{by } \log(1 - z) = -\sum_{n \geq 1} \frac{1}{n} z^n, \quad \forall |z| < 1. \quad \text{Thus}$$

$$\leq \sum_p \sum_{e \geq 1} |p^{-s}|^e = \sum_p (|p|^s - 1)^{-1} < \infty$$

$\prod_p (1 - p^{-s})^{-1}$ is absolutely convergent on $\text{Re}(s) > 1$.

Theorem 4.1.2 (Analytic Continuation I). *For $\text{Re}(s) > 1$,*

$$\zeta(s) = \frac{1}{s-1} + \phi(s)$$

where $\phi(s)$ is a holomorphic function on $\operatorname{Re}(s) > 0$. Thus $\zeta(s)$ extends to a meromorphic function on $\operatorname{Re}(s) > 0$ that has a simple pole at $s = 1$ with residue 1 and no other poles.

pf. For $\operatorname{Re}(s) > 1$, $\zeta(s) - \frac{1}{s-1} = \sum n^{-s} - \int_1^\infty x^{-s} ds = \sum (n^{-s} - \int_n^{n+1} x^{-s} dx) = \sum \int_n^{n+1} (n^{-s} - x^{-s}) dx$. For each $n \geq 1$, $\phi_n(s) \equiv \int_n^{n+1} (n^{-s} - x^{-s}) dx$ is holomorphic on $\operatorname{Re}(s) > 0$. For each fixed s in $\operatorname{Re}(s) > 0$, $\forall x \in [n, n+1]$, $|n^{-s} - x^{-s}| = |\int_n^x st^{-s-1} dt| \leq \int_n^x \frac{|s|}{|t^{s+1}|} dt = \int_n^x \frac{|s|}{t^{1+\operatorname{Re}(s)}} dt \leq \frac{|s|}{n^{1+\operatorname{Re}(s)}} \rightarrow |\phi_n(s)| \leq \int_n^{n+1} |n^{-s} - x^{-s}| dx \leq \frac{|s|}{n^{1+\operatorname{Re}(s)}}$. For any s_0 in $\operatorname{Re}(s) > 0$, $\left\{ \begin{array}{l} \epsilon \equiv \operatorname{Re}(s_0)/2 \\ U \equiv B_{<\epsilon}(s_0) \end{array} \right., \forall n \geq 1, \sup_{s \in U} |\phi_n(s)| \leq \frac{|s_0| + \epsilon}{n^{1+\epsilon}} \equiv M_n$ and $\sum M_n = (|s_0| + \epsilon)\zeta(1 + \epsilon)$ converges. The series $\sum \phi_n$ thus converges locally uniformly on $\operatorname{Re}(s) > 0$. By the Weierstrass test $\sum_n \phi_n$ converges to a function $\phi(s) = \zeta(s) - \frac{1}{s-1}$ which is holomorphic on $\operatorname{Re}(s) > 0$.

Lemma 4.1.1 (Mertens). $\forall x, y \in \mathbb{R}$ with $z > 1$,

$$|\zeta(x)^2 \zeta(x + iy)^4 \zeta(x + 2iy)| \geq 1$$

pf. $\log |\zeta(s)| = -\sum_p \log |1 - p^{-s}|$ by the Euler product
 $= -\sum \operatorname{Re} \log(1 - p^{-s})$ by $\log |z| = \operatorname{Re} \log z$. Let $s = x + iy$. Then
 $= \sum_p \sum_{n \geq 1} \frac{\operatorname{Re}(p^{-ns})}{n}$
 $\log |\zeta(x + iy)| = \sum \sum \frac{\cos(ny \log p)}{np^{nx}} = \sum \sum \frac{2(1 + \cos ny \log p)^2}{np^{nx}} \geq 0$ by the trigonometric identity.
 So the lemma follows.

Corollary 4.1.1. $\zeta(s)$ has no zeros on $\operatorname{Re}(s) \geq 1$.

pf. By the analytic continuity I, $\zeta(s)$ has no zeros on $\operatorname{Re}(s) > 1$. Suppose $\zeta(1 + iy) = 0$ for some $y \in \mathbb{R}$. Then $y \neq 0$ since when $s = 1$, $\zeta(s)$ has a simple pole. Thus $\lim_{x \rightarrow 1} |\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| = 0$, a contradiction to the previous lemma, and the corollary follows.

The prime counting function $\pi : \mathbb{R} \rightarrow \mathbb{Z}_+$ counts the number of primes upto x . Define the log-weighted prime counting function $\vartheta(x) \equiv \sum_{p \leq x} \log p$.

Theorem 4.1.3 (Chebyshev). $\pi(x) \sim \frac{x}{\log x}$ iff $\vartheta(x) \sim x$.

pf. $0 \leq \vartheta(x) \leq \pi(x) \log x \rightarrow \frac{\vartheta(x)}{x} \leq \frac{\pi(x) \log x}{x}$. Then we have $\forall \epsilon \in (0, 1)$,
 $\vartheta(x) \geq \sum_{x^{1-\epsilon} < p \leq x} \log p \geq (1 - \epsilon)(\log x)(\pi(x) - \pi(x^{1-\epsilon})) \geq (1 - \epsilon)(\log x)(\pi(x) - x^{1-\epsilon}) \Leftrightarrow$
 $\pi(x) \leq \frac{1}{1-\epsilon} \frac{\vartheta(x)}{\log x} + x^{1-\epsilon}$. Thus $\forall \epsilon \in (0, 1)$, $\frac{\vartheta(x)}{x} \leq \frac{\pi(x) \log x}{x} \leq (\frac{1}{1-\epsilon}) \frac{\vartheta(x)}{x} + \frac{\log x}{x^\epsilon}$. Then
 $\lim_{x \rightarrow \infty} \frac{\log x}{x^\epsilon} = 0$. By choosing ϵ sufficiently small, we can make the ratios of $\vartheta(x)$ to x and $\pi(x)$ to $x/\log x$ arbitrarily close together as $x \rightarrow \infty$. Thus the lemma follows.

Lemma 4.1.2 (Chebyshev). *For $x \geq 1$*

$$\vartheta(x) \leq (4 \log 2)x$$

pf. For any positive integer n , $2^{2n} = (1+1)^{2n} = \sum_{m=0}^{2n} \binom{2n}{m} \geq \binom{2n}{n} = \frac{(2n)!}{n!n!} \geq \prod_{n < p \leq 2n} p - \exp(\vartheta(2n) - \vartheta(n))$ since $(2n)!$ is divisible by every prime $p \in (n, 2n]$, but $n!$ is not. Taking logarithms on both sides yields $\vartheta(2n) - \vartheta(n) \leq 2n \log 2$, $\forall n$. For any integer $m \geq 1$, $\vartheta(2^m) = \sum_{n=1}^m (\vartheta(2^n) - \vartheta(2^{n-1})) \leq \sum 2^n \log 2 \leq 2^{m+1} \log 2$. For any real $x \geq 1$, $\exists m \in \mathbb{Z}_{++}$ s.t. $2^{m-1} \leq x < 2^m$, which implies $\vartheta(x) \leq \vartheta(2^m) \leq 2^{m+1} \log 2 = (4 \log 2)2^{m-1} = (4 \log 2)x$ and the lemma follows.

$\rightarrow \vartheta(x) = O(x)$.

Lemma 4.1.3. *Let $f : \mathbb{R}_{\geq 1} \rightarrow \mathbb{R}$ be a nondecreasing function. If the integral $\int_1^\infty \frac{f(t)-t}{t^2} dt$ converges, then $f(x) \sim x$.*

pf. Let $F(x) \equiv \int_1^x \frac{f(t)-t}{t^2} dt$. By the hypothesis, $\lim_{x \rightarrow \infty} F(x)$ exists. Then $\forall \lambda > 1$, $\forall \epsilon > 0$, $|F(\lambda x) - F(x)| < \epsilon$ for sufficiently large x . Fix $\lambda > 1$ and suppose there is an unbounded sequence (x_n) s.t. $f(x_n) \geq \lambda x_n$ for all n . For each x_n ,

$$\begin{aligned} F(\lambda x_n) - F(x_n) &= \int_{x_n}^{\lambda x_n} \frac{f(t)-t}{t^2} dt \geq \int \frac{\lambda x_n - t}{t^2} dt \quad \text{by } f \text{ being nondecreasing} \\ &= \int_1^\lambda \frac{\lambda - t}{t^2} dt = c > 0 \text{ constant} \end{aligned}$$

$\epsilon < c$. Then $|F(\lambda x_n) - F(x_n)| = c > \epsilon$ for arbitrarily large x_n , a contradiction. Thus $f(x) < \lambda x$ for all sufficiently large x . Similarly we can get $f(x) > \frac{1}{\lambda}x$ for all sufficiently large x . These inequalities hold for all $\lambda > 1$, by letting λ sufficiently close to 1, we get $\lim f(x)/x = 1$, i.e. $f(x) \sim x$.

$\rightarrow$ for $f = \vartheta$, $H(t) = \vartheta(e^t)e^{-t} - 1$ gives $\int_1^\infty \frac{\vartheta(t)-t}{t^1} dt$ converges $\Leftrightarrow \int_0^\infty H(u)du$ converges where $t = e^u$.

Definition 4.1.2. *Let $h : \mathbb{R}_{++} \rightarrow \mathbb{R}$ be a piecewise continuous function. The Laplace transform $\mathcal{L}h$ is the complex function defined by*

$$\mathcal{L}h(s) \equiv \int_0^\infty e^{-st} h(t) dt$$

which is holomorphic on $\operatorname{Re}(s) > c$ for any $c \in \mathbb{R}$ for which $h(t) = O(e^{ct})$.

Define the auxiliary function $\Phi(x) \equiv \sum_p p^{-s} \log p$.

Lemma 4.1.4. $\mathcal{L}\vartheta(e^t)(x) = \frac{\Phi(s)}{s}$ is holomorphic on $\operatorname{Re}(s) > 1$.

pf. By the Chebyshev lemma, $\vartheta(e^t) = O(e^t)$, so $\mathcal{L}(\vartheta(e^t))$ is holomorphic on $\operatorname{Re}(s) > 1$. Let p_n be the n th prime and put $p_0 = 0$. The function $\vartheta(e^t)$ is constant on $t \in (\log p_n, \log p_{n+1}) \rightarrow \int_{\log p_n}^{\log p_{n+1}} e^{-st} \vartheta(e^t) dt = \vartheta(p_n) \int e^{-st} dt = \frac{1}{s} \vartheta(p_n) (p_n^{-s} - p_{n+1}^{-s})$. Thus we get $(\mathcal{L}\vartheta(e^t))(s) = \int_0^\infty e^{-st} \vartheta(e^t) dt = \frac{1}{s} \sum_{n=1}^\infty \vartheta(p_n) (p_n^{-s} - p_{n+1}^{-s}) = \frac{1}{s} \sum \vartheta(p_n) p_n^{-s} -$

$$\begin{aligned} \frac{1}{s} \sum \vartheta(p_{n-1}) p_n^{-s} &= \frac{1}{s} \sum (\vartheta(p_n) - \vartheta(p_{n-1})) p_n^{-s} = \frac{1}{s} \sum p_n^{-s} \log p_n = \frac{\Phi(s)}{s}. \\ \rightarrow \text{for } H(t) &\equiv \vartheta(e^t) e^{-t} - 1, \end{aligned}$$

$$\mathcal{L}H(s) = \mathcal{L}(\vartheta(e^t) e^{-t})(s) - \mathcal{L}1(s) = \mathcal{L}(\vartheta(e^t)(s+1)) - \frac{1}{s} = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

Lemma 4.1.5. *The function $\Phi(s) - \frac{1}{s-1}$ extends to a meromorphic function on $\text{Re}(s) > 1/2$ that is holomorphic on $\text{Re}(s) \geq 1$.*

pf. $\zeta(s)$ extends to a meromorphic function on $\text{Re}(s) > 0$, which we also denote $\zeta(x)$ with one simple pole at $s = 1$ and no zeros on $\text{Re}(s) \geq 1$. The logarithmic derivative $\zeta'(s)/\zeta(s)$ is meromorphic on $\text{Re}(s) > 0$ with no zeros on $\text{Re}(s) \geq 1$ and only one simple pole at $s = 1$ with residue -1 . Then by the Euler product, for $\text{Re}(s) > 1$, $-\frac{\zeta'(s)}{\zeta(s)} = (-\log \zeta(x))' = (-\log \prod (1 - p^{-s})^{-1})' = (\sum \log(1 - p^{-s}))' = \sum \frac{p^{-s} \log p}{1 - p^{-s}} = \sum \frac{\log p}{p^s - 1} = \sum (\frac{1}{p^s} + \frac{1}{p^s(p^s - 1)}) \log p = \Phi(s) + \sum \frac{\log p}{p^s(p^s - 1)}$. Then $\sum \frac{\log p}{p^s(p^s - 1)}$ converges absolutely and locally uniformly to a holomorphic function on $\text{Re}(s) > 1/2$. $\frac{\zeta'(s)}{\zeta(s)}$ is meromorphic on $\text{Re}(s) > 0$ and on $\text{Re}(s) \geq 1$, it has only a simple pole at $s = 1$ with residue 1 . Thus $\Phi(s) - \frac{1}{s-1}$ extends to a meromorphic function on $\text{Re}(s) > 1/2$ that is holomorphic on $\text{Re}(s) \geq 1$.

Corollary 4.1.2. *The function $\Phi(s+1) - \frac{1}{s}$ and $(\mathcal{L}H)(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$ both extends to meromorphic functions on $\text{Re}(s) > -1/2$ that are holomorphic on $\text{Re}(s) \geq 0$.*

pf. By the previous lemma, the first stmt follows immediately. $\frac{\Phi(s+1)}{s+1} - \frac{1}{s} = \frac{1}{s+1}(\Phi(s+1) - \frac{1}{s}) - \frac{1}{s+1}$ is meromorphic on $\text{Re}(s) > -1/2$ and holomorphic on $\text{Re}(s) \geq 0$ since it is a sum of products of such functions.

Theorem 4.1.4. *Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a bounded piecewise continuous function and suppose its Laplace transform extends to a holomorphic function $g(s)$ on $\text{Re}(s) \geq 0$. Then the integral $\int_0^\infty f(t) dt$ converges and is equal to $g(0)$.*

pf. Wlog assume $f(t) \leq 1, \forall t \geq 0$. For $\tau \in \mathbb{R}_{++}$, define $g_\tau(s) \equiv \int_0^\tau f(t) e^{-st} dt$. Then since $\int_0^\infty f(t) dt - \lim_{\tau \rightarrow \infty} g_\tau(0)$, if $\lim_{\tau \rightarrow \infty} g_\tau(0) = g(0)$, the theorem follows. For $r > 0$, let γ_r be the boundary of the region $\{s : |s| \leq r, \text{Re}(s) \geq -\delta\}$ with $\delta_r > 0$, chosen so that g is holomorphic on γ_r . Such δ_r exists because g is holomorphic on $\text{Re}(s) \geq 0$, hence on some open ball $B_{\leq 2\delta(y)}(iy), \forall y \in [-r, r]$ we may take $\delta_r \equiv \inf\{\delta(y) : y \in [-r, r]\}$ which is positive, given $[-r, r]$ being compact. Each γ_r is a simple closed curve and $\forall \tau > 0, h(s) \equiv (g(s) - g_\tau(s)) e^{s\tau} (1 + \frac{s^2}{r^2})$ is holomorphic on a region containing γ_r . By the Cauchy's integral formula, $g(0) - g_\tau(0) = h(0) = \frac{1}{2\pi i} \int_{\gamma_r} (g(s) - g_\tau(s)) e^{s\tau} (\frac{1}{s} + \frac{s}{r^2}) ds$. Set $r = 3/\epsilon > 0$.

- let γ_r^+ denote the part of γ_r in $\text{Re}(s) > 0$, a semicircle of radius r .

The integrand is absolutely bounded by $1/r$ on γ_r^+ , since $|s| = r$ and $\operatorname{Re}(s) > 0$, $|g(s) - g_\tau(s)| \cdot |e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})| = |\int_\tau^\infty f(t)e^{-st}dt| \cdot \frac{e^{\operatorname{Re}(s)\tau}}{r} \cdot |\frac{r}{s} + \frac{s}{r}| \leq \int_\tau^\infty e^{-\operatorname{Re}(s)t}dt \cdot \frac{e^{\operatorname{Re}(s)\tau}}{r} \cdot \frac{2\operatorname{Re}(s)}{r} = \frac{e^{-\operatorname{Re}(s)\tau}}{r} \cdot \frac{e^{\operatorname{Re}(s)\tau}}{r} \cdot \frac{2\operatorname{Re}(s)}{r} = 2/r^2 \rightarrow |\frac{1}{2\pi i} \int_{\gamma_r^+} (g(s) - g_\tau(s))e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})ds| \leq \frac{1}{2\pi} \cdot \pi r \cdot \frac{2}{r^2} = \frac{1}{r}$.

- let γ_r^- be the part of γ_r in $\operatorname{Re}(s) < 0$, a truncated semicircle.

For any fixed r , the first term $g(s)e^{s\tau}(s^{-1} + sr^{-2})$ in the integrand tends to 0 as $\tau \rightarrow \infty$ for $\operatorname{Re}(s) < 0$ and $|s| \leq r$. For the second term, since $g_\tau(s)$ is holomorphic on $\mathbb{C}$, it makes no difference if we instead integrate over the semicircle of radius r in $\operatorname{Re}(s) < 0$. For $\begin{cases} |s| = r \\ \operatorname{Re}(s) < 0 \end{cases}$, $|g_\tau(s)e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})| = |\int_0^\tau f(t)e^{-st}dt| \cdot \frac{e^{\operatorname{Re}(s)\tau}}{r} \cdot \frac{2\operatorname{Re}(s)}{r} = (1 - \frac{e^{-\operatorname{Re}(s)\tau}}{\operatorname{Re}(s)}) \frac{e^{\operatorname{Re}(s)\tau}}{r} \cdot \frac{2\operatorname{Re}(s)}{r} = \frac{2}{r^2} \cdot (1 - e^{\operatorname{Re}(s)\tau} \operatorname{Re}(s))$ where $\lim_{\tau \rightarrow \infty} (1 - e^{\operatorname{Re}(s)\tau} \operatorname{Re}(s)) = 1$ since $\operatorname{Re}(s) < 0$. Thus we obtain the bound $\frac{1}{r} + o(1)$ when we replace γ_r^+ with γ_r^- under γ_r^+ case above.

Then $g(0) - g_\tau(0) = h(0) = \frac{1}{2\pi i} \int_{\gamma_r} (g(s) - g_\tau(s))e^{s\tau}(\frac{1}{s} + \frac{s}{r^2})ds < \frac{2}{r} + o(1)$ as $\tau \rightarrow \infty$. Thus $\forall \epsilon > 0$, $\forall r = 3/\epsilon > 0$, $|g(0) - g_\tau(0)| < 3/r = \epsilon$ for all sufficiently large τ . So the theorem follows.

→ for a piecewise continuous function $f : \mathbb{R}_+ \rightarrow \mathbb{R}$, its Laplace transform $\mathcal{L}f(s) \equiv \int_0^\infty e^{-st}f(t)dt$ is typically not defined on $\operatorname{Re}(s) \leq c$ where c is the least c for which $f(t) = O(e^{ct})$. But $\mathcal{L}f$ may have an analytic continuation to a larger domain. The theorem says when $\mathcal{L}f$ extends to a holomorphic function on the entire half plane $\operatorname{Re}(s) \geq 0$, its value at $s = 0$ is exactly what we would get by simply plugging 0 into the integral defining $\mathcal{L}f$.

Theorem 4.1.5 (Prime Number Theorem).

$$\pi(x) \sim \frac{x}{\log x}$$

pf. $H(t) = \vartheta(e^t)e^t - 1$ is piecewise continuous and bounded by the Chebyshev lemma. Its Laplace transform extends to a holomorphic function on $\operatorname{Re}(s) \geq 0$ by the previous corollary. Then by the previous theorem, $\int_0^\infty H(t)dt - \int_0^\infty (\vartheta(e^t)e^{-t} - 1)dt$ converges. Replacing t with $\log x$ gives $\int_1^\infty (\vartheta(x)\frac{1}{x} - 1)\frac{dx}{x} = \int_1^\infty \frac{\vartheta(x) - x}{x^2}dx$ converges. Then by the lemma 4.1.3, $\vartheta(x) \sim x \Leftrightarrow \pi(x) \sim \frac{x}{\log x}$ by the Chebyshev theorem.

4.1.2 Dirichlet Theorem

Theorem 4.1.6 (Mertens). As $x \rightarrow \infty$, $\begin{cases} i) & \sum_{p \leq x} \frac{\log p}{p} = \log x + R(x), |R(x)| \leq 2 \\ ii) & \sum_{p \leq x} \frac{1}{p} = \log \log x + B + O(\frac{1}{\log x}) \\ iii) & \sum_{p \leq x} \log(1 - \frac{1}{p}) = -\log \log x - \gamma + O(\frac{1}{\log x}) \end{cases}$.

pf.

-
-
-

Definition 4.1.3. A function $f : \mathbb{Z} \rightarrow \mathbb{C}$ is called an arithmetic function. The function f is multiplicative if $f(1) = 1$ and $f(mn) = f(m)f(n)$ for all $m \perp n \in \mathbb{Z}$. It is totally multiplicative if $f(1) = 1$ and $f(mn) = f(m)f(n)$, $\forall m, n \in \mathbb{Z}$. For $m \in \mathbb{Z}_{++}$, we say f is m -periodic if $f(n+m) = f(n)$, $\forall n \in \mathbb{Z}$ and we call m the period of f if it is the least $m > 0$ for which this holds.

Definition 4.1.4. A Dirichlet character is a periodic totally multiplicative arithmetic function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$.

→ the image of a Dirichlet character is a finite multiplicative closed subset of $\mathbb{C}$, hence the union of a finite subgroup of $U(1)$ and a subset of $\{0\}$.

e.g. the trivial Dirichlet character is $\mathbb{1}(n) \equiv 1$, which is the unique Dirichlet character of period 1.

Each m -periodic Dirichlet character χ restricts to a group character $\chi \in (\mathbb{Z}/m\mathbb{Z})^\times$ and each group character χ of $(\mathbb{Z}/m\mathbb{Z})^\times$ can be extended to a Dirichlet character by defining $\chi(n) = 0$, $\forall n \not\perp m$, i.e. by the extension by zero.

Definition 4.1.5. A Dirichlet character of modulus m is an m -periodic Dirichlet character χ that is the extension by zero of a group character on $(\mathbb{Z}/m\mathbb{Z})^\times$, i.e. an m -periodic Dirichlet character for which $\chi(n) \neq 0 \Leftrightarrow m \perp n$.

Lemma 4.1.6. Let χ be a Dirichlet character of period m . Then χ is a Dirichlet character of modulus m' iff $m|m'm^k$ for some k .

pf.

- Let $m|m'm^k$ for some k . Suppose $\chi(n) \neq 0$ with $m \not\perp n$. Let $p = \gcd(m, n)$. Then $\chi(p)\chi(n/p) = \chi(n) \neq 0 \rightarrow \chi(p) \neq 0$. For any $r \in \mathbb{Z}$, $\chi(r)\chi(p) = \chi(rp) = \chi(rp+m) = \chi(r+m/p)\chi(p)$ so χ is (m/p) -periodic, a contradiction to the m -period hypothesis. Thus $\chi(n) \neq 0 \rightarrow m \perp n$. For any $n \perp m$, $\exists a = n^e \equiv 1 \pmod m$ so that $\chi(1) = \chi(a) = \chi(n^e) = \chi(n)^e \neq 0$ in which case $\chi(n) \neq 0$. Thus $n \perp m \rightarrow \chi(n) \neq 0$. Thus χ is a Dirichlet character of modulus m . If $m|m'm^k$, the prime divisors of m' coincides with those of m . Thus $n \perp m' \Leftrightarrow n \perp m \Leftrightarrow \chi(n) \neq 0$ and therefore χ is also m' -periodic, so χ is a Dirichlet character of modulus m' .

- If χ is a Dirichlet character of modulus m' , then χ is m' -periodic and therefore $m|m'$ by the definition of the period. Since χ is a Dirichlet character of modulus m and of modulus m' , $\forall p$, prime, $p|m \Leftrightarrow \chi(p) = 0 \Leftrightarrow p|m'$ thus the prime divisors of m, m' coincide and m' must divide some power m^k of m .

Definition 4.1.6. Let χ_1, χ_2 be Dirichlet characters of modulus m_1, m_2 respectively with $m_1|m_2$. If $\chi_2(n) = \chi_1(n)$, $\forall n \in (\mathbb{Z}/m_2\mathbb{Z})^\times$, then χ_2 is induced by χ_1 . A Dirichlet character that is not induced by any character other than itself is called primitive.

Lemma 4.1.7. A Dirichlet character χ_2 of modulus m_2 is induced by a Dirichlet character of modulus $m_1|m_2$ iff χ_2 is constant on residue classes in $(\mathbb{Z}/m_2\mathbb{Z})^\times$ that are congruent modulo m_1 . When this holds, the Dirichlet character χ_1 of modulus m_1 that induces χ_2 is uniquely determined.

pf.

- if χ_2 is induced by χ_1 , it must be constant on the residue classes in $(\mathbb{Z}/m_2\mathbb{Z})^\times$ that are congruent modulo m_1 since χ_1 is.
- there is a surjective ring homomorphism $\mathbb{Z}/m_2\mathbb{Z} \rightarrow \mathbb{Z}/m_1\mathbb{Z}$ given by reduction modulo m_1 . Suppose $u_1 \in \mathbb{Z} \bmod m_1$. Let a be the product of all primes dividing m_2/m_1 but not u_1 . Then $u_2 = u_1 + m_1a$ is not divisible by any prime $\begin{cases} p|m_1 & \text{since } u_1 \text{ is not} \\ p|(m_2/m_1) & \text{since such } p \text{ divides exactly one of } u_1 \text{ and } m_1 \end{cases}$. Thus u_2 is a unit modulo m_2 that reduces to u_1 modulo m_1 and therefore $\pi : (\mathbb{Z}/m_2\mathbb{Z})^\times \rightarrow (\mathbb{Z}/m_1\mathbb{Z})^\times$ induced by the above surjective ring homomorphism is also surjective. If χ_2 is a Dirichlet character of modulus m_2 constant on fibers of π , we can define a Dirichlet character χ_1 of modulus m_1 via $\chi_1(n_1) \equiv \chi_2(n_2)$, $\forall n_1 \in (\mathbb{Z}/m_1\mathbb{Z})^\times$ with $n_2 \in \pi^{-1}(n_1)$. Thus χ_1 induces χ_2 .

If $\exists \chi'_1$ also induces χ_2 , it must satisfy the same condition $\chi'_1(n_1) = \chi_2(n_2)$, which gives the uniqueness.

Definition 4.1.7. A Dirichlet character χ induced by $\mathbb{1}$ is called principal. For $m \in \mathbb{Z}_{++}$, $\mathbb{1}_m$ denotes the principal Dirichlet character of modulus m ; it corresponds to the trivial character of $(\mathbb{Z}/m\mathbb{Z})^\times$.

Lemma 4.1.8. Let χ be a Dirichlet character of modulus m . Then

$$\sum_{n \in \mathbb{Z}/m\mathbb{Z}} \chi(n) \neq 0 \Leftrightarrow \chi = \mathbb{1}_m$$

pf. $\chi(n) = 0$, $\forall n \notin (\mathbb{Z}/m\mathbb{Z})^\times$ and the sum over $(\mathbb{Z}/m\mathbb{Z})^\times$ is nonzero iff χ restricts to the trivial character on $(\mathbb{Z}/m\mathbb{Z})^\times$ by the orthogonality of characters.

$\rightarrow$ the principal Dirichlet characters $\mathbb{1}_m$ and $\mathbb{1}_{m'}$ necessarily coincide when $m|m'|m^k$.

Theorem 4.1.7. *Every Dirichlet character χ is induced by a primitive Dirichlet character $\tilde{\chi}$ that is uniquely determined by χ .*

pf. Define a partial ordering $\preceq$ on the set of all Dirichlet characters by defining $\chi_1 \preceq \chi_2$ iff χ_1 induces χ_2 . Then it is reflexive and transitive by the lemma 4.1.7. Let χ be a Dirichlet character of period m and consider the set $X = \{\chi' : \chi' \preceq \chi\}$. Each $\chi' \in X$ necessarily has period $m'|m$ and there is at most one χ' of period m' for each divisor m' of m by the lemma 4.1.7. Thus X is finite and nonempty ($\chi \in X$). Suppose $\chi_1, \chi_2 \in X$ with period m_1, m_2 respectively. Then $m_1, m_2 | m$ as does $m_3 = \gcd(m_1, m_2)$. Then we have a commutative diagram of surjective unit

$$\begin{array}{ccc} (\mathbb{Z}/m\mathbb{Z})^\times & \longrightarrow & (\mathbb{Z}/m_1\mathbb{Z})^\times \\ \downarrow & & \downarrow \\ (\mathbb{Z}/m_2\mathbb{Z})^\times & \longrightarrow & (\mathbb{Z}/m_2\mathbb{Z})^\times \end{array} \quad \text{by pushout. By the lemma 4.1.7., } \chi \text{ is constant}$$

on residue classes on $(\mathbb{Z}/m\mathbb{Z})^\times$ that are congruent modulo m_1, m_2 and therefore it is constant on residue classes in $(\mathbb{Z}/m\mathbb{Z})^\times$ congruent modulo m_3 as are χ_1, χ_2 . Thus there is a unique Dirichlet character χ_3 of modulus m_3 that induces χ, χ_1, χ_2 . This implies for every pair $\chi_1, \chi_2 \in X$ has a lower bound χ_3 under the partial ordering $\preceq$ that is compatible with the total ordering of X by period. Thus X contains a unique element $\tilde{\chi}$ that is minimal both w.r.t. $\preceq$ and w.r.t. the total ordering by period. By the transitivity of $\preceq$, it must be primitive.

Definition 4.1.8. *The conductor of a Dirichlet character χ is the period of the unique primitive Dirichlet character $\tilde{\chi}$ that induces χ .*

Corollary 4.1.3. *For a Dirichlet character χ of modulus m we have $\sum_{n \in \mathbb{Z}/m\mathbb{Z}} \chi(n) \neq 0$ iff χ has conductor 1.*

Corollary 4.1.4. *Let $M(m)$ be the set of Dirichlet characters of modulus m , $X(m)$ be the set of primitive Dirichlet characters of conductor dividing m and $\hat{G}(m)$ be the character group of $(\mathbb{Z}/m\mathbb{Z})^\times$. Then we have canonical bijections*

$$\begin{array}{ccccc} M(m) & \rightarrow & X(m) & \rightarrow & \hat{G}(m) \\ \chi & \mapsto & \tilde{\chi} & \mapsto & (n \mapsto (\tilde{\chi}(n))) \end{array}$$

pf. By the previous theorem, $\chi \mapsto \tilde{\chi}$ is injective. Also $\forall \tilde{\chi} \in X(m)$, it induces the character $\chi \in M(m)$ by setting $\chi(n) \equiv \tilde{\chi}(n), \forall n \in (\mathbb{Z}/m\mathbb{Z})^\times$ and extending by zero, i.e. it is surjective. The map $\begin{array}{ccc} M & \rightarrow & \hat{G}(m) \\ \chi & \mapsto & (m \mapsto \chi(m)) \end{array}$ defines a bijection and it factors through the map $\chi \mapsto \tilde{\chi}$ since $\tilde{\chi} = \chi(n), \forall n \in (\mathbb{Z}/m\mathbb{Z})^\times$.

Definition 4.1.9. *The Dirichlet L-function associated to a Dirichlet character χ is*

$$L(s, \chi) \equiv \prod_p (1 - \chi(p)p^{-s})^{-1} = \sum_{n \geq 1} \chi(n)n^{-s}$$

→ summand converges absolutely on $\text{Re}(s) > 1$ since $|\chi(n)| \leq 1$ and therefore $L(s, \chi)$ is holomorphic on $\text{Re}(s) > 1$.

e.g. for the trivial Dirichlet character $\mathbb{1}$, $L(s, \mathbb{1}) = \zeta(s)$ and for the principal character $\mathbb{1}_m$ of modulus m , induced by $\mathbb{1}$, $\zeta(x) = L(s, \mathbb{1}_m) \prod_{p|m} (1 - p^{-s})^{-1}$.

→ $L(s, \mathbb{1}_m)$ has a simple pole at $s = 1$ with residue $\text{res}_{s=1} L(s, \mathbb{1}_m) = \lim_{s \rightarrow 1^+} (s - 1) \zeta(s) \prod_{p|m} (1 - p^{-s}) = \prod_{p \nmid m} (1 - p^{-1}) = \frac{\phi(m)}{m}$.

Proposition 4.1.1. *Let χ be a nonprincipal Dirichlet character of modulus m . Then $L(s, \chi)$ extends to a holomorphic function on $\text{Re}(s) > 0$.*

pf. Define the function $T : \mathbb{R}_+ \rightarrow \mathbb{C}$ by $T(x) = \sum_{0 < n \leq x} \chi(n)$. For any $x \in \mathbb{R}_+$, $T(x + m) - T(x) = \sum_{x < n \leq x+m} \chi(n) = \sum_{x \in \mathbb{Z}/m\mathbb{Z}} \chi(n) = 0$, given χ being nonprincipal. Thus $T(x)$ is periodic modulo m and therefore bounded. Then

$L(s, \chi) = \sum_{n \geq 1} \chi(n) n^{-s} = \int_0^\infty x^{-s} dT(x)$ by writing as the Stieljtes integral
 $= x^{-s} T(x)|_0^\infty - \int_0^\infty T(x) d(x^{-s}) = 0 - \int_0^\infty T(x) (-s x^{-s-1}) dx = s \int_0^\infty T(x) x^{-s-1} dx$
 where $s \int_0^\infty T(x) x^{-s-1} dx$ is holomorphic on $\text{Re}(s) > 0$ since it is the limit of the uniformly converging sequence of functions $\phi_n(s) \equiv s \int_0^n T(x) x^{-s-1} dx$ and is thus the analytic continuation of $L(s, \chi)$ on $\text{Re}(s) > 0$.

Definition 4.1.10. *The Dedekind zeta function of a number field K is defined as*

$$\zeta_K(s) \equiv \sum_{\alpha} N(\alpha)^{-s} = \prod_{\mathbf{p}} (1 - N(\mathbf{p})^{-s})^{-1}$$

where α ranges over nonzero $\mathcal{O}_K$ -ideals and $\mathbf{p}$ ranges over nonzero prime ideals.

→ it converges absolutely on $\text{Re}(s) > 1$ since $\{\mathbf{p}|p\} \leq n \equiv [K : \mathbb{Q}]$ and $N(\mathbf{p}) \equiv [\mathcal{O}_K : \mathbf{p}] \geq p$ imply $\sum_{\mathbf{p}} |\log(1 - N(\mathbf{p})^{-s})| \leq n \sum_p |\log(1 - p^{-s})|$ and the RHS converges on $\text{Re}(s) > 1$.

Definition 4.1.11. *Let X, Y be metric spaces. A function $f : X \rightarrow Y$ is Lipschitz continuous if $\exists c \in \mathbb{R}_{++}$ s.t. $\forall x_1, x_2 \in X$, $d(f(x_1), f(x_2)) \leq c \cdot d(x_1, x_2)$.*

→ every Lipschitz continuous function is uniformly continuous, but not vice versa.

Definition 4.1.12. *A set B in a metric space X is d -Lipschitz parametrizable if it is the union of the images of a finite number of Lipschitz continuous functions $f_i : [0, 1]^d \rightarrow X$.*

Lemma 4.1.9. *Let $S \subseteq \mathbb{R}^n$ be a measurable set whose boundary $\partial S \equiv \bar{S} \setminus S^0$ is $(n-1)$ -Lipschitz parametrizable. Then $\#(tS \cap \mathbb{Z}^n) = \mu(S)t^n + O(t^{n-1})$ as $t \rightarrow \infty$, where μ is the standard Lebesgue measure on $\mathbb{R}^n$.*

pf. Partition $\mathbb{R}^n$ as the disjoint union of half-open cubes of the form $C(a_1, \dots, a_n) = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_i \in [a_i, a_i + 1)\}$, with $(a_1, \dots, a_n) \in \mathbb{Z}^n$. Let $\mathcal{C}$ be the set of all such half-open cubes C . For each $t > 0$, define
$$\begin{cases} B_0(t) = \#\{C \in \mathcal{C} : C \subseteq tS\} \\ B_1(t) \equiv \#\{C \in \mathcal{C} : C \cap tS \neq \emptyset\} \end{cases}.$$
 Then $\forall t > 0$, $B_0(t) \leq \#(tS \cap \mathbb{Z}^n) \leq B_1(t)$. We can bound $B_1(t) - B_0(t)$ by noting that each $C(a_1, \dots, a_n)$ counted by this difference contains a point $(a_1, \dots, a_n) \in \mathbb{Z}^n$ within a distance $\sqrt{n} = O(1)$ of a point in $\partial tS = t\partial S$. Let $f_1, \dots, f_m$ be Lipschitz functions $[0, 1]^{n-1} \rightarrow \partial S$ whose images cover ∂S and let $c_1, \dots, c_m$ be constants s.t. $d(f_i(x_1), f_i(x_2)) \leq c_i d(x_1, x_2)$, $\forall x_1, x_2 \in [0, 1]^{n-1}$. Then $\forall y \in \partial S$, $\exists i$ s.t. $y = f_i(x_1, \dots, x_{n-1})$. If we put $r_j = \lfloor tx_j \rfloor \in \mathbb{Z}$ so that $0 \leq x_j - r_j/t \leq 1/t$, $d(y, f_i(\frac{r_1}{t}, \dots, \frac{r_{n-1}}{t})) \leq c_i d((x_1, \dots, x_{n-1}), (\frac{r_1}{t}, \dots, \frac{r_{n-1}}{t})) < c_i \sqrt{n}/t \leq c/t$ where $c \equiv \max_i \sqrt{n} c_i$. Thus $\forall y \in \partial S$ lies within a distance c/t of a point in the set $\mathcal{P} = \{f_i(\frac{r_1}{t}, \dots, \frac{r_{n-1}}{t}) : 0 \leq r_i \leq t, 1 \leq i \leq m\}$ which has cardinality $m(t+1)^{n-1} = O(t^{n-1})$. Thus every point of $t\partial S$ is within a distance c of one of the $O(t^{n-1})$ points in $t\mathcal{P}$. The number of integer lattice points within a distance $\sqrt{n}$ of a point in $t\partial S$ is also $O(t^{n-1})$ and therefore $B_1(t) - B_0(t) = O(t^{n-1})$. Also $B_0(t) \leq \mu(tS) = t^n \mu(S) \leq B_1(t)$ and therefore the lemma follows.

Corollary 4.1.5. *Let Λ be a lattice in an $\mathbb{R}$ -vector space $V \cong \mathbb{R}^n$ and let $S \subseteq V$ be a measurable set whose boundary is $(n-1)$ -Lipschitz parametrizable. Then*

$$\#(tS \cap \Lambda) = \frac{\mu(S)}{\text{covol}(\Lambda)} t^n + O(t^{n-1})$$

pf.

- the case $\Lambda \subseteq \mathbb{Z}^n$ is followed by the previous lemma.
- for general Λ , $\exists x > 0$ s.t. $s\Lambda$ is arbitrarily close to an integer lattice, for example take s to be the lcm of all denominators appearing in rational approximations of the coordinates of the basis for Λ . This necessarily have a finite index. Then by the previous case, the corollary hold for $s\Lambda$. But $tS \cap s\Lambda = (t/s)S \cap \Lambda$ and therefore the corollary holds for any Λ .

$\rightarrow \text{covol}(\Lambda) = \mu(F)$ for any fundamental region F for Λ , so the ratio $\mu(S)/\text{covol}(\Lambda) = \mu(S)/\mu(F)$ and therefore it does not depend on the normalisation of the Haar measure μ .

By the chapter 3, the unit group $K_{\mathbb{R}}^{\times} \equiv K \otimes_{\mathbb{Q}} \mathbb{R}$ is the locally compact group $K_{\mathbb{R}}^{\times} \cong \prod_{v|\infty} K_v^{\times} \cong \prod_{v|\infty, \text{real}} \mathbb{R}^{\times} \times \prod_{v|\infty, \text{complex}} \mathbb{C}^{\times}$. There is a continuous homomorphism
$$\begin{aligned} \text{Log} : K_{\mathbb{R}}^{\times} &\rightarrow \mathbb{R}^{r+s} \\ (x_v) &\mapsto (\log \|x_v\|_v) \end{aligned}$$
 and we have an exact sequence of abelian groups

$1 \longrightarrow \mu_K \longrightarrow \mathcal{O}_K^\times \xrightarrow{\text{Log}} \Lambda_K \longrightarrow 0$ in which Λ_K is a lattice in the trace-zero hyperplane $\mathbb{R}_0^{r+s} \equiv \{x \in \mathbb{R}^{r+s} : T(x) = 0\}$. The regulator $R_K = \text{covol}(\Lambda_K)$ in $\mathbb{R}_0^{r+s}$ where it is equipped with the Euclidean measure induced by any coordinate projection $\mathbb{R}^{r+1} \rightarrow \mathbb{R}^{r+s-1}$. Then by the Dirichlet unit theorem, $\mathcal{O}_K^\times = U \otimes \mu_K$ where $U \subseteq \mathcal{O}_K^\times$ is free of rank $r + s - 1$.

Consider the principal ideals of $\mathcal{O}_K$. Then $\forall \alpha, \alpha' \in K$. $(\alpha) = (\alpha')$ iff $\alpha/\alpha' \in \mathcal{O}_K^\times$ and therefore $\{(\alpha) : N(\alpha) \leq t\} = \{\alpha \in K^\times \cap \mathcal{O}_K : N(\alpha) \leq t\}/\mathcal{O}_K^\times$. If we define $K_{\mathbb{R}, \leq t}^\times \equiv \{x \in K_\mathbb{R}^\times : N(x) \leq t\} \subseteq K_\mathbb{R}^\times \subseteq K_\mathbb{R}$, the cardinality of $\{\alpha \in K^\times \cap \mathcal{O}_K : N(\alpha) \leq t\}/\mathcal{O}_K^\times$ is equivalent to the cardinality of the finite set $(K_{\mathbb{R}, \leq t}^\times \cap \mathcal{O}_K)/\mathcal{O}_K^\times$ where the intersection takes place in $K_\mathbb{R}$ and produces a subset of $K_\mathbb{R}^\times$ that we partition into equivalence classes modulo $\mathcal{O}_K^\times$. We have a w_K to 1 map $(K_{\mathbb{R}, \leq t}^\times \cap \mathcal{O}_K)/U \rightarrow (K_{\mathbb{R}, \leq t}^\times \cap \mathcal{O}_K)/\mathcal{O}_K^\times$ where U is defined above. Then we can get the cardinality by getting the one for $(K_{\mathbb{R}, \leq t}^\times \cap \mathcal{O}_K)/U$ and divide it by w_K . For $x = (x_v) \in K_\mathbb{R}^\times$, the norm map

$$N : K_\mathbb{R}^\times \rightarrow \mathbb{R}_{++}^\times \quad x \mapsto \prod_{v \text{ real}} |x_v|_\mathbb{R} \prod_{v \text{ complex}} |x_v|_\mathbb{C}^2 \quad \text{satisfies } T(\text{Log}(x)) = \log N(x), \quad \forall x \in K_\mathbb{R}^\times.$$

Define a surjective homomorphism $\nu : K_\mathbb{R}^\times \rightarrow K_{\mathbb{R},1}^\times$ by $x \mapsto xN(x)^{-1/n}$. Then the image of $K_{\mathbb{R},1}^\times$

under the Log map is precisely the trace zero hyperplane $\mathbb{R}_0^{r+s}$ in $\mathbb{R}^{r+s}$ in which $\text{Log}(U) = \text{Log}(\mathcal{O}_K^\times) = \Lambda_K$ is a lattice. Fix a fundamental domain F for the lattice Λ_K in $\mathbb{R}_0^{r+2}$ so that $S \equiv \nu^{-1}(\text{Log}^{-1}(F))$ is a set of unique coset representatives for the quotient $K_\mathbb{R}^\times/U$. If we define $S_{\leq t} \equiv \{x \in S : N(x) \leq t\} \subset K_\mathbb{R}$, we want to estimate the cardinality of the finite set $S_{\leq t} \cap \mathcal{O}_K$. Here $tS_{\leq 1} = S_{\leq t^n}$ so we can estimate the cardinality of $S_{\leq t} = t^{1/n}S_{\leq 1}$ via the previous corollary with $S = S_{\leq 1}$ and $\Lambda = \mathcal{O}_K$ by replacing t with $t^{1/n}$, provided that the boundary of $S_{\leq 1}$ is $(n-1)$ -Lipschitz parametrisable.

CLAIM : $\partial S_{\leq 1}$ is $(n-1)$ -Lipschitz parametrisable.

pf. The Kernel of the Log map is $\{\pm 1\}^r \times U(1)^s$, where $U(1) \equiv \{z \in \mathbb{C} : z\bar{z} = 1\}$ is the unit circle in $\mathbb{C}$. Thus $K_\mathbb{R}^\times = (\mathbb{R}^\times)^r \times (\mathbb{C}^\times)^s \rightarrow \mathbb{R}^{r+s} \times \{\pm 1\}^r \times [0, 2\pi)^s$

$$x \mapsto (\text{Log} x) \times (\text{sgn} x_1, \dots, \text{sgn} x_n) \times (\arg z_1, \dots, \arg z_s)$$

is a continuous isomorphism of locally compact groups where

$$\begin{cases} \rightarrow \mathbb{R}^{r+s} & \text{Log map} \\ \rightarrow \{\pm 1\}^r & \text{vector of signs of the real components} \\ \rightarrow [0, 2\pi)^s & \text{the vector of angles s.t. } z/|z| = e^{i \arg z} \end{cases} \quad \text{The set } S_{\leq 1} \text{ consists of } 2^r$$

connected components, one for each element of $\{\pm 1\}^r$. We can parametrise each of these using n real parameters; $r + s - 1$ parameters in $[0, 1)$ that encode a point in F as an $\mathbb{R}$ -linear combination of $\text{Log} \epsilon_1, \dots, \text{Log} \epsilon_{s+r-1}$ where $\{\epsilon_i\}$ is a basis for U , and a parameter in $(0, 1]$ that encodes the n th root of the norm. Additionally $\exists s$ parameters in $[0, 1)$ that encode an element of $U(1)^s$. These define a continu-

ously differentiable bijection from the set $C = [0, 1]^{n-1} \times (0, 1] \subseteq [0, 1]^n$ s.t. ∂C is the boundary of the unit n -cube and therefore it is $(n-1)$ -Lipschitz parametrisable. Thus each component of $S_{\leq 1}$ and itself is clearly $(n-1)$ -Lipschitz parametrisable. Thus by the previous corollary, $\#S_{\leq t} \cap \mathcal{O}_K = \frac{\mu(S_{\leq 1})}{\text{covol}(\mathcal{O}_K)} (t^{1/n})^n + O((t^{1/n})^{n-1}) = \frac{\mu(S_{\leq 1})}{|D_K|^{1/2}} t + O(t^{1-1/n})$.

$$\bullet \mu(S_{\leq 1}) = 2^r (2\pi)^s R_K$$

We want to get μ on $K_{\mathbb{R}}$ the Haar measure with the parametrisation. In

terms of the standard Lebesgue measure $\begin{cases} dx & \text{on } \mathbb{R} \\ dA & \text{on } \mathbb{C} \end{cases}, \mu = (dx)^r (2dA)^s$.

For each factor of $K_{\mathbb{R}}^{\times} \cong (\mathbb{R}^{\times})^r \times (\mathbb{C}^{\times})^s$, define the maps

$$\begin{array}{ll} \mathbb{R}^{\times} & \rightarrow \mathbb{R} \times \{\pm 1\} \\ x & \mapsto (\log|x|, \text{sgn}x) \\ \pm e^l & \mapsto (l, \pm 1) \\ dx & \mapsto e^l dl \mu_{\pm 1} \end{array}$$

$$\begin{array}{ll} \mathbb{C}^{\times} & \rightarrow \mathbb{R} \times [0, 2\pi) \\ z & \mapsto (2\log|z|, \arg z) \\ e^{l/2+i\theta} & \mapsto (l, \theta) \\ adA & \mapsto 2e^{l/2}(de^{l/2})d\theta = d^l d; d\theta \end{array} \quad \text{where}$$

$$\begin{cases} dl \text{ is the Lebesgue measure on } \mathbb{R} \\ \mu_{\pm 1} \text{ is the counting measure on } \{\pm 1\} \\ d\theta \text{ is the Lebesgue measure on } [0, 2\pi) \end{cases} \quad \text{Thus there exists a new parametrisation}$$

$$\begin{array}{ll} \mathbb{R}^{r+s} & \rightarrow \mathbb{R}^{r+s-1} \times \mathbb{R} \\ x & \mapsto (x_1, \dots, x_{r+s-1}, y \equiv T(x)) \\ e^{T(x)} \mu_{\mathbb{R}^{r+s}} & \mapsto e^y \mu_{\mathbb{R}^{r+s-1}} dy \end{array}$$

denote the coordinate projection, then the Lebesgue measure of $\pi_0(F)$ in $\mathbb{R}^{r+s-1}$ is the regulator R_K . The Log map gives a bijection

$$\begin{array}{ll} R_{\leq 1} & \rightarrow F + (-\infty, 0] \left(\frac{1}{n}, \dots, \frac{1}{n}, \frac{2}{n}, \dots, \frac{2}{n} \right) \\ x = N(x)^{1/n} v(x) & \mapsto \text{Lod} v(x) + \log N(x) \left(\frac{1}{n}, \dots, \frac{1}{n}, \frac{2}{n}, \dots, \frac{2}{n} \right) \end{array}$$

Since $y = T(\text{Log}x) = \log N(x)$, we can view $S_{\leq 1}$ as an infinite union of cosets of $\text{Log}^{-1}(F)$ parametrised by $e^y = N(x) \in (0, 1]$, Then under the change of coordinate,

$$\begin{array}{ll} K_{\mathbb{R}}^{\times} & \rightarrow \mathbb{R}^{r+s-1} \times \{\pm 1\}^r \times [0, 2\pi)^s \\ S_{\leq 1} & \rightarrow \pi_0(F) \times (-\infty, 0] \times \{\pm 1\}^r \times [0, 2\pi)^s \end{array} \quad \mu(S_{\leq 1}) = \int_{-\infty}^0 e^y R_K 2^r (2\pi)^s dy = 2^r (2\pi)^s R_K.$$

$$\rightarrow \#(S_{\leq t} \cap \mathcal{O}_K) = \frac{\mu(S_{\leq 1})}{\text{covol}(\mathcal{O}_K)} (t^{1/n})^n + O((t^{1/n})^{1-\frac{1}{n}}) = \left(\frac{\mu(S_{\leq 1})}{|D_K|^{1/2}} \right) t + O(t^{1-1/n}).$$

Theorem 4.1.8. *Let K be a number field of degree n . As $t \rightarrow \infty$, the number of nonzero $\mathcal{O}_K$ -ideals α of absolute norm $N(\alpha) \leq t$ is*

$$\left(\frac{2^r (2\pi)^s h_K R_K}{w_K |D_K|^{1/2}} \right) t + O(t^{1-1/n})$$

where r, s are the number of real and complex places of K , respectively, $h_K \equiv \#cl\mathcal{O}_K$ is the class number, R_K is the regulator, $w_K = \#\mu_K$ is the number of roots of unity and $D_K \equiv disc\mathcal{O}_K$.

pf. Group the nonzero $\mathcal{O}_K$ -ideals by ideal class.

- For the trivial class, we just need to count nonzero principal ideals (α) , equivalently, the number of nonzero $\alpha \in \mathcal{O}_K$ with $N(\alpha) \leq t$, mod $\mathcal{O}_K^\times$. Dividing $\#(S_{\leq t} \cap \mathcal{O}_K) = (\frac{2^r(2\pi)^s R_K}{|D_K|^{1/2}})t + O(t^{1-1/n})$ by w_K to account for the w_K to 1 map $S_{\leq t} \cap \mathcal{O}_K \rightarrow (K_{\mathbb{R}, \leq t}^\times \cap \mathcal{O}_K)/\mathcal{O}_K^\times$, we get $\#\{(\alpha) \subseteq \mathcal{O}_K : N(\alpha) \leq t\} = (\frac{2^r(2\pi)^s R_K}{w_K |D_K|^{1/2}})t + O(t^{1-1/n})$.
- For a general ideal class $[\alpha]$, fix one with $\alpha \subseteq \mathcal{O}_K$ nonzero. Multiplication by α gives a bijection $\{\text{ideals } \beta \in [\alpha^{-1}] : N(\beta) \leq t\} \rightarrow \{\text{nonzero principal ideals } (\alpha) \subseteq \alpha : N(\alpha) \leq tN(\alpha)\}$ where the RHS is equivalent to the set $\{\text{nonzero } \alpha \in \alpha : N(\alpha) \leq tN(\alpha)\}/\mathcal{O}_K^\times$. Let $S_{[\alpha], \leq t}$ denote the set of the RHS. The above estimate and the previous corollary gives $\#S_{[\alpha], \leq t} = (\frac{2^r(2\pi)^s R_K}{w_K \text{covol}(\alpha)})tN(\alpha) + O(t^{1-1/n}) = (\frac{2^r(2\pi)^s R_K}{w_K \text{covol}(\mathcal{O}_K)N(\alpha)})tN(\alpha) + O(t^{1-1/n}) = (\frac{2^r(2\pi)^s R_K}{w_K |D_K|^{1/2}})t + O(t^{1-1/n})$ since $\text{covol}(\alpha) = N(\alpha)\text{covol}(\mathcal{O}_K)$. Then the RHS does not depend on the ideal class $[\alpha]$. Summing over the ideal classes gives $\#\{\text{nonzero ideals } \beta \subseteq \mathcal{O}_K : N(\beta) \leq t\} = \sum_{[\alpha] \in cl(\mathcal{O}_K)} \#S_{[\alpha], \leq t} = (\frac{2^r(2\pi)^s h_K R_K}{w_K |D_K|^{1/2}})t + O(t^{1-1/n})$.

Lemma 4.1.10. *Let $a_1, a_2, \dots$ be a sequence of complex numbers and let σ be a real number. Suppose $\sum_{i=1}^t a_i = O(t^\sigma)$ as $t \rightarrow \infty$. Then the Dirichlet series $\sum a_n n^{-s}$ defines a holomorphic function on $\text{Re}(s) > \sigma$.*

pf. Let $A(x) \equiv \sum_{0 < n \leq x} a_n$. Writing the Dirichlet sum as a Stieljtes integral gives $\sum a_n n^{-s} = \int_{1^-}^\infty x^{-s} dA(x) = \frac{A(x)}{x^s} \Big|_{1^-}^\infty - \int_{1^-}^\infty A(x) dx^{-s} = (0 - 0) - \int_{1^-}^\infty A(x)(-sx^{-s-1})dx = s \int_{1^-}^\infty \frac{A(x)}{x^{s+1}} dx$ converges locally uniformly on $\text{Re}(s) > \sigma$ and the lemma follows.

Lemma 4.1.11. *Let $a_1, a_2, \dots$ be a sequence of complex numbers that satisfies $\sum_{i=1}^t a_i = \rho t + O(t^\sigma)$ as $t \rightarrow \infty$ for some $\sigma \in [0, 1)$ and $\rho \in \mathbb{C}^\times$. The Dirichlet series $\sum a_n n^{-s}$ converges on $\text{Re}(s) > 1$ and has a meromorphic continuation to $\text{Re}(s) > \sigma$ that is holomorphic except for a simple pole at $s = 1$ with residue ρ .*

pf. Define $b_n \equiv a_n - \rho$. Then $b_1 + \dots + b_t = O(t^\sigma)$ and $\sum a_n n^{-s} = \rho \sum n^{-s} + \sum b_n n^{-s} = \rho \zeta(s) + \sum b_n n^{-s}$. Here $\zeta(s)$ is holomorphic on $\text{Re}(s) > 1$ and has a meromorphic continuation to $\text{Re}(s) > \sigma$ that is holomorphic except for a simple pole at 1 with residue 1. By the previous lemma, $\sum b_n n^{-s}$ is holomorphic on $\text{Re}(s) > \sigma$ and since $\sigma < 1$, it is holomorphic at $s = 1$. SO the RHS has a meromorphic continuation to $\text{Re}(s) > \sigma$ that is holomorphic except for the simple pole at $s = 1$ coming from $\zeta(s)$ and the residue at $s = 1$ is $\rho \cdot 1 + 0 = \rho$.

Theorem 4.1.9 (Analytic Class Number Formula). *Let K be a number field of degree n . The Dedekind zeta function $\zeta_K(z)$ extends to a meromorphic function on $\text{Re}(z) > 1 - \frac{1}{n}$ that is holomorphic except for a simple pole at $z = 1$ with residue*

$$\lim_{z \rightarrow 1^+} (z-1)\zeta_K(z) = \frac{2^r (2\pi)^s h_K R_K}{w_K |D_K|^{1/2}}$$

where $\begin{cases} r & \text{the number of real places of } K \\ s & \text{the number of complex places of } K \end{cases}$, R_K is the regulator, $h_K \equiv \#cl\mathcal{O}_K$, $w_K \equiv \#\mu_K$, $D_K \equiv disc\mathcal{O}_K$.

pf. $\zeta_K(z) = \sum_{\alpha} N(\alpha)^{-z} = \sum_{t \geq 1} a_t t^{-z}$ where α ranges over nonzero ideals of $\mathcal{O}_K$ and $a_t \equiv \#\{\alpha : N(\alpha) = t\}$ with $t \in \mathbb{Z}_{++}$. If we define $\rho_K \equiv \frac{2^r (2\pi)^s h_K R_K}{w_K |D_K|^{1/2}}$, by the previous theorem, $\sum a_i = \#\{\alpha : N(\alpha) \leq t\} = \rho_K t + O(t^{1-1/n})$ as $t \rightarrow \infty$. By the previous lemma with $\sigma = 1 - 1/n$, $\zeta_K(z) = \sum a_t t^{-z}$ extends to a meromorphic function on $\text{Re}(s) > 1 - 1/n$ that is holomorphic except for a simple pole at $z = 1$ with residue ρ_K .

Given any finite set of primitive Dirichlet characters, if we let m be the lcm of their conductors and consider the subgroup H of $X(m)$ they generate, we may associate to H the subfield $K \equiv \mathbb{Q}(\zeta_m)^{\phi(H)}$ where $\phi(H) \equiv \{\sigma \in \text{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) : \chi(\sigma) = 1, \forall \chi \in H\}$. Then we may regard H as the character group of $\text{Gal}(K/\mathbb{Q})$. The same applies if we replace m with any multiple m' , since $H \subseteq X(m) \subseteq X(m')$ for all $m|m'$ and we will get the same field $K \subseteq \mathbb{Q}(\zeta_m) \subseteq \mathbb{Q}(\zeta_{m'})$. Conversely, for each subfield K of a cyclotomic field $\mathbb{Q}(\zeta_m)$, there is a corresponding subgroup $H \equiv \{\chi \in X(m) : \chi(\sigma) = 1, \forall \sigma \in \text{Gal}(\mathbb{Q}(\zeta_m)/K)\}$ for which $K = \mathbb{Q}(\zeta_m)^{\phi(H)}$. Then Since $\text{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ is abelian, $K/\mathbb{Q}$ is Galois and we may view H as the character group of $\text{Gal}(K/\mathbb{Q})$.

→ **There is an one-to-one correspondence between subgroups $H \subseteq X(m)$ and subfields of $K \subseteq \mathbb{Q}(\zeta_m)$ in which H corresponds to the character group of $\text{Gal}(K/\mathbb{Q})$ and $K = \mathbb{Q}(\zeta_m)^{\phi(H)}$.**

Proposition 4.1.2. *Let p be a prime, m be a positive integer and $m' = m/p^{v_p(m)}$. Then $\mathbb{Q}(\zeta_{m'})$ is the maximal extension of $\mathbb{Q}$ in $\mathbb{Q}(\zeta_m)$ unramified at p . In particular, if p does not divide m , then $\mathbb{Q}(\zeta_m)$ is unramified at p .*

pf. The extension $\mathbb{Q}_p(\zeta_{m'})/\mathbb{Q}_p$ is unramified. Thus $\mathbb{Q}(\zeta_{m'})/\mathbb{Q}$ is unramified at p . This applies to any prime $q \neq p$ dividing m . Thus the extension $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ is ramified only at p . Then there are no nontrivial unramified extensions of $\mathbb{Q}$, so every subfield of $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ that properly contains $\mathbb{Q}$ is ramified at p . Now $\mathbb{Q}(\zeta_m)$ is the compositum of $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ and $\mathbb{Q}(\zeta_{m'})$, which intersect in $\mathbb{Q}$, so any nontrivial

extension $\mathbb{Q}(\zeta_{m'})$ in $\mathbb{Q}(\zeta_m)$ contains a subfield of $\mathbb{Q}(\zeta_{p^{v_p(m)}})$ properly containing $\mathbb{Q}$ which must be ramified at p . Thus the proposition follows.

Theorem 4.1.10. *Let $H \subseteq X(m)$ be a group of primitive Dirichlet characters and let $K = \mathbb{Q}(\zeta_m)^{\phi(H)}$ be the corresponding subfield of $\mathbb{Q}(\zeta_m)$ with $\phi(H)$ defined as above. Then*

$$\zeta_K(x) = \prod_{\chi \in H} L(s, \chi)$$

pf. $\left\{ \begin{array}{l} \zeta_K(x) = \prod_{\mathbf{p}} (1 - N(\mathbf{p})^{-s})^{-1} = \prod_p \prod_{\mathbf{p}|p} (1 - N(\mathbf{p})^{-s})^{-1} \\ \prod_{\chi \in H} L(s, \chi) = \prod_{\chi \in H} \prod_p (1 - \chi(p)p^{-s})^{-1} = \prod_p \prod_{\chi \in H} (1 - \chi(p)p^{-s})^{-1} \end{array} \right.$. Since $K/\mathbb{Q}$ is Galois, $[K : \mathbb{Q}] = e_p f_p g_p$. $\prod_{\mathbf{p}|p} (1 - N(\mathbf{p})^{-s}) = (1 - (p^{f_p})^{-s})^{g_p} = (1 - (p^{-s})^{f_p})^{g_p}$ which does not change if we replace K with the maximal subfield K' of K in which p is unramified. Also on the $\prod_{\chi \in H} (1 - \chi(p)p^{-s})$, any $\chi \in H$ with conductor divisible by p , $\chi(p) = 0$ and therefore we can replace H with the subgroup H' of Dirichlet characters with conductors prime to p . Then by the previous proposition, $K' = \mathbb{Q}(\zeta_m)^{\phi(H')}$. Thus wlog assume $p \nmid m$ so K is unramified at p and $\#H = [K : \mathbb{Q}] = f_p g_p$. Since $K/\mathbb{Q}$ is abelian, and unramified at p , the Artin map gives a Frobenius element σ_p corresponding to the Frobenius automorphism $x \mapsto x^p$ of the residue field, which by definition has order f_p , so σ_p has order f_p in $\text{Gal}(K/\mathbb{Q})$. Viewing H as a character group of $\text{Gal}(K/\mathbb{Q})$, the map $\chi \mapsto \chi(\sigma_p)$ defines a surjective homomorphism from H to the group of f_p th roots of unity $\alpha \in U(1)$ and the kernel of this map has cardinality $\#H/f_p = g_p$. Thus $\prod_{\chi \in H} (1 - \chi(p)p^{-s}) = \prod_{\alpha^{f_p}=1} (1 - \alpha p^{-s})^{g_p} = (1 - (p^{-s})^{f_p})^{g_p}$ by the identity $\prod_{\alpha^{f_p}=1} (1 - \alpha T) = 1 - T^{f_p} \in \mathbb{C}[T]$. Thus they are equivalent.

Theorem 4.1.11. *Let ψ be any nonprincipal Dirichlet character. Then $L(1, \psi) \neq 0$.*

pf. Let ψ be a nonprincipal Dirichlet character, say of modulus m . Then ψ is induced by a nontrivial primitive Dirichlet character $\tilde{\psi}$ of conductor $\tilde{m}$ dividing m . The L -functions of $\psi, \tilde{\psi}$ differ at only finitely many Euler factors $(1 - \psi(p)p^{-s})^{-1}$, which corresponds to primes p dividing $m/\tilde{m}$, and these factors are clearly nonzero at $s = 1$, given $p > 1$. Thus assume wlog $\psi = \tilde{\psi}$ is primitive. Let K be the m th cyclotomic field $\mathbb{Q}(\zeta_m)$. Then by the previous theorem, $\zeta_K(x) = \prod_{\chi} L(s, \chi)$ where χ ranges over the primitive Dirichlet characters of conductor dividing m , including ψ . By the analytic class number formula, the LHS has a simple pole at $s = 1$, and the same must be true for the RHS. Then $\text{ord}_{s=1} \zeta_K(s) = \text{ord}_{s=1} \prod_{\chi} L(s, \chi) \Leftrightarrow -1 = \text{ord}_{s=1} L(s, \mathbb{1}) \prod_{\chi \neq \mathbb{1}} L(s, \chi) = \text{ord}_{s=1} \zeta(s) \prod_{\chi \neq \mathbb{1}} L(s, \chi) = -1 + \sum_{\chi \neq \mathbb{1}} \text{ord}_{s=1} L(s, \chi)$. For nonprincipal χ , $L(s, \chi)$ is holomorphic on $\text{Re}(s) > -$ thus $\text{ord}_{s=1} L(s, \chi) \geq 0$, $\forall \chi$ appearing in the sum, which can therefore be zero iff every term is zero. So $L(1, \chi) \neq 0$ for every nontrivial primitive Dirichlet character χ of conductor dividing m , including ψ .

Theorem 4.1.12 (Dirichlet). *For all coprime integers a, m , there are infinitely many primes $p \equiv a \pmod{m}$.*

pf. Construct the indicator function $\frac{1}{\phi(m)} \sum_{\chi \in X(m)} \chi(p/a) = \begin{cases} 1 & p \equiv a \pmod{m} \\ 0 & \text{otherwise} \end{cases}$ where p/a is computed modulo m and χ ranges over primitive Dirichlet characters of conductor dividing m , which we identify with the character group of $(\mathbb{Z}/m\mathbb{Z})^\times$ via the corollary 4.1.4. As $s \rightarrow 1^+$,

$$\begin{aligned} \sum_{p \equiv a \pmod{m}} p^{-s} &= \sum_p p^{-s} \frac{1}{\phi(m)} \sum_{\chi \in X(m)} \chi(p/a) = \sum_{\chi \in X(m)} \frac{\chi(1/a)}{\phi(m)} \sum_p \chi(p) p^{-s} \\ &= \sum_{\chi \in X(m)} \frac{\chi(1/a)}{\phi(m)} (\log L(s, \chi) + O(1)) = \frac{\log \zeta(s)}{\phi(m)} + \sum_{\chi \in X(m), \chi \neq 1} \frac{\chi(1/a)}{\phi(m)} \log L(s, \chi) + O(1) \end{aligned}$$

So long as χ is not principal, $L(1, \chi) \neq 0$ by the previous theorem. Then $\log L(s, \chi) = O(1)$ as $s \rightarrow 1^+$ and therefore $\sum_{p \equiv a \pmod{m}} p^{-s} = \frac{\log \zeta(s)}{\phi(m)} + O(1)$ is unbounded as $s \rightarrow 1^+$, since $\zeta(s)$ is. Moreover, by the Merten's second theorem implies $\sum_{p \leq x, p \equiv a \pmod{m}} \frac{1}{p} \sim \frac{\log \log x}{\phi(m)}$ which proves there are infinitely many primes $p \equiv a \pmod{m}$.

Definition 4.1.13. *The Dirichlet density of a set of primes S is given by*

$$d(S) \equiv \lim_{s \rightarrow 1^+} \frac{\sum_{p \in S} p^{-s}}{\sum_p p^{-s}}$$

defined whenever this limit exists.

$$\Leftrightarrow \delta(S) \equiv \lim_{x \rightarrow \infty} \frac{\#\{p \leq x: p \in S\}}{\#\{p \leq x\}} \text{ the natural density.}$$

The Dirichlet density of the set $S = \{p \equiv a \pmod{m}\}$ is $d(S) = \lim_{s \rightarrow 1^+} \frac{\sum_{p \in S} p^{-s}}{\sum_p p^{-s}} = \lim_{s \rightarrow 1^+} \frac{\log \zeta(s)/\phi(m)}{\log \zeta(s)} = \frac{1}{\phi(m)}$, thus the primes are equidistributed modulo m in the sense that $\forall a \perp m$, $\sum_{p \leq x, p \equiv a \pmod{m}} \frac{1}{p} \sim \frac{1}{\phi(m)} \sum_{p \leq x} \frac{1}{p} \sim \frac{1}{\phi(m)} \log \log x$.

4.1.3 Kronecker-Weber Theorem

Suppose $K/\mathbb{Q}$ is any arbitrary finite abelian extension. Then the character group of $\text{Gal}(K/\mathbb{Q})$ correspond to a group of Dirichlet characters and we can factor the Dedekind zeta function $\zeta_K(x)$ as a product of Dirichlet L -functions since every finite abelian extension of $\mathbb{Q}$ lies in a cyclotomic field by the Kronecker-Weber theorem.

Theorem 4.1.13 (Kronecker-Weber). *Every abelian extension of $\mathbb{Q}$ is contained in $\mathbb{Q}(\zeta_n)$ for some n , where ζ_n is a primitive n th root of unity.*

pf.

CLAIM : Every finite abelian extension of $\mathbb{Q}_p$ lies in a cyclotomic field $\mathbb{Q}_p(\zeta_m)$.

pf.

- When $K/\mathbb{Q}_p$ of prime power degree l , $l \neq p$

Let F be the maximal unramified extension of $\mathbb{Q}_p$ in K . Then $F = \mathbb{Q}_p(\zeta_n)$ for some $n \in \mathbb{Z}_{\geq 1}$. The extension K/F is totally ramified, and it must be tamely ramified, given the hypothesis $l \neq p$. Then $K = F(\pi^{1/e})$ for some uniformiser π , $e = [K : F]$. We may assume $\pi = -pu$, $u \in \mathcal{O}_K^\times$ since $F/\mathbb{Q}_p$ is unramified. If $q|p$ is the maximal of $\mathcal{O}_F$, the valuation v_q extends v_p with index 1, so $v_q(-pu) = v_p(-p) = 1$. Then the field $K = F(\pi^{1/e})$ lies in the compositum of $F((-p)^{1/e})$ and $F(u^{1/e})$. The extension $F(u^{1/e})/F$ is unramified since $v_q(\text{disc}(x^e - u)) = 0$, $p \nmid e$, so $F(u^{1/e})/\mathbb{Q}_p$ is unramified and $F(u^{1/e}) = \mathbb{Q}_p(\zeta_k) \rightarrow K(u^{1/e}) = K \cdot \mathbb{Q}_p(\zeta_k)$ is a compositum of abelian extension so $K(u^{1/e})/\mathbb{Q}_p$ is abelian and it contains the subextension $\mathbb{Q}_p((-p)^{1/e})/\mathbb{Q}_p$ which must be Galois, since it lies in an abelian extension and totally ramified, since it is an Eisenstein extension. The field $\mathbb{Q}_p((-p)^{1/e})$ contains ζ_e and is totally ramified, but $\mathbb{Q}_p(\zeta_e)/\mathbb{Q}_p$ is unramified, given $p \nmid e$, so $\mathbb{Q}_p(\zeta_e) = \mathbb{Q}_p$. Thus $e|(p-1)$ and $\mathbb{Q}_p((-p)^{1/e}) \subseteq \mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\zeta_p)$. Thus $F((-p)^{1/e}) = F \cdot \mathbb{Q}_p((-p)^{1/e}) \subseteq \mathbb{Q}_p(\zeta_n) \cdot \mathbb{Q}_p(\zeta_p) \subseteq \mathbb{Q}_p(\zeta_{np})$. Thus $K \subseteq \mathbb{Q}_p(\zeta_{knp})$.

For any prime p , $\mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\zeta_p)$.

pf. Let $\alpha = (-p)^{1/(p-1)}$. Then it is a root of the Eisenstein polynomial $x^{p-1} + p$, so extension $\mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\alpha)$ is totally ramified of degree $p-1$ and α is a uniformizer. Let $\pi = \zeta_p - 1$. The minimal polynomial of π is $f(x) \equiv \frac{(x+1)^p - 1}{x} = x^{p-1} + px^{p-2} + \dots + p$, which is Eisenstein, so $\mathbb{Q}_p(\pi) = \mathbb{Q}_p(\zeta_p)$ is also totally ramified of degree $p-1$ and π is a uniformizer. We have $u \equiv -\pi^{p-1}/p \equiv 1 \pmod{\pi}$ so u is a unit in $\mathbb{Q}_p(\zeta_p)$. Let $g(x) = x^{p-1} - u$, $g(1) \equiv 0 \pmod{\pi}$, $g'(1) \equiv p-1 \not\equiv 0 \pmod{\pi}$, so by Hensel's Lemma I, we can lift 1 to a root β of $g(x)$ in $\mathbb{Q}_p(\zeta_p)$. Then $p\beta^{p-1} = pu = -\pi^{p-1} \rightarrow (\pi/\beta)^{p-1} + p = 0$ and therefore $\pi/\beta \in \mathbb{Q}_p(\zeta_p)$ is a root of the minimal polynomial of α . Since $\mathbb{Q}_p(\zeta_p)$ is Galois, this implies $\alpha \in \mathbb{Q}_p(\zeta_p)$, given $\mathbb{Q}_p(\alpha), \mathbb{Q}_p(\zeta_p)$ having degree $p-1$ s.t. two fields coincide.

- $l = p > 2$

There are two candidates for K , the cyclotomic field $\mathbb{Q}_p(\zeta_{p^{p^r}-1})$ which is an unramified extension of degree p^r and the index $p-1$ subfield of the cyclotomic field $\mathbb{Q}_p(\zeta_{p^{r+1}})$ which is a totally ramified extension of degree p^r . If K is contained in the compositum of these two fields, then $K \subseteq \mathbb{Q}_p(\zeta_m)$, $m = (p^{p^r} - 1)p^{r+1}$ and the claim follows. Otherwise, the field $K(\zeta_m)$ is a Galois

extension of $\mathbb{Q}_p$ with $\text{Gal}(K(\zeta_m)/\mathbb{Q}_p) \cong \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/p^s\mathbb{Z}$, where the first factor comes from the Galois group of $\mathbb{Q}_p(\zeta_{p^{p^r-1}})$, the second two factors come from the Galois group of $\mathbb{Q}_p(\zeta_{p^{r+1}})$ and the last factor comes from the hypothesis $K \not\subseteq \mathbb{Q}_p(\zeta_m)$, so $\text{Gal}(K(\zeta_m)/\mathbb{Q}_p(\zeta_m))$ is nontrivial and must have order $p^s, s \geq 1$. The abelian group $\text{Gal}(K(\zeta_m)/\mathbb{Q}_p)$ has a quotient isomorphic to $(\mathbb{Z}/p\mathbb{Z})^3$ and the subfield of $K(\zeta_m)$ corresponding to this quotient is an abelian extension of $\mathbb{Q}_p$ with Galois group $(\mathbb{Z}/p\mathbb{Z})^3$, which is impossible, given it is not cyclic.

For odd p , every totally wildly ramified Galois extension of $\mathbb{Q}_p$ is cyclic.

pf.

If $\text{Gal}(K/\mathbb{Q}_p) \cong (\mathbb{Z}/p\mathbb{Z})^3$, $G \equiv \text{Gal}(K/\mathbb{Q}_p)$ as the internal direct sum of the inertial subgroup $I \leq G$ and a cyclic subgroup $H \leq G$, since L^I is an unramified, and hence cyclic extension of $\mathbb{Q}_p$ with Galois group isomorphic to $G/I \cong H$. But then L^H is totally wildly ramified abelian extension of $\mathbb{Q}_p$ whose Galois group G/H is not cyclic, a contradiction.

- $l = p = 2$

The unramified cyclotomic field $\mathbb{Q}_2(\xi_{2^{2^r-1}})$ has Galois group $\mathbb{Z}/2^r\mathbb{Z}$, and the totally ramified cyclotomic field $\mathbb{Q}_2(\xi_{2^{r+2}})$ has Galois group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^r\mathbb{Z}$. Let $m = (2^{2^r} - 1)2^{r+2}$. If K is not contained in $\mathbb{Q}_2(\xi_m)$, $\text{Gal}(K(\xi_m)/\mathbb{Q}_2) \cong \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}/2^r\mathbb{Z})^2 \times \mathbb{Z}/2^s\mathbb{Z} \quad 1 \leq s \leq r$
 $(\mathbb{Z}/2^r\mathbb{Z})^2 \times \mathbb{Z}/2^s\mathbb{Z} \quad 2 \leq s \leq r$ Thus it admits a quotient isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$ or $(\mathbb{Z}/4\mathbb{Z})^3$. Then since no extension of $\mathbb{Q}_2$ has either of these Galois groups, K must lie in $\mathbb{Q}_2(\xi_m)$.

*** No extension of $\mathbb{Q}_2$ has Galois group isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$ or $(\mathbb{Z}/4\mathbb{Z})^3$**

pf. There are 7 quadratic extensions of $\mathbb{Q}_2$, which has no Galois group $(\mathbb{Z}/2\mathbb{Z})^4$ since this group has 15 subgroups of index 2, whose fixed fields would yield 15 distinct quadratic extensions of $\mathbb{Q}_2$.

Also there are only finitely many extensions of $\mathbb{Q}_2$ of any fixed degree d and these can be enumerated by considering Eisenstein polynomials in $\mathbb{Q}_2[x]$ of degrees dividing d up to an equivalence relation by Krasner's lemma. Then no extension of $\mathbb{Q}_2$ has Galois group $(\mathbb{Z}/4\mathbb{Z})^3$, since it has 28 subgroups whose fixed fields would yield 28 distinct cyclic quartic extensions of $\mathbb{Q}_2$.

4.2 Cohomologies and Local Class Field Theory

4.2.1 Cohomologies

Lemma 4.2.1. For each $x \in L$,
$$\begin{array}{ccc} H_x : L & \rightarrow & L \\ y & \mapsto & x \cdot \sigma(y) \end{array}$$
 which is a linear map of K -vector spaces. Then the characteristic polynomial of H_x is $t^n - N(x) \in K[t]$, where we have normalised the definition of the characteristic polynomial to be monic.

pf. H_x^n corresponds to multiplication by $N(x)$ since $H_x^n(y) = x \cdot \sigma(x \cdot \sigma(x \dots \sigma(y) \dots)) = x \cdot \sigma(x) \dots \sigma^{n-1}(x) \sigma^n(y) = N(x)y$ for any $y \in L$. Thus the minimal polynomial of H_x divides $t^n - N(x)$. The minimal polynomial of the linear operator T always divides its characteristic polynomial, which has degree n . Since $\lambda \in \bar{K}$, $L \rightarrow L \otimes_K \bar{K} \cong \prod_{g \in G} \bar{K}$ so that we can work on this. We can compute the characteristic polynomial after extension of scalars
$$\begin{array}{ccc} L \otimes_K \bar{K} & \rightarrow & \prod_{i=0}^{n-1} \bar{K} \\ a \otimes b & \mapsto & ((\sigma^i a) \cdot b)_i \end{array}$$
. This extends noncanonically to an automorphism of $\bar{K}$. The map $\sigma \otimes id : L \otimes_K \bar{K} \rightarrow L \otimes_K \bar{K}$ corresponds to the permutation of coordinates and there exists a map
$$\begin{array}{ccc} \mu_x \otimes id : L \otimes_K \bar{K} & \rightarrow & L \otimes_K \bar{K} \\ (y_0, \dots, y_{n-1}) & \mapsto & (xy_0, \dots, \sigma^{n-1}xy_{n-1}) \end{array}$$
 where μ_x denote the multiplication by x . Then for any eigenvalue $\lambda \in \bar{K}$ of H_x with corresponding eigenvector $(y_0, \dots, y_{n-1})$, $H_x(y) = (xy_1, (\sigma x)y_2, \dots, (\sigma^{n-1}x)y_0) = (\lambda y_0, \dots, \lambda y_{n-1}) \Leftrightarrow y_i = \frac{\lambda^i}{x \dots \sigma^{i-1}x} y_0$ i.e. all coordinates are uniquely determined by y_0 , so $(y_0, \dots, y_{n-1}) = y_0(1, \frac{\lambda}{x}, \dots, \frac{\lambda^{n-1}}{\prod \sigma^i x})$. So the eigenspace has dimension one as desired and this is valid if $\frac{\lambda^n}{x \dots \sigma^{n-1}x} = \frac{\lambda^n}{N(x)} = 1$ which is indeed the case.

Definition 4.2.1. A cochain complex X of abelian groups is a sequence

$$\dots \longrightarrow X^{-1} \xrightarrow{d^{-1}} X^0 \xrightarrow{d^0} X^1 \xrightarrow{d^1} \dots \text{ s.t. the differential satisfies } d^{i+1}d^i = 0, \forall i.$$

Definition 4.2.2. A map f s.t. the diagram

$$\begin{array}{ccccccc} \dots & \longrightarrow & X^{-1} & \xrightarrow{d^{-1}} & X^0 & \xrightarrow{d^0} & X^1 \longrightarrow \dots \\ & & & & \downarrow f^0 & & \downarrow f^1 \\ \dots & \longrightarrow & Y^{-1} & \xrightarrow{d^{-1}} & Y^0 & \xrightarrow{d^0} & Y^1 \longrightarrow \dots \end{array}$$

commutes is a map of complexes. Then f induces a map of cohomologies because both the kernel and image of the differentials in $X^\bullet$ are preserved in $Y^\bullet$ by commu-

tativity. A map h as in the following diagram

$$\begin{array}{ccccccc} \dots & \longrightarrow & X^{-1} & \xrightarrow{d^{-1}} & X^0 & \xrightarrow{d^0} & X^1 \longrightarrow \dots \\ & \swarrow & \downarrow f^{-1} & \swarrow h^0 & \downarrow f^0 & \swarrow h^1 & \downarrow f^1 \\ \dots & \longrightarrow & Y^{-1} & \xrightarrow{d^{-1}} & Y^0 & \xrightarrow{d^0} & Y^1 \longrightarrow \dots \end{array}$$

s.t. $dh + hd = f$ is a nullhomotopy of f .

Definition 4.2.3. The homotopy cokernel or cone $hCoker(f) = Cone(f)$ has the universal property that maps of chain complexes $hCoker(f) \rightarrow Z$ are equivalent to

$$\left\{ \begin{array}{l} Y \rightarrow Z \text{ a map} \\ X \rightarrow Z \text{ a nullhomotopy} \end{array} \right. \text{ s.t. } \begin{array}{ccc} X & \xrightarrow{\quad} & Z \\ \downarrow f & \nearrow & \\ Y & & \end{array} \text{ commutes.}$$

CLAIM : If $f : X \rightarrow Y$ is a map of complexes, then $hCoker(f)$, characterised by the universal property that maps $hCoker(f) \rightarrow Z$ of chain complexes are equivalent to $\left\{ \begin{array}{l} g : Y \rightarrow Z \text{ a map} \\ h = g \circ f : Z \rightarrow Z \text{ a nullhomotopy} \end{array} \right.$ exist.

$$\text{pf. } hCoker(f) \text{ is } \dots X^0 \oplus Y^{-1} \longrightarrow X^1 \oplus Y^0 \longrightarrow \dots \text{ s.t. } \begin{array}{ccc} d : X^{i+1} \oplus Y^i & \rightarrow & X^{i+2} \oplus Y^{i+1} \\ \begin{bmatrix} x \\ y \end{bmatrix} & \mapsto & \begin{bmatrix} -dx \\ f(x) + dy \end{bmatrix} \end{array}$$

$$\rightarrow \begin{bmatrix} -d & 0 \\ f & d \end{bmatrix} \begin{bmatrix} -d & 0 \\ f & d \end{bmatrix} = \begin{bmatrix} d^2 & 0 \\ -fd + df & d^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ implies } d^2 = 0.$$

$$\begin{array}{ccccccc} \dots & \longrightarrow & X^{i+1} \oplus Y^i & \longrightarrow & X^{i+2} \oplus Y^{i+1} & \longrightarrow & \dots \\ & & \downarrow & & \downarrow & & \\ \dots & \longrightarrow & Z^i & \longrightarrow & Z^{i+1} & \longrightarrow & \dots \end{array}$$

Suppose we have a map $hCoker(f) \rightarrow Z$ s.t.

commutes. If we write such a map as $(x, y) \mapsto h(x) + g(y)$, then $dh(x) + dg(y) = d(h(x) + g(y)) = h(-dx) + g(f(x) + dy) = -hdx + gfx + gdy$. Taking $x = 0$ gives $dg = gd$ so $hd + dh = g \circ f$ a nullhomotopy of $g \circ f$ as desired and the universal property satisfied.

$\rightarrow X \rightarrow Y \rightarrow hCoker(f)$ is **canonically null-homotpic**.

e.g. Let $\begin{cases} X \equiv (\dots \rightarrow 0 \rightarrow A \rightarrow 0 \rightarrow \dots) \\ Y \equiv (\dots \rightarrow 0 \rightarrow B \rightarrow 0 \rightarrow \dots) \end{cases}$ for finite abelian groups A, B in degree 0.

Let $f : A \rightarrow B$. Then $hCoker(f) = (\dots \rightarrow A \xrightarrow{f} B \rightarrow 0 \rightarrow \dots)$ with B in degree 0. Then

$$\begin{cases} H^0 hCoker(f) = Coker(f) \\ H^{-1} hCoker(f) = \ker(f) \end{cases}.$$

Lemma 4.2.2. For all maps $f : X \rightarrow Y$, the sequence $H^i X \rightarrow H^i Y \rightarrow H^i hCoker(f)$ is exact for all i .

pf. The composition is zero since $X \rightarrow Y \rightarrow hCoker(f)$ is nullhomotopic. Let $y \in Y^i$ s.t. $dy = 0$ and suppose its image in $H^i hCoker(f)$ is zero so that $\begin{bmatrix} 0 \\ y \end{bmatrix} =$

$$\begin{bmatrix} -d & 0 \\ f & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -d\alpha \\ f(\alpha) + d\beta \end{bmatrix}, \exists \left\{ \begin{array}{l} \alpha \in X^i \text{ with } d\alpha = 0 \\ \beta \in Y^{i-1} \end{array} \right. . \text{ Then } f(\alpha) + d\beta = y \text{ implies } f(\alpha) = y \text{ in } H^i Y \text{ so the exactness follows.}$$

There is a notion of the homotopy kernel $\mathbf{hKer}(f)$ defined by the universal property that maps $Z \rightarrow \mathbf{hKer}(f)$ are equivalent to

$\left\{ \begin{array}{l} Z \rightarrow X \\ Z \rightarrow X \rightarrow Y \end{array} \right\}$ a map $\rightarrow X \rightarrow_f Y \rightarrow \mathbf{hCoker}(f)$ nullhomotopy of the composition. In particular, $\mathbf{hKer}(f) = \mathbf{hCoker}(f)[-1]$. $\rightarrow X \rightarrow_f Y \rightarrow \mathbf{hCoker}(f)$ is nullhomotopic so there exists a canonical map $X \rightarrow \mathbf{hKer}(Y \rightarrow \mathbf{hCoker}(f))$ where we use the latter term as the mapping cylinder.

Definition 4.2.4. The coinvariants of A are $A_G \equiv A / \sum_{g \in G} (g - 1)A$, which satisfies the universal property that it is the maximal quotient of A with $gx = x$ holding for all $x \in A$, $g \in G$.

Definition 4.2.5. Let X, Y be complexes of A -modules. Then the mapping complex $\underline{\mathbf{hom}}_A(X, Y)$ is the complex of abelian groups defined by $\underline{\mathbf{hom}}_A^i(X, Y) \equiv \prod_{j \in \mathbb{Z}} \mathbf{hom}_A(X^j, Y^{j+i})$, with differential $d^i f \equiv df - (-1)^i fd$. This can be visualised as

$$\begin{array}{ccccccc} \dots & \longrightarrow & \prod_{i \in \mathbb{Z}} \mathbf{hom}(X^i, Y^{i-1}) & \xrightarrow{d^i} & \prod_{i \in \mathbb{Z}} \mathbf{hom}(X^i, Y^i) & \xrightarrow{d^i} & \prod_{i \in \mathbb{Z}} \mathbf{hom}(X^i, Y^{i+1}) & \longrightarrow & \dots \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim & & \\ & & \underline{\mathbf{hom}}^{-1}(X, Y) & & \underline{\mathbf{hom}}^0(X, Y) & & \underline{\mathbf{hom}}^1(X, Y) & & \end{array}$$

$\rightarrow \forall X, Y$ complexes of A -modules, $\underline{\mathbf{hom}}_A(X, Y)$ is a complex.

Definition 4.2.6. A complex P of A -modules is projective if for every acyclic complex Y of A -modules, $\underline{\mathbf{hom}}_A(P, Y)$ is also acyclic.

Definition 4.2.7. An A -module P^i is projective as an A -module if any of the following holds

1. $\mathbf{hom}_A(P^i, -)$ preserves surjections
2. $\mathbf{hom}_A(P^i, -)$ is an exact functor
3. P^i is a direct summand of a free module
4. given any surjection $N \rightarrow M$, every map $P^i \rightarrow M$ of A -modules lifts to a map

$$\begin{array}{ccccc} & & N & & \\ & \nearrow & \downarrow & & \\ P^i \rightarrow N \text{ s.t. } & P^i & \longrightarrow & M & \text{commutes} \\ & & & \downarrow & \\ & & & 0 & \end{array}$$

An A -module P is projective as an A -module iff it is projective as a complex in degree 0.

pf.

- if P is projective as a complex in degree 0, let $f : N \rightarrow M$ be a surjection and form the acyclic complex $X \equiv (\cdots \rightarrow 0 \rightarrow \ker f \rightarrow N \rightarrow M \rightarrow 0 \rightarrow 0)$. Then $\underline{\text{hom}}_A(P, X)$ is $\cdots \rightarrow 0 \rightarrow \text{hom}_A(P, \ker f) \rightarrow \text{hom}_A(P, N) \rightarrow \text{hom}_A(P, M) \rightarrow 0 \rightarrow \cdots$ so $\text{hom}_A(P^i, -)$ preserves surjections and therefore P is projective as an A -module.
- if $X \equiv (\cdots \rightarrow X^1 \rightarrow X^0 \rightarrow X^{-1} \rightarrow \cdots)$ is an acyclic complex and P is projective as an A -module, then $\underline{\text{hom}}_A(P, X) = (\cdots \rightarrow \text{hom}_A(P, X^{-1}) \rightarrow \text{hom}_A(P, X^0) \rightarrow \text{hom}_A(P, X^1) \rightarrow \cdots)$ which is acyclic (if $X^{i-1} \rightarrow \ker(X^i \rightarrow X^{i+1})$ is a surjection, then $\text{hom}(P, X^{i-1}) \rightarrow \text{hom}(P, \ker(X^i \rightarrow X^{i+1})) = \ker(\text{hom}(P, X^i) \rightarrow \text{hom}(P, X^{i+1}))$ is a surjection since $\text{hom}(P, -)$ is exact so it preserves kernels by assumption. Thus $\underline{\text{hom}}_A(P, X)$ is also acyclic and P is projective as a complex in degree 0.

Lemma 4.2.3. *If P is a complex of A -modules with $P^i = 0$ for all $i \gg 0$ and P^i is projective as an A -module for all i , then P is projective as a complex.*

pf. Let Y be an acyclic complex of A -modules.

- **CLAIM : Every map $P \rightarrow Y$ is nullhomotopic.**

pf. Let $f : P \rightarrow Y$. Construct a nullhomotopy h of f by descending induction.

For $i \gg 0$, $P^i = 0$ so $f^i = 0 \rightarrow h^i \equiv 0$.

Assume we have maps $h : P^j \rightarrow Y^{j-1}$, $\forall j > i$ so that

$$\begin{array}{ccccccc} \cdots & \xrightarrow{d} & P^{i-1} & \xrightarrow{d} & P^i & \xrightarrow{d} & P^{i+1} \xrightarrow{d} \cdots \\ & & \downarrow f & \swarrow ? & \downarrow f & \swarrow h & \downarrow f \\ \cdots & \xrightarrow{d} & Y^{i-1} & \xrightarrow{d} & Y^i & \xrightarrow{d} & Y^{i+1} \xrightarrow{d} \cdots \end{array} \quad . \forall x \in P^i, (d(f-hd))(x) = ((df -$$

$(f-hd)d)(x) = (df - fd)(x) = 0$ by the inductive hypothesis. Since $(f-hd) : P^i \rightarrow \ker(d : Y^i \rightarrow Y^{i+1})$ by the above statement. There is a surjection $d : Y^{i-1} \rightarrow \ker(d : Y^i \rightarrow Y^{i+1})$. Therefore the map $f-hd$ lifts to a map $h : P^i \rightarrow Y^{i-1}$ s.t. $dh = f-hd$ as P^i is projective by hypothesis. Thus $hd + df = f$ and h defines a nullhomotopy of f for i .

By induction, the claim follows.

Then $H^0(\underline{\text{hom}}_A(P, A)) = \{P \rightarrow Y\} = 0$ by the previous claim and $H^i = (\underline{\text{hom}}_A(P, Y)) = H^0(\underline{\text{hom}}_A(P, Y[i])) = 0$ as $Y[i]$ is also acyclic for each i . Thus the cohomologies vanish for each i and therefore $\underline{\text{hom}}_A(P, Y)$ is acyclic, i.e. P is projective.

Definition 4.2.8. *A projective resolution of X , an A -module complex, is $P \xrightarrow{qis} X$ where P is a projective complex.*

$$\rightarrow \underline{\text{hom}}_A^{der}(X, Y) \equiv \underline{\text{hom}}_A(P, Y) \text{ for } P \xrightarrow{qis} X \text{ a projective resolution.}$$

e.g. When $A \equiv \mathbb{Z}[G]$, choose any projective resolution $P \xrightarrow{qis} \mathbb{Z}$, and define $X^{hG} \equiv \underline{\text{hom}}_G(P, X)$. Then if X is acyclic, X^{hG} is acyclic and X^{hG} commutes with cones. Also if $X = (\cdots \rightarrow 0 \rightarrow M \rightarrow 0 \rightarrow \cdots)$ is in degree 0 only, then $H^0(X^{hG}) = M^G$.

Proposition 4.2.1. *Let X be a complex of A -modules and suppose X is bounded above. Then there exists a projective resolution $P \xrightarrow{qis} X$.*

pf. Wlog, assume X is bounded above at index 0. Let P^0 be a free module surjecting onto X^0 via a map α^0 . Then take P^{-1} to be a free module surjecting

$$\begin{array}{ccccccc} P^{-1} & \longrightarrow & P^0 & \longrightarrow & 0 & \longrightarrow & \cdots \\ \vdots & & \downarrow \alpha_0 & & \downarrow & & \\ X^{-1} & \longrightarrow & X^0 & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

onto $P^0 \times_{X^0} X^{-1}$ so that commutes. This pre-

serves cohomologies as $H^0 X = X^0 / \text{im}(X^{-1}) = P^0 / P^{-1} = H^0 P$ since P^{-1} surjects onto $\ker \alpha^0$ and has image in $P^0 / \ker \alpha^0 \cong X^0$ equal to X^{-1} as $P^0 \rightarrow X^0$ is a surjection. Since $P^{-1} \rightarrow X^{-1}$ is a surjection, we may iterate the process to construct a projective resolution P of X as desired.

Lemma 4.2.4. *Suppose P_1, P_2 are projective resolutions of a complex X of A -modules. Then there exists a homotopy equivalence γ s.t.*

$$\begin{array}{ccc} P_1 & \xrightarrow{\gamma} & P_2 \\ & \searrow \alpha & \swarrow \beta \\ & X & \end{array}$$

commutes upto homotopy, i.e. $\beta\gamma \cong \alpha$.

pf. Consider the diagram

$$\begin{array}{ccc} P_1 & \xrightarrow{\gamma} & P_2 \\ & \searrow \alpha & \downarrow \beta \\ & & X \\ & \searrow \delta & \downarrow \\ & & \text{hCoker}(\beta) \end{array}$$

. Since β is a quasi-isomorphism,

$\text{hCoker}(\beta)$ is acyclic. Then the composition $\delta = P \rightarrow X \rightarrow \text{hCoker}(\beta)$ is nullhomotopic, hence there is a canonical map $P_1 \xrightarrow{\gamma} \text{hKer}(X \rightarrow \text{hCoker}(\beta)) \cong P_2$ via homotopy equivalence. By symmetry, such a map exists in the opposite direction, hence γ is a homotopy equivalence and the diagram commutes upto homotopy. $\rightarrow \gamma$ is all homotopic.

Definition 4.2.9. *The i th Ext group of two chain complexes of A -modules M, N is defined as $\text{Ext}_A^i(M, N) \equiv H^i \underline{\text{hom}}(P, N)$, where P is some projective resolution of M .*

Define the derived functor $\underline{\text{hom}}^{der}(X, -)$ via

$$\underline{\text{hom}}^{der}(X, Y) \equiv \lim_{P \rightarrow_{qis} X, \text{projective}} \underline{\text{hom}}(P, Y)$$

where P are ordered by maps of chain complexes $P' \xrightarrow{\quad} P \xrightarrow{qis} X$ which forcibly removes the choice P .

CLAIM : Suppose we have a map of chain complexes $f : X_1 \rightarrow X_2$, which have projective resolutions P_1, P_2 respectively. Then we have a map $\phi : P_1 \rightarrow P_2$ s.t. the diagram below commutes. Moreover such a map is unique

$$\begin{array}{ccc} X_1 & \xrightarrow{f} & X_2 \\ qis \uparrow & & \uparrow qis \\ P_1 & \xrightarrow{\phi} & P_2 \end{array}$$

upto homotopy.

pf. Since the derived functor preserves quasi-isomorphisms, the induced map of complexes of maps $\underline{\text{hom}}(P_1, P_2) \rightarrow \underline{\text{hom}}(P_1, X_2)$ is a quasi-isomorphism. So there exists a map $P_1 \rightarrow X_1 \rightarrow X_2$, which is killed by the differential since it is a map of chain complexes. Thus it defines a cohomology class in $H^0 \underline{\text{hom}}(P_1, X_2)$. This implies there is some cohomology class in $H^0 \underline{\text{hom}}(P_1, P_2)$ which is a lift of the map through P_2 .

→ for every chain complex Y , $\underline{\text{hom}}_A^{der}(X_2, Y) \rightarrow \underline{\text{hom}}_A^{der}(X_1, Y)$ by pullbacking along ϕ .

CLAIM : If $H \subseteq G$ is a subgroup and X is a complex of G -modules, then we get a restriction map $X^{hG} \rightarrow X^{hH}$.

pf. Consider $\mathbb{Z}[G/H] = \{f : G/H \rightarrow \mathbb{Z} \mid f \text{ nonzero at finitely many points}\}$, which has a G -action via translations and is equivalent to the induced module from H to G by the trivial module $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} \mathbb{Z}$. Let $P_H \rightarrow \mathbb{Z}$ be a projective resolution for $\mathbb{Z}$ as a H -modules. Then we have an induced G -module $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} P_H$. Then $\mathbb{Z}[G]$ is free as a $\mathbb{Z}[H]$ -module, as choosing coset representatives for G/H yields a basis. Therefore $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} -$ preserves quasi-isomorphisms. This implies $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} P_H \rightarrow \mathbb{Z}[G/H]$ is a projective resolution for $\mathbb{Z}[G/H]$ as a $\mathbb{Z}[G]$ -modules. This is because both $\mathbb{Z}[G]$ and P_H are bounded so it will be bounded and inducing upto $\mathbb{Z}[G]$ preserves projective modules as we will obtain a direct summand of a free module. Then $\underline{\text{hom}}_G^{der}(\mathbb{Z}[G/H], X) \equiv \underline{\text{hom}}_G(\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} P_H, X) = \underline{\text{hom}}_H(P_H, X) \equiv X^{hH}$

by definition. This implies $\exists \begin{array}{ccc} \epsilon : \mathbb{Z}[G/H] & \rightarrow & \mathbb{Z} \\ \sum_{g_i \in G/H} n_i g_i & \mapsto & \sum n_i \end{array}$ gives a map $X^{hG} \rightarrow X^{hH}$, which is clearly G -equivariant when we equip $\mathbb{Z}$ with the trivial action. Then $X^{hG} = \underline{\text{hom}}_G^{der}(\mathbb{Z}, X) \rightarrow \underline{\text{hom}}_G^{der}(\mathbb{Z}[G/H], X) = X^{hH}$ is well-defined upto homotopy.

If $H \leq G$ is finite index, we have a G -equivariant map $\mathbb{Z} \xrightarrow{\kappa} \mathbb{Z}[G/H] \xrightarrow{\epsilon} \mathbb{Z}$ defined as $1 \mapsto \sum_{g \in G/H} g$. Then it corresponds to multiplication by the index

$[G : H]$, which gives an **inflation map** $X^{hH} \rightarrow X^{hG}$ s.t. the composition $X^{hG} \rightarrow X^{hH} \rightarrow X^{hG}$ is homotopic to multiplication by $[G : H]$.

Definition 4.2.10. $H^i(G, X) \equiv H^i(X^{hG})$ is the hypercohomology of G with coefficients in X .

CLAIM : If X is only in nonnegative degrees, then $H^0(G, X) = H^0(X)^G$ and $H^i(G, X) = 0, \forall i < 0$.

pf. Choose any projective resolution P of $\mathbb{Z}$ as a G -module where it has only nonpositive degrees only. By definition, $H^0(G, X)$ is equivalent to the homotopy classes of maps s.t. $f : P \rightarrow X$

$$\begin{array}{ccccccccccc} \dots & \longrightarrow & P^{-2} & \longrightarrow & P^{-1} & \xrightarrow{d} & P^0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \dots \\ & & \downarrow & & \downarrow & & \downarrow f & & \downarrow & & \downarrow & & \\ \dots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & X^0 & \xrightarrow{d} & X^1 & \longrightarrow & X^2 & \longrightarrow & \dots \end{array} \quad \text{commutes. Then}$$

any homotopy of f is 0 and $H^i(G, X) = 0, \forall i < 0$. By commutativity, $df = fd = 0$ and therefore such maps f are equivalent to G -equivariant maps $\mathbb{Z} = P^0/dP^{-1} = \text{Coker}(P^{-1} \rightarrow P^0) \rightarrow \ker(X^0 \rightarrow X^1) = H^0(X)$ by quasi-isomorphism. That is equivalent to a G -invariant vector in the cohomology $H^0(X)$.

• A Projective Resolution of $\mathbb{Z}$ as a G -module

Let $G \equiv \mathbb{Z}/n\mathbb{Z}$ with generator σ . If M is a G -module, then M^{hG} is quasi-isomorphic to the complex

$$\dots \longrightarrow 0 \longrightarrow 0 \longrightarrow M \xrightarrow{1-\sigma} M \xrightarrow{\Sigma \sigma^i} M \xrightarrow{1-\sigma} \dots$$

where the first M is in degree 0.

Definition 4.2.11. Let A be a commutative ring, and B an A -algebra. Then the bar complex $\text{Bar}_A(B)$ is

$$\dots \rightarrow B \otimes_A B \otimes_A B \xrightarrow{b_1 \otimes b_2 \otimes b_3 \mapsto b_1 b_2 \otimes b_3 - b_1 \otimes b_2 b_3} B \otimes_A B \xrightarrow{b_1 \otimes b_2 \mapsto b_1 b_2} B \rightarrow 0 \rightarrow \dots$$

with B in degree 0. In general, $\text{Bar}_A^{-n}(B) \equiv B^{\otimes_A n+1}$ with differential $b_1 \otimes \dots \otimes b_{n+1} \mapsto b_1 b_2 \otimes b_3 \otimes \dots \otimes b_{n+1} - b_1 \otimes b_2 b_3 \otimes \dots \otimes b_{n+1} + \dots$

→ **homotopy equivalent to zero.**

pf. $\begin{array}{ccccccccccc} \dots & \longrightarrow & B \otimes_A B \otimes_A B & \longrightarrow & B \otimes_A B & \longrightarrow & B & \longrightarrow & 0 & \longrightarrow & \dots \\ & \swarrow h^2 & \downarrow id & \swarrow h^1 & \downarrow id & \swarrow h^0 & \downarrow id & & \downarrow & & \\ \dots & \longrightarrow & B \otimes_A B \otimes_A B & \longrightarrow & B \otimes_A B & \longrightarrow & B & \longrightarrow & 0 & \longrightarrow & \dots \end{array} \quad \text{Define}$

$\left\{ \begin{array}{l} h^0(b) \equiv 1 \otimes b \\ h^1(b_1 \otimes b_2) \equiv 1 \otimes b_1 \otimes b_2 \end{array} \right.$. Then $(dh^1 + h^0 d)(b_1 \otimes b_2) = d(1 \otimes b_1 \otimes b_2) + 1 \otimes b_1 b_2 = b_1 \otimes b_1 - 1 \otimes b_1 b_2 + 1 \otimes b_1 b_2 = b_1 \otimes b_2$. We can construct h^n in the similar fashion, and such h is a nullhomotopy we wanted.

$$\begin{array}{ccccccc}
 \dots & \longrightarrow & B \otimes_A B \otimes_A B & \longrightarrow & B \otimes_A B & \longrightarrow & 0 \longrightarrow \dots \\
 & & \downarrow & & \downarrow d & & \downarrow \\
 \dots & \longrightarrow & 0 & \longrightarrow & b & \longrightarrow & 0 \longrightarrow \dots
 \end{array}$$

where d is the multiplication map in the differential. Then its cone is $\text{Bar}_A(B)$ and therefore it is a homotopy equivalent. Given a B -module M , we can tensor with M over B , which yields a diagram

$$\begin{array}{ccccccc}
 \dots & \longrightarrow & B \otimes_A B \otimes_A M & \longrightarrow & B \otimes_A M & \longrightarrow & 0 \longrightarrow \dots \\
 & & \downarrow & & \downarrow b \otimes m \mapsto bm & & \downarrow \\
 \dots & \longrightarrow & 0 & \longrightarrow & M & \longrightarrow & 0 \longrightarrow \dots
 \end{array}$$

that is also a ho-

motopy equivalence. The differentials are the same except the last term is replaced with an element of m . In words, M is canonically equivalent to a complex where every term is of the form $B \otimes_A N$, where N stands for $B \otimes_A \dots \otimes_A \otimes_A M$.

Let $A = \mathbb{Z}$, $B \equiv \mathbb{Z}[G]$, $M \equiv \mathbb{Z}$. Then $\begin{array}{ccccc} B \otimes_A M & \rightarrow & M & \rightarrow & B \otimes_A M \\ b \otimes m & \mapsto & bm & \mapsto & 1 \otimes bm \end{array}$ s.t. $1 \otimes bm \neq b \otimes m = b(1 \otimes m)$ since the action is on B not on M and it is not an action of B -modules. Thus we obtain a canonical projective resolution of $\mathbb{Z}$ by free

$$\begin{array}{ccccccc}
 \dots & \longrightarrow & \mathbb{Z}[G^3] & \longrightarrow & \mathbb{Z}[G \times G] & \longrightarrow & \mathbb{Z}[G] \longrightarrow 0 \longrightarrow \dots \\
 \text{G-modules} & & \downarrow & & \downarrow & & \downarrow \epsilon \\
 \dots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} \longrightarrow 0 \longrightarrow \dots
 \end{array}$$

with

differentials $(g_1, g_2) \mapsto g_1 g_2 - g_1$ and so forth, since G acts trivially on $\mathbb{Z}$. Here $\mathbb{Z}[G \times G] \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}} \mathbb{Z}[G]$ since both have a basis by the elements of the product group. Then we have a complex of the form

$$\dots \rightarrow 0 \rightarrow M \rightarrow \mathbb{Z}[G] \otimes M \rightarrow \mathbb{Z}[G \times G] \otimes M \rightarrow \dots$$

with M in degree 0 and G finite.

Definition 4.2.12. Elements in $\mathbb{Z}[G] \otimes M$ in the kernel of the differential are called group n -cocycles with coefficients in M ; elements in the image of differential are called n -coboundaries.

Denote P_G^{can} for the resolution we defined. Then $X^{hG} \cong \underline{\text{hom}}_G(P_G^{can}, X)$.

For every G -module M , $\cdots \rightarrow 0 \rightarrow M \xrightarrow{(gm - m)\Pi_{g \in G}} \Pi_{g \in G} M \rightarrow \Pi_{g, h \in G} M \rightarrow \cdots$ where $\{\phi : G \rightarrow M \in \Pi_{g \in G} M$. Then ϕ is killed by this differential if it is a 1-cocycle, that is $\phi(gh) = \phi(g) + g \cdot \phi(h)$, $\forall g, h \in G$ via the group action. Similarly ϕ is a 1-coboundary if $\exists m \in M$ s.t. $\phi(g) = g \cdot m - m$, $\forall g \in G$. Then

$$H^1(G, M) \equiv H^1(M^{hG}) = \{1\text{-cocycles}\} / \{1\text{-coboundaries}\}$$

If M is a G -module, $M_G = M \otimes_{\mathbb{Z}[G]} \mathbb{Z}$. Define $I_G \equiv \ker \epsilon$ so that we have a s.e.s.

$$0 \longrightarrow I_G \longrightarrow \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0$$

where $\epsilon : \mathbb{Z}[G] \rightarrow \mathbb{Z}$
 $\sum n_i g_i \mapsto \sum n_i$.

Definition 4.2.13. If X, Y are chain complexes, then

$$(X \otimes Y)^i \equiv \bigoplus_{j \in \mathbb{Z}} X^j \otimes Y^{i-j}, \quad d(x \otimes y) \equiv dx \otimes y + (-1)^j x \otimes dy$$

If X is a complex of right A -modules, and Y is a complex of left A -modules, then $X \otimes_A Y$ is defined similarly.

Definition 4.2.14. A complex F of left A -modules is flat if for every acyclic complex Y of right A -modules, $Y \otimes_A F$ is also acyclic, i.e. $- \otimes_A F$ preserves injections.

CLAIM : Any project complex is flat.

pf.

- F is in degree 0 only and F^0 is flat.

For every complex $Y = Y^\bullet$, $\cdots \rightarrow Y^i \otimes_A F \xrightarrow{d^i \otimes id} Y^{i+1} \otimes_A F \rightarrow \cdots$ for $Y \otimes_A F$. Since F is flat, $H^i(Y \otimes_A F) = H^i(Y) \otimes_A (F)$, $\forall i$, so if Y is acyclic, so is $Y \otimes_A F$.

- F is in degree 0 and -1 only and F^0, F^{-1} are flat.

Then $F^\bullet = \text{hCoker}(F^{-1} \rightarrow F^0)$. Then since tensor products commute with homotopy cokernels, $Y \otimes_A F = \text{hCoker}(Y \otimes_A F^{-1} \rightarrow Y \otimes_A F^0)$ so by the previous case, if Y is acyclic, so are $Y \otimes_A F^0, Y \otimes_A F^{-1}$. Thus $Y \otimes_A F$ is as well by l.e.s. on cohomology. Similarly, for any bounded F , if F^i are flat for all i , so is F .

- General

$$\begin{array}{ccccccccccc}
F_0 & \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & F^0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
\downarrow & & & \downarrow & & \downarrow & & \downarrow & & \downarrow id & & \downarrow & & \\
F_1 & \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & F^1 & \xrightarrow{d} & F^0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
\downarrow & & & \downarrow & & \downarrow & & \downarrow id & & \downarrow id & & \downarrow & & \\
F_2 & \cdots & \longrightarrow & 0 & \longrightarrow & F^2 & \xrightarrow{d} & F^1 & \xrightarrow{d} & F^0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
\downarrow & & & \downarrow & & \downarrow id & & \downarrow id & & \downarrow id & & \downarrow & & \\
\vdots & . & & \cdots & & \cdots & & \cdots & & \cdots & & \cdots & & .
\end{array}$$

Then all squares in the diagram commutes, hence there are all morphisms of complexes and $F = \lim F_i$. Since direct limits commute with tensor products, $Y \otimes_A F = \lim Y \otimes_A F_i$. Then by the previous case, $Y \otimes_A F_i$ is acyclic $\forall i$. Since cohomology commutes with direct limits, if Y is acyclic, so is $Y \otimes_A F$.

→ for any A -modules Y , choose a projective resolution $F \rightarrow Y$. Then since F is projective, F is flat. Define $Y \otimes_A^{der} X \equiv F \otimes_A X$, which is well-defined upto quasi-isomorphism.

Definition 4.2.15. The i th torsion group of Y against X is

$$Tor_i^A(Y, X) \equiv H^{-i}(Y \otimes_A^{der} X)$$

Definition 4.2.16. The homotopy coinvariants of a chain complex X is the complex $X_{hG} \equiv X \otimes_{\mathbb{Z}[G]}^{der} \mathbb{Z} \cong X \otimes_{\mathbb{Z}[G]} P_G$.

→ if X is acyclic, so is X_{hG} and it commute with cones. Also $H^0(X_{hG}) = M_G = \mathbb{Z} \otimes_{\mathbb{Z}[G]} M = M / \sum_{g \in G} (g - 1)M$.

Definition 4.2.17. $H_i(G, X) \equiv H^{-i}(X_{hG})$.

CLAIM : $H_1(G, \mathbb{Z}) = G^{ab}$ where G^{ab} denotes the abelianisation of G .

pf. $0 \rightarrow I_G \rightarrow \mathbb{Z}[G] \rightarrow_{\epsilon} \mathbb{Z} \rightarrow 0$ is a s.e.s. which gives a l.e.s. on cohomologies $H_1(G, \mathbb{Z}[G]) \rightarrow H_1(G, \mathbb{Z}) \rightarrow H_0(G, I_G) \rightarrow H_0(G, \mathbb{Z}[G]) \rightarrow H_0(G, \mathbb{Z})$. Then $H_0(G, \mathbb{Z}[G]) = H^0(\mathbb{Z}[G])_G = \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} \mathbb{Z} = \mathbb{Z}$ and $H_1(G, \mathbb{Z}[G]) = 0$ since $\mathbb{Z}[G]_{hG} \equiv \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} P_G = P_G \cong \mathbb{Z}$ is a quasi-isomorphism. Thus the above exact sequence becomes $0 \rightarrow H_1(G, \mathbb{Z}) \rightarrow H_0(G, I_G) \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}$. We have an isomorphism $H_0(G, I_G) \cong H_1(G, \mathbb{Z}) = (I_G)_G = I_G / I_G^2$ since $M_G = M / I_G \cdot M$. Then $\begin{array}{ccc} \mathbb{Z}[G] / I_G^2 & \rightarrow & G^{ab} \otimes \mathbb{Z} \\ g & \mapsto & (\bar{g}, 1) \end{array} : \forall [g] \in \mathbb{Z}[G],$

$$[g] + [h] \mapsto (\bar{g}\bar{h}, 2)$$

$g \mapsto (\bar{g}, 1)$, $\forall g, h \in G$, which is a homomorphism. It has an inverse $h \mapsto (\bar{h}, 1)$

duced by the map $\begin{array}{ccc} G \otimes \mathbb{Z} & \rightarrow & \mathbb{Z}[G] / I_G^2 \\ (g, n) & \mapsto & [g] + n - 1 \end{array}$, a homomorphism $([g] - 1)([h] - 1) = [gh] - [g] - [h] + 1 \in I_G^2 \rightarrow ([g] - 1) + ([h] - 1) \equiv [gh] - 1 \pmod{I_G^2}$. Thus the claim follows.

4.2.2 Tate Cohomologies

Define the norm map

$$N : X_{hG} \rightarrow X^{hG} \quad \text{Then set}$$

$$X_{hG} = X \otimes_{\mathbb{Z}[G]} P_G \rightarrow X \otimes_{\mathbb{Z}[G]} \mathbb{Z} \rightarrow \underline{\text{hom}}_{\mathbb{Z}[G]}(\mathbb{Z}, X) \rightarrow \underline{\text{hom}}_{\mathbb{Z}[G]}(P_G, X) = X^{hG}.$$

$$X^{tG} \equiv \text{hCoker}(N : X_{hG} \rightarrow X^{hG})$$

→ the **Tate cohomology** is defined as $\hat{H}^i(G, X) \equiv H^i(X^{tG})$ and X^{tG} is said to be the **Tate complex**.

For the Tate cohomologies for modules, suppose G acts on an A -module M . Then giving a map $f : N \rightarrow M^G$ for an abelian group N is the same as giving a map $f : N \rightarrow M$ s.t. $g \cdot f(x) = f(x)$, $\forall g \in G$. Dually, giving a map $f : M_G \rightarrow N$ is the

$$\text{same as giving a map } f : M \rightarrow N \text{ s.t. } f(g \cdot x) = f(x). \text{ Then}$$

$$\begin{array}{ccc} M & \xrightarrow{N \cong \sum g} & M \\ \downarrow & & \uparrow \\ M_G & \xrightarrow{N} & M^G \end{array} \quad \text{given}$$

$\begin{cases} N(g \cdot x) = N(x) \\ g \cdot N(x) = N(x) \end{cases}$. If $\#G$ is invertible in M , the norm map is an isomorphism. For G -modules, we have

$$X_{hG} = P \otimes_{\mathbb{Z}[G]} X \rightarrow \mathbb{Z} \otimes_{\mathbb{Z}[G]} X \rightarrow \underline{\text{hom}}_G(\mathbb{Z}, X) \rightarrow \underline{\text{hom}}_G(P, X) = X^{hG}$$

where the last map is via pullback and the middle map is the norm map applied termwise via the norm on modules. This map gives a map we will call $N : X_{hG} \rightarrow X^{hG}$.

Proposition 4.2.2. *Let M be a G -module, thought of as a complex in degree 0.*

- Then*
- i) $\hat{H}^i(G, M) = H^i(G, M) \quad i \geq 1$
 - ii) $\hat{H}^i(G, M) = M^G/N(M) \quad i = 0$
 - iii) $\hat{H}^i(G, M) = \ker N/(g-1) \quad i = -1$
 - iv) $\hat{H}^i(G, M) = H_{-i-1}(G, M) \quad i \leq -2$

pf. $M_{hG} \rightarrow M^{hG} \rightarrow M^{tG} = \text{hCoker}(N)$ yields a l.e.s. on cohomology

$$\cdots \rightarrow H_{-i}(M_{hG}) \rightarrow H^i(M^{hG}) \rightarrow \hat{H}^i(G, M) \rightarrow H_{-i-1}(M_{hG}) \rightarrow \cdots$$

- $i \geq 1$

$H_{-i}(M_{hG}), H_{-i-1}(M_{hG})$ vanish, so $H^i(G, M) \cong \hat{H}^i(G, M)$ by exactness.

- $i = 0, -1$

$H_{-1}(M_{hG}), H^{-1}(G, M)$ vanish, so $\exists$ a s.e.s. $0 \rightarrow \hat{H}^{-1}(G, M) \rightarrow M_G \rightarrow M^G \rightarrow \hat{H}^0(G, M) \rightarrow 0$. Thus ii and iii follow.

- $i \leq -2$

$H^{-i-1}(G, M), H^{-i}(G, M)$ vanish for $i \geq 1$. Thus $\hat{H}^i(G, M) \cong H_{-i-1}(G, M)$ by exactness.

Given a s.e.s. of G -modules $0 \rightarrow M \rightarrow E \rightarrow N \rightarrow 0$, we have l.e.s.

$$\begin{cases} \hat{H}^0(G, M) \rightarrow \hat{H}^0(G, E) \rightarrow \hat{H}^0(G, N) \rightarrow_{\delta} \hat{H}^1(G, M) \rightarrow \dots \\ \hat{H}^1(G, M) \rightarrow \hat{H}^1(G, E) \rightarrow \hat{H}^1(G, N) \rightarrow_{\partial} \hat{H}^2(G, M) \rightarrow \dots \end{cases}, \text{ where}$$

$$\begin{cases} \delta & \text{lifts } x \in \hat{H}^0(N) = N^G/N(N) \text{ to } \tilde{x} \in E \text{ s.t. } (1 - \sigma)\tilde{x} \in \ker N \\ \partial & \text{lifts } x \in \hat{H}^1(G, N) \text{ to } \tilde{x} \in E \text{ s.t. } N\tilde{x} = \sum \sigma^i \tilde{x} \in M^G \end{cases}.$$

Lemma 4.2.5. Suppose $0 \longrightarrow X^0 \xrightarrow{d^1} X^1 \xrightarrow{d^2} \dots \xrightarrow{d^{n-1}} X^{n-1} \xrightarrow{d^n} X^n \longrightarrow 0$ is exact and $\forall X^i$ are finite. Then

$$\#X^0 \cdot \#X^2 \dots = \#X^1 \cdot \#X^3 \dots$$

pf.

- $n = 1$

$0 \rightarrow X^0 \rightarrow X^1 \rightarrow 0$ is exact so $X^0 \cong X^1$ and therefore $\#X^0 = \#X^1$.

- $n > 1$, even

We can separate the exact sequence into two $\begin{cases} 0 \rightarrow X^0 \rightarrow \dots \rightarrow X^{n-1} \rightarrow \text{imd}^{n-1} \rightarrow 0 \\ 0 \rightarrow \text{imd}^{n-1} \rightarrow X^{n-1} \rightarrow X^n \rightarrow 0 \end{cases}$.

Then $\#X^0 \cdot \#X^2 \dots \#X^n = \#X^0 \dots \#X^{n-2} \cdot \frac{\#X^{n-1}}{\#\text{imd}^{n-1}} = \#X^1 \dots \#\text{imd}^{n-1} \cdot \frac{\#X^{n-1}}{\#\text{imd}^{n-1}} = \#X^1 \dots \#X^{n-1}$ by inductive hypothesis.

- similarly for $n > 1$, odd

Definition 4.2.18. The Herbrand quotient or Euler characteristic of a G -module M is

$$\chi(M) \equiv \frac{\#\hat{H}^0(G, M)}{\#\hat{H}^1(G, M)}$$

which is only defined when both are finite.

Lemma 4.2.6. Let $0 \rightarrow M \rightarrow E \rightarrow N \rightarrow 0$ be a s.e.s. of G -modules. If χ is defined for two of the three G -modules, then it is defined for all three, in which case $\chi(M) \cdot \chi(N) = \chi(E)$.

pf. Construct a l.e.s. $0 \rightarrow \ker \alpha \rightarrow \hat{H}^0(M) \rightarrow_{\alpha} \hat{H}^0(E) \rightarrow \hat{H}^0(N) \rightarrow_{\delta} \hat{H}^1(M) \rightarrow \hat{H}^1(E) \rightarrow_{\beta} \hat{H}^1(N) \rightarrow \text{Coker}(\beta) \rightarrow 0$. Since $\hat{H}^1(E) \rightarrow_{\beta} \hat{H}^1(N) \rightarrow_{\delta} \hat{H}^0(M) \rightarrow \hat{H}^0(E)$ is an exact sequence, $\ker \alpha = \text{im } \delta = \hat{H}^1(N)/\ker \delta = \hat{H}^1(N)/\text{im } \beta = \text{Coker}(\beta)$. Then by the previous lemma we get the result.

Theorem 4.2.1 (Hilbert's Theorem 90). *If L/K is a cyclic extension and $x \in L^{\times}$ with $N(x) = 1$, then $x = y/\sigma y$, $\exists y \in L^{\times}$.*

pf. Let $x \in L$ s.t. $N(x) = 1$. Then the characteristic polynomial of $\hat{H}_x$ is $t^n - 1$, implying 1 is a root and hence an eigenvalue of H_x . Thus $\ker((H_x - 1) \otimes_K \bar{K}) \neq 0$. Since tensor products commute with taking kernels for fields, $\ker(H_x - 1) \neq 0$. Thus $\exists y \in L^{\times}$ s.t. $H_x(y) = x \cdot \sigma y = y \Leftrightarrow x = y/\sigma y$.
 $\rightarrow \hat{H}^1(G, K^{\times}) = 0$.

e.g. $L \equiv \mathbb{Q}(i)$, $K = \mathbb{Q}$. Choose $x \in \mathbb{Q}(i)$ with $N(x) = 1$. then $x = a/c + (b/c)i$, $a, b, c \in \mathbb{Z}$ satisfying $a^2 + b^2 = c^2$. Then by the Hilbert theorem 90, $(r - s)^2 + (2rs)^2 = (r + s)^2$.

4.3 Local Class Field Theory

4.3.1 Direct Construction of LCFT

Definition 4.3.1. K^{unr} is the p -adic completion of the maximal unramified extension of $K \subseteq \bar{K}$.

e.g. $K \equiv \begin{cases} \mathbb{F}_q((t)) \\ \mathbb{Q}_p \end{cases} \rightarrow K^{unr} = W(\bar{\mathbb{F}}_p)[1/p]$ so the basic structure of K^{unr} is a local field with residue field $\bar{\mathbb{F}}_q$.

Let π be a uniformiser of K , which will continue to be a uniformiser in each K_n , $\mathcal{O}_{K^{unr}}$ is the π -adic completion of $\cup_n \mathcal{O}_{K_n}$. Then there exists a s.e.s.

$$0 \rightarrow \mathbb{O}_{K^{unr}}^{\times} \rightarrow K^{unr, \times} \rightarrow \mathbb{Z} \rightarrow 0$$

where $v : K^{unr, \times} \rightarrow \mathbb{Z}$ is the normalised valuation on K^{unr} . Then $\text{Gal}(K^{unr}/K) = \text{Aut}_{cts}(K^{unr}/K)$. Letting k denote the residue field of K and σ denote the Frobenius element of $\text{Gal}(\bar{k}/k)$, we have

$$\begin{aligned} \text{Gal}(K^{unr}/K) &\cong \text{Gal}(\bar{k}/k) &\cong \hat{\mathbb{Z}} \\ \sigma &\mapsto 1 \end{aligned}$$

Since $K = \{x \in K^{unr} : x = \sigma x\}$, $0 \rightarrow K \rightarrow K^{unr} \rightarrow K^{unr}$ is the resolution where $1 - \sigma : K^{unr} \rightarrow K^{unr}$. Then

$$0 \rightarrow K^{\times} \rightarrow K^{unr, \times} \rightarrow K^{unr, \times} \rightarrow \mathbb{Z} \rightarrow 0$$

where $\begin{array}{ccc} K^{unr,\times} & \rightarrow & K^{unr,\times} \\ x & \mapsto & x/\sigma x \end{array}$ is an exact sequence.

pf. This is true iff $\forall x \in \mathcal{O}_{K^{unr}}^\times$ can be written as $y/\sigma y$, $\exists y \in \mathcal{O}_{K^{unr}}^\times$. This amounts to showing that $\begin{array}{ccc} \mathcal{O}_{K^{unr}}^\times & \rightarrow & \mathcal{O}_{K^{unr}}^\times \\ x & \mapsto & x/\sigma x \end{array}$ is surjective. By completeness of the filtration by the maximal ideal, we can reduce the case into the associated graded level. This gives maps $\begin{array}{ccc} \bar{\mathbb{F}}_q^\times & \rightarrow & \bar{\mathbb{F}}_q^\times \\ x & \mapsto & x/x^q \end{array}$, $\begin{array}{ccc} \bar{\mathbb{F}}_q & \rightarrow & \bar{\mathbb{F}}_q \\ x & \mapsto & (1-q)x \end{array}$, where the first is surjective given $x^{q-1} = 1/y$ having solution since $\bar{\mathbb{F}}_q$ is algebraically closed. The latter is invertible, as the map is just the identity.

CLAIM : If $x \in K^{unr}$ is algebraic over K , then $K' \equiv K(x)$ is unramified over K .

pf. As a finite algebraic extension of K , K' is a local field, and there exists an embedding $\mathcal{O}_{K'}/p_K \mathcal{O}_{K'} \hookrightarrow \mathcal{O}_{K^{unr}}/p_K \mathcal{O}_{K^{unr}} = \bar{k}$. so $\mathcal{O}_{K'}/p_K \mathcal{O}_{K'}$ is a field, hence uniformisers of K , and K' are identical.

CLAIM : $(K^{unr})^{\sigma=1} = K$, i.e. the element fixed by the Frobenius automorphism $\sigma \in G$ is K .

Theorem 4.3.1 (Vanishing Theorem). *If L/K is totally ramified, then the complex $(L^{unr,\times})^{tG}$ is acyclic.*

Suppose L/K is a general finite Galois extension with $G \equiv \text{Gal}(L/K)$. If L/K is unramified, $L \otimes_K K^{unr} = \prod_{L \hookrightarrow K^{unr}} K^{unr}$ canonically, indexed by such embedding. Here we have an action of $\begin{cases} G & x \otimes y \mapsto gx \otimes y \\ \hat{\mathbb{Z}}\sigma & x \otimes y \mapsto x \otimes \sigma y \end{cases}$ on each factor, which clearly commute. Then the points fixed under σ are $L = L \otimes_K K \hookrightarrow L \otimes_K K^{unr}$. Then similarly,

$$1 \rightarrow L^\times \rightarrow (L \otimes_K K^{unr})^\times \rightarrow (L \otimes_K K^{unr})^\times \rightarrow \mathbb{Z} \rightarrow 0$$

is exact.

pf. If $x \in K^{unr}$ is a unit, so is σx , so the map $\begin{array}{ccc} (L \otimes_K K^{unr})^\times & \rightarrow & (L \otimes_K K^{unr})^\times \\ x & \mapsto & x/\sigma x \end{array}$ is well-defined. Moreover, x is in the kernel iff x is fixed under the action of σ i.e. $x \in K$ and since $L \otimes_K K = L$, we obtain a unit of L , which shows the exactness

$$\begin{array}{ccc} (L \otimes_K K^{unr})^\times & \xrightarrow{\quad N_{L/K} \otimes id \quad} & \mathbb{Z} \\ & \searrow \quad \quad \quad \nearrow v & \\ & K \otimes_K K^{unr,\times} & \end{array} \quad \text{where } N_{L/K}(x) \equiv$$

$\prod_{g \in G} gx$ is the last map. Thus its kernel is $\mathcal{O}_{K^{unr}}^\times$, which is precisely the image of $x \mapsto x/\sigma x$. Also $1 \otimes \pi \mapsto 1$ is surjective and therefore the rest is also exact.

Theorem 4.3.2. *for a cyclic group G , X^{tG} is acyclic iff $\hat{H}^0(G, X) = 0 = \hat{H}^1(G, X)$.*

pf. X^{tG} can be computed to be a 2-periodic complex. Any two values of distinct parity, therefore, would suffice.

Lemma 4.3.1. *Let V be a nonzero $\mathbb{F}_p[G]$ -module. Then $V^G \neq 0$.*

pf. wlog, assume V is finite-dimensional over $\mathbb{F}_p$ since V is clearly the union of its finite-dimensional G -submodules. Then $\#V = p^r$, $0 < r < \infty$. Every G -orbit of V either has size 1 or divisible by p . Since $\{0\}$ is a G -orbit of size 1, there must be another, that is $\exists 0 \neq v \in V^G$.

Proposition 4.3.1. *Let X be a complex of $\mathbb{F}_p[G]$ -modules. If $\hat{H}^0(G, X) = 0$, then X^{tG} is acyclic.*

pf. **CLAIM:** $\hat{H}^0(G, X \otimes V) = 0$ for every finite dimensional G -representation V over $\mathbb{F}_p$.

pf. By the previous lemma, we have a s.e.s. of $\mathbb{F}_p[G]$ -modules $0 \rightarrow V^G \rightarrow V \rightarrow W \rightarrow 0$ where $\dim V > \dim W$. Then $\text{hCoker}(V^G \otimes X \rightarrow V \otimes X) \cong W \otimes X$. Then by the inductive hypothesis $\hat{H}^0(G, W \otimes X) = 0$, and $\hat{H}^0(G, V^G \otimes X) = \hat{H}^0(G, \bigoplus_{\dim V^G} X) = \bigoplus \hat{H}^0(G, X) = 0$ by the hypothesis. This implies $\hat{H}^0(G, V \otimes X) = 0$. By induction the claim follows.

- negative degree

$$0 \rightarrow V_1 \rightarrow \mathbb{F}_p[G] \rightarrow \mathbb{F}_p \rightarrow 0$$

is a s.e.s. where $V_1 = \ker(\epsilon : \mathbb{F}_p[G] \rightarrow \mathbb{F}_p)$. Let $\bar{X}$ be X with the trivial G -action. Then $\begin{array}{ccc} \underline{X} \otimes_{\mathbb{F}_p[G]} & \rightarrow & X \otimes_{\mathbb{F}_p[G]} \\ x \otimes g & \mapsto & gx \otimes g \end{array}$ is a G -equivariant map and a bijection, hence an isomorphism. Since $(\underline{X} \otimes_{\mathbb{F}_p[G]})^{tG}$ is acyclic, so is $(S \otimes_{\mathbb{F}_p[G]})^{tG}$. Then by the l.e.s. on Tate cohomology, $\hat{H}^{i-1}(G, X) \cong \hat{H}^i(G, X \otimes V_1)$ where by the previous claim the RHS is zero for $i = 0$. By iteration it gives the result for negative degrees.

- positive degree

$$0 \rightarrow \mathbb{F}_p \rightarrow \mathbb{F}_p[G] \rightarrow V_2 \rightarrow 0$$

is a s.e.s. where $V_2 = \text{Coker}(1 \mapsto \sum_{g \in G} g)$. Then similar argument above gives $\hat{H}^i(G, X \otimes V_2) \cong \hat{H}^{i+1}(G, X)$ and by the previous claim the LHS is 0 for $i = 0$. By iteration, the result for the positive i follows.

Theorem 4.3.3. *The previous theorem holds if G is a p -group.*

pf. Define $X/p \equiv \text{hCoker}(X \rightarrow_{\times p} X)$. As a complex of $\mathbb{Z}[G]$ -module, X/p is quasi-isomorphic to a complex of $\mathbb{F}_p[G]$ -modules. Since X is only defined upto quasi-isomorphism, we may assume it is projective as a complex of $\mathbb{Z}[G]$ -modules. Thus there exists a quasi-isomorphism $X/p = X \otimes_{\mathbb{Z}[G]} \text{hCoker}(\mathbb{Z}[G] \rightarrow \mathbb{Z}[G]) \cong X \otimes_{\mathbb{Z}[G]}$

$\mathbb{F}_p[G]$ where the RHS computes X modded out by p termwise. Then by a l.e.s. on Tate cohomology, $\rightarrow \hat{H}^i(G, X) \rightarrow_{\times p} \hat{H}^i(G, X) \rightarrow \hat{H}^i(G, X/p) \rightarrow \hat{H}^{i+1}(G, X) \rightarrow_{\times p} \hat{H}^{i+1}(G, X) \rightarrow$ and when $i = 0$, $\hat{H}^0(G, X/p) = 0$ by the hypothesis. Thus $(X/p)^{tG}$ is acyclic by the previous proposition, i.e. $\hat{H}^i(G, X/p) = 0, \forall i$. Therefore multiplication by p is an isomorphism on $\hat{H}^i(G, X)$ for each i . But multiplication by $\#G$ is zero on $\hat{H}^i(G, X)$. Since G is a p -group, this is only possible if $\hat{H}^i(G, X) = 0, \forall i$.

Theorem 4.3.4. *Suppose for every prime p , and every p -Sylow subgroup $G_p \subseteq G$, $\hat{H}^0(G_p, X) = 0 = \hat{H}^1(G_p, X)$. Then X^{tG} is acyclic.*

pf. Since multiplication by $\#G$ is the zero map on $\hat{H}^i(G, X)$, it is $\#G$ -torsion. Thus it suffices to show $\hat{H}^i(G, X)[p] = 0, \forall p$. For every subgroup H of G , there are restriction and inflation maps $\begin{cases} X^{tG} \rightarrow X^{tH} \\ X^{tH} \rightarrow X^{tG} \end{cases}$ whose composition as an endomorphism of X^{tG} is homotopic to the multiplication by the index $[G : H]$. Applying this to a p -Sylow subgroup $H = G_p$, and taking cohomology, $\hat{H}^i(G, X)[p] \subset \hat{H}^i(G, X) \rightarrow \hat{H}^i(G_p, X) \rightarrow \hat{H}^i(G, X)$ whose composition is multiplication by $[G : G_p]$ which is prime to p . Thus it is an isomorphism when restricted to $\hat{H}^i(G, X)[p]$ and in particular, $\hat{H}^i(G, X)[p] \rightarrow \hat{H}^i(G_p, X)$ is injective. But by the previous theorem, $\hat{H}^i(G_p, X) = 0, \forall i$ and the theorem follows.

Proposition 4.3.2. *Let G be an l -group. Then there is a chain of normal subgroups $1G_0 \dots G_m = G$ s.t. G_{i+1}/G_i is cyclic for all i .*

pf. Let G act on itself via the adjoint action, i.e. $g \cdot x = gxg^{-1}, \forall x, g \in G$. Then the size of every G -orbit is either 1 or divisible by l . Since $\sum_{O \in G\text{-orbits}} \#O = \#G = l^n > 1$ and the G -orbit of 1 has order 1. Then l must divide the number of G -orbits of size 1, hence $\#Z(G) \neq 0$. Then choosing a nontrivial $x \in Z(G)$ and $G/\langle x \rangle$ is a normal subgroup of G . Then by induction, we get the proposition.

$\rightarrow L = L_m/L_{m-1}/\dots/L_0 = K$ cyclic extension. So we can reduce the case where G is cyclic of order n .

For each m , let $K \subseteq K_m \subseteq K^{unr}$ denote the degree m unramified extension of K .

CLAIM : Let $x \in K^\times$. Then x is in the image of $N : L_m^\times \rightarrow K_m^\times$.

pf. Here $K^\times/NL^\times = \mathcal{O}_K^\times/N\mathcal{O}_L^\times$. Then $0 \rightarrow \mathcal{O}_L^\times \rightarrow L^\times \rightarrow \mathbb{Z} \rightarrow 0$ yields the exact sequence $0 = \hat{H}^{-1}(\mathbb{Z}) \rightarrow \hat{H}^0(\mathcal{O}_L^\times) \hookrightarrow \hat{H}^0(L^\times) \rightarrow \hat{H}^0(\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$ where $\hat{H}^0(L^\times) \rightarrow \hat{H}^0(\mathbb{Z})$ is the zero map, given L/K being totally ramified. Thus $\hat{H}^0(L^\times) \cong \hat{H}^0(\mathcal{O}_L^\times)$.

$$\begin{array}{ccccccc}
L^\times & \xrightarrow{N} & K^\times & \longrightarrow & K^\times/NL^\times & \longrightarrow & 0 \\
\downarrow & & \downarrow & & \downarrow & & \\
L_m^\times & \xrightarrow{N} & K_m^\times & \longrightarrow & K_m^\times/NL_m^\times & \longrightarrow & 0 \\
\downarrow N_{L_m/L} & & \downarrow N_{K_m/K} & & \downarrow & & \\
L^\times & \longrightarrow & K^\times & \xrightarrow{N} & K^\times/NL^\times & \longrightarrow & 0
\end{array}$$

where $K^\times/NL^\times \rightarrow K^\times/NL^\times$ of induced

maps is raising to the n th power. Here $N_{K_m/K} : K_m^\times/NL_m^\times \rightarrow K^\times/NL^\times$ where both $K^\times/NL^\times, K_m^\times/NL_m^\times$ have order n , hence this map is injective iff it is surjective. Moreover, it is equivalent to the map $N_{K_m/K} : \mathcal{O}_{K_m}^\times/N\mathcal{O}_{L_m}^\times \rightarrow \mathcal{O}_K^\times/N\mathcal{O}_L^\times$. Since $N : \mathcal{O}_{K_m}^\times \rightarrow \mathcal{O}_K^\times$ is surjective, the above map is surjective too, i.e. it is an isomorphism. Therefore the induced map $K^\times/NL^\times \rightarrow K^\times/NL^\times$ is zero and the claim follows.

$\rightarrow N : \cup_m L_m^\times \rightarrow \cup_m K_m^\times$ is surjective.

Lemma 4.3.2. *Let X be a complex of G -modules, and suppose we have an exact sequence $1 \rightarrow H \rightarrow G \rightarrow G/H \rightarrow 1$. If $X^{tH} \cong 0 \cong (X^{hH})^{tG/H}$, then $X^{tG} \cong 0$.*

pf. $X_{hH} \rightarrow X^{hH}$ is a quasi-isomorphism since $X^{tH} \cong 0$. Also $X_{hG} \cong (X_{hH})_{hG/H} \cong (X^{hH})_{hG/H} \rightarrow (X^{hH})^{hG/H} \cong X^{hG}$ is a quasi-isomorphism since $(X^{hH})^{tG/H} \cong 0$. This quasi-isomorphism is given by the composition of two norm maps, which implies $X^{tG} = \text{hCoker}(X_{hG} \rightarrow X^{hG})$ is acyclic as desired.

Theorem 4.3.5 (General Vanishing theorem). *$[(L \otimes_K K^{unr})^\times]^{tG}$ is acyclic.*

pf. Let $L/L_0/K$ be the maximal unramified extension of K inside of L , so that L/L_0 is totally ramified. Let $H \equiv \text{Gal}(L/L_0)$ so that L_0/K is Galois with G/H . $L \otimes_K K^{unr} = L \otimes_{L_0} L_0 \otimes_K K^{unr} = L \otimes_{L_0} \Pi_{L_0 \hookrightarrow K^{unr}} K^{unr} = \Pi L \otimes_{L_0} K^{unr}$ canonically. Then $[(L \otimes_K K^{unr})^\times]^{tH} \cong \Pi_{L_0 \hookrightarrow K^{unr}} [(L \otimes_{L_0} K^{unr})^\times]^{tH} \cong \Pi [(L \otimes_{L_0} L_0^{unr})^\times]^{tH} \cong 0$ by the totally ramified case. Also $\Pi_{L_0 \hookrightarrow K^{unr}} [(L \otimes_{L_0} K^{unr})^\times]^{hH} = \Pi K^{unr, \times} \cong K^{unr, \times} [G/H]$ as a G/H -module. Then Tate cohomology vanishes for induced modules. Then by the previous lemma, the theorem follows.

- if L/K is totally ramified, $L^{unr} = L \otimes_K K^{unr}$, the induced Frobenius automorphisms of L^{unr} and K^{unr} are idenetical as L, K have the same residue field. The norm $N_{L/K} : L^{unr, \times} \rightarrow K^{unr, \times}$ satisfies $v_{K^{unr}} \circ N = v_{L^{unr}}$.

- If L/K is unramified of degree n ,
$$\begin{array}{ccc}
L \otimes_K K^{unr} & \rightarrow & \Pi_{i=0}^{n-1} K^{unr} \\
x \otimes y & \mapsto & ((\sigma^i x) \cdot y)_{i=0}^{n-1}
\end{array}$$
 where the product is taken via the fixed embedding $L \hookrightarrow K^{unr}$. Then for an element $(x_0, \dots, x_{n-1}) \in \Pi K^{unr, \times}$ to be in the image of $y/\sigma y$ iff $\Pi x_i \in \mathcal{O}_{K^{unr}}^\times$.

$$y_0 = x_0 \cdot \sigma y_{n-1}$$

$$\text{pf. } (y_0, \dots, y_{n-1}) \mapsto \left(\frac{y_0}{\sigma y_{n-1}}, \frac{y_1}{\sigma y_0}, \dots \right) \equiv (x_0, x_1, \dots) \rightarrow \vdots$$

$$y_{n-1} = x_{n-1} \cdot \sigma x_{n-2} \dots \sigma^{n-1} x_0 \sigma^n y_{n-1}$$

So we have $\frac{y_{n-1}}{\sigma^n y_{n-1}} = x_{n-1} \cdot \sigma x_{n-1} \dots \sigma^{n-1} x_0$. Then $\sigma^n = id$ if we are in $\mathcal{O}_{K^{unr}}^\times$

Theorem 4.3.6 (Main Theorem of Cohomological LCFT). *Let L/K be an extension of nonarchimedean local fields with finite Galois group G . Then*

$$(L^\times)^{tG} \cong (\mathbb{Z}[-2])^{tG}$$

pf. Assume L/K is totally ramified. Then there exists an exact sequence

$$1 \rightarrow L^\times \rightarrow L^{unr,\times} \rightarrow K^{unr,\times} \rightarrow \mathbb{Z} \rightarrow 0$$

We may rewrite this as

$$\begin{array}{ccccccccccc} A & & \cdots & \longrightarrow & 0 & \longrightarrow & L^\times & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\ \downarrow & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ B & & \cdots & \longrightarrow & 0 & \longrightarrow & L^{unr,\times} & \xrightarrow{1-\sigma} & L^{unr,\times} & \longrightarrow & 0 & \longrightarrow & \cdots \\ \downarrow & & & & \downarrow & & \downarrow & & \downarrow v & & \downarrow & & \\ \text{Coker}(A \rightarrow B) & & \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \end{array} \quad \text{where}$$

$L^\times$ is in degree -1. The final quasi-isomorphism to the homotopy cokernel is obtained from the above exact sequence because $A \rightarrow B$ is an injection. The termwise cokernel yields an injection $L^{unr,\times}/L^\times \hookrightarrow L^{unr,\times}$ since omitting the quotient $x \mapsto x/\sigma x$

$L^\times$ is precisely the kernel of this map. Then we have a quasi-isomorphism $B^{tG} = \text{hCoker}(L^{unr,\times} \rightarrow L^{unr,\times})^{tG} \cong \text{hCoker}((L^{unr,\times})^{tG} \rightarrow (L^{unr,\times})^{tG})$. By the vanishing theorem, $(L^{unr,\times})$ is acyclic, and therefore the homotopy cokernel is as well by the l.e.s. on cohomologies. Thus $(L^\times[2])^{tG} = \text{hCoker}((L^\times[1])^{tG} \rightarrow 0) = \text{hCoker}(A^{tG} \rightarrow B^{tG}) \cong \mathbb{Z}^{tG}$ and the theorem follows.

4.3.2 The Existence Theorem and LCFT

Definition 4.3.2. A subgroup Γ of $K^\times$ is a norm group if $\Gamma = NL^\times$ for some finite extension L/K .

Lemma 4.3.3. Given $\text{char } K = 0$, if $\Gamma \subseteq K^\times$ is a norm subgroup, then Γ is closed and of finite index.

pf. Let L/K be an extension of degree n s.t. $\Gamma = NL^\times$. Then $\Gamma \supseteq N_{L/K}K^\times = (K^\times)^n$, which is a finite index closed subgroup because it contains $1 + \mathfrak{p}_K^n$ for all sufficiently large n . Thus so is Γ . If $\text{char } K > 0$, $(K^\times)^n$ actually has infinite index in $K^\times$.

Lemma 4.3.4. If Γ' is a subgroup of $K^\times$ s.t. $K^\times \supseteq \Gamma' \supseteq \Gamma$ for a norm subgroup Γ , then Γ' is a norm subgroup as well.

pf. Let L/K be a finite extension s.t. $\Gamma = NL^\times$. We may assume L/K is abelian. Then by LCFT, $\Gamma'/\Gamma \subseteq K^\times/NL^\times \cong \text{Gal}(L/K)$ is a normal subgroup as $\text{Gal}(L/K)$ is abelian by assumption. Thus there exists some intermediate extension $L/K'/K$ s.t. $\Gamma'/\Gamma = \text{Gal}(L/K')$ and $K^\times/N(K')^\times = \text{Gal}(K'/K) = \text{Gal}(L/K)/\text{Gal}(L/K') = (K^\times/NL^\times)/(\Gamma'/\Gamma) = K^\times/\Gamma'$ canonically. Thus $\Gamma' = N(K')^\times$.

OR

$$\begin{array}{ccc} \text{Gal}(L/K) & \xrightarrow{\cong} & K^\times/NL^\times \\ \downarrow & & \downarrow \alpha \\ \text{Gal}(K'/K) & \xrightarrow{\cong} & K^\times/N(K')^\times \end{array} \quad \begin{array}{l} \text{by the LCFT. Since the inverse image of } \Gamma'/\Gamma = \ker \alpha \\ \text{in } K^\times \text{ is both } \Gamma' \text{ and } N(K')^\times, \Gamma' = N(K')^\times. \end{array}$$

Given $n \nmid \text{char}(K)$ and the group of n th roots of unity $\mu_n \subseteq K^\times$ has order n , $K^\times/(K^\times)^n \cong \text{hom}_{cts}(\text{Gal}(K), \mu_n)$ where these are group homomorphisms. If K is local,

$$\begin{aligned} K^\times/(K^\times)^n &= \text{hom}_{cts}(\text{Gal}(K), \mu_n) \\ &= \text{hom}_{cts}(\text{Gal}^{ab}(K), \mu_n) \quad \text{factor through the abelisation of } \text{Gal}(K) \\ &= \text{hom}_{cts}(\lim_{L/K} K^\times/NL^\times, \mu_n) \quad \text{by LCFT} \\ &= \lim_{L/K} \text{hom}(K^\times/NL^\times, \mu_n) \quad \text{by duality} \\ &\hookrightarrow \text{hom}(K^\times/(K^\times)^n, \mu_n) = \text{hom}(K^\times/(K^\times)^n, \mu_n) \end{aligned}$$

Let A be an n -torsion finite abelian group. Then $\#A = \#\text{hom}(A, \mathbb{Z}/n\mathbb{Z})$.

pf. A is a direct sum of groups $\mathbb{Z}/d\mathbb{Z}$ for $d|n$, so we may reduce to the case where $A = \mathbb{Z}/d\mathbb{Z}$ for d . Then $\text{hom}(\mathbb{Z}/d\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})[d]$ which has order $d = \#A$ as desired.

→ if $\mu_n \subseteq K$, $(K^\times)^n$ is a norm subgroup.

$$\begin{array}{ccc} \text{Gal}(K) & \xrightarrow{cts} & \text{hom}(K^\times/(K^\times)^n, \mu_n) \\ \downarrow & \nearrow \beta & \\ K^\times & \xrightarrow{\alpha} \lim_{L/K} K^\times/NL^\times \xrightarrow{\cong} & \text{Gal}^{ab}(K) \end{array} \quad \begin{array}{l} \text{pf.} \\ \text{where } \alpha \end{array}$$

is continuous as an open subgroup inside the inverse limit is a norm subgroup, hence its inverse image in $K^\times$ is a finite index and open subgroup. $\ker(\beta \circ \alpha) = (K^\times)^n$, which is open in the inverse limit as the maps are continuous. Then by the previous lemma, $(K^\times)^n$ is a norm subgroup.

Theorem 4.3.7 (Existence Theorem). *Given $\text{char} K = 0$, a subgroup Γ of $K^\times$ is a norm group iff Γ is closed and of finite index.*

pf.

- Let K be a general local field of characteristic 0. Let $L \equiv K(\zeta_n)/K$ where ζ_n denotes the set of primitive n th roots of unity. Since $(L^\times)^n$ is a norm subgroup in $L^\times$ by the previous statement, $N(L^\times)^n = N((L^\times)^n) \subseteq K^\times$ is a

norm subgroup in $K^\times$. But $N(L^\times)^n \subseteq (K^\times)^n \subseteq K^\times$, so by the previous lemma $(K^\times)^n$ is a norm subgroup in $K^\times$.

- For all N , $\exists n$ s.t. $(\mathcal{O}_K^\times)^n = (K^\times)^n \cap \mathcal{O}_K^\times \subseteq 1 + \mathfrak{p}_K^N$. Indeed $(\mathcal{O}_K^\times)^{q-1} \subseteq 1 + \mathfrak{p}_K$, where $q = \#\mathcal{O}_K/\mathfrak{p}_K$, since the reduction mod $\mathfrak{p}_K$ raised to the $(q-1)$ st power must be 1. Thus for sufficiently large $v(n)$, $(\mathcal{O}_K^\times)^{(q-1)n} \subseteq 1 + \mathfrak{p}_K^N$, since $(1+x)^n = 1 + nx + \dots$ and if $v(n) \gg 0$, all terms aside from 1 will be in $\mathfrak{p}_K^N$. As for finite index subgroups in the $\mathbb{Z}$ -direction, that is, where we restrict to the multiples of π^N , it suffices to simply replace n by nN so that only elements of valuation divisible by N are realised. Thus every finite index open subgroup of $K^\times$ contains $(K^\times)^n$ for some n , which is a norm subgroup, hence is itself a norm subgroup by the previous lemma.

Definition 4.3.3. A G -module M is smooth if $\forall x \in M$, $\exists$ a finite index open subgroup $K \subseteq G$ s.t. $K \cdot x = x$.

e.g. if $G \equiv \text{Gal}(\bar{K}/K)$, then G acts on both $\bar{K}, \bar{K}^\times$, both of which are smooth G -modules.

Definition 4.3.4. Let X be a complex of smooth G -modules bounded from below. Then

$$X^{hG} \equiv \lim(X^{K_i})^{hG/K_i}, \quad K_i = \ker(G \rightarrow G_i)$$

where X^{K_i} denotes the vectors stabilised by K_i

→ if X is acyclic, so is X^{hG} .

Proposition 4.3.3. If L/K is a G -Galois extension, then $H^1(G, L^\times) \equiv H^1((L^\times)^{hG}) = 0$.

pf. $L = \cup_i L_i$ where each L_i is a finite G_i -Galois extension of K . Then by definition, $H^1(G, L) = \lim H^1(G_i, L_i^\times) = 0$ by Hilbert's 90 theorem.

Corollary 4.3.1. Let $G \equiv \text{textGal}(\bar{K}/K)$ and n be prime to $\text{char}(K)$. If $\mu_n \subseteq K$, $K^\times/(K^\times)^n \cong \text{hom}_{cts}(G, \mu_n)$.

4.3.3 Brauer Group

4.4 Local Class Field Theory and Global Class Field Theory

4.4.1 Hilbert Symbols and Quadratic Extensions

Definition 4.4.1. Let the Hilbert symbol

$$(\cdot, \cdot)_{K^\times/(K^\times)^2 \times K^\times/(K^\times)^2} \rightarrow \{\pm 1\}$$

$$(a, b) \mapsto \begin{cases} 1 & \exists x, y \in K \text{ s.t. } ax^2 + by^2 = 1 \\ -1 & \text{otherwise} \end{cases}$$

for all $a, b \in K^\times$.

Let $\mathfrak{p} \subseteq \mathcal{O}_K$ be the unique maximal ideal, $k \equiv \mathcal{O}_K/\mathfrak{p}$ be the residue field with $\text{char}(k) = p$, an odd prime and $\pi \in \mathfrak{p}$ a uniformiser.

Proposition 4.4.1. Let $f : A \rightarrow B$ be a homomorphism of completely filtered abelian groups, i.e. $f(F_n A) \subset F_n(B)$, $\forall n \geq 0$. If the induced map $F_n A/F_{n+1} A \rightarrow F_n B/F_{n+1} B$ is surjective/injective, then f is surjective/injective.

pf. Assume the associated graded maps are surjective and both filtrations are complete. $\forall x \in B$, $\exists y_0 \in B$ s.t. $f(y_0) \equiv x \pmod{F_1 B}$ so that $x - f(y_0) \in F_1 B$. Since the associated graded map is surjective, $\exists \epsilon_1 \in F_1 A$ s.t. $f(\epsilon_1) \equiv x - f(y_0) \pmod{F_2 B}$. Since f is a homomorphism, $f(y_1) = f(y_0 + \epsilon_1) = f(y_0) + f(\epsilon_1) \equiv x \pmod{F_2 B}$ where $y_1 \equiv y_0 + \epsilon_1$. Also we can iterate this process to get $y_n \equiv y_{n+1} \pmod{F_{n+1} A}$. Then there is an induced element $y \in \lim A/F_n A = A$ corresponding to $(y_0, y_1, \dots)$ under the inverse limit s.t. $f(y) = x$, given both filtration being complete. So the proposition follows.

Let $x \in K^\times$ and write $x = \pi^{v(x)} y$, $y \in \mathcal{O}_K^\times$. Then TFAE

1. x is a square
2. $v(x)$ is even and y is a square
3. $y \pmod{\mathfrak{p}}$ is a square in $K^\times$

pf. $\begin{cases} 1 \rightarrow 2 \\ 2 \rightarrow 3 \end{cases}$ are trivial. Consider $\sigma : \mathcal{O}_K^\times \rightarrow S \subseteq \mathcal{O}_K^\times$
 $x \mapsto x^2$. If it is surjective, we have the inference $3 \rightarrow 1$. Here $\mathcal{O}_K^\times$ is a filtered abelian group with complete filtration $\mathcal{O}_K^\times \supseteq 1 + \mathfrak{p} \supseteq 1 + \mathfrak{p}^2 \supseteq \dots$ where $1 + \mathfrak{p}^n$ are all open subgroups of $\mathcal{O}_K^\times$. Clearly $\mathcal{O}_K^\times/(1 + \mathfrak{p}) = k^\times$ and similarly $(1 + \mathfrak{p}^n)/(1 + \mathfrak{p}^{n+1}) \cong k$ as $\forall \begin{cases} 1 + a\pi^n \\ 1 + b\pi^n \end{cases} \in 1 + \mathfrak{p}$, where

$a, b \in \mathcal{O}_K/\mathfrak{p}$, $(1 + a\pi^n)(1 + b\pi^n) = 1 + (a + b)\pi^n + ab\pi^{2n} \equiv 1 + (a + b)\pi^n \pmod{1 + \mathfrak{p}^{n+1}}$, given $2n \geq n + 1$, i.e. multiplication corresponds to the addition in k . Then σ acts

$$\begin{array}{ccccccc} \mathcal{O}_K^\times & \supseteq & 1 + \mathfrak{p} & \supseteq & 1 + \mathfrak{p}^2 & \supseteq & \dots \\ \downarrow \sigma & & \downarrow \sigma & & \downarrow \sigma & & \\ \mathcal{O}_K^\times & \supseteq & S & \supseteq & 1 + \mathfrak{p} & \supseteq & 1 + \mathfrak{p}^2 \supseteq \dots \end{array} \quad \text{where } 1 + \mathfrak{p} \subseteq S$$

on the filtration given 1 being trivially a square. Then $\sigma : \mathcal{O}_K^\times/(1 + \mathfrak{p}) = k^\times \rightarrow (k^\times)^2 = S/(1 + \mathfrak{p})$ is surjective. Moreover, $(1 + \mathfrak{p}^n)/(1 + \mathfrak{p}^{n+1}) \rightarrow (1 + \mathfrak{p})/(1 + \mathfrak{p}^{n+1})$ is equivalent to the map $k \rightarrow k$ since $(1 + \pi^n x)^2 = 1 + 2x\pi^n \pmod{\mathfrak{p}^{n+1}}$ and $2n \geq n + 1$. Here the above map is an isomorphism, given $\text{char } k = p \neq 2$. Thus σ is surjective on each graded term, so by the previous proposition, the above map is surjective as desired.

CLAIM : $(a, b) = 1$ iff b is a norm for the extension $K(\sqrt{a})/K$ i.e. there exists an element of $K(\sqrt{a})$ whose norm is b .

pf. If b is a norm, $\exists \alpha, \beta \in K$ s.t. $\alpha^2 - \beta^2 a = N(\alpha + \beta\sqrt{a}) = b \Leftrightarrow \alpha^2 = a\beta^2 + b$.

$$\begin{cases} a(\frac{\beta}{\alpha})^2 + b(\frac{1}{\alpha})^2 = 1 & \alpha \neq 0 \\ a \cdot \frac{1}{4} \frac{(a+1)^2}{a^2} + (-\beta^2 a) \frac{1}{4\beta^2} \frac{(a-1)^2}{a^2} = \frac{(a+1)^2 - (a-1)^2}{4a} = 1 & \alpha \neq 0 \end{cases} \rightarrow (a, b) = 1.$$

The converse holds by iterating the argument above with $\alpha \neq 0$.

CLAIM : Bimultiplicativity and nondegeneracy hold for the Hilbert symbol iff for all quadratic extensions L/K , $N(L^\times) \subseteq K^\times$ is a subgroup of index 2, that is $K^\times/NL^\times = \mathbb{Z}/2\mathbb{Z}$.

pf. Assume for all degree two extensions L/K , $NL^\times \subseteq K^\times$ is a subgroup of index 2. Let $a \in K^\times$. Consider a map $(a, \cdot) : K^\times \rightarrow \{\pm 1\}$. If a is a square, its image is identically 1, so it is a homomorphism. Otherwise, assume $L \equiv K(\sqrt{a})$. Then by the previous claim, $(a, b) = 1$ iff $b \in N(L^\times)$. The Hilbert symbol with a factors as $K^\times \rightarrow K^\times/NL^\times \cong \{\pm 1\}$ where the isomorphism is canonical given both groups having order 2. The projection is trivially a homomorphism. Thus bimultiplicativity follows. Let $a \notin (K^\times)^2$. Then $\exists b \in K^\times$ s.t. it is not a norm of $L = K(\sqrt{a})$ which is true iff $(a, b) = -1$. Then by contrapositive, the map above is surjective, so the non-degeneracy follows.

Conversely, if $a \notin (K^\times)^2$, then the map above is surjective by nondegeneracy and homomorphism by bimultiplicativity. Hence its kernel $N(K(\sqrt{a})^\times) \subseteq K^\times$ must have index $\#\{\pm 1\} = 2$.

Theorem 4.4.1. If L/K is a quadratic extension of local fields with $\text{char } K \neq 2$, then $NL^\times \subseteq K^\times$ is a subgroup of index 2.

pf. Let $L \equiv K(\sqrt{d})$ so that L only depends on d upto multiplication by squares.

- $v(d) = 0 \rightarrow \sqrt{d} \in \mathcal{O}_K^\times$.

– L/K unramified.

$$\begin{array}{ccccccc}
 \mathcal{O}_K^\times & \supseteq & 1 + \mathfrak{p}_L & \supseteq & 1 + \mathfrak{p}_L^\times & \supseteq & \dots \\
 \downarrow N & & \downarrow N & & \downarrow N & & \\
 \mathcal{O}_K^\times & \supseteq & 1 + \mathfrak{p}_K & \supseteq & 1 + \mathfrak{p}_K^\times & \supseteq & \dots
 \end{array}$$

is the diagram of filtrations preserved by the norm homomorphism. On the associated graded terms,

$$\begin{array}{ccc}
 \mathcal{O}_L^\times / (1 + \mathfrak{p}_L) & \xrightarrow{N} & \mathcal{O}_K^\times / (1 + \mathfrak{p}_K) \\
 \downarrow \cong & & \downarrow \cong \\
 k_L^\times & \xrightarrow{N} & k_K^\times
 \end{array}$$

commutes. Here k_L/k_K is a degree two

extension like L/K and the norm map between them is $N: \mathbb{F}_{q^2}^\times \rightarrow \mathbb{F}_q^\times$
 $x \mapsto x^{1+q}$.

If $x \in \mathbb{F}_q^\times$, $x = N(\sqrt{-x})$, $\sqrt{-x} \in \mathbb{F}_{q^2}^\times$ given that it is the unique degree two extension of $\mathbb{F}_q$. Thus it is surjective, so by the commutativity of the diagram, the map on Gr_0 is surjective. On the subsequent terms,

$$\begin{array}{ccc}
 (1 + \mathfrak{p}_L^n) / (1 + \mathfrak{p}_L^{n+1}) & \xrightarrow{N} & (1 + \mathfrak{p}_K^n) / (1 + \mathfrak{p}_K^{n+1}) \\
 \downarrow \cong & & \downarrow \cong \\
 k_L & \xrightarrow{T} & k_K
 \end{array}$$

commutes since $1 + a\pi^n \mapsto$

$(1 + a\pi^n)(1 + \sigma a\pi^n) = 1 + (a + \sigma a)\pi^n + a\sigma a\pi^{2n} \equiv 1 + Ta\pi^n \pmod{\pi^{n+1}}$. Then

the map between residue fields $T: \mathbb{F}_{q^2} \rightarrow \mathbb{F}_q$
 $x \mapsto x + x^q$ is surjective, given

that it is a separable extension. Then by commutativity, the map on the graded term Gr_n is surjective. Then by the previous proposition, the map is surjective. This implies $N(\mathcal{O}_L^\times) = \mathcal{O}_K^\times$. Then $\forall y \in L^\times$, $N(y) = y\sigma y$ and $v(y\sigma y) = 2v(y)$, given that $\text{Gal}(L/K)$ preserves valuations. Also π^2 is a norm. Hence if $v(d) = 0$, $NL^\times \subseteq K^\times$ is a subgroup of index 2 as desired.

–

- $v(d) = 1 \rightarrow$ tamely ramified, given $\text{char} K \neq 2$.

Let $L \equiv K(\sqrt{d})$ where $v(d) = 1$, i.e. d is not a square. Then $N(\alpha + \beta\sqrt{d}) = \alpha^2 - \beta^2 d$, and if $x \in (\mathcal{O}_K^\times)^2$, $x \in N(\sqrt{x})$ so x is a norm. Conversely $v(\alpha^2 - d\beta^2) = 0 = \min(v(\alpha^2), v(\beta^2) + 1) = v(\alpha^2)$ since $v(\alpha^2)$ is even, and the latter is odd, hence the two are unequal. Thus $\alpha, \beta \in \mathcal{O}_K^\times$, $x = \alpha^2 - d\beta^2$ is a square mod $\mathfrak{p}$ and this is true iff x is a square in $\mathcal{O}_K^\times$. Since $-d = N(\sqrt{d})$, there exists a uniformiser of K that is a norm. Then $x \in K^\times$ is a norm for $K(\sqrt{d})$ iff $(-d)^{-v(x)}x$ is a square mod $\mathfrak{p}$, so the group of norms of $L^\times$ has index 2 in $K^\times$.

Proposition 4.4.2. *The Hilbert symbol satisfies the following properties;*

1. *bimultiplicativity; $\forall a, b, c \in K^\times$, $(a, bc) = (a, b) \cdot (a, c)$*
2. *non-degeneracy; $\forall a \in K^\times$, if $(a, b) = 1$, $\forall b \in K^\times$, then $a \in (K^\times)^2$.*

pf. by the previous theorem and claims.

Definition 4.4.2. *The ring of adels is defined as*

$$\mathbb{A}_{\mathbb{Q}} \equiv \lim \Pi_{p \notin S} \mathbb{Z}_p \times \Pi_{p \in S} \mathbb{Q}_p$$

where the S are finite sets of places of $\mathbb{Q}$, ordered by inclusion and $\mathbb{Q}_{\infty} = \mathbb{R}$. The group of ideles is the multiplicative group of the ring of adeles

$$\mathbb{A}_{\mathbb{Q}}^\times = \lim \Pi_{p \notin S} \mathbb{Z}_p^\times \times \Pi_{p \in S} \mathbb{Q}_p^\times$$

Consider the map

$$\begin{aligned} \mathbb{A}_{\mathbb{Q}}^\times \times \mathbb{Q}^\times &\rightarrow \{\pm 1\} \\ (x, y) &\mapsto \Pi_p (x_p, y)_p \end{aligned}$$

where p may be either prime or ∞ , and $x = (x_p)_p \in \mathbb{A}_{\mathbb{Q}}^\times$. Then for all but finitely many p , $p \neq 2, \infty$, $x_p, y \in \mathbb{Z}_p^\times$, and therefore $(x_p, y)_p = 1$. Thus $(x_p, y)_p = 1$ for all but finitely many p .

CLAIM : $\forall x, y \in \mathbb{Q}^\times$, $\Pi_p (x, y)_p = 1$.

pf. Since the map is invariant under multiplication by squares, wlog assume

$$\begin{cases} x = \pm p_1 \dots p_r \\ y = \pm q_1 \dots q_s \end{cases}, \text{ where } p, q \text{ are odd primes.}$$

- $p \neq q$

$\Pi_l (p, q)_l = (p, q)_p (p, q)_q (p, q)_2 = \left(\frac{p}{q}\right) \left(\frac{q}{p}\right) (-1)^{\frac{p-1}{2} \frac{q-1}{2}} = 1$ by the quadratic reciprocity.

- $q, 2$

$\Pi_p (2, q)_p = (2, l)_l (2, l)_2 = \left(\frac{2}{l}\right) (-1)^{\theta(l)}$. For the first part, if ζ_8 is a primitive eighth root of unity, $\zeta_8 = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$, $\zeta_8 + \zeta_8^{-1} = 2\text{Re}(\zeta_8) = \sqrt{2}$ so $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\zeta_8)$ and there exists a character $\chi : (\mathbb{Z}/8\mathbb{Z}) \rightarrow \{\pm 1\}$. Then $\left(\frac{2}{\cdot}\right) = \chi$ given their kernels are the same. Then the above equation becomes 1.

•

4.4.2 The First Inequality

4.4.3 The Second Inequality

Appendix A

Representation Theory

A.1 Abstract Representation Theory

A.1.1 Semisimple Rings

Definition A.1.1. Let R be a ring. A left/right R -module is a commutative group M with a biadditive map $R \times M \rightarrow M$ s.t.

$$(a, x) \mapsto ax/xa$$

$$\left\{ \begin{array}{ll} 1x = x, \forall x \in M & x1 = x, \forall x \in M \\ (ab)x = a(bx), \forall a, b \in R, \forall x \in M & x(ab) = (xa)b, \forall a, b \in R, \forall x \in M \end{array} \right.$$

→ if the ring R is commutative, the notions of left and right R -module coincide.

e.g. the ring R with left/right multiplication by itself is a left/right R -module, called the **left/right R -module**.

Definition A.1.2. Let M be a R -module. A R -submodule of M is a subgroup N of M s.t. $ax \in N, \forall a \in R, x \in N$.

e.g. a submodule of the left R -module $M = R$ is just a left ideal of R .

Definition A.1.3. If M is a R -module and N is a submodule of M , then the quotient group M/N has a structure of R -module given by $a(x + N) = ax + N, \forall a \in R, x \in M$. This is called a quotient R -module.

Definition A.1.4. Let M be a R -module and $(M_i)_{i \in I}$ be a family of submodules of M . Then M is the sum of the M_i and write $M = \sum_{i \in I} M_i$ if $\forall x \in M, \exists x_i \in M_i$, all but finitely many equal to 0, s.t. $x = \sum_{i \in I} x_i$. Additionally M is the direct sum of M_i and write $M = \bigoplus_{i \in I} M_i$ if $\forall x \in M, \exists! x_i \in M_i$ s.t. $x = \sum_{i \in I} x_i$.

Definition A.1.5. Let M, N be R -modules. A R -linear map from M to N is a morphism of abelian groups $\phi: M \rightarrow N$ s.t. $\phi(ax) = a\phi(x), \forall a \in R, x \in M$.

Definition A.1.6. An exact sequence of R -modules is an exact sequence of abelian groups where all the abelian groups are R -modules and all the maps are R -linear.

e.g. for $\phi : M \rightarrow N$ a R -linear map, $0 \rightarrow \ker \phi \rightarrow M \rightarrow N \rightarrow \operatorname{coker} \phi \rightarrow 0$ is an exact sequence of R -modules.

Definition A.1.7. Let R be a ring and M be a R -module. Then M is simple or irreducible if $M \neq 0$ and if the only R -submodules of M are 0 and M . Additionally M is said to be semi-simple or completely reducible if for every submodule N of M , there exists a submodule N' of M s.t. $M = N \oplus N'$.

Lemma A.1.1. If $M \neq 0$ and M is semisimple, then it has a simple submodule.

pf. Let $x \in M \setminus \{0\}$ and set $M' = Rx$. Then M' is semisimple and nonzero, so we may assume $M = M'$. Let X be the set of submodules N of M s.t. $x \notin N$, ordered by inclusion. This set is nonempty because it contains the zero submodule. If $Y \subset X$ is totally ordered, $N = \bigcup_{N' \in Y} N'$ is a submodule of M . Also $x \notin N$ by definition, so N is an element of X , so $\exists$ an upper bound of any totally ordered subset of X . Then by Zorn's lemma, X has a maximal element N . As $x \notin N$, $N \neq M$. As M is semisimple, there exists a submodule N' of M s.t. $M = N \oplus N'$ and $N' \neq 0$ because $N \neq M$. If N' is not simple, choose a submodule $0 \neq N'' \subsetneq N'$, so that $N \oplus N'' \notin X$ by maximality of N in X , so $x \in N \oplus N''$, but then $N \oplus N'' = M$ a contradiction. Thus N' is simple and the lemma follows.

Theorem A.1.1. Let R be a ring and M be a R -module. Then TFAE

1. M is semisimple
2. M is the direct sum of a family of simple submodules
3. M is the sum of a family of simple submodules

pf.

- $2 \rightarrow 3$ direct sums are sums
- $1 \rightarrow 3$

assume M is semisimple. Let M' be the sum of all the simple submodules of M and choose a submodule M'' of M s.t. $M = M' \oplus M''$. If $M' \neq M$, $M'' \neq 0$, so by the previous lemma, M'' has a simple submodule N , but then $N \subset M'$, a contradiction that M', M'' are in direct sum. So $M' = M$.

- $3 \rightarrow 1$

let (M_i) be the family of all simple submodules of M . Assume $M = \sum_{i \in I} M_i$. Let N be a submodule of M . Define the set X be a subset K of I s.t. the number $N + \sum_{k \in K} M_k$ is direct, ordered by inclusion. This set is not empty since $\emptyset \in X$. If $Y \subset X$ is a totally ordered subset, let $K = \cup_{K' \in Y} K'$. Let $n \in N$ and $(m_k)_{k \in K} \in \prod_{k \in K} M_k$ s.t. $n + \sum m_k = 0$. Let $K_0 \subset K$ be a finite subset s.t. $m_k = 0, \forall k \in K \setminus K_0$. Then $\forall k \in K_0, \exists L_k \in Y$ s.t. $k \in L_k$. As Y is totally ordered and K_0 is finite, $\exists L \in Y$ s.t. $\cup_{k \in K_0} L_k \subset L$. Then $L \in X$, so $n + \sum m_k = 0$ implies $n = 0$ and $m_k = 0, \forall k \in K_0$. By the choice of K_0 , we get n and all m_k are 0. So $K \in X$. Then by Zorn's lemma X has a maximal element J . Let $M' = N \oplus \bigoplus_{j \in J} M_j$. Let $i \in I$. If $M_i \not\subset M'$, $M' \cap M_i = 0$ given M_i being simple. Then the sum $M' + M_i = (N + \sum_{j \in J} M_j) + M_i$ is direct, so $J \cup \{i\} \in X$, a contradiction to the maximality of J , so $M = M'$, so M is semisimple.

Definition A.1.8. Let R be a ring and M be a R -module. Then M is said to be finitely generated if there exists a finite family $(x_i)_{i \in I}$ of M s.t. $M = \sum_{i \in I} Rx_i$. We say that M is cyclic, if $\exists x \in M$ s.t. $M = Rx$.

Theorem A.1.2. Let R be a ring. TFAE

1. all s.e.s. of R -modules split
2. all R -modules are semisimple
3. all finitely generated R -modules are semisimple
4. all cyclic R -modules are semisimple
5. the left regular R -module R is semisimple

pf.

- $1 \leftrightarrow 2$ by the definition of semisimple
- $2 \rightarrow 3 \rightarrow 4 \rightarrow 5$ is obvious
- $5 \rightarrow 2$

let M be a R -module. If $x \in M$, $Rx \subset M$ is a quotient of R , so it is semisimple because R is semisimple. As $M = \sum_{x \in M} Rx$, by the previous theorem, M is semisimple.

Definition A.1.9. A R -module P is called *projective* if for every surjective R -linear map $\pi : M \rightarrow N$ and every R -linear map $\phi : P \rightarrow N$, there exists a R -linear map

$$\psi : P \rightarrow M \text{ s.t. } \pi\psi = \phi.$$

$$\begin{array}{ccc} & P & \\ & \downarrow \phi & \\ M & \xrightarrow{\pi} & N \end{array}$$

$\swarrow \psi$

Definition A.1.10. A R -module F is called *free* if there exists a family $(e_i)_{i \in I}$ of elements of F s.t. $F = \bigoplus_{i \in I} Re_i$ and s.t. $\forall i \in I$, the map $\begin{array}{ccc} R & \rightarrow & Re_i \\ a & \mapsto & ae_i \end{array}$ is an isomorphism.

Lemma A.1.2. Let P be a R -module. Then TFAE

1. P is projective
2. P is a direct summand of a free R -module

pf.

- $2 \rightarrow 1$

every free R -module is projective.

pf. if $F = \bigoplus Re_i$, let $\pi : M \rightarrow N$ be a surjective R -linear map and $\phi : P \rightarrow N$ be a R -linear map. $\forall i \in I$ choose $x_i \in M$ s.t. $\pi(x_i) = \phi(e_i)$. Define a R -linear map $\begin{array}{ccc} \psi : P & \rightarrow & M \\ e_i & \mapsto & x_i \end{array}$. Then $\pi\psi = \phi$ and it is projective.

a direct summand of projective module is projective

pf. let P be projective and suppose $P = P' \oplus P''$. Let $\pi : M \rightarrow N$ be surjective R -linear map and $\phi' : P' \rightarrow N$ be a R -linear map. Let $\phi = \phi' + 0 : P = P' \oplus P'' \rightarrow N$. As P is projective, $\exists \psi : P \rightarrow M$ s.t. $\pi\psi = \phi$. Write $\psi = \psi' \oplus \psi''$ where $\begin{cases} \psi' : P' \rightarrow M \\ \psi'' : P'' \rightarrow M \end{cases}$. Then $\pi\psi' = \phi'$ and the stmt follows.

Then free module is a projective module and its direct summand is also projective and the lemma follows.

- $1 \rightarrow 2$

suppose P is projective. Let $M = \bigoplus_{x \in P} R$ and define $\begin{array}{ccc} \phi : M & \rightarrow & P \\ (a_x)_{x \in P} & \mapsto & \sum_{x \in P} a_x x \end{array}$. Then ϕ is R -linear and surjective, so there exists a R -linear map $\psi : P \rightarrow M$ s.t. $\phi\psi = id_P$. So $M \cong P \oplus \ker \phi$ and we have found a free R -module M s.t. P is a direct summand of M .

Theorem A.1.3. Let R be a ring. TFAE

1. R is a semisimple ring.
2. all R -modules are projective
3. all finitely generated R -modules are projective
4. all cyclic R -modules are projective

pf.

- $1 \leftrightarrow 2$ by the definition
- $2 \rightarrow 3 \rightarrow 4$ is obvious
- $4 \rightarrow 1$

suppose all cyclic R -modules are projective. Let $I \subset R$ be a left ideal, then R/I is a cyclic R -module, hence projective. So there exists a s.e.s. $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$ which splits. So R is semisimple.

Theorem A.1.4 (Schur's Lemma). *Let R be a ring, M, N be R -modules and $u : M \rightarrow N$ be a R -linear map.*

1. if M is simple, then $u = 0$ or u is injective
2. if N is simple, then $u = 0$ or u is surjective

pf. since $\ker u \subseteq M$, given M being simple, $\ker u$ is either M or 0 and the first part of the theorem follows. Similarly, $\operatorname{Im} u \subseteq N$, given N being simple, $\operatorname{Im} u$ is either N or 0 and the last part follows.

Definition A.1.11. *Let R be a ring. Write $S(R)$ for the set of isomorphism classes of simple R -modules.*

Lemma A.1.3. *Let M be a R -module and suppose there exists a simple R -module S and a set I and an isomorphism $\phi : M \rightarrow \bigoplus_{i \in I} S$. Then every simple R -submodule of M is isomorphic to S .*

pf. Let N be a simple R -submodule of M and suppose N is not isomorphic to S . Then $\forall i \in I$, the composition of the projection on the i th summand $\bigoplus S \rightarrow S$ and of ϕ is a R -linear map $N \rightarrow S$ which has to be zero by Schur's lemma. Then ϕ is zero on N , which implies $N = 0$, a contradiction to the fact that N is simple.

Theorem A.1.5. *Let R be a ring and M be a semisimple R -module. $\forall S \in S(R)$, let M_S be the sum of all the submodules of M that are isomorphic to S . Then*

$$\left\{ \begin{array}{l} M = \bigoplus_{S \in S(R)} M_S \\ \exists I_S \text{ s.t. } M_S \cong \bigoplus_{i \in I_S} S, \forall S \in S(R) \end{array} \right.$$
Let N be a submodule of M . If we write $N \cong \bigoplus_{S \in S(R)} N_S$ and $N_S \cong \bigoplus_{i \in J_S} S$ as in the previous statements, then $\forall S \in S(R)$, $N_S = M_S \cap N$, and we can find an injection $J_S \hookrightarrow I_S$.

pf.

- $\sum_{S \in S(R)} M_S$ is the sum of all the simple submodules of M . As M is semisimple, $M = \sum_{S \in S(R)} M_S$. Let $S \in S(R)$ and $X = S(R) \setminus \{S\}$. Also, let $N \equiv M_S \cap (\sum_{S' \in X} M_{S'})$. As N is a submodule of M , it is semisimple. So if it is nonzero, it is a simple submodule of M_S , hence isomorphic to S and also a simple submodule of $\sum_{S' \in X} M_{S'}$, hence isomorphic to an element of X , a contradiction. So $N = 0$ and the stmt follows.
- As M_S is a submodule of the semisimple R -module M , it is semisimple. Then M_S is the direct sum of a family of simple submodules by the theorem A.1.1. But every simple submodules of M_S is isomorphic to S .
- Let $S \in S(R)$. Then N_S is a sum of simple R -modules isomorphic to S , so $N_S \subset M_S$. On the other hand, $N \cap M_S$ is a direct sum of simple submodules of N and all these simple modules have to be isomorphic to S , so $N \cap M_S \subset N_S$. So we may assume $\begin{cases} M = M_S \\ N = N_S \end{cases}$ for some $S \in S(R)$. Then we can reduce the case into the one s.t. if S is simple R -module and I, J are two sets s.t. we have an injective R -linear map $u : N \equiv \bigoplus_{j \in J} S \hookrightarrow M \equiv \bigoplus_{i \in I} S$, then there exists an injection $J \hookrightarrow I$.

For any R -modules M_1, M_2 , let $\text{hom}(M_1, M_2) \subset \text{hom}_R(M_1, M_2)$ be the subgroup of R -linear maps that are 0 outside a finite length submodule of M_1 . Then $\begin{cases} \text{hom}(N, S) = \bigoplus_{j \in J} \text{End}_R(S) \subset \text{hom}_R(N, S) = \prod_{j \in J} \text{End}_R(S) \\ \text{hom}(M, S) = \bigoplus_{i \in I} \text{End}_R(S) \subset \text{hom}_R(M, S) = \prod_{i \in I} \text{End}_R(S) \end{cases}$. Here $\mathbb{D} \equiv \text{End}_R(S)$ is a division ring by Schur's lemma, and $\text{hom}_R(M, S), \text{hom}_R(N, S)$ are naturally $\mathbb{D}$ -modules and $\text{hom}(M, S), \text{hom}(N, S)$ are $\mathbb{D}$ -submodules. Moreover, $\exists \begin{matrix} \phi : \text{hom}(M, S) & \rightarrow & \text{hom}(N, S) \\ f & \mapsto & f \circ u \end{matrix}$ is $\mathbb{D}$ -linear. Let $g \in \text{hom}(N, S)$. As M is a semisimple R -module, $\exists N'$ a R -submodule of M s.t. $M = u(N) \oplus N'$. Define $f : M \rightarrow S$ by $f(u(x) + y) = g(x)$ if $x \in N, y \in N'$, since u is injective, and clearly R -linear. So $\phi(f) = g$ so it is surjective. Then the stmt follows by the incomplete basis theorem.

→ the nonzero M_S are called the **isotypic components of M** and $M = \bigoplus_{S \in S(R)} M_S$ is called the **isotypic decomposition of M** .

Definition A.1.12. A ring R is called simple if $R \neq 0$ and if the only ideals of R are 0 and R .

Proposition A.1.1. If R is a ring and $n \geq 1$ is an integer, then any ideal I of $M_n(R)$ is of the form $I = M_n(J)$, where J is a uniquely determined ideal of R .

pf. If J is an ideal of R , $M_n(J)$ is an ideal of $M_n(R)$. Also if J, J' are two ideals of R s.t. $M_n(J) = M_n(J')$, then $J = J'$. So we just need to prove any ideal of $M_n(R)$ is of the form $M_n(J)$. Let I be an ideal of $M_n(R)$ and let J be the set of $(1,1)$ -entries of elements of I . Let $x, y \in J$ and let $a \in R$. Choose matrices $X, Y \in I$ s.t. $(1,1)$ -entries of X, Y are x, y respectively. Then $aX, Xa, X+Y$ are in I and their respective $(1,1)$ -entries are $ax, xa, x+y$ so $ax, xa, x+y \in J$. So J is an ideal of R . Let's denote by E_{ij} the elementary matrices in $M_n(R)$. If $X = (x_{ij}) \in M_n(R)$, then $E_{ij}XE_{kl} = x_{jk}E_{il}$. If $X \in I$, $\forall j, k \in \{1, \dots, n\}$, $E_{1j}XE_{k1} = x_{jk}E_{11} \in I$ so $x_{jk} \in J$ so $X \in M_n(J)$. This implies $I \subset M_n(J)$. Let $X = (x_{ij}) \in M_n(J)$. Then $X = \sum_{ij} x_{ij}E_{ij}$. Fix $i, j \in \{1, \dots, n\}$. Choose $Y \in I$ s.t. $(1,1)$ -entry of Y is x_{ij} . Then $E_{i1}YE_{1j} = x_{11}E_{ij} \in I$ so $M_n(J) \subset I$. Thus the proposition follows.

Corollary A.1.1. If R is a simple ring, then so is $M_n(R)$, $\forall n \geq 1$.

Theorem A.1.6. Let $\mathbb{D}$ be a division ring, $n \geq 1$ be an integer and $R = M_n(\mathbb{D})$. Let $V = \mathbb{D}^n$. Then V is viewed as a left R -module by considering $\mathbb{D}^n$ as a space of $n \times 1$ matrices, and also viewed as the right $\mathbb{D}$ -module $\mathbb{D}_{\mathbb{D}}^n$. Then

1. the ring R is simple, semisimple, left Artinian and left Noetherian.
2. R has a unique simple left module which is V . As a left R -module, R and V^n are isomorphic.
3. $\text{End}_{\mathbb{D}}(V) = R$
4. $\text{End}_R(V) = \mathbb{D}$

pf.

- By the previous proposition, R is simple because $\mathbb{D}$ is. As a left $\mathbb{D}$ -vector space, R is finite dimensional. Since left ideals of R are $\mathbb{D}$ -vector subspaces of R , R is left Artinian and left Noetherian. Take a nonzero R -submodule W

of V . Let $w \in W \setminus \{0\}$ and write $w = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$. Choose $i_0 \in \{1, \dots, n\}$ s.t. $w_{i_0} \neq 0$.

Then $\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = (w_{i_0}^{-1}E_{1i_0})w \in W$. If $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$ is any element of V , then $v =$

$\begin{bmatrix} v_1 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ v_n & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in W$ so $W = V$. As the left R -module, $R \cong V^n$, the ring R is semisimple.

- Let M be a simple R -module. Choose $x \in M \setminus \{0\}$. Then the map $\begin{matrix} u : R & \rightarrow & M \\ a & \mapsto & ax \end{matrix}$ is surjective, so M is isomorphic to a quotient of the R -module R . As $R \cong V^n$, $M \cong V$.

- Consider the map $\begin{matrix} \phi : R & \rightarrow & \text{End}_{\mathbb{D}}(V_{\mathbb{D}}) \\ a & \mapsto & (x \mapsto ax) \end{matrix}$. This map is well-defined because for all $a \in R$, $\forall x \in V$, $\lambda \in \mathbb{D}$, $\phi(a)(x\lambda) = a(x\lambda) = (ax)\lambda = (\phi(a)(x))\lambda$, so $\phi(a)$ is $\mathbb{D}$ -linear. Let $a = (a_{ij}) \in R = M_n(\mathbb{D})$ s.t. $\phi(a) = 0$. Then $\forall j \in \{1, \dots, n\}$, if $e_j \in V$ is the element with j th entry equal to 1 and all other entries equal to 0, we have $0 = \phi(a)(e_j) = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix}$ so $a = 0$, which implies ϕ is injective. Let

$u \in \text{End}_{\mathbb{D}}(V_{\mathbb{D}})$. $\forall j \in \{1, \dots, n\}$ if $e_j \in V$ is defined as above, $u(e_j) = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix}$. Let $a = (a_{ij}) \in M_n(\mathbb{D})$. Then $\phi(a)(e_j) = u(e_j)$, $\forall j \in \{1, \dots, n\}$. As $(e_1, \dots, e_n)$ is a basis of $V_{\mathbb{D}}$ over $\mathbb{D}$, this implies $\phi(a) = u$ so ϕ is surjective.

- Let $E = \text{End}_R(V)$. Write the action of the ring E on V on the right, that is, write $x(v) = vx$, $\forall v \in V$, $x \in E$. Let $\begin{matrix} \psi : \mathbb{D} & \rightarrow & E \\ \lambda & \mapsto & (v \mapsto v\lambda) \end{matrix}$. As for λ , this map is well-defined, i.e. $\psi(\lambda)$ is R -linear $\forall \lambda \in \mathbb{D}$ and it is a morphism of rings. Because $\mathbb{D}$ is a division algebra, ψ is injective. Let $x \in E$ and define $\lambda \in \mathbb{D}$ by $e_1x = \lambda e_1 + \sum_{j=2}^n \mu_j e_j$ s.t. $\mu_j \in \mathbb{D}$, $e_1, \dots, e_n \in V$ are as in the proof above.

Let $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in V$ and let $a = \begin{bmatrix} v_1 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ v_n & 0 & \dots & 0 \end{bmatrix} \in M_n(\mathbb{D}) = R$. Then $v = ae_1$ so

$$vx = (ae_1)x = a(e_1x) = a \begin{bmatrix} \lambda \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} = \begin{bmatrix} v_1\lambda \\ \vdots \\ v_n\lambda \end{bmatrix} = v\lambda \text{ so } x = \psi(\lambda) \text{ and } \psi \text{ is surjective.}$$

Corollary A.1.2. *If $\mathbb{D}, \mathbb{D}'$ are two division rings and $n, n' \geq 1$ are two integers s.t. $M_n(\mathbb{D}) \cong M_{n'}(\mathbb{D}')$ as rings, then $\mathbb{D} \cong \mathbb{D}'$ and $n = n'$.*

pf. Let $R = M_n(\mathbb{D}) \cong M_{n'}(\mathbb{D}')$ and let M be the unique simple R -module by the previous theorem 2. Then by the previous theorem 3, $\mathbb{D}' \cong \text{End}_R(M) \cong \mathbb{D}$. Hence $n = \dim_{\mathbb{D}}(M) = \dim_{\mathbb{D}'}(M) = n'$.

Definition A.1.13. If R is a ring, we denote by R^{op} its opposite ring; it is isomorphic to R as an additive group and its multiplication is given by $ab(R^{op}) = ba(R)$.

Theorem A.1.7 (Double Centraliser Property). Let R be a simple ring and I be a nonzero left ideal of R . Let $D = \text{End}_R(I)$ and make I a right D^{op} -module by setting $xu = u(x)$ if $\begin{cases} x \in I \\ u \in D \end{cases}$. Then the map $f: R \rightarrow \text{End}_{D^{op}}(I)$ is an isomorphism of rings.

pf. As $I \neq 0$, $\ker f \neq R$. As $\ker f$ is an ideal of R and R is simple, $\ker f = 0$, i.e. f is injective. Let $E = \text{End}_{D^{op}}(I)$. Make I a left E -module by setting $hx = h(x)$, $\forall x \in I, h \in E$. Then $\forall x \in I, \forall h \in E, hf(x) = f(hx)$. Indeed if $a \in I$, $\begin{matrix} r_a: I \rightarrow I \\ y \mapsto ya \end{matrix}$ is in D so $h(xa) = h(r_a(x)) = h(xr_a) = h(x)r_a = h(x)a$ so $(hf(x))(a) = h(xa) = h(x)a = f(h(x))(a)$. Thus $Ef(I) \subset f(I)$. Since I is a nonzero left ideal of R , IR is a nonzero ideal of R , hence $IR = R$ as R is simple. Then $f(I)f(R) = f(R)$ and $E = Ef(R) = Ef(I)f(R) \subset f(I)f(R) = f(R)$, so f is surjective.

Corollary A.1.3. If R is a simple ring with a minimal nonzero left ideal, then there exists a unique division ring $\mathbb{D}$ and a unique integer $n \geq 1$ s.t. $R \cong M_n(\mathbb{D})$. In particular, a simple ring is left Artinian iff it is of the form $M_n(\mathbb{D})$ with $\mathbb{D}$ a division ring and $n \in \mathbb{Z}_{++}$.

pf. Let $I \subset R$ be a minimal nonzero left ideal. Then I is simple R -module, so $\mathbb{D} \equiv \text{End}_R(I)$ is a division ring by Schur's lemma. Make I a right $\mathbb{D}^{op}$ -module as in the previous theorem. Then $R \cong \text{End}_{\mathbb{D}^{op}}(I)$. Let $E = \text{End}_{\mathbb{D}^{op}}(I)$ and let $E_f = \{u \in E \mid rk(u) < \infty\}$ where $rk(u)$ is the dimension over $\mathbb{D}^{op}$ of the image of u . Then $E_f \neq 0$ and E_f is an ideal of E . Since $E \cong R$ is a simple ring, $E_f = E$, hence the identity of I is in E_f and I is a finite dimensional right $\mathbb{D}^{op}$ -vector space. The uniqueness follows by the previous corollary.

Proposition A.1.2. Let $R_1, \dots, R_n$ be rings and let $R = R_1 \times \dots \times R_n$. Then every R -module is of the form $M = M_1 \times \dots \times M_n$ with M_i a R_i -module for $1 \leq i \leq n$.

pf. In R , write $1 = e_1 + \dots + e_n$, $\forall e_i \in R_i$. As an element of $R_1 \times \dots \times R_n$, e_i is the n -tuple with i th entry equal to $1 \in R_i$ and 0 for all other entries. Then all e_i are central in R . Let M be a R -module, set $M_i = e_i M$. Then R acts on M_i through the obvious projection $R \rightarrow R_i$, so we need to show $M \cong M_1 \times \dots \times M_n$. Consider the map $M_1 \times \dots \times M_n \rightarrow M$ which is the R -linear map. If $x \in M$, $x = xe_1 + \dots + xe_n$

with $xe_i \in M_i$, $\forall i$ so u is surjective. If $\sum x_i = 0$, $x_i \in M_i$ then $\forall j \in \{1, \dots, n\}$, $0 = e_j(x_1 + \dots + x_n) = e_j x_j = x_j$ so u is injective.

Corollary A.1.4. *Suppose $R = R_1 \times \dots \times R_n$ as in the previous proposition.*

1. *R is semisimple iff all the R_i are semisimple*
2. *Every simple R -module is of the form $0 \times \dots \times 0 \times M_i \times 0 \times \dots \times 0$, $i \in \{1, \dots, n\}$ and M_i a simple R_i -module*

Corollary A.1.5. *Let $\mathbb{D}_1, \dots, \mathbb{D}_r$ be division rings, and $n_1, \dots, n_r \geq 1$ be integers. Then $R \equiv M_{n_1}(\mathbb{D}_1) \times \dots \times M_{n_r}(\mathbb{D}_r)$ is a semisimple ring and its simple modules are $\mathbb{D}_1^{n_1}, \dots, \mathbb{D}_r^{n_r}$.*

Let R be a ring and I be a left ideal of R . Denote $\mathcal{J}_I \subset R$ the sum of all the left ideals I' of R that are isomorphic to I as R -modules. Then it is a left ideal of R .

Lemma A.1.4. *Let R be a ring and I be a minimal nonzero left ideal of R . Then $\mathcal{J}_I$ is an ideal of R . Moreover if I, J are two nonisomorphic minimal nonzero left ideals of R , then $\mathcal{J}_I \mathcal{J}_J = 0$.*

pf. Let I' be a left ideal of R s.t. $I' \cong I$ as R -modules, and let $a \in R$. Then we have a surjective R -linear map $\begin{matrix} I' & \rightarrow & I'a \\ x & \mapsto & xa \end{matrix}$ and I' is a simple R -module, so $I'a = 0$ or $I'a \cong I'$. If $I'a = 0$, $I'a \subset \mathcal{J}_I$, otherwise $I'a$ is another left ideal of R isomorphic to I , so $I'a \subset \mathcal{J}_I$. thus $\mathcal{J}_I$ is a right ideal.

Let J be another minimal nonzero left ideal of R , and suppose $\mathcal{J}_I \mathcal{J}_J \neq 0$. Then there are left ideals I', J' of R and an element $a \in J'$ s.t. $I' \cong I$, $J' \cong J$, $I'a \neq 0$. As J' is a simple R -module and $I'a \subset J'$ is a nonzero submodule of J' , $I'a = J'$. As I' is a simple R module, the surjective map $\begin{matrix} I' & \rightarrow & I'a \\ x & \mapsto & xa \end{matrix}$ is an isomorphism. So $I \cong I' \cong I'a = J' \cong J$.

Theorem A.1.8 (Artin-Wedderburn). *Let R be a semisimple ring. Let $(I_i)_{i \in A}$ be a set of representatives of the isomorphism classes of minimal nonzero left ideals of R , and let $R_i = \mathcal{J}_{I_i} \subset R$. Then A is finite so we choose an identification $A = \{1, \dots, r\}$. Moreover, all the R_i are rings with units, $R \cong R_1 \times \dots \times R_r$ as rings and $\forall i \in \{1, \dots, r\}$, there exists a unique division ring $\mathbb{D}_i$ and a unique integer $n_i \geq 1$ s.t. $R_i \cong M_{n_i}(\mathbb{D}_i)$.*

pf. As R is a semisimple ring, R is a direct sum of simple submodules by the theorem A.1.1. So $R = \bigoplus_{i \in A} R_i$ by the theorem A.1.5. Then A is finite.

By the previous lemma, every $\mathcal{J}_i$ is an ideal of R . As $R = \bigoplus_{i=1}^r R_i$, all the R_i are rings with units and $R \cong R_1 \times \dots \times R_r$ as rings. Fix $i \in \{1, \dots, r\}$. Let $J \neq 0$ be an ideal of R_i . As J is also an ideal of R , it contains a minimal nonzero left

ideal I of R . By definition of $I_1, \dots, I_r$, $\exists j \in \{1, \dots, r\}$ s.t. $I \cong I_j$ as R -modules. As $I \subset J \subset R_i$, $j = i$. So $R_i = \mathcal{I}_{I_i} = \mathcal{I}_I$. Hence it suffices to show if I' is a left ideal of R s.t. $I \cong I'$, then $I' \subset J$. Fix such a I' and let $\phi : I \rightarrow I'$ be an isomorphism of R -modules. As R is a semisimple ring, there exists a left ideal I'' of R s.t. $R = I \oplus I''$. Write $1 = e + e''$, $e \in I$, $e'' \in I''$. Then $e^2 = e$, $I = Re$. So $I' = \phi(I) = \phi(Re) = \phi(Re^2) = \phi((Re)e) = \phi(Ie) = I\phi(e) \subset J$.

e.g. let K be a field and suppose R is a semisimple K -algebra that is finite dimensional as a K -vector space. Then its simple factors R_i are also K -algebras and so are the division rings $\mathbb{D}_i$ with $\dim_K \mathbb{D}_i < \infty$. In particular, if K is algebraically closed, all the $\mathbb{D}_i$ are equal to K , so $R \cong M_{n_1}(K) \times \dots \times M_{n_r}(K)$.

Definition A.1.14. Let R be a ring. The Jacobson radical of R is the intersection of all the maximal left ideals of R , denoted by $\text{rad}(R)$.

A.1.2 The Representation Ring

Definition A.1.15. Let R be a ring. Define $K(R)$ to be the quotient of the free abelian group on the basis elements $[M]$, for M a finite length R -module, by all the relations of the form $[M] = [M'] + [M'']$, where $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an exact sequence of R -modules. It is called the Grothendieck group of the category of finite length R -modules. Define $PK(R)$ to be the quotient of the free abelian group on the basis elements $[P]$, for P a finite length projective R -module, by all the relations of the form $[P] = [P'] + [P'']$ where $0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0$ is an exact sequence of R -modules. It is called the Grothendieck group of the category of finite length projective R -modules.

$\rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow 0$ gives $[0] + [0] = [0]$ in $K(R), PK(R)$ so $[0] = 0$. Then for an isomorphism of R -modules $M \rightarrow M'$, $0 \rightarrow M \rightarrow M' \rightarrow 0$ gives $[M] = [M'] + [0] = [M']$ in $K(R)$, and similarly for $PK(R)$. So if $0 = M_n \subset M_{n-1} \subset \dots \subset M_0 = M$, $[M] = \sum_{i=1}^n [M_{i-1}/M_i]$ in $K(R)$.

Proposition A.1.3. As a group, $K(R)$ is the free abelian group with basis $\{[M] : M \in S(R)\}$, where $S(R)$ is the set of isomorphism classes of simple R -modules.

pf. Let S be the free abelian group on the set $S(R)$, and denoted by $(e_M)_{M \in S(R)}$ its canonical basis. Define $\phi : S \rightarrow K(R)$ $\psi : K(R) \rightarrow S$ where

$$e_M \mapsto [M], \quad M \mapsto \sum e_{M_{i-1}/M_i}$$
where $M = M_0 \supset M_1 \supset \dots \supset M_n = 0$. If $0 \rightarrow M' \rightarrow M \xrightarrow{u} M'' \rightarrow 0$ is a s.e.s. of finite length R -modules, choose Jordan-Hölder series $\begin{cases} M' = M'_0 \supset \dots \supset M'_n = 0 \\ M'' = M''_0 \supset \dots \supset M''_m = 0 \end{cases} \cdot \forall 0 \leq$

$j \leq m$, $M_j = u^{-1}(M'_j)$. Then $M = M_0 \supset \cdots \supset M_m = M'_0 \supset \cdots \supset M'_n = 0$ is a Jordan-Hölder series for M , which gives the desired equality. So ψ is well-defined and ψ, ϕ are inverses of each other, so the proposition follows.

Proposition A.1.4. *If R is a semisimple ring, the $K(R) = PK(R)$ and for all finite length R -modules M, M' , $[M] = [M']$ in $K(R)$ iff $M \cong M'$.*

pf. If R is semisimple, every R -module is projective, so $K(R) = PK(R)$.

Let M, M' be two R -modules of finite length. As R is semisimple, $\begin{cases} M \cong \bigoplus_{N \in S(R)} N^{\oplus r_N} \\ M' \cong \bigoplus_{N \in S(R)} N^{\oplus r'_N} \end{cases}$.

Then $[M] = \sum r_N [N]$, $[M'] = \sum r'_N [N]$. Then by the previous proposition, $[M] = [M']$ iff $r_N = r'_N$, $\forall N \in S(R)$, which is equivalent to $M \cong M'$.

Definition A.1.16. *The representation ring of G over k , denoted by $R_k(G)$ is defined as $R_k(G) = K(k[G])$. Also $P_k(G) = PK(k[G])$, $S_k(G) = S(k[G])$.*

$\rightarrow \forall [M], [M'] \in R_k(G)$, $[M][M'] = [M \otimes_k M']$ defines a biadditive map $R_k(G) \otimes R_k(G) \rightarrow R_k(G)$ with unit element equal to $[1]$. By the properties of tensor products, it is commutative.

Proposition A.1.5. *If M is a $k[G]$ -module and N is a projective $k[G]$ -module, then $k[G]$ -module $M \otimes_k N$ is also projective.*

pf. As N is projective over $k[G]$, it is a direct summand of some free $k[G]$ -module $k[G]^{\oplus I}$ and then $M \otimes_k N$ is a direct summand of $M \otimes_k k[G]^{\oplus I} = (M \otimes_k k[G])^{\oplus I}$. So we can reduce into the case where $M \otimes_k k[G]$. Let $\underline{M}$ be the k -vector space M , considered as a representation of G with trivial action. Define a k -linear map

$$u : \underline{M} \otimes_k k[G] \rightarrow M \otimes_k k[G] \\ \text{map} \quad x \otimes g \mapsto gx \otimes g. \text{ Then } k\text{-linear map}$$

$$v : M \otimes_k k[G] \rightarrow \underline{M} \otimes_k k[G] \\ x \otimes g \mapsto g^{-1}x \otimes g \text{ is an inverse of } u, \text{ so } u \text{ is an isomorphism of } k\text{-}$$

modules. Also $\forall g, h \in G$, $\forall x \in \underline{M}$, $\begin{cases} h \cdot u(x \otimes g) = h \cdot (gx \otimes g) = (hgx) \otimes (hg) \\ u(h \cdot (x \otimes g)) = u(x \otimes (hg)) = (hgx) \otimes (hg) \end{cases}$ gives u is G -equivariant. Since $\underline{M} \otimes_k k[G]$ is a direct sum of copies of $k[G]$, the $k[G]$ -module $M \otimes_k k[G]$ is free, and the proposition follows.

e.g. $G = \mathbb{Z}/n\mathbb{Z}$, k a field containing all the n th root of unity, and $\text{char}(k) \nmid n$. Then $R_k(G) \cong \mathbb{Z}^n$ as a group and $R_k(G)$ is isomorphic to the group algebra $\mathbb{Z}[\mathbb{Z}/n\mathbb{Z}]$ as a ring.

A.1.3 Induction and Frobenius Reciprocity

Let R be a ring and $\alpha : H \rightarrow G$ be a morphism of groups. Then α gives a morphism of rings $\phi : R[H] \rightarrow R[G]$.

Definition A.1.17. If M is a $R[G]$ -module, we can see it as a $R[H]$ -module by making $R[H]$ act via the morphism $\phi : R[H] \rightarrow R[G]$. The resulting $R[H]$ -module is denoted by $\text{Res}_H^G M$ and called the restriction of M to H .

→ if $u : M \rightarrow N$ is a morphism of $R[G]$ -modules, $\text{Res}_H^G(u) : \text{Res}_H^G M \rightarrow \text{Res}_H^G N$ for the same map u , seen as a morphism of $R[H]$ -modules.

Definition A.1.18. If M is a $R[H]$ -module, set $\text{Ind}_H^G M = R[G] \otimes_{R[H]} M$ where $R[H]$ acts on the left on $R[G]$ via the morphism ϕ . This is called the induction of M from H to G .

→ if $u : M \rightarrow N$ is a morphism of $R[H]$ -modules, $\text{Ind}_H^G(u) : \text{Ind}_H^G M \rightarrow \text{Ind}_H^G N$ for the $R[G]$ -linear map $\text{id} \otimes u$.

Definition A.1.19. If M is a $R[H]$ -module, we set $\text{CoInd}_H^G M = \text{hom}_{R[H]}(R[G], M)$ where $R[G]$ is seen as a left $R[H]$ -module via $\phi : R[H] \rightarrow R[G]$. We make this into a left $R[G]$ -module using the right regular action of $R[G]$ on itself, i.e. $\forall x \in R[G], u \in \text{CoInd}_H^G M, (x \cdot u)(y) = u(yx), \forall y \in R[H]$. The $R[G]$ -module $\text{CoInd}_H^G M$ is called the coinduction of M from H to G .

→ if $u : M \rightarrow N$ is a morphism of $R[H]$ -modules, we write $\text{CoInd}_H^G(u) : \text{CoInd}_H^G M \rightarrow \text{CoInd}_H^G N$ for the $R[G]$ -linear map sending $v \in \text{hom}_{R[H]}(R[G], M)$ to $u \circ v$.

Let M be a $R[H]$ -module. Then restriction map $R[G] \rightarrow M$ along the inclusion $G \subset R[G]$ induces an isomorphism of R -modules

$$\psi : \text{CoInd}_H^G M \rightarrow \{f : G \rightarrow M \mid \forall h \in H, \forall g \in G, f(\alpha(h)g) = hf(g)\}$$

pf. Because G generates the $R[H]$ -module $R[G]$, so a $R[H]$ -linear map $u : R[G] \rightarrow M$ is uniquely determined by its restriction to G . So ψ is injective. Let $f : G \rightarrow M$ satisfying the condition in the formula above, and define a R -linear map $R[G] \rightarrow M$. Then $\forall h \in H, x = \sum_{g \in G} c_g g \in R[G], u(hx) = u(\sum c_g \alpha(h)g) = \sum c_g f(\alpha(h)g) = \sum c_g hf(g) = hu(x)$, i.e. u is $R[H]$ -linear. Then $\psi(u) = f$, so ψ is surjective.

Proposition A.1.6. If $\alpha : H \rightarrow G$ is injective, then $\phi : R[H] \rightarrow R[G]$ makes $R[G]$ into a free $R[H]$ -module. More precisely, let $(g_i)_{i \in I}$ be a complete set of representatives of $\alpha(H) \subset G$ in G , then $(g_i)_{i \in I}$ is a basis of $R[G]$ as a left/right $R[H]$ -module.

Definition A.1.20. Suppose $G = \{1\}$. Then for every $R[H]$ -module M , $\text{Ind}_H^1 M$ is called the R -module of coinvariants of M under H and denote by M_H .

→ $M_H = M/(hm - m)$ the quotient of M by the R -submodule generated by all $hm - m$.

Lemma A.1.5. *Let M be a $R[H]$ -module.*

1. *if $\alpha : H \rightarrow G$ is surjective, then $\text{Ind}_H^G M = M_{\ker \alpha}$, with the following action of G ; $\forall g \in G, x \in M_{\ker \alpha}$, choose preimage $h \in H$ of g by α and $m \in M$ of x by the obvious quotient map, and then gx is the image in $M_{\ker \alpha}$ of hm .*
2. $\text{Ind}_H^G M = \text{Ind}_{\alpha(H)}^G M_{\ker \alpha}$

pf.

- Let $u : \text{Ind}_H^G M = R[G] \otimes_{R[H]} M \rightarrow M_{\ker \alpha}$ be defined as follows; $\forall a \in R[G], m \in M$, choose $b \in R[H]$ s.t. $\phi(b) = a$ and take for $u(a \otimes m)$ the image of bm in $M_{\ker \alpha}$. Because $\ker \phi$ is the $R[H]$ -submodule of $R[H]$, generated by $h-1, h \in \ker \alpha$ this does not depend on the choice of h . Since the formula $u(a \otimes m)$ is additive in a and m and takes the same value on $(a\phi(x), m)$ and (a, xm) if $x \in R[H]$, the map is well defined on $R[G] \otimes_{R[H]} M$.

$$\begin{array}{ccc} v' : M & \rightarrow & R[G] \otimes_{R[H]} M \\ m & \mapsto & 1 \otimes m \end{array}$$
 If $m \in M, h \in \ker \alpha, v'(hm - m) = 1 \otimes hm - 1 \otimes m = \alpha(h) \otimes m - 1 \otimes m = 0$ so v' defines a R -linear map $v : M_{\ker \alpha} \rightarrow \text{Ind}_H^G M$. Here u and v' are inverses to each other, so the statement follows.

- $\text{Ind}_H^G M = \text{Ind}_{\alpha(H)}^G \text{Ind}_H^{\alpha(H)} M$ transitivity of $\otimes$
 $= \text{Ind}_{\alpha(H)}^G M_{\ker \alpha}$ statement 1

Theorem A.1.9. *For every exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ of $R[H]$ -modules, the sequence*

$$\text{Ind}_H^G M' \rightarrow \text{Ind}_H^G M \rightarrow \text{Ind}_H^G M'' \rightarrow 0$$

is exact. Moreover $0 \rightarrow \text{Ind}_H^G M' \rightarrow \text{Ind}_H^G M \rightarrow \text{Ind}_H^G M'' \rightarrow 0$ is exact if

$$\left\{ \begin{array}{l} \alpha : H \rightarrow G \text{ is injective} \\ \ker \alpha < \infty, R \text{ is semisimple, } |\ker \alpha| \text{ is invertible in } R \end{array} \right.$$

pf. By the property of tensor product the first stmt follows.

The right $R[H]$ -module $R[G]$ is free by the previous proposition and therefore the tensor products by $R[G]$ over $R[H]$ preserves exact sequence.

By the previous lemma, $\text{Ind}_H^G M = \text{Ind}_{\alpha(H)}^G M_{\ker \alpha}$ for every $R[H]$ -module M . Let $f : M \rightarrow N$ be an injective $R[G]$ -linear map. As $R[G]$ is semisimple, there exists a $R[G]$ -module M' of N s.t. $N = M \oplus M'$. Then $N_G = M_G \oplus M'_G$ and $M_G \rightarrow N_G$ is injective. So coinvariants preserves injection and therefore we have the exact sequence preserved.

Definition A.1.21. *Suppose $G = \{1\}$. Then for every $R[H]$ -module M , $\text{CoInd}_H^1 M$ is called the R -module invariants of M under H and denoted by M^H .*

$$\rightarrow M^H = \{m \in M \mid \forall h \in H, hm = m\}.$$

Lemma A.1.6. *Let M be a $R[H]$ -module.*

1. *if $\alpha : H \rightarrow G$ is surjective, $\text{CoInd}_H^G M = M^{\ker \alpha}$ with the following action of G ; $\forall g \in G, \forall m \in M^{\ker \alpha}$, choose a preimage $h \in H$ of g by α and then gm is defined to be hm .*
2. $\text{CoInd}_H^G M = \text{CoInd}_{\alpha(H)}^G M^{\ker \alpha}$

pf.

- Let $u : \text{CoInd}_H^G M = \text{hom}_{R[H]}(R[G], M) \rightarrow M^{\ker \alpha}$ be defined as follows; if $f : R[G] \rightarrow M$ is a $R[H]$ -linear map, take $u(f) = f(1)$. Then it is well-defined because $\forall h \in \ker \alpha, hf(1) = f(\alpha(h)1) = f(1)$. Let $v : M^{\ker \alpha} \rightarrow \text{hom}_{R[H]}(R[G], M)$ be the map defined follows; if $m \in M, a \in R[G]$, choose $b \in R[H]$ s.t. $\phi(b) = a$ and set $v(m)(a) = bm$. This does not depend on the choice of b because $\ker \phi$ is the $R[H]$ -submodule of $R[H]$ generated by all $h - 1, h \in \ker \alpha$. Then u and v are inverses to each other and the stmt follows.

$$\begin{aligned} \bullet \quad \text{CoInd}_H^G M &= \text{CoInd}_{\alpha(H)}^G \text{CoInd}_H^{\alpha(H)} M && \text{properties of homomorphisms} \\ &= \text{CoInd}_{\alpha(H)}^G M^{\ker \alpha} && \text{by the statement 1} \end{aligned}$$

Theorem A.1.10. *For every exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ of $R[H]$ -modules, the sequence $0 \rightarrow \text{CoInd}_H^G M' \rightarrow \text{CoInd}_H^G M \rightarrow \text{CoInd}_H^G M''$ is exact. Moreover $0 \rightarrow \text{CoInd}_H^G M' \rightarrow \text{CoInd}_H^G M \rightarrow \text{CoInd}_H^G M'' \rightarrow 0$ if*

$$\left\{ \begin{array}{l} \alpha : H \rightarrow G \text{ is injective} \\ \ker \alpha < \infty, R \text{ is semisimple, } |\ker \alpha| \text{ is invertible in } R \end{array} \right.$$

pf. By the left-exactness of hom , the first stmt follows.

By the previous proposition, the left $R[H]$ -module $R[G]$ is free, so taking $\text{hom}_{R[H]}(R[G], \cdot)$ preserves exact sequences.

By the previous lemma, $\text{CoInd}_H^G M = \text{CoInd}_{\alpha(H)}^G M^{\ker \alpha}$ for every $R[H]$ -module M . Let $M \rightarrow N$ be a surjective $R[G]$ -linear map. As $R[G]$ is semisimple, there exists a $R[G]$ -submodule N of M s.t. $M = N \oplus N'$ and by the definition of invariants, $M^G = N^G \oplus N'^G$ and $M^G \rightarrow N^G$ is surjective. So invariants preserves surjection and therefore we get a s.e.s.

Definition A.1.22. *Let M be a $R[H]$ -module. We denote by*

$$\begin{array}{ccc} \epsilon_M : \text{Res}_H^G \text{CoInd}_H^G M & \rightarrow & M \\ u & \mapsto & u(1) \end{array},$$

where $u \in \text{hom}_{R[H]}(R[G], M)$. Additionally denote by

$$\begin{array}{ccc} \eta_M : M & \rightarrow & \text{Res}_H^G \text{Ind}_H^G M \\ m & \mapsto & 1 \otimes m \end{array}.$$

$\rightarrow \forall x \in R[H], \forall u \in \text{hom}_{R[H]}(R[G], M), \epsilon_M(x \cdot u) = (x \cdot u)(1) = u(x) = xu(1)$ so it is $R[H]$ -linear. η_M is clearly $R[H]$ -linear.

Theorem A.1.11 (Frobenius Reciprocity). *Let M be a $R[G]$ -module and N be a $R[H]$ -module.*

1. *the morphism of groups*

$$\begin{aligned} \Phi : \text{hom}_{R[G]}(M, \text{CoInd}_H^G N) &\rightarrow \text{hom}_{R[H]}(\text{Res}_H^G M, N) \\ u &\mapsto \epsilon_N \circ \text{Res}_H^G(u) \end{aligned}$$

is an isomorphism.

2. *the morphism of groups*

$$\begin{aligned} \Phi' : \text{hom}_{R[G]}(\text{Ind}_H^G N, M) &\rightarrow \text{hom}_{R[H]}(N, \text{Res}_H^G M) \\ u &\mapsto \text{Res}_H^G(u) \circ \eta_N \end{aligned}$$

is an isomorphism.

pf. Let $R_1 = R[H]$, $R_2 = R[G]$ for simplicity.

•

$$\begin{aligned} \Psi : \text{hom}_{R[H]}(\text{Res}_H^G M, N) = \text{hom}_{R_1}(M, N) &\rightarrow \text{hom}_{R[G]}(M, \text{CoInd}_H^G N) = \text{hom}_{R_2}(M, \text{hom}_{R_1}(R_2, N)) \\ u : M \rightarrow N &\mapsto m \mapsto (a \mapsto u(am)) \end{aligned}$$

Let u be a R_1 -linear map from M to N . Then $\Psi(u) : M \rightarrow \text{hom}_{R_1}(R_2, N)$ is the R_2 -linear map $m \mapsto (a \mapsto u(am))$ and $\Phi\Psi(u) : M \rightarrow N$ is the R_1 -linear map sending $m \in M$ to $(\Psi(u)(m))(1) = u(m)$. So $\Phi\Psi(u) = u$. Conversely, let v be a R_2 -linear map from M to $\text{hom}_{R_1}(R_2, N)$. Then $\Phi(v)$ is the R_1 -linear map $m \mapsto v(m)(1)$ and $\forall m \in M, a \in R_2, (\Psi\Phi(v)(m))(a) = \Phi(v)(am) = (v(am))(1) = (a \cdot v(m))(1) = v(m)(a)$, $\Psi\Phi(v) = v$. Thus they are inverses to each other, so isomorphism.

•

$$\begin{aligned} \Psi' : \text{hom}_{R[H]}(N, \text{Res}_H^G M) = \text{hom}_{R_1}(N, M) &\rightarrow \text{hom}_{R[G]}(\text{Ind}_H^G N, M) = \text{hom}_{R_2}(R_2 \otimes_{R_1} N, M) \\ u : N \rightarrow M &\mapsto a \otimes n \mapsto au(n) \end{aligned}$$

Let u be a R_1 -linear map from N to M . Then $\forall n \in N, \Phi'\Psi'(u)(n) = \Psi'(u)(1 \otimes n) = u(n)$, $\Phi'\Psi'(u) = u$. Let v be a R_2 -linear map from $R_2 \otimes_{R_1} N$ to M . Then $\forall a \in R_2, \forall n \in N, \Psi'\Phi'(v)(a \otimes n) = a\Phi'(v)(n) = av(1 \otimes n) = v(a \otimes n)$, $\Psi'\Phi'(v) = v$. Thus they are inverses to each other, so isomorphism.

A.2 Finite Group Representation

A.2.1 Introduction

Let R be a ring and G be a group. A $R[G]$ -module is a R -module M with a R -linear action of G , i.e. a morphism of monoids $\rho : G \rightarrow \text{End}_R(M)$. This is called a R -linear representation of G on the R -module M , denoted by (M, ρ) . The representation (M, ρ) is called **irreducible** if the $R[G]$ -module M is simple and it is called **faithful** if ρ is injective. A $R[G]$ -linear map is also called a G -equivariant map. A sub- $R[G]$ -module is also called a **subrepresentation**. **The regular representation of G** is the representation corresponding to the left regular $R[G]$ -module.

Let $\epsilon : R[G] \rightarrow R$ be called the augmentation map. It is a surjective R -linear ring homomorphism, and its kernel is called the augmentation ideal of $R[G]$.

Theorem A.2.1. *Let R be a ring and G be a finite group. Then $R[G]$ is a semisimple ring iff R is a semisimple ring and $|G|$ is invertible in R .*

pf.

- Let M be a $R[G]$ -module and let N be a $R[G]$ -submodule of M . As R is semisimple, $\exists N'$, a R -submodule of M s.t. $M = N \oplus N'$, so there exists a surjective R -linear map $f : M \rightarrow N$ s.t. $f|_N = \text{id}_N$. Define
$$F : M \rightarrow N$$
$$v \mapsto |G|^{-1} \sum_{g \in G} g^{-1} f(gv).$$
 Then it is R -linear and $\forall g \in G, \forall v \in M, F(gv) = |G|^{-1} \sum_{h \in G} h^{-1} f(hgv) = |G|^{-1} g \sum_{h \in G} (hg)^{-1} f(hgv) = gF(v)$, i.e. $F(gv) = gF(v)$. Thus $F(v) = |G|^{-1} \sum g^{-1} f(gv) = |G|^{-1} \sum g^{-1} gv = v$, i.e. $F|_N = \text{id}_N$. Here $\forall v \in M, F(v) \in N$ and therefore $FF(v) = F(v)$, hence $F(v - F(v)) = 0, v = (v - F(v)) + F(v) \in \ker F + N$. If $v \in N \cap \ker F, v = F(v) = 0$. Thus $M = N \oplus \ker F$.
- Assume $R[G]$ is semisimple. Then there exists the augmentation map ϵ . It makes R into a $R[G]$ -module, which is automatically semisimple by assumption. As $R[G]$ acts on R through its quotient $R[G]/\ker \epsilon \cong R$, the R -module R is semisimple, so R is a semisimple ring. It remains to show $|G|$ is invertible in R ; by the remark below it suffices to show $|G| \neq 0$ in each $\mathbb{D}_i$ appearing in the Artin-Wedderburn decomposition of R . Suppose $|G| = 0$ in $\mathbb{D} \equiv \mathbb{D}_i$. As $M_{n_i}(\mathbb{D})[G] \cong M_{n_i}(\mathbb{D}[G])$ is a quotient of $R[G]$, it is semisimple, hence so is $\mathbb{D}[G]$ by the proposition on ideals of matrix rings. Let $c = \sum_{g \in G} g \in \mathbb{D}[G]$. Then $c \neq 0, c$ is central, and $c^2 = |G|c = 0$, so the left ideal $I = \mathbb{D}[G]c$ satisfies $Ic = \mathbb{D}[G]c^2 = 0$. By semisimplicity write $\mathbb{D}[G] = I \oplus J$ and $1 = e + f$ with $e \in I, f \in J$. Then $c = ec + fc = cf \in J$, using $ec \in Ic = 0$ and the centrality of c , while $c = 1c \in I$, so $c \in I \cap J = 0$, a contradiction.

By the Artin-Wedderburn theorem, $R = M_{n_1}(\mathbb{D}_1) \times \cdots \times M_{n_r}(\mathbb{D}_r)$, where $\mathbb{D}_j$ are division rings and $n_j \geq 1, \forall j$. For every i , let $R_i = M_{n_i}(\mathbb{D}_i)$. Then we have a surjective morphism of rings $R[G] \rightarrow R_i$ and therefore R_i is also semisimple. Also $|G|$ is invertible in R iff it is invertible in every R_i , iff it is invertible in every $\mathbb{D}_i$, iff it is nonzero in $\mathbb{D}_i$.

Theorem A.2.2. 1. *There are uniquely determined k -division algebras $\mathbb{D}_1, \dots, \mathbb{D}_r$ and integers $n_1, \dots, n_r \geq 1$ s.t. $\dim_k(\mathbb{D}_i)$ is finite for every i and*

$$k[G]/\text{rad}(k[G]) \cong M_{n_1}(\mathbb{D}_1) \times \cdots \times M_{n_r}(\mathbb{D}_r)$$

A complete set of representatives of the isomorphism classes of irreducible representations of G is given by $V_1 \equiv \mathbb{D}_1^{n_1}, \dots, V_r \equiv \mathbb{D}_r^{n_r}$ and we have a G -equivariant isomorphism

$$k[G]/\text{rad}(k[G]) \cong V_1^{n_1} \oplus \cdots \oplus V_r^{n_r}$$

2. *If k is algebraically closed, then $\mathbb{D}_i = k, \forall i$, and*

$$\sum_V (\dim V)^2 = \sum_{i=1}^r \dim(V_i)^2 = \sum_{i=1}^r n_i^2 = \dim(k[G]/\text{rad}(k[G])) \leq \dim(k[G]) = |G|$$

where the first sum is taken over the isomorphism classes of irreducible representations of G . The inequality is an equality iff the characteristic of k does not divide $|G|$.

Definition A.2.1. *If R is a ring and G is a group, denote by $S_R(G)$ the set of isomorphism classes of irreducible representations of G on R -modules.*

e.g.

- Fix R, G as above. The trivial representation of G over R is the representation of G on R given by the augmentation map $\epsilon : R[G] \rightarrow R$ by the trivial action of G on R , denoted by $\mathbb{1}$.
- If G is the symmetric group $\mathcal{G}_n$ and R is any ring, we denote by sgn the representation of G on R given by the sign morphism $G \rightarrow \{\pm 1\}$ composed with the obvious map $\{\pm 1\} \rightarrow R$.
- Let G be the quaternion group $Q = \{\pm 1, \pm i, \pm j, \pm k\} \subset \mathbb{H}$ with the multiplication given by that of $\mathbb{H}$. Then $\mathbb{R}[G] \cong \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{H}$ so $S_{\mathbb{R}}(G)$ has 5 elements.

Proposition A.2.1. *If G is a finite abelian group and k is an algebraically closed field, then every simple $k[G]$ -module is of dimension 1 over k .*

pf. Let V be a simple $k[G]$ -module. Then any nonzero element $v \in V$ gives a surjective k -linear map $k[G] \rightarrow V$ sending $x \in k[G]$ to xv . Then V is a finite dimensional vector space, and so is $\text{End}_{k[G]}(V)$. The finite dimensional k -algebra $\text{End}_{k[G]}(V)$ is also a k -division algebra by the Schur's lemma, $\text{End}_{k[G]}(V) = k$. Moreover, as G is abelian, $k[G]$ is a commutative ring, so the action of any element of $k[G]$ on V is $k[G]$ -linear. Thus the action of $k[G]$ on V is given by a map of k -algebras $k[G] \rightarrow \text{End}_{k[G]}(V) = k \subset \text{End}_k(V)$. So $\forall v \in V$, the subspace kv is a $k[G]$ -submodule of V . As V is irreducible, $\dim V = 1$.

e.g. Let $G = \mathbb{Z}/n\mathbb{Z}$, $n \geq 1$, k be an algebraically closed field.

- $\text{char } k \nmid n$: fix a primitive n th root ζ_n of 1 in k . The k -algebra $k[G]$ is semisimple, so $k[G] = k \times \cdots \times k$, where n factors k correspond to the n irreducible representations of G given by the maps

$$\begin{array}{ccc} G = \mathbb{Z}/n\mathbb{Z} & \rightarrow & k^\times \\ i & \mapsto & \zeta_n^i \end{array}.$$
- $\text{char } k | n$: let $p = \text{char}(k)$ and write $n = p^r m$ with $(m, p) = 1$. Then $G = \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$. $(1, 0), (0, 1) \in \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ have images $a, b \in k^\times$ under the map $G \rightarrow k^\times$ corresponding to the representation. Then $a^{p^r} = 1$ so $a = 1$ given $\text{char}(k) = p$. Also b can be any of the m solutions of the equation $x^m = 1$ in k . So we can get m irreducible representations of G , and $k[G]/\text{rad}(k[G]) \cong k^m$.

A.2.2 Characters in $\text{char}(k) = 0$

Definition A.2.2. A function $f : G \rightarrow k$ is called central if $\forall g, h \in G$, $f(gh) = f(hg)$. We write $\mathcal{C}(G, k)$ for the k -algebra of central functions from G to k .

Definition A.2.3. Let (V, ρ) be a representation of G . The character of V is the function

$$\begin{array}{ccc} \chi_V : G & \rightarrow & k \\ g & \mapsto & \text{Tr}(\rho(g)) \end{array}$$

Proposition A.2.2. Let $(V, \rho_V), (W, \rho_W)$ be representations of G . Then

$$\left\{ \begin{array}{l} \chi_V(1) = \dim_k V \\ \chi_V \in \mathcal{C}(G, k) \\ \chi_{V \oplus W} = \chi_V + \chi_W \\ \chi_{V \otimes_k W} = \chi_V \chi_W \end{array} \right.$$

pf. Choose k -bases $\left\{ \begin{array}{c} (e_1, \dots, e_n) \\ (f_1, \dots, f_m) \end{array} \right\} \begin{array}{c} V \\ W \end{array}$. Let $g \in G$. In the chosen bases of V, W , write $\left\{ \begin{array}{c} \rho_V(g) \\ \rho_W(g) \end{array} \right\}$ as matrices $\left\{ \begin{array}{c} (x_{ij}) \\ (y_{ij}) \end{array} \right\}$. Then $(e_i \otimes f_j)$ is a basis for $V \otimes_k W$, and the matrix $\rho_V(g) \otimes \rho_W(g)$ in this basis is $(x_{ii'} y_{jj'})$. Hence $\chi_{V \otimes_k W}(g) = \sum_{i=1}^n \sum_{j=1}^m x_{ii} y_{jj} = (\sum_i x_{ii})(\sum_j y_{jj}) = \chi_V(g) \chi_W(g)$.

Corollary A.2.1. *The map $V \mapsto \chi_V$ induces a morphism of rings $R_k(G) \rightarrow \mathcal{C}(G, k)$.*

For K/k an extension of fields, given every representation of G over k , $\chi_{V \otimes_k K} =$

$$\begin{array}{ccc} R_k(G) & \longrightarrow & \mathcal{C}(G, k) \\ \downarrow & & \downarrow \\ R_K(G) & \longrightarrow & \mathcal{C}(G, K) \end{array}$$

where the left vertical arrow is given by $[V] \mapsto [V \otimes_k K]$ and the right vertical arrow is the obvious inclusion.

Definition A.2.4. *Let V, W be representations of G . The k -vector space $\text{hom}_k(V, W)$ becomes a representation of G if we make $g \in G$ act by $(g \cdot f)(v) = gf(g^{-1}v)$, $\forall f \in \text{hom}_k(V, W)$ and $\forall v \in V$. In particular, $V^* \equiv \text{hom}_k(V, k)$ is a representation of G and $(g \cdot f)(v) = f(g^{-1}v)$, $\forall g \in G$, $f \in V^*$, $v \in V$.*

Definition A.2.5. *For every representation V of G , we set*

$$V^G = \{v \in V \mid \forall g \in G, gv = v\}$$

This is the space of invariants of G in V .

$$\rightarrow \text{hom}_G(V, W) = \text{hom}_k(V, W)^G$$

Proposition A.2.3. *Let V, W be representations of G . Then the map*

$$\begin{array}{ccc} \phi : V^* \otimes_k W & \rightarrow & \text{hom}_k(V, W) \\ v \otimes w & \mapsto & (v \mapsto f(v)w) \end{array} \text{ is a } G\text{-equivariant isomorphism.}$$

pf. It is well-defined, because the map $\begin{array}{ccc} V \times W & \rightarrow & \text{hom}_k(V, W) \\ (v, w) & \mapsto & (v \mapsto f(v)w) \end{array}$ is k -bilinear.

Let $f \in V^*$, $w \in W$, $g \in G$, $v \in V$. Then

$$\begin{cases} (\phi(g(f \otimes w)))(v) = (\phi((gf) \otimes (gw)))(v) = (gf)(v)gw = f(g^{-1}v)gw \\ (g\phi(f \otimes w))(v) = g(\phi(f \otimes w)(g^{-1}v)) = gf(g^{-1}v)w \end{cases} \text{ so } \phi \text{ is } G\text{-equivariant.}$$

Let (e_i) be a basis of W as a k -vector space. Then we have k -linear isomorphisms

$$\begin{array}{ccc} u : \oplus_i \text{hom}_k(V, k) & \rightarrow & \text{hom}_k(V, W) \\ (f_i) & \mapsto & (v \mapsto \sum f_i(v)e_i) \end{array} \quad \begin{array}{ccc} v : \oplus V^* & \rightarrow & V^* \otimes_k W \\ (f_i) & \mapsto & \sum f_i \otimes e_i \end{array} . \text{ Then } \phi = u \circ v^{-1}$$

is an isomorphism.

Proposition A.2.4. *Let V, W be representations of G . Then $\forall g \in G$,*

$$\begin{cases} \chi_{V^*}(g) = \chi_V(g^{-1}) \\ \chi_{\text{hom}_k(V, W)}(g) = \chi_V(g^{-1})\chi_W(g) \end{cases} .$$

pf. Write ρ for the action morphism $G \rightarrow \text{End}_k(V)$. Choose a basis $\mathcal{B}$ of V as a k -vector space, and let M be the matrix of $\rho(g)^{-1}$ in $\mathcal{B}$. Then g acts by the matrix M' on V^* , so the result follows. Then the first stmt implies the second stmt.

Theorem A.2.3 (Schur's Lemma). *Let V, W be irreducible representations of G . Then $\text{hom}_G(V, W) = 0$ unless $V \cong W$, and $\text{End}_G(V)$ is a finite-dimensional k -division algebra. If moreover k is algebraically closed, then $\text{End}_G(V) = k$.*

pf. The first part follows from the first Schur's lemma.

We have a map of rings $\begin{array}{ccc} K & \rightarrow & \text{End}_G(V) \\ \lambda & \mapsto & \lambda \cdot 1 \end{array}$ which is injective, given k a field. Let

$a \in R$. Then $\begin{array}{ccc} m_a : \text{End}_G(V) & \rightarrow & \text{End}_G(V) \\ x & \mapsto & ax \end{array}$ is k -linear. Here $\text{End}_G(V)$ is finite dimensional k -vector space and k is algebraically closed, so m_a has at least one eigenvalue, i.e. $\exists \lambda \in k$ s.t. $\ker(m_a - \lambda \cdot \text{Id}) = \ker(m_{a-\lambda}) \neq 0$. This implies $a - \lambda$ is not invertible. Since $\text{End}_G(V)$ is a division algebra, $a - \lambda = 0 \Leftrightarrow a = \lambda \in k$.

Lemma A.2.1. *Suppose k is algebraically closed and let (V, ρ) be an irreducible representation of G . Then*

$$\sum_{g \in G} \chi_V(g) = \begin{cases} 0 & V \not\cong \mathbb{1} \\ |G| & V \cong \mathbb{1} \end{cases}$$

pf. Extend $\rho : G \rightarrow \text{End}_k(V)$ to $\rho : k[G] \rightarrow \text{End}_k(V)$ and let $c = \sum_{g \in G} g \in k[G]$. Then c is central in $k[G]$, so $\rho(c) \in \text{End}_k(V)$ is a G -equivariant endomorphism of V . Then by the Schur's lemma, $\exists \lambda \in k$ s.t. $\rho(c) = \lambda \text{id}_V$, so $\sum_{g \in G} \chi_V(g) = T(\rho(c)) = \lambda \dim V$. Moreover $\forall h \in G$, $\lambda \rho(h) = \rho(c) \rho(h) = \rho(ch) = \rho(c) = \lambda \text{id}_V$. So if $\lambda \neq 0$, $V \cong \mathbb{1}$, $\sum \chi_V(g) = \sum 1 = |G|$.

Lemma A.2.2. *Let V be a representation of G . Then*

$$\sum_{g \in G} \chi_V(g) = |G| \dim_k(V^G)$$

pf. Assume wlog k is algebraically closed (we has seen $\chi_{V \otimes_k K} = \chi_V$ and we can make V algebraically closed by tensoring with algebraically closed K). Write $V = \oplus_{W \in S_k(G)} W^{\oplus n_W}$. Then $\sum_{g \in G} \chi_V(g) = \sum_{W \in S_k(G)} n_W \sum_{g \in G} \chi_W(g) = n_{\mathbb{1}} |G|$ by the previous lemma.

If $W \in S_k(G)$, $W \not\cong \mathbb{1}$, $W^G = 0$ since W^G is a subrepresentation of W . Then $V^G = \mathbb{1}^{n_{\mathbb{1}}}$, $\dim_k(V^G) = n_{\mathbb{1}}$.

Theorem A.2.4. *Let V, W be representations of G . Then*

$$\frac{1}{|G|} \sum_{g \in G} \chi_V(g^{-1}) \chi_W(g) = \frac{1}{|G|} \sum_{g \in G} \chi_{V^*}(g) \chi_W(g) = \dim_k(\text{hom}_G(V, W))$$

pf. $\frac{1}{|G|} \sum \chi_V(g^{-1}) \chi_W(g) = \frac{1}{|G|} \sum \chi_{V^*}(g) \chi_W(g) = \frac{1}{|G|} \sum \chi_{\text{hom}_k(V, W)}(g) = \dim_k(\text{hom}_k(V, W)^G) = \dim_k(\text{hom}_G(V, W))$ by the previous lemma.

Corollary A.2.2. *If V, W are irreducible and nonisomorphic, then $\sum_{g \in G} \chi_{V^*}(g) \chi_W(g) = 0$.*

pf. by the previous theorem and the Schur's lemma.

Corollary A.2.3. *Suppose k is algebraically closed and let V be a representation of G . Then V is irreducible iff $\sum_{g \in G} \chi_{V^*}(g) \chi_V(g) = |G|$.*

pf. By the previous theorem $\sum_{g \in G} \chi_{V^*}(g) \chi_V(g) = |G| \dim_k(\text{End}_G(V))$. If V is irreducible, $\text{End}_G(V) = k$ by Schur's lemma so the one direction follows. Conversely, suppose $\sum_{g \in G} \chi_{V^*}(g) \chi_V(g) = |G|$. Write $v = \oplus_{i \in I} V_i^{\oplus n_i}$ where V_i are irreducible and pairwise nonisomorphic. Then by the previous corollary, $\sum \chi_{V^*}(g) \chi_V(g) = \sum n_i n_j \sum \chi_{V_i^*}(g) \chi_{V_j}(g) = |G| \sum n_i^2$. So only one of the n_i can be nonzero, and moreover it has to be equal to 1, hence V is irreducible.

Corollary A.2.4. *The family $(\chi_V)_{V \in S_k(G)}$ is linearly independent in $\mathcal{C}(G, k)$.*

pf. Suppose $\sum_{W \in S_k(G)} \alpha_W \chi_W = 0$, with $\alpha_W \in k$. Then $\forall V \in S_k(G)$, $0 = \sum_{g \in G} (\sum_{W \in S_k(G)} \alpha_W \chi_W(g)) \chi_{V^*}(g) = \sum \alpha_W \sum \chi_W(g) \chi_{V^*}(g) = \alpha_V |G| \rightarrow \alpha_V = 0$, so LI follows.

Corollary A.2.5. *For any representations V, V' of G , $V \cong V'$ iff $\chi_V = \chi_{V'}$.*

pf. Write $\begin{cases} V = \oplus_{W \in S_k(G)} W^{\oplus n_W} \\ V' = \oplus_{W \in S_k(G)} W^{\oplus n'_W} \end{cases}$. Then $\begin{cases} \chi_V = \sum n_W \chi_W \\ \chi_{V'} = \sum n'_W \chi_W \end{cases}$. By the previous corollary, $\chi_V = \chi_{V'}$ iff $n_W = n'_W$, $\forall W \in S_k(G) \Leftrightarrow V \cong V'$.

Lemma A.2.3. *If $f \in \mathcal{C}(G, k)$ is s.t. $\sum_{g \in G} f(g) \chi_{W^*}(g) = 0$, $\forall W \in S_k(G)$, then $f = 0$.*

pf. If $\rho : G \rightarrow \text{End}_k(V)$ is a representation of G , we set $\rho(f) = \sum f(g) \rho(g) \in \text{End}_k(V)$. Then $\forall g \in G$, $\rho(g) \rho(f) = \sum_{h \in G} f(h) \rho(gh) = \sum_{h \in G} f(h) \rho(ghg^{-1}g) = \sum f(ghg^{-1}) \rho(ghg^{-1}) \rho(g)$ because f is central. So if V is irreducible, $\rho(f) \in \text{End}_G(V) = k$ by Schur's lemma, hence $\rho(f) = \lambda \text{id}_V$, $\lambda \in k$. So $\lambda \dim(V) = T(\rho(f)) = \sum_{g \in G} f(g) \chi_V(g) = 0 \rightarrow \lambda = 0$ and $\rho(f) = 0$. By the semisimplicity of $k[G]$, $\rho(f) = 0$ for any representation ρ of G . Then $0 = \rho_{\text{reg}}(f)1 = \sum f(g)g$ in $k[G]$, so $f(g) = 0$, $\forall g \in G$.

Theorem A.2.5. *Suppose that the field k is algebraically closed. Then the family $(\chi_W)_{W \in S_k(G)}$ is a basis of $\mathcal{C}(G, k)$.*

pf. They are L.I by the corollary A.2.4. so we just need to show it spans $\mathcal{C}(G, k)$. Let $f \in \mathcal{C}(G, k)$. Then $\forall W \in S_k(G)$, $\alpha_W = \frac{1}{|G|} \sum_{g \in G} f(g) \chi_{W^*}(g)$. Set $f' = f - \sum_{W \in S_k(G)} \alpha_W \chi_W$. Then $\forall W \in S_k(G)$, $\sum_{g \in G} f'(g) \chi_{W^*}(g) = \sum f(g) \chi_{W^*}(g) - \alpha_W \sum \chi_W(g) \chi_{W^*}(g) = 0 \rightarrow f' = 0$ and f corresponds to χ_W by the previous corollaries. So it spans.

Corollary A.2.6. *Suppose that k is algebraically closed. Then $|S_k(G)| = \dim_k \mathcal{C}(G, k)$ and this is also equal to the number of conjugacy classes in G .*

Corollary A.2.7. Suppose k is algebraically closed. Then the map $R_k(G) \rightarrow \mathcal{C}(G, k)$
 $V \mapsto \chi_V$
induces an isomorphism of k -algebra $R_k(G) \otimes_{\mathbb{Z}} k \rightarrow \mathcal{C}(G, k)$.

Let G_1, G_2 be two groups. If V_1/V_2 is a representation of G_1/G_2 , then $V_1 \otimes_k V_2$ becomes a representation of $G_1 \times G_2$ with action $(g_1, g_2)(v_1 \otimes v_2) = (g_1 v_1) \otimes (g_2 v_2)$.

Theorem A.2.6. Suppose k is algebraically closed, and let V_1/V_2 be a representation of G_1/G_2 . The representation $V_1 \otimes_k V_2$ of $G_1 \times G_2$ is irreducible iff V_1, V_2 are both irreducible. Every irreducible representation of $G_1 \times G_2$ is of the form $V_1 \otimes_k V_2$.

pf. The character of $V \equiv V_1 \otimes_k V_2$ is the map $(g_1, g_2) \mapsto \chi_{V_1}(g_1)\chi_{V_2}(g_2)$. So $\langle V, V \rangle_G = \frac{1}{|G_1 \times G_2|} \sum_{(g_1, g_2)} \chi_{V_1}(g_1)\chi_{V_2}(g_2) = \langle V_1, V_1 \rangle_{G_1} \langle V_2, V_2 \rangle_{G_2}$. All three brackets are nonnegative integers, $\langle V, V \rangle_G = 1$ iff $\langle V_1, V_1 \rangle_{G_1} = \langle V_2, V_2 \rangle_{G_2} = 1$. So if V_1, V_2 are irreducible representation of G_1/G_2 , then $V_1 \otimes V_2$ is an irreducible representation of G . Thus we can define a map $S_k(G_1) \times S_k(G_2) \rightarrow S_k(G_1 \times G_2)$. Since the representations $V_1 \otimes V_2 \sim V'_1 \otimes V'_2$ iff $V_1 \sim V'_1, V_2 \sim V'_2$, the map is injective. Let $X(G_1)$ be the set of conjugacy classes in G_1 , and similarly for $X(G_2), X(G_1 \times G_2)$. Then we have a surjective map $u : G_1 \times G_2 \rightarrow X(G_1 \times G_2)$. Then $(g_1, g_2), (g'_1, g'_2) \in G_1 \times G_2$ are conjugate iff $g_i \sim g'_i$ are conjugate in G_i . So u induces a bijection $X(G_1) \times X(G_2) \rightarrow X(G_1 \times G_2)$. As $|S_k(G)| = |X(G)|, \forall G \in \{G_1, G_2, G_1 \times G_2\}$, so $|S_k(G_1) \times S_k(G_2)| = |S_k(G_1 \times G_2)|$. So source and target have the same cardinality and the map is a bijection and therefore the theorem follows.

Fix a finite group G and we assume k is any field s.t. $\text{char}(k) \nmid |G|$. If $f_1, f_2 \in \mathcal{C}(G, k)$, write $\langle f_1, f_2 \rangle_G = \frac{1}{|G|} \sum_{g \in G} f_1(g)f_2(g^{-1}) = \frac{1}{|G|} \sum_{g \in G} f_1(g^{-1})f_2(g)$. Then the previous theorem can be expressed as; for any representations V, W of G , we have $\langle \chi_V, \chi_W \rangle = \dim_k(\text{hom}_G(V, W))$.

Definition A.2.6. Let H be a subgroup of G . If $f \in \mathcal{C}(H, k)$, define

$$\begin{aligned} \text{Ind}_H^G f : G &\rightarrow k \\ g &\mapsto \frac{1}{|H|} \sum_{s \in G | s^{-1}gs \in H} f(s^{-1}gs) \end{aligned}$$

Theorem A.2.7. $\forall f \in \mathcal{C}(H, k), \text{Ind}_H^G f \in \mathcal{C}(G, k)$
 $\text{Ind}_H^G \chi_V = \chi_{\text{Ind}_H^G V}$ for a representation V of H .

pf. Let (V, ρ_V) be a representation of H and write $(W, \rho_W) = \text{Ind}_H^G(V, \rho_V)$. Let $g_1, \dots, g_r$ be a system of representatives of G/H . Then $k[G] = \bigoplus_{i=1}^r g_i k[H]$, so $W = k[G] \otimes_{k[H]} V = \bigoplus_{i=1}^r W_i$ with $W_i = g_i k[H] \otimes_{k[H]} V$. Let $g \in G$. Then $\exists \sigma \in \mathcal{G}_r$ and $h_1, \dots, h_r \in H$ s.t. $gg_i = g_{\sigma(i)}h_i, \forall i$. Then $\rho_W(g)(W_i) = W_{\sigma(i)}, \forall i$. Choose a basis $(e_1, \dots, e_n)$ of V . Then $\forall i, (g_i \otimes e_1, \dots, g_i \otimes e_n)$ is a basis of W_i , and $\rho_W(g)(g_i \otimes e_s) = g_{\sigma(i)} \otimes \rho_V(h_i)e_s$. Thus $T(\rho_W(g)) = \sum_{i | \sigma(i)=i} T(\rho_V(h_i)) = \sum_{i | g_i^{-1}gg_i \in H} f(g_i^{-1}gg_i)$. But

$G = \coprod_{i=1}^r g_i H$ and if $s \in g_i H$, $\begin{cases} s^{-1}gs \in H \text{ iff } g_i^{-1}g g_i \in H \\ f(s^{-1}gs) = f(g_i^{-1}g g_i) \end{cases}$. So the second stmt follows. The first stmt follows by the first one.
 $\rightarrow$ if $f \in \mathcal{C}(G, k)$, $\text{Res}_H^G f$ for $f|_H$. Then $\text{Res}_H^G \chi_V = \chi_{\text{Res}_H^G V}$.

Theorem A.2.8. If $\begin{cases} f_1 \in \mathcal{C}(H, k) \\ f_2 \in \mathcal{C}(G, k) \end{cases}$, then $\langle f_1, \text{Res}_H^G f_2 \rangle_H = \langle \text{Ind}_H^G f_1, f_2 \rangle_G$.

pf. $\langle f_1, \text{Res}_H^G f_2 \rangle_H = \frac{1}{|H|} \sum_{h \in H} f_1(h) f_2(h)$
 $\langle \text{Ind}_H^G f_1, f_2 \rangle_G = \frac{1}{|G|} \sum_{g \in G} \frac{1}{|H|} \sum_{s \in G|_{s^{-1}gs \in H}} f_1(s^{-1}gs) f_2(g) = \frac{1}{|G|} \sum \frac{1}{|H|} \sum f_1(s^{-1}gs) f_2(s^{-1}gs) =$
 $\frac{1}{|G|} \frac{1}{|H|} \sum_{h \in H} \sum_{s \in G} f_1(h) f_2(h) = \frac{1}{|H|} \sum_{h \in H} f_1(h) f_2(h)$. They are equivalent.

Let R be a ring, G be a finite group and H, K be subgroups of G . Let (V, ρ) be a representation of H and $g_1, \dots, g_r$ be a system of representatives of the double classes in $K \setminus G/H$, i.e. $G = \coprod_{i=1}^r K g_i H$. For every $i \in \{1, \dots, r\}$, let $H_i = g_i H g_i^{-1} \cap K$ and let (V_i, ρ_i) be the representation of H_i on V , given by $\rho_i(h) = \rho(g_i^{-1} h g_i)$.

Theorem A.2.9 (Mackey's Formula). We have an isomorphism of $R[K]$ -modules

$$\text{Res}_K^G \text{Ind}_H^G V \cong \bigoplus_{i=1}^r \text{Ind}_{H_i}^{K} V_i$$

pf. $\forall j \in \{1, \dots, r\}$, let $(x_i)_{i \in I_j}$ be a system of representatives of K/H_j . Then $\{x_i g_j, 1 \leq j \leq r, i \in I_j\}$ is a system of representatives of G/H ($G = \coprod_{j=1}^r K g_j H = \coprod_{j=1}^r \coprod_{i \in I_j} x_i H_j g_j H = \coprod_{j=1}^r \coprod_{i \in I_j} x_i g_j (g_j^{-1} H_j g_j) H = \coprod_{j=1}^r \coprod_{i \in I_j} x_i g_j H$). So $W \equiv \text{Ind}_H^G V = \bigoplus_{j=1}^r W_j$, $W_j = \bigoplus_{i \in I_j} x_i g_j k [H] \otimes_{k[H]} V$. Fix j and consider the R -linear maps $\begin{matrix} \phi : \text{Ind}_{H_j}^K V_j & \rightarrow & W_j \\ \sum_{i \in I_j} x_i \otimes v_i & \mapsto & \sum (x_i g_j) \otimes v_i \end{matrix}$
and $\begin{matrix} \psi : W_j & \rightarrow & \text{Ind}_{H_j}^K V_j \\ \sum_{i \in I_j} (x_i g_j) \otimes v_i & \mapsto & \sum_{i \in I_j} x_i \otimes v_i \end{matrix}$. Then ϕ, ψ are inverses to each other. Since ϕ is K -linear ($\forall y \in K, \exists \sigma \in \mathcal{G}_{I_j}, h_i \in H_j, i \in I_j$ s.t. $yx_i = x_{\sigma(i)} h_i, \forall i \in I_j$, $\phi(y(\sum x_i \otimes v_i)) = \phi(\sum (yx_i) \otimes v_i) = \phi(\sum (x_{\sigma(i)} h_i) \otimes v_i) = \phi(\sum x_{\sigma(i)} \otimes (\rho_j(h_i) v_i)) = \phi(\sum x_{\sigma(i)} \otimes (\rho(g_j^{-1} h_i g_j) v_i)) = \sum (x_{\sigma(i)} g_j) \otimes (\rho(g_j^{-1} h_i g_j) v_i) = \sum (x_{\sigma(i)} h_i g_j) \otimes v_i = \sum (yx_i g_j) \otimes v_i = y \phi(\sum x_i \otimes v_i)$), the theorem follows.

$$\langle V, W \rangle_G = \langle \chi_V, \chi_W \rangle_G = \dim_k \text{hom}_G(V, W)$$

Let H be a subgroup of G and (V, ρ) be a representation of H . For every $g \in G$, $H_g = g H g^{-1} \cap H$ and define two representations $\begin{cases} \rho_g(h) = \rho(g^{-1} h g) \\ \text{Res}_g(\rho)(h) = \rho(h) \end{cases}$.

Theorem A.2.10. TFAE

1. $W \equiv \text{Ind}_H^G V$ is irreducible

2. V is irreducible, and $\forall g \in G \setminus H, \text{hom}_{H_g}(\rho^g, \text{Res}_g(\rho)) = 0$

pf. $\langle W, W \rangle_G = \langle V, \text{Res}_H^G W \rangle_H$ by Frobenius reciprocity
 $= \langle V, \bigoplus_{g \in H \setminus G/H} \text{Ind}_{H_g}^H(\rho^g) \rangle_H$ Mackey's formula
 $= \sum_{g \in H \setminus G/H} \langle V, \text{Ind}_{H_g}^H(\rho^g) \rangle_H = \sum \langle \text{Ind}_{H_g}^H(\rho^g), V \rangle_H$ where all the summands are ≥ 0 . If $g \in H, H_g = H$ and $\rho_G \cong \text{Res}_g(\rho) = \rho$, so $\langle \rho^g, \text{Res}_g(V) \rangle_H = \langle V, V \rangle_H$. Then W is irreducible $\Leftrightarrow \langle W, W \rangle_G = 1 \Leftrightarrow \langle V, V \rangle_H = 1$
 $\langle \rho^g, \text{Res}_g(\rho) \rangle_H = 0, \forall g \in G \setminus H \Leftrightarrow V$ is irreducible and $\text{hom}_{H_g}(\rho^g, \text{Res}_g(\rho)) = 0, \forall g \in G \setminus H$.

Proposition A.2.5. Write $R(G)_{\mathbb{Q}} = R(G) \otimes_{\mathbb{Z}} \mathbb{Q}$. Then the $\mathbb{Z}$ -linear map $\langle \cdot, \cdot \rangle_G: R(G) \times R(G) \rightarrow \mathbb{Z}$ induces a $\mathbb{Q}$ -linear isomorphism

$$\begin{array}{ccc} R(G)_{\mathbb{Q}} & \rightarrow & R(G)_{\mathbb{Q}}^* \equiv \text{hom}_{\mathbb{Q}}(R(G)_{\mathbb{Q}}, \mathbb{Q}) \\ x & \mapsto & (y \mapsto \langle x, y \rangle_G) \end{array}$$

Let H be a subgroup of G . If we use the isomorphism above to identify $R(G)_{\mathbb{Q}}$ and $R(H)_{\mathbb{Q}}$ with their duals, then the transpose of Ind_G^H is Res_H^G .

pf.

- Denote by $\begin{cases} (e_V)_{V \in S_k(G)} & \text{the basis } ([V])_{V \in S_k(G)} \text{ of } R(G)_{\mathbb{Q}} \\ (e_V^*)_{V \in S_k(G)} & \text{the dual basis} \end{cases}$. Then by the theorem A.2.4, and the corollary A.2.2., the map above sends each e_V to a nonzero multiple of e_V^* . If k is algebraically closed, the map sends each e_V to e_V^* itself by the corollary A.2.3., and it actually induces an isomorphism $R(G) \rightarrow \text{hom}_{\mathbb{Z}}(R(G), \mathbb{Z})$.
- by the Frobenius reciprocity

Theorem A.2.11 (Artin). Let X be a set of subgroups of G . Consider the map $\text{Ind}_X = \bigoplus_{H \in X} \text{Ind}_H^G: \bigoplus_{H \in X} R(H) \rightarrow R(G)$. Then TFAE

1. $G = \bigcup_{H \in X} \bigcup_{g \in G} gHg^{-1}$
2. $\text{Ind}_X \otimes_{\mathbb{Z}} \mathbb{Q}$ is surjective, that is $\forall x \in R(G), \exists d \geq 1, \exists x_H \in R(H), H \in X$, st. $dx = \sum_{H \in X} \text{Ind}_H^G x_H$.

pf.

- Let $S = \bigcup_{H \in X} \bigcup_{g \in G} gHg^{-1}$. Then if $H \in X, x_H \in R(H)_{\mathbb{Q}}, \text{Ind}_H^G x_H$ is zero on $G \setminus S$. Then by 2, this implies $x = 0$ on $G \setminus S, \forall x \in R(G)_{\mathbb{Q}}$. Then by A.2.5., $\forall f \in \mathcal{C}(G, k), f|_{G \setminus S} = 0$. Hence $G \setminus S = \emptyset$.

- By the previous proposition, the transpose of $\text{Ind}_X \otimes_{\mathbb{Z}} \mathbb{Q}$ is $\bigoplus_{H \in X} \text{Res}_H^G : R(G)_{\mathbb{Q}} \rightarrow \bigoplus_{H \in X} R(H)_{\mathbb{Q}}$. By 1, the sum of the restriction maps $\mathcal{C}(G, k) \rightarrow \bigoplus_{H \in X} \mathcal{C}(H, k)$ is injective, and therefore the map above is injective and 2 follows.

Corollary A.2.8. *Take for X the set of cyclic subgroups of G . Then $\text{Ind}_X \otimes_{\mathbb{Z}} \mathbb{Q}$ is surjective.*

pf. Because $\forall g \in G$, it is an element of the cyclic subgroup that it generates, by the Artin's theorem, the corollary follows.

Definition A.2.7. *Let p be a prime number. A finite group H is called p -elementary if $H = C \times P$, with C a cyclic group of order prime to p and P a p -group.*

For every prime number p , let $X(p)$ be the set of p -elementary subgroups of G . Let X be the union of all the $X(p)$ for p prime.

Theorem A.2.12. *Let p be a prime number, write $|G| = p^r m$ with $(p, m) = 1$, and let V_p be the image of*

$$\text{Ind}_p \equiv \bigoplus_{H \in X(p)} \text{Ind}_H^G : \bigoplus_{H \in X(p)} R(H) \rightarrow R(G)$$

Then $m \in V_p$. In particular, $R(G)/V_p$ is a finite group of order prime to p .

pf. Let $n = |G|$ and $\mathcal{O} = \mathbb{Z}[1, \zeta_n, \dots, \zeta_n^{n-1}] \subset k$, where ζ_n is a primitive n th root of 1 in k . Then $\forall x \in R(G)$, x takes values in $\mathcal{O}$. Indeed for every representation (V, ρ) of G and $\forall g \in G$, $\rho(g)$ has eigenvalues in $\mathcal{O}$ by the previous proposition, so $\chi_V(g) \in \mathcal{O}$. We have $\mathcal{O} \cap \mathbb{Q} = \mathbb{Z}$ since $\forall \alpha \in \mathcal{O} \cap \mathbb{Q}$, $f \in \mathbb{Q}[T]$ be its minimal polynomial over $\mathbb{Q}$, then $f \in \mathbb{Z}[T]$ as α is integral over $\mathbb{Z}$, $\deg f = 1$ as $\alpha \in \mathbb{Q}$. Then the map $\mathcal{O}/\mathbb{Z} \rightarrow (\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Q})/\mathbb{Q}$ is injective, so $\mathcal{O}/\mathbb{Z}$ is torsion free, so it is free as a $\mathbb{Z}$ -module, so $\mathcal{O}$ has a $\mathbb{Z}$ -basis of the form $(1, \alpha_1, \dots, \alpha_c)$. The image of $\mathcal{O} \otimes \text{Ind}_p : \bigoplus_{H \in X} \mathcal{O} \otimes_{\mathbb{Z}} R(H) \rightarrow \mathcal{O} \otimes_{\mathbb{Z}} R(G)$ is $\mathcal{O} \otimes_{\mathbb{Z}} V_p$ and we have $(\mathcal{O} \otimes_{\mathbb{Z}} V_p) \cap R(G) = V_p$ since $\mathcal{O} \otimes_{\mathbb{Z}} V_p = V_p \oplus \bigoplus_{i=1}^c \alpha_i V_p \subset \mathcal{O} \otimes_{\mathbb{Z}} R(G) = R(G) \oplus \bigoplus \alpha_i R(G)$ and if $y = x + \sum \alpha_i x_i \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$, $x \in R(G)$ iff $\forall x_i = 0$. Thus the theorem is reduced to the case when $m \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$.

$$\theta_C : C \rightarrow \mathbb{Z}$$

If C is a cyclic group of G of order c , $\exists$ $x \mapsto \begin{cases} c & C = \langle x \rangle \\ 0 & \text{otherwise} \end{cases}$. Let $\theta'_C =$

$\text{Ind}_C^G \theta_C$. Let $x \in G$. Then $\theta'_C(x) = \frac{1}{c} \sum_{s \in G |_{sxs^{-1} \in C}} \theta_C(sxs^{-1}) = \sum_{s \in G |_{\langle sxs^{-1} \rangle = C}} 1$. $\forall s \in G$, sxs^{-1} generates exactly one cyclic subgroup of G . Hence $\sum_{C \subset G, \text{cyclic}} \theta'_C(x) = \sum_{s \in G} 1 = |G|$. Let C be any cyclic subgroup of G .

- **CLAIM :** $\theta_C \in R(C)$

pf. Let $c \equiv |C|$. If $c = 1$, $\theta_C \in R(C)$.

When $c > 1$, $c = \sum_{B \subset C, \text{cyclic}} \text{Ind}_B^C \theta_B = \theta_C + \sum_{B \subsetneq C, \text{cyclic}} \text{Ind}_B^C \theta_B$. Since $c \in R(C)$ and $\forall \theta_B \in R(B)$, $B \subsetneq C$ by the inductive hypothesis, so $\theta_C \in R(C)$.

Let $f \in \mathcal{C}(G, \mathbb{Z})$ s.t. $f(G) \subset n\mathbb{Z}$. Then write $f = nf'$, $f' \in \mathcal{C}(G, \mathbb{Z})$. $n = \sum_{C \subset G, \text{cyclic}} \text{Ind}_C^G \theta_C$, hence $f = \sum_{C \subset G, \text{cyclic}} \text{Ind}_C^G (\theta_C) f' = \sum \text{Ind}_C^G (\theta_C \text{Res}_C^G f')$. Write $f_C = \theta_C \text{Res}_C^G f'$. Then $f_C \in \mathcal{C}(C, \mathbb{Z})$ and $f_C(C) \subset |C|\mathbb{Z}$. So $\forall \chi \in R(C)$, $\langle f_C, \chi \rangle_C = \frac{1}{|C|} \sum_{x \in C} f_C(x) \chi(x) \in \mathcal{O}$ and therefore $f_C = \sum_{W \in S_k(C)} \langle f_C, \chi_W \rangle_C \chi_W \in \mathcal{O} \otimes_{\mathbb{Z}} R(C)$. So $f = \sum_{C \subset G, \text{cyclic}} \alpha_C \text{Ind}_C^G x_C$, where $\begin{cases} \alpha_C \in \mathcal{O} \\ x_C \in R(C) \end{cases}$.

Definition A.2.8. $x \in G$ is called p -unipotent if its order is a power of p . $x \in G$ is called p -regular if its order is prime to p .

- **CLAIM :** $\forall x \in G$, $\exists (x_r, x_u)$ where $x_r, x_u \in G$ s.t.

$$\begin{cases} x_r \text{ is } p\text{-regular, and } x_u \text{ is } p\text{-unipotent} \\ x = x_r x_u = x_u x_r \end{cases}.$$

pf. If x_r, x_u satisfies the conditions above, they have to be powers of x ,

since if $\begin{cases} a = |x_r| \\ b = |x_u| \end{cases}$, $(a, b) = 1$ so there exists an integer $N \geq 1$ s.t. $a|N$

and $N \equiv 1 \pmod{b}$ and then $\begin{cases} x^N = x_r^N x_u^N = x_u \\ x^{1-N} = x x_u^{-1} = x_r \end{cases}$. Suppose we have two such

pairs $(x_r, x_u), (x'_r, x'_u)$ satisfying the above conditions. Then $\exists N \geq 1$ s.t. $x_u^{p^N} = (x'_u)^{p^N} = 1$, $x_r^{p^N} = x_r$, $(x'_r)^{p^N} = x'_r$. Then $x^{p^N} = x_r = x'_r$ and therefore $x_u = x'_u$, so the uniqueness follows. Assume G is generated by x , hence G is cyclic. So we may assume $G = \mathbb{Z}/n\mathbb{Z}$ and $x = 1$. Write $n = p^r m$ with $p \nmid m$. Then by CRT, $G \cong \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ and we can take $x_r = (0, 1)$ and $x_u = (1, 0)$ so such x_r, x_u exist.

- **CLAIM :** Let $\chi \in \mathcal{O} \otimes_{\mathbb{Z}} R(G)$ be s.t. $\chi(G) \subset \mathbb{Z}$, and $x \in G$. Write $x = x_r x_u$. Then $\chi(x) = \chi(x_r) \pmod{p}$.

pf. Assume $G = \langle x \rangle$. Let $\chi_1, \dots, \chi_n$ be the characters of the irreducible representations of G over k , which are all 1-dimensional. Write $\chi = \sum_{i=1}^n a_i \chi_i$ with $a_i \in \mathcal{O}$. Let $q = p^r$ be the order of x_i . Then $x^q = x_r^q$ so $\forall i$, $\chi_i(x)^q = \chi_i(x_r)^q$. So $\chi(x)^q = (\sum a_i \chi_i(x))^q = \sum a_i^q \chi_i(x)^q = \sum a_i^q \chi_i(x_r)^q = \chi(x_r)^q \pmod{p\mathcal{O}}$.

As $\begin{cases} \chi(x), \chi(x_r) \in \mathbb{Z} \\ p\mathcal{O} \cap \mathbb{Z} = p\mathbb{Z} \end{cases}$, $\chi(x)^q = \chi(x_r)^q \pmod{p}$. Also $a^p \equiv a \pmod{p}$, $\forall a \in \mathbb{Z}$ implies $\chi(x) \equiv \chi(x_r) \pmod{p}$.

- **CLAIM :** Let $x \in G$ be p -regular. Let $H = C \times P$ be p -elementary with $C = \langle x \rangle$ and P a Sylow p -subgroup of $Z_G(x)$. Then $\exists \psi \in \mathcal{O} \otimes_{\mathbb{Z}} R(H)$ s.t. $\psi(H) \subset \mathbb{Z}$ and if $\psi' = \text{Ind}_H^G \psi$,

$$\begin{cases} \psi'(x) \neq 0 \pmod{p} \\ \psi'(s) = 0, \forall s \in G, \text{ a } p\text{-regular element that is not conjugate to } x \end{cases}.$$

pf. Let $\begin{cases} c = |C| \\ p^r = |P| \end{cases}$. Let $\begin{matrix} \psi_C : C & \rightarrow & \mathbb{Z} \\ y & \mapsto & \begin{cases} c & y = x \\ 0 & \text{otherwise} \end{cases} \end{matrix}$. As $\psi_C(C) \subset c\mathbb{Z}$, $\psi_C \in$

$\mathcal{O} \otimes_{\mathbb{Z}} R(C)$ by the previous claim. Let $\begin{matrix} \psi : H = C \times P & \rightarrow & \mathbb{Z} \\ (x, y) & \mapsto & \psi_C(x) \end{matrix}$. Then $\psi \in$
 $\mathcal{O} \otimes_{\mathbb{Z}} R(H)$. Let $s \in G$ be p -regular. Then $\psi'(x) = \frac{1}{cp^r} \sum_{y \in G |_{ysy^{-1} \in H}} \psi(ysy^{-1})$. Let
 $y \in G$. If $ysy^{-1} \in H$, $ysy^{-1} \in C$ so $\psi(ysy^{-1}) \neq 0$ iff $ysy^{-1} = x$. Hence $\psi'(s) = 0$
 if s is not conjugate to x . Also $\psi'(x) = \frac{1}{cp^r} \sum_{y \in G |_{yxy^{-1} = x}} \psi(x) = \frac{1}{p^r} \sum_{y \in Z_G(x)} 1 =$
 $\frac{1}{p^r} |Z_G(x)| \neq 0 \pmod p$.

- $\exists \psi \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$ s.t. $\psi(G) \subset \mathbb{Z}$ and $\psi(x) \neq 0 \pmod p, \forall x \in G$.

Write $n \equiv |G| = p^r m$ with $(p, m) = 1$. Choose a $\psi \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$ as previously. Let $N = \psi(p^r) = |(\mathbb{Z}/p^r\mathbb{Z})^\times|$. Then $\forall l \in \mathbb{Z}$ prime to p , $l^N = 1 \pmod{p^r}$. So $\forall x \in G$, $\psi(x)^N = 1 \pmod{p^r}$, so $m(\psi^N - 1) \in \mathcal{C}(G, \mathbb{Z})$ takes its values in $n\mathbb{Z}$. Then this implies $m(\psi^N - 1) \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$. AS it is an ideal of $\mathcal{O} \otimes_{\mathbb{Z}} R(G)$, $m\psi^N \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$. Then $m = m\psi^N - m(\psi^N - 1) \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$ and the theorem follows.

Corollary A.2.9. *For every representation V of G , there exist monomial representations $V_1, \dots, V_r$ of G and integers $n_1, \dots, n_r \in \mathbb{Z}$ s.t. in $R(G)$, $[V] = \sum_{i=1}^r n_i [V_i]$.*

Theorem A.2.13 (Brauer). *Let*

$$\text{Ind}_X = \bigoplus_{H \in X} \text{Ind}_H^G : \bigoplus_{H \in X} R(H) \rightarrow R(G)$$

Then it is surjective.

pf. $\text{Im}(\text{Ind}_X) = \sum_{p \text{ prime}} V_p$ by the previous theorem, so $R(G)/\text{Im}(\text{Ind}_X)$ is a finite group of order prime to every prime number, i.e. the trivial group.

Definition A.2.9. *A representation V of G is called monomial if there exists a subgroup H of G and a 1-dimensional representation W of H s.t. $V \cong \text{Ind}_H^G W$.*

Lemma A.2.4. *Let P be a p -group and suppose P is not abelian. Denote by $Z(P)$ the center of P . Then there exists an abelian normal subgroup A of P s.t. $Z(P) \subsetneq A$.*

pf. The quotient $P/Z(P)$ is a nontrivial p -group, so its center is nontrivial. Choose $A' \subset Z(P/Z(P))$ cyclic of order p , and let A be its inverse image in P . Clearly $Z(P) \subsetneq A$ and A is normal in P because it is the inverse image of a normal subgroup of $P/Z(P)$. Also A is abelian because it is generated by $Z(P)$ and by a lift of a generator of A' .

Proposition A.2.6. *Let H be a p -elementary group. Then every irreducible representation of H is monomial.*

pf. Write $H = C \times P$ with C cyclic of order prime to p and P a p -group. Then irreducible representations of H are all of the form $V_1 \otimes_k V_2$ by the theorem A.2.6. where V_1, V_2 is an irreducible representation of C, P respectively. Then V_1 is 1-dimensional, so we just need to show V_2 is monomial. Assume $H = P$ is a p -group. If P is abelian, every irreducible representation of P is 1-dimensional so assume P is not abelian. Let (V, ρ) be an irreducible representation of P . If $\ker \rho \neq \{0\}$, by the inductive hypothesis to $P/\ker \rho$, ρ is monomial. So assume ρ is faithful. Then by the previous lemma, there exists a normal abelian subgroup A of P s.t. $Z(P) \subsetneq A$ where $Z(P)$ is the center of P . Let $V = V_1 \oplus \cdots \oplus V_n$ be the isotypic decomposition of $\text{Res}_A^P V$. Because A is abelian, A acts on each V_i through a morphism of groups $\rho_i : A \rightarrow k^\times$. We cannot have $V_1 = V$ because otherwise $\rho(A)$ would be contained in the center of $\rho(P)$, so A would be contained in the center of P , a contradiction. Let $g \in P$, $i \in \{1, \dots, n\}$. Then if $v \in V_i$, $y \in A$, $\rho(y)\rho(g)v = \rho(g)\rho(g^{-1}yg)v = \rho_i(g^{-1}yg)\rho(g)v$, so $\rho(g)$ sends V_i bijectively to the isotypic component of $\text{Res}_A^P V$ corresponding to the map
$$\begin{array}{ccc} A & \rightarrow & k^\times \\ y & \mapsto & \rho_i(g^{-1}yg) \end{array}$$
, i.e. the action of P on V permutes the V_i , so we get an action of P on the set $\{V_1, \dots, V_n\}$. As V is irreducible, this action is transitive. Hence all the V_i are isomorphic as k -vector space and so $\dim_k V_1 = \cdots = \dim_k V_n = \frac{1}{n} \dim_k V$. Let $Q = \{g \in P \mid \rho(g)V_1 = V_1\}$, then Q is a subgroup of P and $|P/Q| = n > 1$. So $Q \neq P$. As Q stabilises V_1 , we get a representation of Q on V_1 , denoted by V_Q . Define
$$\begin{array}{ccc} \phi : \text{Ind}_Q^P V_Q & \rightarrow & V \\ g \otimes v & \mapsto & \rho(g)v \end{array}$$
, which is well-defined by the definition of Q . It is P -equivariant and surjective given $V = \sum_{g \in P} \rho(g)V_1$. Moreover, $\dim_k \text{Ind}_Q^P V_Q = |P/Q| \dim_k V_Q = n \dim_k V_1 = \dim_k V$ so ϕ is an isomorphism and $V \cong \text{Ind}_Q^P V_Q$ as representations of P . Also since V is irreducible, so is V_Q . By the inductive hypothesis, V is monomial.

A.2.3 Characters in $\text{char}(k) = p$

Let R be a ring.

Definition A.2.10. If M is a R -module, we write $\text{rad}(M) = \text{rad}(R)M$. This is a submodule of M .

$\rightarrow$ for R left-Artinian, R -module M is semisimple iff $\text{rad}(M) = 0$

pf. If $\text{rad}(M) = 0$, M is $R/\text{rad}(R)$ -module, so it is semisimple because $R/\text{rad}(R)$ is a semisimple ring.

If M is semisimple R -module, $M = \bigoplus_{i \in I} M_i$ with all the M_i simple. On each M_i , R acts through $R/\text{rad}(R)$ so $\text{rad}(R)M = 0$.

Definition A.2.11. A R -module M is called indecomposable if for every direct sum decomposition $M = M' \oplus M''$, we have either $M' = 0$ or $M'' = 0$.

Definition A.2.12. A ring S is called *local* if $S \neq \{0\}$ and S has a unique maximal left ideal.

Theorem A.2.14. Let S be a ring. TFAE

1. S is local
2. S has a unique maximal right ideal
3. $\text{rad}(S)$ is a maximal left ideal of S
4. $\text{rad}(S)$ is a maximal right ideal of S
5. $S \neq \{0\}$ and for every $x \in S$, either x or $1 - x$ is invertible
6. $S/\text{rad}(S)$ is a division algebra
7. $S \neq \{0\}$ and every $x \in S \setminus \text{rad}(S)$ is invertible

pf.

- $i \leftrightarrow iii$ follows from the definition of $\text{rad}(S)$
 $ii \leftrightarrow iv$ follows since $\text{rad}(R) = \text{rad}(R^{op})$.
 $vii \leftrightarrow v$ follows since $x \in \text{rad}(R)$ and $\forall y, z \in R$, $1 - yxz$ is invertible are equivalent, when $R, R/\text{rad}(R)$ have the same simple modules.
 $vii \rightarrow vi$ trivially
- $iii \rightarrow vii$

pf. Let $x \in S \setminus \text{rad}(S)$. Then $Sx \not\subset \text{rad}(S)$. As $\text{rad}(S)$ is a maximal left ideal of S , $Sx = S$ so $\exists y \in S$ s.t. $yx = 1$. If $xy \in \text{rad}(S)$, $1 - yxy = 0$ is invertible, a contradiction. Thus $xy \notin \text{rad}(S)$. Then $\exists z \in S$ s.t. $zxy = 1$ so y is a left and right invertible, hence invertible and $x = y^{-1}$ is also invertible.

- $iv \rightarrow vii$ follows by the previous argument
- $v \rightarrow iii$, $v \rightarrow iv$.

pf. Let $x \in S \setminus \text{rad}(S)$. Then $\exists y, z \in S$ s.t. $1 - yxz$ is not invertible. Then by v , this implies yxz is invertible, i.e. $x = y^{-1}z^{-1}$ is also invertible. So any left/right ideal of S that strictly contains $\text{rad}(S)$ contains an invertible element, hence is equal to S .

- $vi \rightarrow iii$

pf. Let $x \in S \setminus \text{rad}(S)$. Then $\exists y \in S$ s.t. $\begin{cases} yx \in 1 + \text{rad}(S) \\ xy \in 1 + \text{rad}(S) \end{cases}$. Then xy, yx are invertible hence x, y are invertible. So any left ideal strictly containing $\text{rad}(S)$ contains an invertible element, which gives iii .

Proposition A.2.7. *Let S be a left Artinian ring. Then $\text{rad}(S)^n = 0$ for n big enough. In fact, if $n = \text{lg}(S)$, $\text{rad}(S)^n = 0$.*

pf. Let $S = I_0 \supset \dots \supset I_n = 0$ be a Jordan-Hölder series for S . Then I_i are left ideals of S and I_i/I_{i+1} is a simple S -module $\forall i$. As $\text{rad}(S)$ annihilates every simple S -module, $\text{rad}(S)I_i \subset I_{i+1}$, $\forall i$ and therefore $\text{rad}(S)^n = \text{rad}(S)^n I_0 \subset I_n = 0$.

Corollary A.2.10. *Let S be a left Artinian ring. Then TFAE*

1. S is local
2. every element of S is nilpotent or invertible

pf.

- $i \rightarrow ii$

pf. Let $x \in S$ and suppose x is not nilpotent. Then $x \notin \text{rad}(S)$ by the previous proposition, so x is invertible by the previous theorem

- $ii \rightarrow i$

pf. If $x \in S$ is nilpotent, $\sum_{n \geq 0} x^n$ is finite, hence defines an element of S , and this element is the inverse of $1 - x$. So ii implies x or $1 - x$ is invertible $\forall x \in S$, and by the previous theorem S is local.

Proposition A.2.8 (Fitting's Lemma). *Let R be a ring and M be a R -module of finite length. Let $f \in \text{End}_R(M)$. Then for n big enough,*

$$M = \ker f^n \oplus \text{Im} f^n$$

pf. As M has finite length, every non-increasing or non-decreasing sequence of R -submodules of M has to stabilise. Then for the non-decreasing sequence $\ker f^n$ and the non-increasing sequence $\text{Im} f^n$, we find an integer $N \geq 0$ s.t. $\forall n \geq N$, $\begin{cases} \ker f^n = \ker f^{n+1} \\ \text{Im} f^n = \text{Im} f^{n+1} \end{cases}$. Let $n \geq N$, if $x \in M$, $f^n(x) \in \text{Im} f^{2n}$, so there exists $y \in M$ s.t. $f^n(x) = f^{2n}(y)$ and so $x - f^n(x) \in \ker f^n$ and $x = f^n(x) + (x - f^n(x)) \in \text{Im} f^n + \ker f^n$. So $M = \text{Im} f^n + \ker f^n$. If $\exists 0 \neq x \in \ker f^n \cap \text{Im} f^n$, $x = f^n(y)$, $\exists y \in M$. Then $y \in \ker f^{2n} = \ker f^n$ so $x = f^n(y) = 0$. So $M = \ker f^n \oplus \text{Im} f^n$.

Corollary A.2.11. *Let R, M be as in the Fitting's lemma. Then M is indecomposable iff $\text{End}_R(M)$ is local.*

pf. Suppose M is indecomposable, and let $f \in \text{End}_R(M)$. Then there exists an integer $n \geq 1$ s.t. $M = \ker f^n \oplus \text{Im} f^n$. As M is indecomposable, either

$$\begin{cases} \text{Im} f^n = 0 & f \text{ is nilpotent} \\ \ker f^n = 0 & f^n \text{ is invertible} \end{cases}.$$

Suppose M is not indecomposable and write $M = M' \oplus M''$ with $M', M'' \neq 0$. Let $\pi : M \rightarrow M'$ be the projection with kernel M'' . Then π is not invertible because $\ker \pi = M'' \neq 0$ and $1 - \pi$ is not invertible because $\ker(1 - \pi) = M' \neq 0$ so $\text{End}_R M$ is not local.

Theorem A.2.15 (Krull-Schmidt-Remak Theorem). *Let M be a R -module of finite length. Then $M = M_1 \oplus \cdots \oplus M_r$ with the M_i indecomposable. Moreover, the M_i are uniquely determined up to reordering.*

pf.

- existence

pf. if $lg(M) = 1$, M is simple and the existence follows trivially.

if $lg(M) \geq 2$, and M is not indecomposable, write $M = M' \oplus M''$ with $M', M'' \neq 0$. Then $lg(M'), lg(M'') < lg(M)$ and therefore by the inductive hypothesis, we get the decompositions for M', M'' and therefore for M .

- uniqueness

pf. Assume the contrary. Then $M = M_1 \oplus \cdots \oplus M_r = M'_1 \oplus \cdots \oplus M'_s$ with all the M_i, M'_j are indecomposable. If $r = 1$, $M_1 = M'_1$ with $s = 1$ is obvious. Suppose $r \geq 2$. Then $\forall j \in \{1, \dots, s\}$, let $u_j \in \text{End}_R(M_1)$ be the composition $M_1 \hookrightarrow M \rightarrow M'_j \hookrightarrow M \rightarrow M_1$, where all maps are obvious injections or projections. Then $id_{M_1} = \sum_{j=1}^s u_j$, so $\exists j$ s.t. $u_j \notin \text{rad}(\text{End}_R(M_1))$. Wlog assume $j = 1$. As M_1 is indecomposable, $\text{End}_R(M_1)$ is local by the previous corollary, so u_1 is invertible by the previous theorem. Write $u_1 = vw$ where w is the composition $M_1 \hookrightarrow M \rightarrow M'_1$ and v is the composition $M'_1 \hookrightarrow M \rightarrow M_1$. Then $(u_1^{-1}v)w = id_{M_1}$ so w is injective and $M'_1 = w(M_1) \oplus \ker(u_1^{-1}v)$. As M'_1 is indecomposable and $w(M_1) \neq 0$, $\ker(u_1^{-1}v) = 0$, so w is an isomorphism, and so is v . Let $x \in M_1 \cap (\bigoplus_{j \geq 2} M'_j)$. Then the projection of x on M'_1 is 0 so $w(x) = 0$ so $x = 0$ given w being injective. Hence M_1 and $\bigoplus_{j \geq 2} M'_j$ are direct sums. Moreover if $x \in M$, then the surjectivity of w implies $\exists x_1 \in M_1$ s.t. $x - x_1 \in \sum_{j \geq 2} M'_j$ so $M = M_1 \oplus \sum_{j \geq 2} M'_j$. We get $M = M_1 \oplus \sum_{j \geq 2} M'_j$ and $M_1 \cong M'_1$, by the inductive hypothesis, $M/M_1 \cong \bigoplus_{j=2}^r M_j \cong \bigoplus_{j=2}^s M'_j$.

Proposition A.2.9. *Let P be a projective R -module, M be a R -module and I be an ideal of R . Then the reduction modulo I map*

$$\text{hom}_R(P, M) \rightarrow \text{hom}_R(P/IP, M/IM) = \text{hom}_{R/I}(P/IP, M/IM)$$

is surjective.

pf. Denote by $\begin{cases} \pi : P \rightarrow P/IP \\ \pi' : M \rightarrow M/IM \end{cases}$ the projections. Then $\forall u \in \text{hom}_R(P/IP, M/IM)$,

$$\begin{array}{ccc} P & \xrightarrow{u'} & M \\ \downarrow \pi & \searrow u\pi & \downarrow \pi' \\ P/IP & \xrightarrow{u} & M/IM \end{array} \quad . \text{ As } \pi' \text{ is surjective and } P \text{ is projective, } \exists u' : P \rightarrow M \text{ s.t.}$$

$\pi' u' = u\pi$.

Lemma A.2.5. *If M is a nonzero R -module of finite length, $\text{rad}(M) \neq M$.*

pf. Let $M' \subsetneq M$ be a maximal proper submodule. Then M/M' is simple so $\text{rad}(R)$ acts trivially on M/M' so $\text{rad}(M) = \text{rad}(R)M \subset M'$.

Corollary A.2.12. *Assume R is left-Artinian. Let P be a projective R -module of finite length. Then P is indecomposable iff $P/\text{rad}(P)$ is simple.*

pf. If P is not indecomposable, $P = P_1 \oplus P_2$, with $P_1, P_2 \neq 0$. Then $P/\text{rad}(P) = P_1/\text{rad}(P_1) \oplus P_2/\text{rad}(P_2)$ and each factor is not zero by the previous lemma. So $P/\text{rad}(P)$ is not simple.

Assume $M \equiv P/\text{rad}(P)$ is not simple. As R is left-Artinian, $R/\text{rad}(R)$ is semisimple, so M is a semisimple module and therefore $M = M_1 \oplus M_2$ with $M_1, M_2 \neq 0$. Let $\pi_1 \in \text{End}_R(M_1)$ be the composition $M \rightarrow M_1 \hookrightarrow M$ where the maps are the obvious projection and inclusion. Then by the previous proposition, $\exists \pi \in \text{End}_R(P)$ s.t. $\pi \bmod \text{rad}(P) = \pi_1$. Then neither π nor $1 - \pi$ are invertible so $\text{End}_R(P)$ is not local, so P cannot be indecomposable by the corollary A.2.11.

Write $PI(R)$ for the set of isomorphism classes of finite length projective indecomposable R -modules and $S(R)$ for the set of isomorphism classes of simple R -modules.

Proposition A.2.10. *Suppose R is left-Artinian and left-Noetherian. Then the map $P \mapsto P/\text{rad}(P)$ induces a bijection $PI(R) \rightarrow S(R)$.*

pf.

- Well-definedness follows by the previous corollary.

- Let M be a simple R -module. Any $x \in M \setminus \{0\}$ gives a surjective map $\begin{matrix} R & \rightarrow & M \\ a & \mapsto & ax \end{matrix}$. As R is a R -module of finite length, by the Krull-Schmidt-Remak Theorem, we can write $R = P_1 \oplus \cdots \oplus P_n$ where P_i are indecomposable module that are automatically projective as direct summands of a free module. Then $R/\text{rad}(R) = \bigoplus P_i/\text{rad}(P_i)$ surjects to M and every $P_i/\text{rad}(P_i)$ is simple by the previous corollary. So M is isomorphic to one of $P_i/\text{rad}(P_i)$ by the Schur's lemma. The map above is surjective.
- Let P, P' be two projective indecomposable modules of finite length, and suppose we have a R -module isomorphism $u : P/\text{rad}(P) \rightarrow P'/\text{rad}(P')$. Then by the previous proposition, there exists a R -module map $u' : P \rightarrow P'$ s.t. $u = u' \bmod \text{rad}(R)$. Let $N \subset P'$ be a proper maximal submodule. Then P'/N is simple so $\text{rad}(R)(P'/N) = 0$ so $\text{rad}(P') = \text{rad}(R)P' \subset N$. As $P'/\text{rad}(P')$ is simple, $N = \text{rad}(P')$ and $\text{rad}(P')$ is the unique proper maximal submodule of P' . As $u'(P) \not\subset \text{rad}(P')$, $u'(P) = P'$ so u' is surjective. As P' is projective, $P \cong P' \oplus \ker u'$ but since P is indecomposable, $\ker u' = 0$ and u' is an isomorphism.

If M is a simple R -module, its inverse image in $PI(R)$ is a minimal projective R -module s.t. P surjects onto M . This is called a **projective envelope** or **projective cover of M** .

Theorem A.2.16. *Let S be a ring, and I be an ideal of S s.t. every element of I is nilpotent. Let $\bar{e} \in S/I$ be idempotent. Then there exists $e \in S$ idempotent s.t. $\bar{e} = e \bmod I$.*

pf. $\bar{f} = 1 - \bar{e}$ is also idempotent and $\bar{e}\bar{f} = \bar{f}\bar{e} = 0$. Let e be any lift of $\bar{e}$ and let $f = 1 - e$. Then $ef = fe \in I$ and $e + f = 1 \bmod I$. Then by the assumption on I , $\exists k \geq 1$ s.t. $(ef)^k = e^k f^k = 0$. Here $e^k = \bar{e}^k = \bar{e} \bmod I$. Let $\begin{cases} e' = e^k \\ f' = f^k \end{cases}$. Then $e'f' = f'e' = 0$ and $e' + f' = e^k + f^k = \bar{e} + \bar{f} = 1 \bmod I$. Let $x = 1 - (e' + f')$. As $x \in I$, $\exists n \geq 1$ s.t. $x^n = 0$. So $u \equiv 1 + x + \cdots + x^{n-1}$ is an inverse of $1 - x = e' + f'$ and it commutes with e', f' . Let $\begin{cases} e'' = ue' \\ f'' = uf' \end{cases}$. Then $e'' \equiv \bar{e} \bmod I$ and $\begin{cases} e''f'' = f''e'' = 0 \\ f'' = uf' \end{cases}$. Then $e'' = \bar{e} \bmod I$ so $e''f'' = f''e'' = 0$ and $e'' + f'' = u(e' + f') = 1$. So $(e'')^2 = (e'')^2 + e''f'' = e''(e'' + f'') = e''$ and we have the idempotent lift of $\bar{e}$.

Corollary A.2.13. *Let S be a ring and I be an ideal of S . Suppose that the obvious map $S \rightarrow \hat{S} \equiv \lim S/I^n$ is an isomorphism. Then any idempotent of S/I lifts to an idempotent of S .*

pf. Let $\bar{e} \in S/I$ be idempotent. Then by the previous theorem and induction on $n \geq 1$, we can construct a sequence of idempotents $e_n \in S/I^n$ s.t. $e_1 = \bar{e}$ and e_{n+1} lifts to e_n , $\forall n$. Then $e \equiv (e_n) \in \hat{S}$ and its preimage in S is an idempotent lifting $\bar{e}$.

Corollary A.2.14. 1. $PK(R)$ is a free $\mathbb{Z}$ -module with basis $([P])_{P \in PI(R)}$.

2. If P, P' are projective R -modules of finite length, then $P \cong P'$ as R -modules iff $[P] = [P']$ in $PK(R)$.

pf. Write $\begin{cases} P = P_1 \oplus \cdots \oplus P_r \\ P' = P'_1 \oplus \cdots \oplus P'_s \end{cases}$ with P_i, P'_j indecomposable by the Krull-Schmidt-Remak theorem. Then P_i, P'_j are projective since it is a direct summand of projective modules, and therefore they are in $PI(R)$. Then by the same theorem, $P \cong P'$ iff $\exists \sigma : \{1, \dots, r\} \rightarrow \{1, \dots, s\}$ s.t. $P_i \cong P'_{\sigma(i)}$, $\forall i$. Then by i , this is equivalent to $[P] = [P']$.

Let Λ be a commutative ring and G be a finite group. Assume every $\Lambda[G]$ -module is of finite type over Λ .

Definition A.2.13. We denote by $P_\Lambda(G)$ the quotient of the free $\mathbb{Z}$ -module on all the $[P]$, for P projective $\Lambda[G]$ -module that is of finite type as a Λ -module, by the relations $[P] = [P'] + [P'']$, for every exact sequence $0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0$.

$\rightarrow \Lambda \rightarrow \Lambda'$ a morphism of commutative rings, then $P \mapsto P \otimes_\Lambda \Lambda'$ induces a morphism of groups $P_\Lambda(G) \rightarrow P_{\Lambda'}(G)$ (if P is a projective $\Lambda[G]$ -module, it is a direct summand of some free $\Lambda[G]$ -module F and then $P \otimes_\Lambda \Lambda'$ is a direct summand of the free $\Lambda'[G]$ -module $F \otimes_\Lambda \Lambda'$ and the morphism of commutative rings give Λ' as Λ -modules).

Proposition A.2.11. Let P be a $\Lambda[G]$ -module. Then TFAE

1. P is a projective $\Lambda[G]$ -module
2. P is projective as a Λ -module and there exists $u \in \text{End}_\Lambda(P)$ s.t. $\forall x \in P$, $\sum_{g \in G} gu(g^{-1}x) = x$.

pf. Write P_0 for P seen as a Λ -module. Let $Q = \Lambda[G] \otimes_\Lambda P_0$. Then there exists a surjective map $\begin{matrix} q : Q & \rightarrow & P \\ x \otimes y & \mapsto & xy \end{matrix}$.

- **CLAIM :** $\begin{matrix} \phi : \text{End}_\Lambda(P_0) & \rightarrow & \text{hom}_{\Lambda[G]}(P, Q) \\ u & \mapsto & \sum_{g \in G} g \otimes ug^{-1} \end{matrix}$ is an well-defined isomorphism of Λ -modules.

pf. $\phi(u)$ is $\Lambda[G]$ -linear $\forall u \in \text{End}_\Lambda(P_0)$ and therefore ϕ is well-defined.

Let $u \in \text{End}_\Lambda(P_0)$. Here $Q = \bigoplus_{g \in G} g \otimes P_0$ as a Λ -module. So if $x \in P$ s.t. $0 = \phi(u)(x) = \sum g \otimes u(g^{-1}x)$, $u(g^{-1}x) = 0$, $\forall g \in G$ and in particular $u(x) = 0$. Hence $u = 0$ if $\phi(u) = 0$ and ϕ is injective.

Let $v \in \text{hom}_{\Lambda[G]}(P, Q)$. Then $v(x) = \sum_{g \in G} g \otimes u_g(x)$, $\forall x \in P$ with $u_g \in \text{End}_{\Lambda}(P_0)$. By $\Lambda[G]$ -linearity of v , $\forall h \in G$, $\forall x \in P$, $v(h^{-1}x) = \sum g \otimes u_g(h^{-1}x) = \sum (h^{-1}g) \otimes u_g(x)$ so $u_g(x) = u_{h^{-1}g}(h^{-1}x)$, $\forall h, g \in G$, $\forall x \in P$. In particular, $u_g(x) = u_1(g^{-1}x)$, $\forall g \in G$, $\forall x \in P$ and therefore $v = \phi(u_1)$ and ϕ is surjective.

- $i \rightarrow ii$

pf. If P is projective, it is a direct summand of some free module $\Lambda[G]^{(I)}$. As $\Lambda[G]$ is a free Λ -module, $\Lambda[G]^{(I)}$ is also a free Λ -module, so P_0 is projective. Also as $q : Q \rightarrow P$ is surjective and Λ -linear, $\exists \Lambda$ -linear map $s : P \rightarrow Q$ s.t. $qs = id_P$. Write $s = \phi(u)$ with $u \in \text{End}_{\Lambda}(P_0)$. Then $id_P = q \sum gu g^{-1}$ which gives ii .

- $ii \rightarrow i$

pf. If P_0 is projective Λ -module, Q is projective $\Lambda[G]$ -module. Also $s \equiv \phi(u) : P \rightarrow Q$ satisfies $qs = id_P$, so P is a direct summand of Q , hence is also a projective $\Lambda[G]$ -module.

Theorem A.2.17. *Suppose Λ is a discrete valuation ring with residue field k and maximal ideal $\mathfrak{m}$.*

1. *if P is a $\Lambda[G]$ -module that is free of finite type over Λ , then P is projective $\Lambda[G]$ -module iff $\bar{P} \equiv P \otimes_{\Lambda} k$ is a projective $k[G]$ -module.*
2. *if P, P' are projective $\Lambda[G]$ -modules, then $P \cong P'$ as $\Lambda[G]$ -module iff $P \otimes_{\Lambda} k \cong P' \otimes_{\Lambda} k$ as $k[G]$ -modules.*
3. *suppose the DVR Λ is complete. If $\bar{P}$ is a projective $k[G]$ -module, then there exists a unique projective $\Lambda[G]$ -module P s.t. $\bar{P} \cong P \otimes_{\Lambda} k$.*

pf.

- If P is projective as a $\Lambda[G]$ -module, then $\bar{P}$ is projective as a $k[G]$ -module. Suppose $\bar{P}$ is a projective $k[G]$ -module. Then by the previous proposition, $\exists \bar{u} \in \text{End}_k(\bar{P})$ s.t. $\sum g \bar{u} g^{-1} = id_{\bar{P}}$. As P is a projective Λ -module, $\exists u \in \text{End}_{\Lambda}(P)$ lifting $\bar{u}$ by the proposition A.2.16. Then $u' \equiv \sum gu g^{-1} \in \text{End}_{\Lambda[G]}(P)$ is equal to $id_{\bar{P}}$ modulo $\mathfrak{m}$. So $\det(u') = 1 \pmod{\mathfrak{m}}$, so $\det(u') \in \Lambda^{\times}$ so u' is invertible and $\sum g(u(u')^{-1})g^{-1} = id_P$. Then by the previous proposition, P is projective $\Lambda[G]$ -module.
- Let $\bar{u} : P \otimes_{\Lambda} k \rightarrow P' \otimes_{\Lambda} k$ be an isomorphism of $k[G]$ -modules. Then by the previous proposition A.2.26., there exists a $\Lambda[G]$ -module map $u : P \rightarrow P'$ lifting $\bar{u}$. We know P, P' are projective Λ -modules of finite type, hence they

are free Λ -modules of finite type. Their ranks are equal, because they are equal to the dimension over k of $P \otimes_{\Lambda} k \cong P' \otimes_{\Lambda} k$. If we choose Λ -bases of P, P' , then the matrix A of u is square, and $\det(A) \neq 0 \pmod{\mathfrak{m}}$ because $\bar{u}$ is invertible. Hence $\det(A) \in \Lambda^{\times}$ and u is an isomorphism of Λ -modules.

- The uniqueness follows from *ii*. For existence, write $\bar{F} = \bar{P} \oplus \bar{P}'$ with $\bar{F} = k[G]^{\oplus n}$, and let $F = \Lambda[G]^{\oplus n}$, and $B = \text{End}_{\Lambda[G]}(F)$. Then the map $B \rightarrow \text{End}_{k[G]}(\bar{F})$ is surjective by the proposition A.2.26., so it identifies $\text{End}_{k[G]}(\bar{F})$ with $B/\mathfrak{m}B$. Let $\bar{e} \in \text{End}_{k[G]}(\bar{F})$ be the projection on $\bar{P}$ with kernel $\bar{P}'$. Then there exists an idempotent $e \in B$ lifting $\bar{e}$. Then $F = \text{Im}(e) \oplus \ker(e)$ and $P \equiv \text{Im}(e)$ is a projective $\Lambda[G]$ -module s.t. $P \otimes_{\Lambda} k = \text{Im}(\bar{e}) = \bar{P}$.

Fix a complete DVR Λ with uniformizer ω , maximal ideal $\mathfrak{m} = (\omega)$, residue field k

and fraction field K and assume $\text{char}(K) = 0$.

$$\begin{array}{ccc} P_k(G) & \xrightarrow{c} & R_k(G) \\ & \searrow e & \nearrow d \\ & R_K(G) & \end{array}$$

where d is the decomposition $P_k(G) \xrightarrow{\psi^{-1}} P_{\Lambda}(G) \xrightarrow{\otimes_{\Lambda} K} P_K(G) = R_K(G)$ and c is the obvious map. Let V be a $K[G]$ -module. Let $M_1 \subset V$ be a Λ -lattice, i.e. a finite type Λ -submodule of V s.t. $KM_1 = V$. After replacing M_1 by $\sum_{g \in G} gM_1$, assume M_1 is stable by G . Then $\bar{M}_1 \equiv M_1 \otimes_{\Lambda} k$ is a $k[G]$ -module.

Theorem A.2.18. $[\bar{M}_1] \in R_k(G)$ only depends on V .

pf. Let M_2 be another G -stable Λ -lattice of V .

- $\omega M_1 \subset M_2 \subset M_1$

let $N = M_1/M_2$, which is a $k[G]$ -module because $\omega M_1 \subset M_2$. From $\omega M_2 \subset \omega M_1 \subset M_2 \subset M_1$, there exists an exact sequence of $k[G]$ -modules $0 \rightarrow N \rightarrow \bar{M}_2 \equiv M_2 \otimes_{\Lambda} k = M_2/\omega M_2 \rightarrow \bar{M}_1 \rightarrow N \rightarrow 0$. So in $R_k(G)$, $[N] - [\bar{M}_2] + [\bar{M}_1] - [N] = 0 \rightarrow [\bar{M}_1] = [\bar{M}_2]$

- general case

multiplying M_2 by a high enough power of ω , we may assume $M_2 \subset M_1$. There also exists $n \geq 1$ s.t. $\omega^n M_1 \subset M_2$. Then when $n=1$, by the previous statement the theorem follows. If $n \geq 2$, let $M_3 = \omega^{n-1} M_1 + M_2$. Then $\omega^{n-1} M_1 \subset M_3 \subset M_1$ so $[M_3 \otimes_{\Lambda} k] = [\bar{M}_1]$ by the inductive hypothesis. Also $\omega M_3 \subset M_2 \subset M_3$ so $[M_3 \otimes_{\Lambda} k] = [M_2 \otimes_{\Lambda} k]$ by the previous statement. Then the theorem holds for n .

Theorem A.2.19. Assume $p = \text{char}(k)$ and $p \nmid |G|$.

1. every $k[G]$ -module is semisimple.
2. every $\Lambda[G]$ -module that is projective as a Λ -module is projective as a $\Lambda[G]$ -module.
3. the map $d : R_K \rightarrow R_k(G)$ is an isomorphism and so are c, e . Also d induces a bijection $S_K(G) \rightarrow S_k(G)$.

pf.

- i is trivial by the chapter 1 Maschke's theorem
- ii follows from i and the previous theorem A.2.17.
- by i , $P_k(G) = R_k(G)$ and c is just the identity morphism. Also $de = id_{R_k(G)}$ and $e([S_k(G)]) \subset S_K(G)$. Let V, V' be two $K[G]$ -modules s.t. $d([V] - [V']) = 0$, let $\begin{cases} M \subset V \\ M' \subset V' \end{cases}$ be G -stable Λ -lattices and let $\begin{cases} \bar{M} = M \otimes_{\Lambda} k \\ \bar{M}' = M' \otimes_{\Lambda} k \end{cases}$. Then $d([V] - [V']) = [\bar{M}] - [\bar{M}'] = 0 \in R_k(G)$. As $p \nmid |G|$, $\bar{M} \cong \bar{M}'$ as $k[G]$ -modules, so $M \cong M'$ as $\Lambda[G]$ by the previous theorem and therefore $V \cong V'$ and $[V] - [V'] = 0$.

CLAIM : Let G be a finite group and p a prime number. If k is algebraically closed field of characteristic p and G is a p -group and V is an irreducible representation of G on a finite dimensional k -vector space, V is the trivial representation.

pf.

- Suppose G is abelian. Let V be an irreducible representation of V . Then by Schur's lemma, $\text{End}_{k[G]}(V) = k$. As G is abelian, $k[G]$ is commutative, so the action of any element of $k[G]$ on V is in $\text{End}_{k[G]}(V)$, so there exists a k -algebra map
$$\begin{array}{ccc} u : k[G] & \rightarrow & k \\ x & \mapsto & (v \mapsto u(x)v) \end{array}$$
. In particular, every k -subspace of V is a subrepresentation; as V is irreducible $\dim V = 1$. Giving u is the same as giving a morphism of groups $\rho : G \rightarrow k^{\times}$. But the only element of $k^{\times}$ that has order a power of p is 1, so ρ is trivial so G acts trivially on V .
- Assume G is not abelian. Let Z be the center of G which is a nontrivial abelian p -group, so the inductive hypothesis applies to it. Let V be an irreducible representation of G . Denote by V^Z the k -subspace of $v \in V$ s.t. $gv = v$, $\forall g \in Z$. If $g \in G$, $v \in V^Z$, then $\forall h \in Z$, $h(gv) = (hg)v = (gh)v = g(hv) = gv \in V^Z$. Thus V^Z is a subrepresentation of V . Let $V = V_0 \supset \dots \supset V_n = 0$ be a Jordan-Hölder series for V seen as a $k[Z]$ -module. Then V_{n-1} is a simple $k[Z]$ -module, so it is the trivial representation of Z by the inductive hypothesis. This means $V_{n-1} \subset V^Z$ so $V^Z \neq 0$. As V is irreducible $V^Z = V$ so the

action of G on V factors G/Z and we can see V as an irreducible representation of G/Z . By the inductive hypothesis, this is the trivial representation of G/Z and so V is the trivial representation of G .

Theorem A.2.20. *The maps*

$$\begin{cases} \text{Ind} \equiv \bigoplus_{l, \text{ prime}} \bigoplus_{H \in X(l)} \text{Ind}_H^G : \bigoplus_{l \neq p, \text{ prime}} \bigoplus_{H \in X(l)} R_k(H) \rightarrow R_k(G) \\ \text{Ind} \equiv \bigoplus_{l, \text{ prime}} \bigoplus_{H \in X(l)} \text{Ind}_H^G : \bigoplus_{l \neq p, \text{ prime}} \bigoplus_{H \in X(l)} P_k(H) \rightarrow P_k(G) \end{cases}$$

are both surjective.

pf. Let $\begin{cases} \mathbb{1}_K & \text{be the unit element in } R_K(G) \\ \mathbb{1}_k & \text{be the unit element in } R_k(G) \end{cases}$. Then $d(\mathbb{1}_K) = \mathbb{1}_k$. By the Brauer's theorem, $\mathbb{1}_K = \sum_{l, \text{ prime}} \sum_{H \in X(l)} \text{Ind}_H^G x_H$, $\exists x_H \in R_K(H)$. Applying d to this equality gives $\mathbb{1}_k = \sum_{l, \text{ prime}} \sum_{H \in X(l)} \text{Ind}_H^G x'_H$ with $x'_H = d(x_H)$ in $R_k(H)$. So if $y \in \begin{cases} R_k(G) \\ P_k(G) \end{cases}$, $y = y\mathbb{1}_k = \sum_{l, \text{ prime}} \sum_{H \in X(l)} \text{Ind}_H^G (x'_H \text{Res}_H^G y)$ where $x'_H \text{Res}_H^G(y) \in P_k(H)$ if $y \in P_k(G)$ given Res_H^G sending $P_k(G)$ to $P_k(H)$.

→ given k algebraically closed, when $l = p$, for any l -elementary group $G = H \times P$, and P acts simple $k[G]$ -module, so each is pulled back from H . Since H is cyclic of order prime to p , it is l' -elementary for any prime $l' \mid |H|$; hence every class $\text{Ind}_H^G[M]$ already lies in the image without $l \neq p$, so domain can be taken as above.

Lemma A.2.6. *Suppose $G = P \times H$ with P a p -group and H of order prime to p . Then P acts trivially on every semisimple $k[G]$ -module.*

pf. Let M be a simple $k[G]$ -module. As P is a p -group its only irreducible representation over k is the trivial representation, so $M^P \neq 0$. As $G = P \times H$, the action of G preserves M^P . As M is simple, $M^P = M$ and so P acts trivially on M .

Corollary A.2.15. *If k is algebraically closed and K contains all the $|G|$ th roots of the unity in $\bar{K}$, then $d : R_K(G) \rightarrow R_k(G)$ is surjective.*

pf. By the previous theorem, we may assume G is l -elementary, for some prime l . Write $G = C \times G'$, with C cyclic of order prime to l and G' a l -group. Let $\begin{cases} l = p, & P = G', H = C \\ l \neq p & P = C_p, H = C^p \times G' \end{cases}$ where $C = C_p \times C^p$, C_p a p -group, and C^p of order prime to p . Then $G = H \times P$ with P a p -group and H of order prime to p . Let M be a simple $k[G]$ -module. Then by the lemma, P acts trivially on M . So M is a simple $k[H]$ -module. Then by the previous theorem A.2.17., there exists $K[H]$ -module V s.t. $d[V] = [M]$. We see V as a $K[G]$ -module by making P act trivially, and then we still have $d[V] = [M]$.

Proposition A.2.12. *If G is a finite p -group, then $\begin{cases} P_k(G) \cong \mathbb{Z} \\ R_k(G) \cong \mathbb{Z} \end{cases}$ and the map $P_k(G) \rightarrow R_k(G)$ corresponds to multiplication by $p^n \equiv |G|$.*

pf. The $\mathbb{Z}$ -rank of $P_k(G)$ and $R_k(G)$ is 1, and there exists an isomorphism $R_k(G) \xrightarrow{\quad} \mathbb{Z}$ and $k[G]/\text{rad}(k[G]) = k$. Thus $k[G]$ is local and every element of $k[G]$ is nilpotent or invertible. In particular, the only idempotents are 0 and 1. Let $k[G] \equiv M = M_1 \oplus M_2$ with $\begin{cases} M_1 = MI_1 \\ M_2 = MI_2 \end{cases}$ two left ideals $I_1, I_2 \subset k[G]$. Then we get two idempotents $e_1, e_2 \in k[G]$ s.t. $I_1 = k[G]e_1, I_2 = k[G]e_2$, but there are only 0,1 for idempotents, this implies $k[G]$ is a projective indecomposable module, i.e. there is the unique projective indecomposable finite length $k[G]$ -module that surjects to $\mathbb{1}$. Since in $R_k(G)$, $[k[G]] = \dim_k(k[G])\mathbb{1} = p^n\mathbb{1}$, the proposition follows.

Theorem A.2.21. *Assume k is algebraically closed. Then $c : P_k(G) \rightarrow R_k(G)$ is injective and its image contains $p^n R_k(G)$ where p^n is the biggest power of p dividing $|G|$.*

pf. By the Brauer's theorem we wlog assume G is l -elementary for some prime number l . Then we can write $G = H \times P$, with P a p -group, H of order prime to p . Then the trivial $k[H]$ -module is projective given $k[H]$ being a semisimple ring. Thus with trivial action of H , $k[P]$ is a projective $k[G]$ -module and by the previous proposition $k[P] = |P|\mathbb{1} \in R_k(G)$ is in the image of c . As $\text{Im}(c)$ is an ideal of $R_k(G)$, $p^n R_k(G) \subset \text{Im}(c)$.

Since $R_k(G)/\text{Im}(c)$ is a torsion group and $P_k(G), R_k(G)$ are free $\mathbb{Z}$ -modules of the same finite rank, c needs to be injective.

Corollary A.2.16. *The map $e : P_k(G) \rightarrow R_K(G)$ is injective.*

pf. by the injectivity of c from the previous theorem and the surjectivity of d and commutative diagram above.

Definition A.2.14. *An element $x \in G$ is called p -singular if x is not p -regular, i.e. if p divides the order of x .*

A.3 Compact Group Representation

Definition A.3.1. *A topological group is a topological space G with a group structure s.t. the maps $\begin{matrix} G & \rightarrow & G & G \times G & \rightarrow & G \\ x & \mapsto & x^{-1} & (x, y) & \mapsto & xy \end{matrix}$ are both continuous.*

Let G be a Hausdorff locally compact topological group. By a measure on G , we mean a measure on the Borel σ -algebra of B .

Definition A.3.2. A nonzero measure μ on G is called a left Haar measure if for every measurable function $f : G \rightarrow \mathbb{C}$, $\forall g \in G$,

$$\int_G f(x) d\mu = \int_G f(gx) d\mu$$

Denote by $\mathcal{C}$ the space of continuous functions with compact support $G \rightarrow \mathbb{C}$, equipped with the compact open topology (generated by the sets $\{f \in \mathcal{C} \mid f(K) \subset U\}$ for $\begin{cases} K \subset G & \text{compact} \\ U \subset \mathbb{C} & \text{open} \end{cases}$). Let $\mathcal{C}^*$ be the space of continuous linear maps $\mathcal{C} \rightarrow \mathbb{C}$, equipped with the weak topology (generated by $\{\lambda \in \mathcal{C}^* : |(\lambda - \lambda_1)(f_1)| < r_1, \dots, |(\lambda - \lambda_n)(f_n)| < r_n\}$, $\forall \lambda_i \in \mathcal{C}^*$, $\forall f_i \in \mathcal{C}$, $\forall r_n \in \mathbb{R}_{++}$). If G is compact and metrisable, for every probability measure μ on G , the linear form $f \mapsto \int_G f(x) d\mu$ on $\mathcal{C}$ is continuous, hence in $\mathcal{C}^*$. This implies we can the set $\mathcal{P}(G)$ of probability measures on G with a subset of $\mathcal{C}^*$.

- **The compact open topology on $\mathcal{C}$ is the same as the topology of uniform convergence, i.e. the topology induced by the norm $\|\cdot\|_\infty$.**

pf. Let $f_0 \in \mathcal{C}$

Assume the open nhod of f_0 is of the form $X \equiv \{f \in \mathcal{C} : f(K_1) \subset U_1, \dots, f(K_m) \subset U_m\}$ where $\forall K_i \subset G$ are compact and $\forall U_j \subset \mathbb{C}$ are open. Then $\forall \epsilon > 0$, $V_\epsilon \equiv \{f \in \mathcal{C} : \|f - f_0\|_\infty < \epsilon\}$. Let $i \in \{1, \dots, m\}$. As $f_0(K_i) \subset U_i$ is compact, $\exists \epsilon_i > 0$ s.t. $\{x \in \mathbb{C} : \exists y \in f_0(K_i), |x - y| < \epsilon_i\} \subset U_i$. Let $\epsilon = \min_i \epsilon_i$. Then $V_\epsilon \subset X$.

Let $\epsilon > 0$ and V_ϵ be defined as above. Let $\eta = \epsilon/2$. Then $\forall x \in G$, $U_x \equiv \{y \in G : |f_0(x) - f_0(y)| < \eta\} \subset K_x \equiv \{y \in G : |f_0(x) - f_0(y)| \leq \eta\}$. Then U_x is open and K_x is compact. As $G = \cup_{x \in G} U_x$ is compact, $\exists x_1, \dots, x_n \in G$ s.t. $G = \cup_{i=1}^n U_{x_i} = \cup_{i=1}^n K_{x_i}$. $\forall i$, $K_i = K_{x_i}$ for simplicity and let $U_i = \{a \in \mathbb{C} : |f_0(x_i) - a| < \eta\}$. Then $X \equiv \{f \in \mathcal{C} : \forall i, f(K_i) \subset U_i\} \subset V_\epsilon$.

- **Let ρ be the representation of G on $\mathcal{C}$ defined by $(\rho(g)f)(x) = f(g^{-1}x)$, $\forall f \in \mathcal{C}$, $x, g \in G$. Then $\begin{matrix} G \times \mathcal{C} & \rightarrow & \mathcal{C} \\ (g, f) & \mapsto & \rho(g)f \end{matrix}$ is continuous.**

pf. Denote by μ the map $G \times \mathcal{C} \rightarrow \mathcal{C}$. Let $\begin{cases} g_0 \in G \\ f_0 \in \mathcal{C} \end{cases}$. As G is compact, $f_0 : G \rightarrow \mathbb{C}$ is uniformly continuous, so there exist two collections $\begin{cases} U_1, \dots, U_n \\ V_1, \dots, V_n \end{cases}$ of

open subsets of G s.t. $\begin{cases} K_i \equiv \overline{U_i} \subset V_i, \forall i \\ G = \cup_{i=1}^n U_i \text{ and } \forall i, \forall x, y \in V_i, |f_0(x) - f_0(y)| < \epsilon/2 \end{cases}$.

Let $i \in \{1, \dots, n\}$. $\forall x \in K_i$, $\exists j$ s.t. $g_0^{-1}x \in V_j$ and open nhod $\begin{cases} U_x \text{ of } x \text{ in } G \\ W_x \text{ of } g_0 \text{ in } G \end{cases}$ s.t. $\forall g \in W_x$, $\forall y \in U_x$, $g^{-1}y \in V_j$. As K_i is compact, we can choose $x_1, \dots, x_m$ s.t. $K_i \subset U_{x_1} \cup \dots \cup U_{x_m}$. Then the open nhod $W_i \equiv W_{x_1} \cap \dots \cap W_{x_m}$ of g_0 has the property that $\forall g \in W_i$, $\forall x \in K_i$, $\exists j \in \{1, \dots, n\}$ s.t. $g_0^{-1}x, g^{-1}x \in V_j$. Define $W \equiv \bigcap_{i=1}^n W_i$. Let $g \in W$ and $f \in \mathcal{C}$ s.t. $\|f - f_0\|_\infty < \epsilon/2$. Then $\forall x \in G$, choose i s.t. $x \in K_i$. Then $\exists j \in \{1, \dots, n\}$ s.t. $g_0^{-1}x, g^{-1}x \in V_j$. Then $|(gf - g_0f_0)(x)| = |f(g^{-1}x) - f_0(g_0^{-1}x)| = |(f(g^{-1}x) - f_0(g^{-1}x)) + (f_0(g^{-1}x) - f_0(g_0^{-1}x))| \leq |(f(g^{-1}x) - f_0(g^{-1}x))| + |(f_0(g^{-1}x) - f_0(g_0^{-1}x))| < \epsilon/2 + \epsilon/2 = \epsilon$, so it is continuous.

- **The contragredient representation ρ^* s.t. $\rho^*(g)\lambda = \lambda \circ \rho(g^{-1})$, $\forall g \in G$, $\lambda \in \mathcal{C}^*$, is continuous.**

pf. Let $g_0 \in G$, $\lambda_0 \in \mathcal{C}^*$, $f \in \mathcal{C}$, $r \in \mathbb{R}_{++}$. Let $\|\lambda_0\| = \sup\{\|\lambda_0(f_1)\|_\infty, f_1 \in \mathcal{C}, \|f_1\|_\infty = 1\}$. Then $\|\lambda_0\| < \infty$ since λ_0 is continuous and $\|\lambda_0(f_1)\|_\infty \leq \|\lambda_0\| \|f_1\|_\infty$, $\forall f_1 \in \mathcal{C}$. Then by the previous statement, there exists a nhod W of g_0 in G s.t. $\forall g \in W$, $\|g^{-1}f - g_0^{-1}f\|_\infty < \frac{r}{2(1+\|\lambda_0\|)}$. Let $U \equiv \{\lambda \in \mathcal{C}^* : |(\lambda - \lambda_0)(g^{-1}f)| < r/2\}$. Then U is a nhod of λ_0 in $\mathcal{C}^*$ and $\forall g \in W$, $\forall \lambda \in U$, $|(g\lambda - g_0\lambda_0)(f)| = |\lambda(g^{-1}f) - \lambda_0(g^{-1}f)| + |\lambda_0(g^{-1}f) - \lambda_0(g_0^{-1}f)| < r$.

- **$\mathcal{P}(G)$ is a convex compact G -invariant subset of $\mathcal{C}^*$.**

pf. Denote by B its unit ball of $\mathcal{C}$ for the sup-norm. Let B_+ be the set of $f \in B$ s.t. $f(G) \subset \mathbb{R}_+$. Then B_+ generates $\mathcal{C}$ as a vector space so the map $\begin{matrix} \mathcal{C}^* & \rightarrow & \Pi_{B_+}\mathbb{C} \\ \lambda & \mapsto & (\lambda(f))_{f \in B_+} \end{matrix}$ is injective, and we can use it to identify $\mathcal{C}^*$ to a subspace of $\Pi_{B_+}\mathbb{C}$. By the definition, the weak topology on $\mathcal{C}^*$ is the topology induced by the product topology on $\Pi_B\mathbb{C}$. Let $K = \Pi_B[0, 1] \subset \Pi_B\mathbb{C}$. Then by Tychonoff's theorem, K is compact. By the Riesz representation theorem, $\mathcal{C}^* \cap K$ is closed in K . Indeed, if $(a_f)_{f \in B_+}$ is in the closure of $\mathcal{C}^* \cap K$, $\begin{matrix} B_+ & \rightarrow & \mathbb{C} \\ f & \mapsto & a_f \end{matrix}$ extends to a linear map $\mathcal{C} \rightarrow \mathbb{C}$ and this linear map is positive, given $a_f \in \mathbb{R}_+$. Thus it is of the form $f \mapsto \int f d\mu$, where μ is a regular Borel measure on G and hence it is continuous, i.e. an element of $\mathcal{C}^*$. So $K \cap \mathcal{C}^*$ is compact and therefore $\mathcal{P}(G)$ is compact given $\mathcal{P}(G) = \{\lambda \in K \cap \mathcal{C}^* | \lambda(1) = 1\}$ is closed in $L \cap \mathcal{C}^*$. The other properties are trivial.

- **G has a left Haar measure.**

pf. $\mathcal{P}(G)$ is nonempty because $f \mapsto f(1)$ is in $\mathcal{P}(G)$. $\mathcal{C}^*$ is locally convex since $\{\lambda \in \mathcal{C}^* : |\lambda(f_1)| < r_1, \dots, |\lambda(f_s)| < r_s\}$, $\forall f_i \in \mathcal{C}$, $\forall r_i \in \mathbb{R}_{++}$ form a basis of nhods of 0 in $\mathcal{C}^*$ and they are convex. Also the action of G on $\mathcal{P}(G)$ is

equicontinuous, i.e. for every nhod of W of 0 in $\mathcal{C}^*$, there exists a nhod V of 0 in $\mathcal{C}^*$ s.t. $\forall \mu_1, \mu_2 \in \mathcal{P}(G)$ s.t. $\mu_1 - \mu_2 \in V$, $g \cdot (\mu_1 - \mu_2) \in W$, $\forall g \in G$. Then by the Markov-Kakutani fixed point theorem, $\exists \mu \in \mathcal{P}(G)$ s.t. $g\mu = \mu$, $\forall g \in G$. This is a left Haar measure.

Theorem A.3.1. *Let G be a compact Hausdorff topological group. Let $\mathcal{C}(G, \mathbb{C})$ be the $\mathbb{C}$ -algebra of continuous functions from G to $\mathbb{C}$ with the norm $\|\cdot\|_\infty$, the supremum norm. Then there exists a unique $\mathbb{C}$ -linear map $\lambda : \mathcal{C}(G, \mathbb{C}) \rightarrow \mathbb{C}$ s.t.*

$$\left\{ \begin{array}{l} \lambda \text{ is positive, i.e. } \lambda(f) \geq 0, \forall f \in \mathcal{C}(G, \mathbb{C}) \\ \lambda \text{ is left invariant, i.e. } \lambda(f) = \lambda(f(g \cdot)), \forall f \in \mathcal{C}(G, \mathbb{C}), \forall g \in G \\ \lambda \text{ is right invariant, i.e. } \lambda(f) = \lambda(f(\cdot g)), \forall f \in \mathcal{C}(G, \mathbb{C}), \forall g \in G \\ \lambda(1) = 1 \end{array} \right. \text{Moreover, if } \lambda \text{ is}$$

continuous, there exists a unique probability measure dg on G s.t. $\forall f \in \mathcal{C}(G, \mathbb{C})$, $\lambda(f) = \int_G f(g)dg$ and this measure satisfies

$$\int_G f(g)dg = \int_G f(g^{-1})dg \quad \forall f : G \rightarrow \mathbb{C}, \text{measurable.}$$

$\rightarrow dg$ is called a bi-invariant probability Haar measure on G .

There exists a function $c : G \rightarrow \mathbb{R}_{++}$ s.t. $\forall g \in G$, $\forall f \in \mathcal{C}(G, \mathbb{C})$, $\int_G f(xg)d\mu(x) = c(g)\int_G f(x)d\mu(x)$.

pf. Let $g \in G$. Then $d\mu(x)$, $d\mu(xg^{-1})$ are both left Haar measures on G so $\exists c(g) \in \mathbb{R}_{++}$ s.t. $d\mu(xg^{-1}) = c(g)d\mu(x)$ i.e. $\forall f \in \mathcal{C}(G, \mathbb{C})$, by the previous theorem uniqueness and therefore the above equation holds.

$\rightarrow \forall g_1, g_2 \in G$, $c(g_1g_2)d\mu(x) = d\mu(x(g_1g_2)^{-1}) = d\mu(xg_2^{-1}g_1^{-1}) = c(g_1)d\mu(xg_2^{-1}) = c(g_1)c(g_2)d\mu(x)$ so $c(g_1g_2) = c(g_1)c(g_2)$, hence $c : G \rightarrow (\mathbb{R}_{++}, \times)$ is a morphism of groups. Let $\epsilon > 0$. Choose $f \in \mathcal{C}(G, \mathbb{R}_+)$ s.t. $\int_G f(x)d\mu(x) = 1$. Let U be a nhod of 1 in G s.t. $\forall x \in G$, $\forall y \in U$, $|f(x) - f(xy)| \leq \epsilon$ and let K be the suport of f . Then if $g, g' \in G$ s.t. $g^{-1}g' \in U$, $|c(g') - c(g)| = |(c(g') - g(g)) \int_G f(x)d\mu(x)| = |\int_G f(xg')d\mu(x) - \int_G f(xg)d\mu(x)| \leq \epsilon\mu(K)$ hence it is continuous.

Definition A.3.3. *The function $c : G \rightarrow (\mathbb{R}_{++}, \times)$ is called the modular function of G and we say G is unimodular if $c(G) = 1$.*

If G is compact, it is unimodular.

pf. If G is compact, $c(G)$ is a compact subgroup of $\mathbb{R}_{++}$. The only compact subgroup of $\mathbb{R}_{++}$ is $\{1\}$, so $c(G) = \{1\}$, i.e. G is unimodular.

Let $f \in \mathcal{C}(G, \mathbb{C})$. Then $\int_G f(x^{-1})d\mu(x) = \int_G c(x)f(x)d\mu(x)$.

pf. $\int_G \int_G g(yx)c(x)^{-1}f(x^{-1})d\mu(x)d\mu(y)$
 $= \begin{cases} \int_G f(x^{-1})c(x)^{-1}(\int_G g(yx)d\mu(y))d\mu(x) = \int_G f(x^{-1})d\mu(x) \int_G g(y)d\mu(y) & \text{by the definition of } c \\ \int_G \int_G g(z)c(y^{-1}z)^{-1}f(z^{-1}y)d\mu(z)d\mu(y) = \int_G g(z)d\mu(z) \int_G c(t)f(t)d\mu(t) & \text{by } z = yx, t = z^{-1}y \end{cases}$
 So the equivalence follows by choosing $\int_G g(x)d\mu(x) = 1$.

If G is just locally compact, we can find nonzero $\mathbb{C}$ -linear map satisfying the first and second or the first and third and they are unique upto multiplication by a positive real number. These are also continuous and the corresponding measures on G are called left-invariant or right-invariant Haar measure. Then

$\int_G f(g)d_r g = \int_G f(g^{-1})d_l g$, $\begin{cases} d_l & \text{left invariant} \\ d_r & \text{right invariant} \end{cases}$ for every measurable function $f : G \rightarrow \mathbb{C}$.

e.g.

- a finite group G with the discrete topology is a topology group. A left and right-invariant Haar measure on G is given by $d_g(A) = |A|/|G|$.
- $G = U(1) \equiv \{z \in \mathbb{C} : |z| = 1\}$ with the topology induced by that of $\mathbb{C}$. Then it is a topological group and we have an isomorphism of topological groups $\phi : \mathbb{R}/\mathbb{Z} \rightarrow U(1)$
 $t \mapsto e^{2i\pi t}$. Then $\int_G f(g)d_g = \int_0^1 (f \circ \phi)(t)dt = \int_0^1 f(e^{2i\pi t})dt$ where dt is the Lebesgue measure on $\mathbb{R}$.

Definition A.3.4. Let G be a topological group and V be a normed $\mathbb{C}$ -vector space. Then a continuous representation of G on V is an abstract representation of G on V s.t. the action map $\begin{matrix} G \times V & \rightarrow & V \\ (g, v) & \mapsto & gv \end{matrix}$ is continuous.

→ let $\text{End}(V)$ be the $\mathbb{C}$ -algebra of continuous endomorphisms of V . Put the operator norm on $\text{End}(V)$ and consider a continuous representation of G on V . Then the action of each $g \in G$ on V is continuous endomorphism of V , so $G \rightarrow \text{End}(V)$ exists. But it is not necessarily continuous.

Definition A.3.5. If V is a normed $\mathbb{C}$ -vector space, we denote by $\text{End}(V)$ the $\mathbb{C}$ -algebra of continuous endomorphisms of V and we put on it the topology given by the operator norm. Write $GL(V)$ for $\text{End}(V)^\times$, with the topology induced by that of $\text{End}(V)$.

Proposition A.3.1. Let V be a normed $\mathbb{C}$ -vector space and $\rho : G \rightarrow GL(V)$ be a morphism of groups. Consider the following conditions:

1. the map $\begin{matrix} G \times V & \rightarrow & V \\ (g, v) & \mapsto & \rho(g)v \end{matrix}$ is continuous

2. $\forall v \in V$, the map $\begin{matrix} G & \rightarrow & V \\ g & \mapsto & \rho(g)(v) \end{matrix}$ is continuous

3. the map $\rho : G \rightarrow GL(V)$ is continuous

Then $3 \rightarrow 1 \rightarrow 2$. If moreover V is finite-dimensional, then all three conditions are equivalent.

pf.

• $1 \rightarrow 2$ is trivial.

• $3 \rightarrow 1$

Let $\begin{cases} g_0 \in G \\ v_0 \in V \end{cases}$ and $\epsilon > 0$. Choose a δ s.t. $0 < \delta \leq \frac{\epsilon}{2\|\rho(g_0)\|}$ and let U be a nhod of g_0 in G s.t. $g \in U$ implies $\|\rho(g) - \rho(g_0)\| < \frac{\epsilon}{2(\|v_0\| + \delta)}$. Then if

$$\begin{cases} g \in U \\ \|v - v_0\| < \delta \end{cases},$$

$$\begin{aligned} \|\rho(g)(v) - \rho(g_0)(v_0)\| &\leq \|\rho(g)(v) - \rho(g_0)(v)\| + \|\rho(g_0)(v) - \rho(g_0)(v_0)\| && \text{by the triangular inequality} \\ &\leq \|\rho(g) - \rho(g_0)\| \|v\| + \|\rho(g_0)\| \|v - v_0\| \end{aligned}$$

$$\begin{aligned} &< \frac{\epsilon}{2(\|v_0\| + \delta)} (\|v_0\| + \delta) + \|\rho(g_0)\| \delta && \text{since } \begin{cases} \|v\| \leq \|v_0\| + \delta \\ \text{the choice of } g, \delta \end{cases} \\ &\leq \epsilon/2 + \epsilon/2 = \epsilon \end{aligned}$$

• given V finite-dimensional, $2 \rightarrow 3$

Let $(e_1, \dots, e_n)$ be a basis of V and let $\|\cdot\|$ be the supremum norm. We use the corresponding operator norm on $\text{End}(V)$ and denote it by $\|\cdot\|$. Let $g_0 \in G$ and $\epsilon > 0$. For every i , the function $\begin{matrix} G & \rightarrow & V \\ g & \mapsto & \rho(g)(e_i) \end{matrix}$ is continuous by 2, so there exists a nhod U_i of g_0 in G s.t. $g \in U_i$ implies $\|\rho(g)(e_i) - \rho(g_0)(e_i)\| \leq \epsilon/n$. Let $U = \cap_{i=1}^n U_i$. Then if $g \in U$, then for all $v = \sum_{i=1}^n x_i e_i \in V$, $\|\rho(g)(v) - \rho(g_0)(v)\| \leq \sum |x_i| \|\rho(g)(e_i) - \rho(g_0)(e_i)\| \leq \sum |x_i| \epsilon/n \leq \epsilon \|v\|$ so $\|\rho(g) - \rho(g_0)\| \leq \epsilon$.

A complex Hilbert space is a $\mathbb{C}$ -vector space V with a Hermitian inner product s.t. V is complete for the corresponding norm. If V is a finite dimensional $\mathbb{C}$ -vector space with a Hermitian inner product, it is complete and therefore a Hilbert space. Write $U(V)$ for the group of unitary endomorphisms of V .

Proposition A.3.2. *If V is a Hilbert space and $\rho : G \rightarrow U(V)$ is a morphism of groups, then TFAE*

1. the map $\begin{matrix} G \times V & \rightarrow & V \\ (g, v) & \mapsto & \rho(g)(v) \end{matrix}$ is continuous

2. $\forall v \in V$, the map $\begin{matrix} G & \rightarrow & V \\ g & \mapsto & \rho(g)(v) \end{matrix}$ is continuous.

pf.

- by the previous proposition, $2 \rightarrow 1$.

- Let $\begin{cases} g_0 \in G \\ v_0 \in V \end{cases}$ and $\epsilon > 0$. Choose a nhood U of g_0 in G s.t. $g \in U$ implies $\|\rho(g)(v_0) - \rho(g_0)(v_0)\| < \epsilon/2$ and take $\delta = \epsilon/2$. Then if $g \in U$
 $\|\rho(g)(v) - \rho(g_0)(v_0)\| \leq \|\rho(g)(v) - \rho(g)(v_0)\| + \|\rho(g)(v_0) - \rho(g_0)(v_0)\|$
 $\|v - v_0\| < \delta, \quad < \|\rho(g)\| \|v - v_0\| + \epsilon/2, > < \epsilon/2 + \epsilon/2 = \epsilon$
 since $\rho(g) \in U(V)$.

CLAIM : if $\rho : G \rightarrow U(V)$ is a unitary representation of G on a Hilbert space $(V, \langle \cdot, \cdot \rangle)$, for every G -invariant subspace W of V , the subspace $W^\perp$ is also G -invariant.

pf. if $w \in W^\perp$, $g \in G$, then $\forall v \in W$, $\langle v, \rho(g)w \rangle = \langle \rho(g)^{-1}v, w \rangle = 0$ since $\rho(g)^{-1}v \in W$ by the G -invariance of W . This implies $\rho(g)W \in W^\perp$ and $W^\perp$ is G -invariant.

$\rightarrow$ if W is a closed G -invariant subspace of V , $V = W \oplus W^\perp$ with $W^\perp$ a closed G -invariant subspace. If V is finite dimensional, every subspace is closed and therefore every unitary representation of G on V is semisimple.

Theorem A.3.2. Assume that the group G is compact Hausdorff. Let V be a finite-dimensional $\mathbb{C}$ -vector space and $\rho : G \rightarrow GL(V)$ be a continuous representation of G on V . Then there exists a Hermitian inner product on V that makes ρ a unitary representation.

pf. Let $\langle \cdot, \cdot \rangle_0$ be a Hermitian inner product on V . Define
 $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$
 $(v, w) \mapsto \int_G \langle \rho(g)v, \rho(g)w \rangle_0 dg$. Then $\langle \rho(g)v, \rho(g)w \rangle = \langle v, w \rangle$, $\forall v, w \in V$ and $\forall g \in G$. It is a Hermitian product trivially. Let $v \in V \setminus \{0\}$. Then the function $\begin{matrix} G & \rightarrow & \mathbb{R} \\ g & \mapsto & \langle \rho(g)v, \rho(g)v \rangle_0 \end{matrix}$ is continuous and takes positive values. As G is compact, $\exists \epsilon > 0$ s.t. $\langle \rho(g)v, \rho(g)v \rangle_0 > \epsilon$, $\forall g \in G$. Then $\langle v, v \rangle \geq \epsilon > 0$ and it is positive definite.

Corollary A.3.1. If G is compact Hausdorff, then every finite-dimensional continuous representation of G is semisimple.

Definition A.3.6. Let V, W be Hermitian inner product spaces. If $v \in V$, $w \in W$, we define the $\mathbb{C}$ -linear map $\begin{matrix} T_{v,w}^0 : V & \rightarrow & W \\ x & \mapsto & \langle x, v \rangle w \end{matrix}$.

$$\begin{array}{ccc} V & \rightarrow & \text{hom}_{\mathbb{C}}(V, W) \\ v & \mapsto & T_{v,w}^0 \end{array} \text{ is semi-linear.}$$

Proposition A.3.3. 1. $\forall v \in V, \forall w \in W, (T_{v,w}^0)^* = T_{w,v}^0$

2. $\forall v \in V, \forall w \in W, T_{v,w}^0$ generate $\text{hom}_{\mathbb{C}}(V, W)$ as a $\mathbb{C}$ -vector space

3. if $V = W, \forall v, w \in V, T(T_{v,w}^0) = \langle w, v \rangle$

4. $\forall v_1, v_2 \in V, \forall w_1, w_2 \in W, T((T_{v_1,w_1}^0 T_{v_2,w_2}^0)^*) = \langle w_1, w_2 \rangle \langle v_2, v_1 \rangle$

pf.

- $\forall x \in V, y \in W, \langle T_{v,w}^0(x), y \rangle = \langle \langle x, v \rangle w, y \rangle = \langle x, v \rangle \langle w, y \rangle = \langle x, v \rangle \overline{\langle y, w \rangle} = \langle x, \langle y, w \rangle v \rangle = \langle x, T_{w,v}^0(y) \rangle$
- if we choose the orthonormal bases (e_i) of V and (f_j) of $W, \forall i, j$, the matrix of T_{e_i, f_j}^0 in these bases is the $m \times n$ with (i, j) -entry equal to 1 and all other entries equal to 0. These clearly generate the $\mathbb{C}$ -vector space $M_{mn}(\mathbb{C})$.
- let $(v_1, \dots, v_n)$ be an orthogonal basis of V s.t. $v_1 = v$. Then $T_{v,w}^0(v_i) = \begin{cases} \langle v_1, v_1 \rangle w & i = 1 \\ 0 & \text{otherwise} \end{cases}$. As $w = \sum \frac{\langle w, v_i \rangle}{\langle v_i, v_i \rangle} v_i$, the proposition follows
- $\forall y \in W, T_{v_1, w_1}^0 (T_{v_2, w_2}^0)^* = T_{v_1, w_1}^0 T_{w_2, v_2}^0(y) = T_{v_1, w_1}^0(\langle y, w_2 \rangle v_2) = \langle y, w_2 \rangle \langle v_2, v_1 \rangle w_1$. Choose an orthogonal basis $(y_1, \dots, y_n)$ of W s.t. $y_1 = w_2$. Then $T_{v_1, w_1}^0 (T_{v_2, w_2}^0)^*(y_i) = \begin{cases} \langle y_1, y_1 \rangle \langle v_2, v_1 \rangle w_1 & i = 1 \\ 0 & \text{otherwise} \end{cases}$. As $w_1 = \sum \frac{\langle w_1, y_i \rangle}{\langle y_i, y_i \rangle} y_i$, the proposition follows.

Let G be a Hausdorff compact group.

Definition A.3.7. Let $\rho : G \rightarrow GL(V)$ be a continuous representation of G on a finite dimensional complex vector space. Remember that the map $\begin{array}{ccc} \chi_V : G & \rightarrow & \mathbb{C} \\ g & \mapsto & T(\rho(g)) \end{array}$ is called the character of the representation (V, ρ) .

$\rightarrow \chi_V : G \rightarrow \mathbb{C}$ is a continuous map by the previous proposition A.3.1.

Theorem A.3.3. Let V, W be continuous representations of G on finite-dimensional complex vector spaces. Assume V, W are both irreducible. Choose G -invariant Hermitian inner products on V, W that will both be denoted by $\langle \cdot, \cdot \rangle$. Then $\begin{cases} \forall v_1, v_2 \in V \\ \forall w_1, w_2 \in W \end{cases}$,

$$\int_G \langle g v_1, v_2 \rangle \overline{\langle g w_1, w_2 \rangle} dg = \begin{cases} \frac{1}{\dim_{\mathbb{C}} V} \langle v_1, w_1 \rangle \overline{\langle v_2, w_2 \rangle} & V \cong W \\ 0 & \text{otherwise} \end{cases}$$

pf. If $v \in V, w \in W$, let $T_{v,w} = \int_G g T_{v,w}^0 g^{-1} dg \in \text{hom}_{\mathbb{C}}(V, W)$ where $T_{v,w}^0$ is defined above. Then $T_{v,w}$ is G -equivariant, so by the Schur's lemma, $T_{v,w} = \begin{cases} 0 & V \not\cong W \\ c(v,w) id_V & V \cong W \end{cases}$ with $c(v,w) \in \mathbb{C}$. Suppose $V \cong W$, then $c(v,w) \dim_{\mathbb{C}} V = T(T_{v,w}) = \int_G T(g T_{v,w}^0 g^{-1}) dg = T(T_{v,w}^0)$ so by the previous proposition, $c(v,w) = \frac{1}{\dim_{\mathbb{C}} V} \langle w, v \rangle$. Let $I = \int_G \langle g v_1, v_2 \rangle \overline{\langle g w_1, w_2 \rangle} dg$. Then $I = \int_G \langle g v_1, v_2 \rangle \langle g^{-1} w_2, w_1 \rangle dg = \int_G \langle g^{-1} w_2, w_1 \rangle \langle g v_1, v_2 \rangle dg = \int_G \langle g T_{w_1, v_1}^0 g^{-1} w_2, v_2 \rangle dg = \langle T_{w_1, v_1}(w_2), v_2 \rangle$ so $I = \begin{cases} 0 & V \not\cong W \\ c(w_1, v_1) \langle w_2, v_2 \rangle & V \cong W \end{cases}$ and the theorem follows.

Corollary A.3.2 (Schur's Orthogonality). *Let V, W be continuous representations of G on finite dimensional complex vector spaces. Assume V, W are both irreducible. Then*

$$\int_G \chi_V(g) \overline{\chi_W(g)} dg = \begin{cases} 0 & V \not\cong W \\ 1 & V \cong W \end{cases}$$

pf. Choose G -invariant Hermitian inner products on V, W and fix orthonormal bases $\begin{cases} (v_1, \dots, v_n) & \text{for } V \\ (w_1, \dots, w_n) & \text{for } W \end{cases}$. Then for any $\mathbb{C}$ -linear endomorphism u of V or W , we have $T(u) = \sum \langle u(v_i), v_i \rangle$ or $\sum \langle u(w_j), w_j \rangle$. In particular, $\forall g \in G$, $\begin{cases} \chi_V(g) = \sum \langle g v_i, v_i \rangle \\ \chi_W(g) = \sum \langle g w_j, w_j \rangle \end{cases}$ and therefore $\int_G \chi_V(g) \overline{\chi_W(g)} dg = \sum_{i,j} \int_G \langle g v_i, v_i \rangle \overline{\langle g w_j, w_j \rangle} dg = \begin{cases} 0 & V \not\cong W \\ \frac{1}{\dim_{\mathbb{C}} V} \sum_{i,j} \langle v_i, v_j \rangle \overline{\langle v_i, v_j \rangle} = 1 & V \cong W \end{cases}$ and the corollary follows.

Definition A.3.8. $L^2(G)$ is the quotient

$$\{f : G \rightarrow \mathbb{C}, \text{ measurable} \mid \int_G |f(g)|^2 dg < \infty\} / \{f : G \rightarrow \mathbb{C}, \text{ measurable} \mid \int_G |f(g)|^2 dg = 0\}$$

This is a Hilbert space for the Hermitian inner product given by $\langle f_1, f_2 \rangle = \int_{g \in G} f_1(g) \overline{f_2(g)} dg$.

Definition A.3.9. If $x \in G$, and f is a function from G to $\mathbb{C}$, define

$$\begin{array}{ccc} L_x f : G & \rightarrow & \mathbb{C} & R_x f : G & \rightarrow & \mathbb{C} \\ g & \mapsto & f(xg) & g & \mapsto & f(gx) \end{array}$$

Proposition A.3.4. $\forall x, y \in G$, the operators L_x, R_x induce endomorphisms L_x, R_x of $L^2(G)$ with the following properties; $L_x R_y = R_y L_x$, $R_x R_y = R_{xy}$, $L_x L_y = L_{yx}$, $R_x^* = R_x^{-1} = R_{x^{-1}}$, $L_x^* = L_x^{-1} = L_{x^{-1}}$.

In particular, R_x, L_x are unitary endomorphisms of $L^2(G)$.

pf.

- Let $f_1, f_2 : G \rightarrow \mathbb{C}$ be measurable functions. Then

$$\begin{cases} \langle R_x f_1, R_x f_2 \rangle = \int_G f_1(gx) \overline{f_2(gx)} dg = \int_G f_1(g) \overline{f_2(g)} dg = \langle f_1, f_2 \rangle & \text{by the right invariance} \\ \langle L_x f_1, L_x f_2 \rangle = \int_G f_1(xg) \overline{f_2(xg)} dg = \int_G f_1(g) \overline{f_2(g)} dg = \langle f_1, f_2 \rangle & \text{by the left-invariance} \end{cases}$$
 Thus R_x, L_x preserves both spaces in the quotient defining $L^2(G)$ and therefore they induce endomorphisms of $L^2(G)$.
- Let $f \in L^2(G)$, $g \in G$. Then

$$\begin{cases} (L_x R_y f)(g) = (R_y f)(xg) = f(xgy) = (L_x f)(gy) = (R_y L_x f)(g) \\ (L_x L_y f)(g) = (L_y f)(xg) = f(yxg) = (L_{yx} f)(g) \\ (R_x R_y f)(g) = (R_y f)(gx) = f(gyx) = (R_{yx} f)(g) \end{cases}$$
- by the first stmt, L_x, R_x are unitary endomorphisms and by the second stmt, $L_x L_{x^{-1}} = L_{xx^{-1}} = L_1 = Id \rightarrow L_x^{-1} = L_{x^{-1}}$ and similarly for R_x .

Definition A.3.10. If f is a function $G \rightarrow \mathbb{C}$, define a function $\tilde{f} : G \rightarrow \mathbb{C}$ by $\tilde{f}(g) = \overline{f(g^{-1})}$.

Proposition A.3.5. The above function defines a unitary endomorphism $f \mapsto \tilde{f}$ of $L^2(G)$.

pf. Let $f : G \rightarrow \mathbb{C}$ be a measurable function. Then, $\int_G |\overline{f(g^{-1})}|^2 dg = \int_G |f(g)|^2 dg$ because G being compact implies the Haar measure dg is equal to its own pullback by $g \mapsto g^{-1}$. Then the proposition follows.

Definition A.3.11. If $f_1, f_2 \in L^2(G)$, define $f_1 * f_2 : G \rightarrow \mathbb{C}$ by $(f_1 * f_2)(g) = \int_G f_1(gx) f_2(x^{-1}) dx$, called the convolution product of f_1, f_2 .

Proposition A.3.6. 1. $\forall f_1, f_2 \in L^2(G)$, $f_1 * f_2$ is a bounded function on G and we have $\|f_1 * f_2\|_\infty \leq \|f_1\|_2 \|f_2\|_2$

2. the convolution product is associative and distributive w.r.t. addition

pf.

- $\forall g \in G$, $|f_1 * f_2(g)| = |\int_G f_1(gx) f_2(x^{-1}) dx| = |\int_G (L_g f_1) \overline{\tilde{f}_2(x)} dx| \leq \|L_g f_1\|_2 \|\tilde{f}_2\|_2 = \|f_1\|_2 \|f_2\|_2$ by CS inequality.
- $\forall f_1, f_2, f_3 \in L^2(G)$, $\forall g \in G$

$$\begin{cases} ((f_1 * f_2) * f_3)(g) = \int_G (f_1 * f_2)(gx) f_3(x^{-1}) dx = \int_G (\int_G f_1(gxy) f_2(y^{-1}) dy) f_3(x^{-1}) dx \\ (f_1 * (f_2 * f_3))(g) = \int_G f_1(gz) (f_2 * f_3)(z^{-1}) dz = \int_G f_1(gz) (\int_G f_2(z^{-1}t) f_3(t^{-1}) dt) dz \end{cases}$$
 By the change of variables $\begin{cases} z \mapsto y \\ z \mapsto y^{-1}x \end{cases}$ and Fubini's theorem and bi-invariance Haar measure dg , they are equivalent, hence it is associative.

$\rightarrow f_1 * f_2$ are continuous for all $f_1, f_2 \in L^2(G)$ since if f_1, f_2 are the limit in $L^2(G)$ of sequences $f_{(1,n)}, f_{(2,n)}$, then the sequence $(f_{1,n} * f_{2,n}) \rightarrow f_1 * f_2$ in $L^\infty(G)$. Since the continuous functions are dense in $L^2(G)$ and the convolution of the two continuous functions is continuous, the continuity follows.

Definition A.3.12. Let $f \in L^2(G)$. Define endomorphisms L_f, R_f of $L^2(G)$ by

$$\begin{cases} L_f(f_1) = f * f_1 \\ R_f(f_1) = f_1 * f \end{cases}.$$

$\rightarrow$ well-defined and continuous by the previous proposition.

Proposition A.3.7. 1. $\forall f \in L^2(G), R_f^* = R_{\bar{f}}, L_f^* = L_{\bar{f}}$.

2. $\forall f \in L^2(G), \forall x \in G, R_x L_f = L_f R_x, R_f L_x = L_x R_f$

pf.

- $\forall f_1, f_2 \in L^2(G),$

$$\begin{aligned} \langle R_f(f_1), f_2 \rangle &= \int_G (f_1 * f) \overline{f_2(g)} dg = \int_G \int_G f_1(gx^{-1}) \overline{f(x) f_2(g)} dx dg \\ &= \int \int f_1(y) \overline{f_2(yx)} \overline{f(x)} dx dy \quad \text{by the change of variables } y = gx^{-1} \\ &= \int \int f_1(y) \overline{f_2(yz^{-1})} \overline{f(z)} dz dy \quad \text{by the change of variables } z = x^{-1} \\ &= \int_G f_1(y) \overline{(f_2 * \tilde{f})(y)} dy = \langle f_1, R_{\tilde{f}} f_2 \rangle \\ &\text{Similarly for } L_f^*. \end{aligned}$$

- $\forall h \in L^2(G), \forall g \in G, \begin{cases} (R_x L_f(h))(g) = (f * h)(gx) = \int_G f(gxy) h(y^{-1}) dy \\ (L_f R_x(h))(g) = (f * (R_x h))(g) = \int_G f(gz) h(z^{-1}x) dz \end{cases}$ are equivalent by the change of variables $z = xy$.
 Similarly for $R_f L_x = L_x R_f$.

Definition A.3.13. If V_1, V_2 are two Hilbert spaces over $\mathbb{C}$ and $T : V_1 \rightarrow V_2$ is a continuous $\mathbb{C}$ -linear map, we say T is compact if the set $\overline{T(B)} \subset V_2$ is compact, where $B = \{v \in V_1 \mid \|v\| = 1\}$.

Lemma A.3.1. 1. every finite rank operator is compact

2. the space of compact operators $T : V_1 \rightarrow V_2$ is closed in $\text{hom}(V_1, V_2)$

pf.

- The closed unit ball of a finite dimensional $\mathbb{C}$ -vector space is compact, and therefore every finite rank operator is compact.

- Let (T_n) be a sequence of compact operators in $\text{hom}(V_1, V_2)$ and suppose it converges to $T \in \text{hom}(V_1, V_2)$. Let (x_n) be a sequence in B . Construct a sequence $x^{(r)} = (x_n^{(r)})$ of subsequences of (x_n) in the following inductive way; take $x^{(0)} = (x_n)$. If $r \geq 1$, suppose $x^{(r-1)}$ is constructed and take for $x^{(r)}$ a subsequence of $x^{(r-1)}$ s.t. $(T_r(x_n^{(r)}))$ is a Cauchy sequence. Set $y_n = x_n^{(n)}$. Let $\epsilon > 0$. Choose $m \geq 1$ s.t. $\|T - T_m\| \leq \epsilon$ and $N \geq m$ s.t. $\forall r, s \geq N$, $\|T_m(y_r) - T_m(y_s)\| \leq \epsilon$. Then if $r, s \geq N$, $\|T(y_r) - T(y_s)\| \leq \|(T - T_m)(y_r)\| + \|T_m(y_r - y_s)\| + \|(T_m - T)(y_s)\| \leq 3\epsilon$, given $y_s, y_r \in B$. Hence $(T(y_n))$ is a Cauchy sequence. Since V_2 is complete, the lemma follows.

Theorem A.3.4. *Let V_1, V_2 be Hilbert spaces. Write $\text{hom}(V_1, V_2)$ for the space of continuous $\mathbb{C}$ -linear maps from V_1 to V_2 and consider the topology on it defined by the operator norm. Then $\forall T \in \text{hom}(V_1, V_2)$, TFAE*

1. T is compact
2. T is a limit of finite rank element of $\text{hom}(V_1, V_2)$

pf.

- $2 \rightarrow 1$ follows by the previous lemma.
- $1 \rightarrow 2$

pf. Let $T \in \text{hom}(V_1, V_2)$ be a compact operator. Let (W_n) be a sequence of finite dimensional subspace of V_2 s.t. $V_2 = \cup W_n$, and $\forall n \geq 0$, $\pi_n : V_2 \rightarrow W_n$ be the orthogonal projection. Set $T_n = \pi_n \circ T$. The $\forall x \in V_1$, $T_n(x) \rightarrow T(x)$ as $n \rightarrow \infty$. Let $\epsilon > 0$. As $\overline{T(B)}$ is compact, $\exists x_1, \dots, x_r \in B$ s.t. $\forall y \in B$, $\exists i \in \{1, \dots, r\}$ s.t. $\|T(x_i) - T(y)\| \leq \epsilon$. Then $\forall n \geq 0$, $\|T_n(y) - T_n(x_i)\| = \|\pi_n(T(y) - T(x_i))\| \leq \|T(y) - T(x_i)\| \leq \epsilon$. Choose $N \geq 0$ s.t. $\forall i$, $\forall n \geq N$, $\|T_n(x_i) - T(x_i)\| \leq \epsilon$. Then fix $n \geq N$, and if $y \in B$ choose $i \in \{1, \dots, r\}$ s.t. $\|T(y) - T(x_i)\| \leq \epsilon$. Then $\|T(y) - T_n(y)\| \leq \|T(y) - T(x_i)\| + \|T(x_i) - T_n(x_i)\| + \|T_n(x_i) - T_n(y)\| \leq 3\epsilon$.

Definition A.3.14. *Let V be a Hilbert space, and T be a continuous endomorphism of V . We say $\lambda \in \mathbb{C}$ is an eigenvalue of T if $\ker(T - \lambda \text{id}_V) \neq \{0\}$. We denote by $\text{Spec}(T) \subset \mathbb{C}$ the set of eigenvalues of T and call it the spectrum of T .*

Theorem A.3.5 (Spectral Theorem). *Let V be a Hilbert space over $\mathbb{C}$, and let $T : V \rightarrow V$ be a continuous endomorphism of V . Assume T is compact and self-adjoint,*

and write $V_\lambda = \ker(T - \lambda \text{id}_V)$, $\forall \lambda \in \mathbb{C}$. Then

$$\left\{ \begin{array}{l} \text{Spec}(T) \subset \mathbb{R} \\ \text{if } \lambda, \mu \in \text{Spec}(T), \lambda \neq \mu, \text{ then } V_\mu \subset V_\lambda^\perp \\ \text{if } \lambda \in \text{Spec}(T) \setminus \{0\}, \dim_{\mathbb{C}} V_\lambda < \infty \\ \text{Spec}(T) \text{ is finite or countable} \\ \bigoplus_{\lambda \in \text{Spec}(T)} V_\lambda \text{ is dense in } V \end{array} \right.$$

Theorem A.3.6. $\forall f \in L^2(G)$, $\exists (f_n)_{n \geq 0}$ a sequence of functions in $L^2(G)$ s.t.

$$\begin{cases} \tilde{f}_n = f_n, \forall n \geq 0 \\ \|f_n\|_2 = 1, \forall n \geq 0 \\ f * f_n \rightarrow f \text{ in } L^2(G) \text{ as } n \rightarrow \infty \end{cases}.$$

pf. Let $\epsilon > 0$. Choose a nhoud U of 1 in G s.t. $U = U^{-1}$ and $\forall x, y \in U$, $\|R_x f - f\|_2 < \epsilon$. Let $h_\epsilon = \frac{1}{\text{vol}(U)} \mathbb{1}_U$. Then $\tilde{h}_\epsilon = h_\epsilon$ and $\|h_\epsilon\|_2 = 1$. Also

$$(f * h_\epsilon)(g) = \int_G f(gx^{-1})h_\epsilon(x)dx = \frac{1}{\text{vol}(U)} \int_U f(gx^{-1})dx$$

$$\rightarrow (f * h_\epsilon)(g) - f(g) = \frac{1}{\text{vol}(U)} \int_U (R_{x^{-1}}(f)(g) - f(g))dx. \text{ This gives}$$

$$\begin{aligned} \|(f * h_\epsilon) - f\|_2^2 &= \int_G |(f * h_\epsilon)(g) - f(g)|^2 dg = \frac{1}{\text{vol}(U)^2} \int_{G \times U \times U} (R_{x^{-1}}f - f)(g) \overline{(R_{y^{-1}}f - f)(g)} dx dy dg \\ &= \frac{1}{\text{vol}(U)^2} \int_{U \times U} \langle R_{x^{-1}}f - f, R_{y^{-1}}f - f \rangle dx dy \leq \frac{1}{\text{vol}(U)^2} \int \|R_{x^{-1}}f - f\|_2 \|R_{y^{-1}}f - f\|_2 dx dy \\ &\leq \frac{1}{\text{vol}(U)^2} \int \epsilon^2 dx dy = \epsilon^2 \text{ by the CS inequality and the choice of } U. \text{ If we take } f_n = h_{1/2^n}, \text{ the theorem follows.} \end{aligned}$$

Lemma A.3.2. If $f_1, f_2 : G \rightarrow \mathbb{C}$, define
$$\begin{aligned} f_1 \otimes f_2 : G \times G &\rightarrow \mathbb{C} \\ (g_1, g_2) &\mapsto f_1(g_1)f_2(g_2). \end{aligned}$$
 This induces a $\mathbb{C}$ -linear map $L^2(G) \otimes_{\mathbb{C}} L^2(G) \rightarrow L^2(G \times G)$ which is injective with dense image.

pf. Take the Hilbert basis $(e_i)_{i \in I}$ of $L^2(G)$. Then the family $(e_i \otimes e_j)$ of $L^2(G \times G)$ is orthonormal and any function on $L^2(G \times G)$ that is orthogonal to all $e_i \otimes e_j$ is 0 a.e. hence 0.

Lemma A.3.3. $\forall K \in L^2(G \times G)$ define
$$\begin{aligned} T_K : L^2(G) &\rightarrow L^2(G) \\ f(x) &\mapsto \int_G K(x, y)f(y)dy. \end{aligned}$$
 Then $\|T_K\| \leq \|K\|_2$.

Theorem A.3.7. $\forall f \in L^2(G)$, the endomorphisms R_f and L_f of $L^2(G)$ are compact.

pf. Consider
$$\begin{aligned} K : G \times G &\rightarrow \mathbb{C} \\ (x, y) &\mapsto f(xy^{-1}). \end{aligned}$$
 Then $K \in L^2(G \times G)$, so by the previous lemma, $\exists (g_i)_{i \in I}, (h_i)_{i \in I}$ in $L^2(G)$ s.t. $\sum g_i \otimes h_i$ converges to K in $L^2(G \times G)$. Here $T_K = L_f$. For every finite subset J of I , let $S_J = \sum_{i, j \in J} g_i \otimes h_j \in L^2(G \times G)$ and $T_J = T_{S_J}$. Then $\forall h \in L^2(G)$, $\forall x \in G$, $(T_J h)(x) = \sum_{(i, j) \in J^2} \int_G h(y)g_i(x)h_j(y)dy = \sum_i \sum_j \int_G h(y)h_j(y)dy g_i(x)$, i.e. $\forall h \in L^2(G)$, $T_J h$ is in the finite-dimensional subspace of $L^2(G)$ spanned by g_i , $i \in J$. Hence the operator T_J has finite rank. By the previous lemma, $\|T_J\|_2 \leq \|S_J\|_2 \rightarrow \|K\|_2$ so L_f is the limit of the operator T_J and therefore L_f is compact by the previous theorem.

Definition A.3.15. We make $G \times G$ act on $L^2(G)$ by $(x, y) \cdot f = L_{x^{-1}}R_y f = R_y L_{x^{-1}} f$, i.e. $((x, y) \cdot f)(g) = f(x^{-1}gy)$, $\forall g \in G$.

Proposition A.3.8. The representation of $G \times G$ on $L^2(G)$ defined above is continuous and so it is a unitary representation.

pf. $L_{x^{-1}}R_y$ is a unitary endomorphism of $L^2(G)$, $\forall x, y \in G$, so if the representation is continuous, the proposition follows. Fix $f_0 \in L^2(G)$ and let $x_0, y_0 \in G$, and $\epsilon > 0$. Choose $f : G \rightarrow \mathbb{C}$ continuous s.t. $\|f_0 - f\|_2 \leq \epsilon/3$. As G is compact and f is continuous, f is uniformly continuous, so there exists a nhoud U of 1 in G s.t. $\forall x, y \in U$, $\forall g \in G$, $|f(x^{-1}gy) - f(g)| \leq \epsilon/3$. If $x \in x_0U$, $y \in y_0U$, $\|L_{x^{-1}}R_yf - L_{x_0^{-1}}R_{y_0}f\|_2 = \sqrt{\int_G |f(x^{-1}gy) - f(x_0^{-1}gy_0)|^2 dg} \leq \epsilon/3$. Then $\|L_{x^{-1}}R_yf_0 - L_{x_0^{-1}}R_{y_0}f_0\|_2 \leq \|L_{x^{-1}}R_yf_0 - L_{x^{-1}}R_yf\|_2 + \|L_{x^{-1}}R_yf - L_{x_0^{-1}}R_{y_0}f\|_2 + \|L_{x_0^{-1}}R_{y_0}f - L_{x_0^{-1}}R_{y_0}f_0\|_2 \leq \epsilon$, so the map
$$\begin{array}{ccc} G \times G & \rightarrow & L^2(G) \\ (x, y) & \mapsto & L_{x^{-1}}R_yf_0 \end{array}$$
 is continuous. Then by the previous proposition A.3.1., the representation is continuous.

Assume G is a compact Hausdorff group.

Definition A.3.16. Denote by $\hat{G}$ the set of isomorphism classes of irreducible continuous representations of G on finite dimensional complex vector spaces.

Let $(V_\rho, \rho) \in \hat{G}$. Fix a G -invariant Hermitian inner product $\langle \cdot, \cdot \rangle$ on V_ρ . Then $G \times G$ acts on $\text{End}_{\mathbb{C}}(V_\rho) \cong V_\rho^* \otimes_{\mathbb{C}} V_\rho$ by $(g_1, g_2) \cdot u = \rho(g_1)u\rho(g_2)^{-1}$.

Definition A.3.17. The Hilbert-Schmidt inner product on $\text{End}_{\mathbb{C}}(V_\rho)$ is given by the formula

$$\langle u, v \rangle_{HS} = T(uv^*)$$

where v^* is the adjoint of v for the chosen G -invariant inner product $\langle \cdot, \cdot \rangle$ on V_ρ .

$\rightarrow$ Hermitian inner product and $G \times G$ -invariant given $\rho(g)$ begin unitary $\forall g \in G$.

Lemma A.3.4. Let V be a Hilbert space with a unitary representation of G and let $V_1, V_2 \subset V$ be finite-dimensional irreducible subrepresentations of G . Then either $V_1 \cong V_2$ as representations of G or $\langle v_1, v_2 \rangle = 0$, $\forall v_1 \in V_1, v_2 \in V_2$.

pf. $V_2^\perp$ is stable by G and $V = V_2 \oplus V_2^\perp$. So the orthogonal projection $\pi : V \rightarrow V_2$ is G -equivariant, hence the composition $V_1 \subset V \rightarrow V_2$ is G -equivariant. If $V_1 \not\cong V_2$, then this G -equivariant map $V_1 \rightarrow V_2$ has to be 0 by Schur's lemma, which means $V_1 \subset V_2^\perp$.

Define
$$\begin{array}{ccc} \iota_\rho : \text{End}_{\mathbb{C}}(V_\rho) & \rightarrow & L^2(G) \\ u & \mapsto & (g \mapsto T(\rho(g)^{-1} \circ u)) \end{array} \quad (\forall u \in \text{End}_{\mathbb{C}}(V_\rho), \text{ the function } g \mapsto T(\rho(g)^{-1} \circ u) \text{ is continuous on } G, \text{ hence on } L^2(G), \text{ given } G \text{ being compact. So it is well-defined}).$$

Theorem A.3.8. 1. $\forall (V_\rho, \rho) \in \hat{G}$, $\text{End}_{\mathbb{C}}(V_\rho)$ is an irreducible representation of $G \times G$. Also if $(V_\rho, \rho) \not\cong (V_{\rho'}, \rho')$, then $\text{End}_{\mathbb{C}}(V_\rho) \not\cong \text{End}_{\mathbb{C}}(V_{\rho'})$

2. $\forall (V_\rho, \rho) \in \hat{G}$, the map $\iota_\rho : \text{End}_{\mathbb{C}}(V_\rho) \rightarrow L^2(G)$ is $G \times G$ -equivariant, the map $\dim(V_\rho)^{1/2} \iota_\rho$ is an isometry, and $(\dim_{\mathbb{C}} V_\rho) \iota_\rho$ sends the composition in $\text{End}_{\mathbb{C}}(V_\rho)$ to the convolution product in $L^2(G)$
3. the map $\iota = \bigoplus_{\rho \in \hat{G}} \iota_\rho : \bigoplus_{\rho \in \hat{G}} \text{End}_{\mathbb{C}}(V_\rho) \rightarrow L^2(G)$ is injective and has dense image, i.e. as a representation of $G \times G$ and as $\mathbb{C}$ -algebra,

$$L^2(G) \cong \bigoplus_{\rho \in \hat{G}} \widehat{\text{End}_{\mathbb{C}}(V_\rho)}$$

pf.

- let $\rho \in \hat{G}$. Then $\chi_{\text{End}_{\mathbb{C}}(V_\rho)}(g_1, g_2) = \chi_{V_\rho}(g_1) \overline{\chi_{V_\rho}(g_2)}$ by the definition of the action of $G \times G$ on $\text{End}_{\mathbb{C}}(V_\rho)$. By the Schur's orthogonality formula, $\int_{G \times G} |\chi_{\text{End}_{\mathbb{C}}(V_\rho)}(g_1, g_2)|^2 dg_1 dg_2 = (\int_G |\chi_{V_\rho}(g_1)|^2 dg_1)(\int_G |\chi_{V_\rho}(g_2)|^2 dg_2) = 1$ so $\text{End}_{\mathbb{C}}(V_\rho)$ is irreducible. Moreover, if $(V_{\rho'}, \rho') \not\cong (V_\rho, \rho)$, then by the Schur's orthogonality, $\langle \chi_{\text{End}_{\mathbb{C}}(V_\rho)}, \chi_{\text{End}_{\mathbb{C}}(V_{\rho'})} \rangle_{L^2(G \times G)} = \langle \chi_{V_\rho}, \chi_{V_{\rho'}} \rangle_{L^2(G)} = 0$ so $\text{End}_{\mathbb{C}}(V_\rho) \not\cong \text{End}_{\mathbb{C}}(V_{\rho'})$.

- fix $(V_\rho, \rho) \in \hat{G}$.

$\forall x, y \in G, \forall u \in \text{End}_{\mathbb{C}}(V_\rho), \forall g \in G,$

$$\iota_\rho(\rho(x)u\rho(y)^{-1})(g) = T(\rho(g)^{-1}\rho(x)u\rho(y)^{-1}) = T(\rho(x^{-1}gy)^{-1}u) = (L_{x^{-1}}R_y\iota_\rho(u))(g) \rightarrow \iota_\rho \text{ is } G \times G\text{-equivariant.}$$

Let $u_1, u_2 \in \text{End}_{\mathbb{C}}(V_\rho)$. $\text{End}_{\mathbb{C}}(V_\rho)$ is spanned by the $T_{v,w}^0, \forall v, w \in V_\rho$, so as-

$$\text{sume } \begin{cases} u_1 = T_{v_1, w_1}^0 \\ u_2 = T_{v_2, w_2}^0 \end{cases}, \exists v_1, v_2, w_1, w_2 \in V_\rho. \text{ Then,}$$

$$\begin{aligned} \langle \iota_\rho(u_1), \iota_\rho(u_2) \rangle &= \int_G T(\rho(g)^{-1}T_{v_1, w_1}^0) \overline{T(\rho(g)^{-1}T_{v_2, w_2}^0)} dg & \rho(g)T_{v,w}^0 &= T_{v, gw}^0 \\ &= \int_G \langle g^{-1}v_1, w_1 \rangle \langle g^{-1}w_2, v_2 \rangle dg \\ &= \int_G \langle gv_2, w_2 \rangle \langle gv_1, w_1 \rangle dg \\ &= \frac{1}{\dim_{\mathbb{C}} V_\rho} \langle v_2, v_1 \rangle \langle w_1, w_2 \rangle & \text{by the theorem A.3.3.} \\ &= \frac{1}{\dim_{\mathbb{C}} V_\rho} \langle u_1, u_2 \rangle_{HS} & \text{by the proposition A.3.3.} \end{aligned}$$

Thus $\dim(V_\rho)^{1/2} \iota_\rho$ is an isometry.

Let $g \in G$. Then $\iota_\rho(u_1 u_2)(g) = T(\rho(g)^{-1}T_{v_1, w_1}^0 (T_{w_2, v_2}^0)^*) = \langle \rho(g)^{-1}w_1, v_2 \rangle \langle w_2, v_1 \rangle$ by the proposition A.3.3. On the other hand,

$$\begin{aligned} (\iota_\rho(u_1) * \iota_\rho(u_2))(g) &= \int_G T(\rho(x)^{-1}\rho(g)^{-1}T_{v_1, w_1}^0) T(\rho(x)T_{w_2, v_2}^0) dx \\ &= \int_G \langle \rho(x)^{-1}\rho(g)^{-1}w_2, v_1 \rangle \langle \rho(x)w_2, v_2 \rangle dx \\ &= \int_G \langle \rho(x)w_2, v_2 \rangle \langle \rho(x)v_1, \rho(g)^{-1}w_1 \rangle dx \\ &= \frac{1}{\dim_{\mathbb{C}}(V_\rho)} \langle \rho(g)^{-1}w_1, v_2 \rangle \langle w_2, v_1 \rangle & \text{by the theorem A.3.3.} \end{aligned}$$

- Let $u = (u_\rho) \in \bigoplus_{\rho \in \hat{G}} \text{End}_{\mathbb{C}}(V_\rho)$ be s.t. $\iota(u) = 0$. Then $\sum \iota_\rho(u_\rho) = 0$. By the previous lemma, and the statement 1, the subspaces $\iota_\rho(\text{End}_{\mathbb{C}}(V_\rho))$ are pairwise orthogonal to each other, so $\forall \rho \in \hat{G}, 0 = \langle \iota_\rho(u_\rho), \iota(u) \rangle = \langle \iota_\rho(u_\rho), \iota_\rho(u_\rho) \rangle =$

$\frac{1}{\dim_{\mathbb{C}} V_{\rho}} \langle u_{\rho}, u_{\rho} \rangle$. Thus all $u_{\rho} = 0$, i.e. $u = 0$ and ι is injective.

CLAIM : Every finite dimensional G -subrepresentation of $L^2(G)$ is contained in $\text{Im}(\iota)$, where we make G act on $L^2(G)$ by the left regular action.

pf. As finite dimensional representations of G are semisimple, if $\forall \rho \in \hat{G}$, $\forall u : V_{\rho} \rightarrow L^2(G)$, G -equivariant map, if $\text{Im}(u) \subset \text{Im}(\iota)$, then the claim follows. Fix $\rho \in \hat{G}$ and $u : V_{\rho} \rightarrow L^2(G)$, G -equivariant map. Let $v \in V_{\rho}$. Then $\forall f \in$

$$\begin{aligned}
 (u(v) * f)(g) &= \int_G u(v)(gx) \overline{f(x^{-1})} dx \\
 &= \int_G L_g(u(v))(x) \overline{\tilde{f}(x)} dx \\
 &= \langle L_g(u(v)), \tilde{f} \rangle_{L^2(G)} \\
 L^2(G), \forall g \in G, &= \langle u(\rho(g)^{-1}v), \tilde{f} \rangle_{L^2(G)} \quad \text{by the } G\text{-equivariance of } u \\
 &= \langle \rho(g)^{-1}v, u^*(\tilde{f}) \rangle_{V_{\rho}} \\
 &= T(\rho(g)^{-1}T_{u^*(\tilde{f}),v}^0) \quad \text{by the definition of } T_{v,w}^0 \\
 &= \iota_{\rho}(T_{u^*(\tilde{f}),v}^0)(g)
 \end{aligned}$$

Thus $u(v) * f \in \text{Im}(\iota_{\rho})$, $\forall f \in L^2(G)$. Then there exists a sequence (f_n) in $L^2(G)$ s.t. $u(v) * f_n \rightarrow u(v)$ as $n \rightarrow \infty$ by the theorem A.3.6. Thus $u(v) \in \overline{\text{Im}\iota_{\rho}}$. As $\text{Im}\iota_{\rho}$ is a finite-dimensional subspace of $L^2(G)$, it is closed in $L^2(G)$ so $u(v) \in \text{Im}\iota_{\rho} \subset \text{Im}\iota$.

Let $f \in \text{Im}(\iota)^{\perp}$. Then f is orthogonal to every finite dimensional G -subrepresentation of $L^2(G)$. Let $h \in L^2(G)$ be s.t. $h = \tilde{h}$. Then the endomorphism R_h of $L^2(G)$ is self-adjoint and compact by the theorem A.3.7. Hence by the spectral theorem, $\ker R_h$ is the orthogonal of the closure of the direct sum of the eigenspaces $\ker(R_h - \lambda \text{id}_{L^2(G)})$, $\lambda \in \mathbb{R}^{\times}$, which are all finite dimensional. As R_h is G -equivariant and commutes with the left regular action, there eigenspaces are left stable by G and so f is orthogonal to all of them, hence $f \in \ker R_h$, $f * h = 0$. Then by the theorem A.3.6., there exists a sequence $(h_n)_{n \geq 0}$ of elements of $L^2(G)$ s.t. $\tilde{h}_n = h_n$, $\forall n \geq 0$ and $f * h_n \rightarrow f$ as $n \rightarrow \infty$. Then $f * h_n = 0$, $\forall n \geq 0$ and therefore $f = 0$. So $\text{Im}\iota$ is dense in $L^2(G)$.